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**Report prepared for the WHO annual consultation on
the composition of influenza vaccine for the Northern
Hemisphere 2020-2021**

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Influenza Epidemiology in the WHO European Region

The epidemiological section of this report was prepared using data reported to the European Surveillance System (TESSy) and presented in Flu News Europe (<http://www.flunewseurope.org>) the joint ECDC-WHO Europe weekly influenza update, with the assistance of members of the Influenza and Other Respiratory Pathogens Group at the WHO European Office and the ECDC Influenza Surveillance Group:

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WHO Vaccine Recommendation Meeting for the 2020-2021 Northern Hemisphere influenza season

WHO CC, London – Final Report for the meeting: 24-27th February 2020

Epidemiology

Influenza activity in the WHO European Region, October 2019 – early February 2020

Timing and spread

The 2019-2020 influenza season (weeks 40/2019 to 06/2020) in the WHO European Region started one week later than the 2016-2017 season, and 1-4 weeks earlier than those of 2015-2016, 2017-2018 and 2018-2019 (**Figure E1-A**). Week 47/2019 was the first week when more than 10% of ILI or ARI specimens from sentinel primary care sites in the Region tested positive for influenza viruses (**Figure E1-B**). The proportion of specimens positive for influenza reached 54% in week 05/2020, compared to peaks of 53-60% in the previous four seasons (**Figure E1-A**). Peak activity, in which $\geq 40\%$ of sentinel specimens tested positive for influenza, has lasted for 5 weeks, beginning in week 02 and remaining elevated at week 06 (**Figure E1-B**). Activity may have peaked or be near peaking for the season. Based on weekly reporting on the intensity of influenza activity by each of the Member States and areas, activity increased in the west of the WHO European Region –notably in Ireland and the United Kingdom (Northern Ireland, Wales, and Scotland) – as well as Latvia and Georgia in the east, prior to other countries (**Figure E2**).

Distribution of viruses

Between week 40/2019 and week 06/2020, nearly 98,000 influenza detections from sentinel and non-sentinel primary care sources were reported to The European Surveillance System (TESSy) (**Table E1-A**), an increase (of approximately 8,500) compared to the 2018-2019 season. Combined data from sentinel and non-sentinel sources showed that the majority of viruses detected (78%) were type A and 22% were type B. This ratio was generally consistent for Southern Europe, Western Europe, Northern Europe and Western Asia, but with a lower proportion of type A virus detections reported in Eastern Europe (58%) and Central Asia (46%) (**Table E1-A**). Combined sentinel and non-sentinel data for type A viruses showed an equal distribution of A(H1N1)pdm09 viruses and A(H3N2) viruses. However, there was greater variation in these proportions when stratified by sub-region. A majority of A(H1N1)pdm09 viruses, ranging from 55% to 98%, occurred in Southern Europe, Western Europe, Eastern Europe, West Asia and Central Asia. In contrast, A(H3N2) viruses predominated (79%) in Northern Europe (**Table E1-A**). Only 11% of type B viruses reported were ascribed to a lineage of which 95-100% were B/Victoria and up to 5% were B/Yamagata.

In general, the regional distribution of influenza by type and subtype remained consistent when analysis was restricted to sentinel source data only, although there were higher proportions of A(H3N2) viruses in Eastern Europe (45% vs. 26%) and Central Asia (82% vs. 45%) compared to combined sentinel and non-sentinel data sources (**Table E1-B**). Of

influenza B virus detections from sentinel source data, the majority (98-100%) were of the B/Victoria-lineage. The subtype distribution proportions for influenza type A viruses detected, from both sentinel and non-sentinel sources, were relatively consistent for most weeks in the season to week 06/2020 (**Figure E1-B/C**).

A dominant virus (sub)type (comprising at least 60% of influenza virus detections) could be assigned by 42 Member States and areas within the WHO European Region, with only eight having insufficient numbers (ten or fewer) of detections. Twenty-six Member States and areas reported a predominantly influenza type A virus season to week 06/2020, while 9 reported predominance of influenza type B viruses, and 7 reported codominance of type A and type B viruses (**Figure E3**). One reported dominance of type A without specifying a subtype, and 25 were able to ascribe dominance (the proportion of detections of an influenza subtype was 60% or greater) or co-dominance (between 41% and 59%) of influenza A virus subtypes to the season: 12 in which A(H1N1)pdm09 was dominant, 7 in which A(H3N2) was dominant, and 6 in which A(H1N1)pdm09 and A(H3N2) were co-dominant. Five countries reported dominance of type B without specifying a lineage, and 4 were able to ascribe a dominant lineage of B/Victoria.

Influenza virus characterisation

Genetic characterisations of 1,955 influenza viruses, collected between weeks 40/2019 to 06/2020 and ascribed to genetic clades as defined at the start of the 2019-2020 season, were reported to TESSy:

- among 656 A(H1N1)pdm09 viruses characterised, 600 (91%) fell in the A/Norway/3433/2018 clade (6B.1A5A), 35 (5%) in the A/Switzerland/3330/2018 clade (6B.1A5B), 13 (2%) in the A/Slovenia/1489/2019 clade (6B.1A7) and 8 in the A/Brisbane/02/2018 clade (6B.1A1);
- among 780 A(H3N2) viruses attributed to a clade, 368 (47.1%) fell in the 3C.2a1b group [subgroups T131K (n = 246: 31.5%), T135K-A (n = 44: 5.6%) and T135K-B (n = 78: 10%)] and 412 (52.8%) in clade 3C.3a;
- All 493 B/Victoria-lineage viruses characterised belonged to clade 1A (represented by B/Brisbane/60/2008), with 15 of these falling in a subclade of viruses with a two amino acid deletion in HA1 (1A.Δ2; represented by the vaccine component B/Colorado/06/2017) and 478 falling in a subclade of viruses with a three amino acid deletion and K136E substitution in HA1 (1A.Δ3B; represented by B/Washington/02/2019, a vaccine component for the southern hemisphere 2020 season);
- the 26 influenza B/Yamagata-lineage viruses belonged to clade 3, represented by the vaccine component B/Phuket/3073/2013.

Neuraminidase inhibitor susceptibility was assessed for 841 influenza viruses with collection dates since week 40/2019: 352 A(H3N2), 319 A(H1N1)pdm09 and 170 type B. Only 3 viruses (<1%) showed any reduction in susceptibility. One A(H3N2) virus carried an R292K substitution in NA, indicative of highly reduced inhibition (HRI) by oseltamivir and reduced inhibition (RI) by zanamivir, which was confirmed by genotypic assay. One A(H1N1)pdm09 virus carried amino acid substitution H275Y in NA indicative of HRI by oseltamivir. One type B virus showed evidence of RI by oseltamivir.

Hospitalised cases

Between week 40/2019 and week 06/2020, 11 Member States reported a total of 7,832 hospitalised laboratory-confirmed influenza cases. Of these, 2,744 cases were admitted to intensive care units (ICUs) with five countries accounting for over 95% of the cases: United Kingdom (England, 58%); Spain (19%); France (11%); Ireland (4%); and Sweden (3%). The remaining 5,088 cases were reported from other hospital wards and two countries accounted for 94% of these cases: Ireland (64%) and Spain (30%). Among cases admitted to ICUs, the majority (93%) of influenza viruses detected were type A and of the 875 viruses subtyped, 50% were A(H3N2) and 50% were A(H1N1)pdm09. More than one third (37%) of the 1,146 influenza-associated ICU admissions with a recorded age were in patients aged 65 years or older, and 50% were aged from 15-64 years (**Figure E4-A**).

Eighteen Member States and areas (Albania, Armenia, Azerbaijan, Belarus, Bosnia and Herzegovina, Georgia, Kazakhstan, Kosovo*, Kyrgyzstan, Montenegro, North Macedonia, Republic of Moldova, Russian Federation, Serbia, Tajikistan, Turkmenistan, Ukraine and Uzbekistan) with surveillance systems for severe acute respiratory infection (SARI) reported a total of 22,420 cases between week 40/2019 and week 06/2020. In most countries, the majority of SARI cases were among children, although higher proportions were observed in adults in Albania, Azerbaijan, Belarus, Bosnia and Herzegovina, Russian Federation, Serbia, and Ukraine (**Figure E4-B**). Of 5,719 SARI cases tested for influenza virus over the period, 1,610 (28%) were positive. Approximately half (55%) were infected with influenza type A viruses and, of the 796 subtyped, 62% were A(H1N1)pdm09 and 38% were A(H3N2). Of the influenza positive cases infected with type B viruses, 317 were ascribed to a lineage, with 98% being B/Victoria lineage. Information on distributions of influenza types and subtypes or lineages by age group were not available for these data.

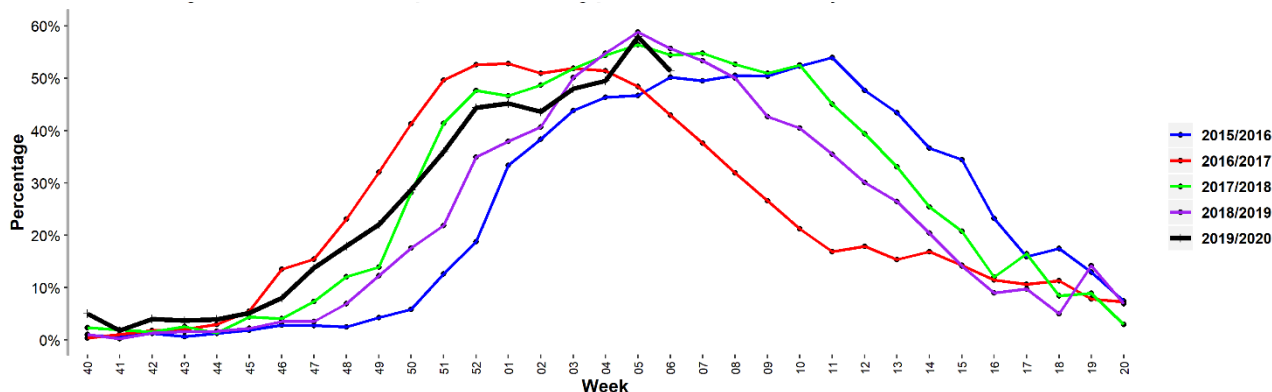
Mortality data

Pooled analyses of all-cause excess and influenza-attributable mortality from 24 European countries or sub-national regions participating in the EuroMOMO network [Austria, Belgium, Berlin (Germany), Denmark, England (United Kingdom (UK)), Estonia, Finland, France, Greece, Hesse (Germany), Hungary, Ireland, Italy, Luxemburg, Malta, the Netherlands, Northern Ireland (UK), Norway, Portugal, Scotland (UK), Spain, Sweden, Switzerland and Wales (UK)] showed a small tendency towards increased excess mortality over the past few weeks in some of the participating countries, particularly Spain and Portugal.

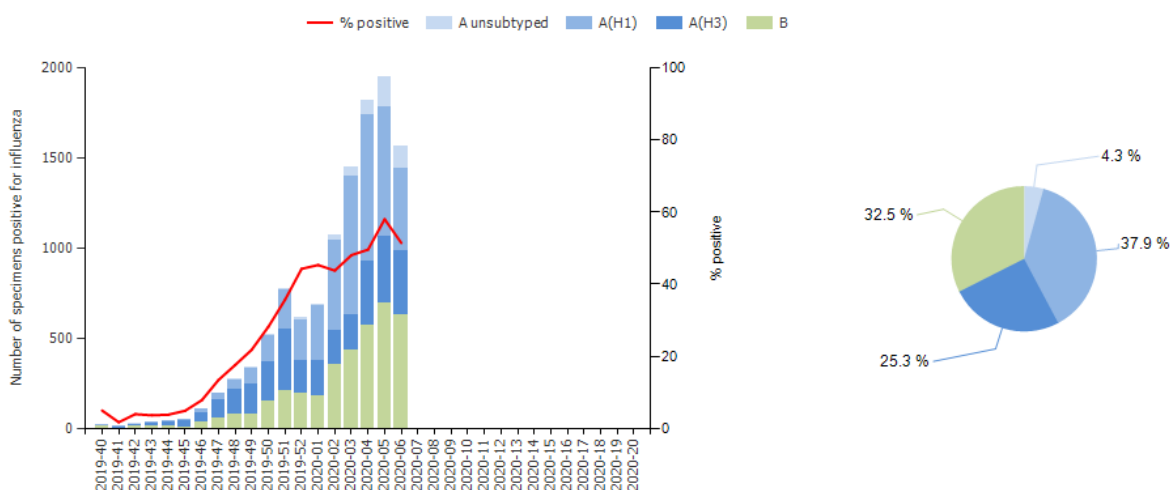
*in accordance with. United Nations Security Council. Resolution 1244 (1999).

Figure E1. Seasonal (weeks 40–06) influenza activity in the WHO European Region

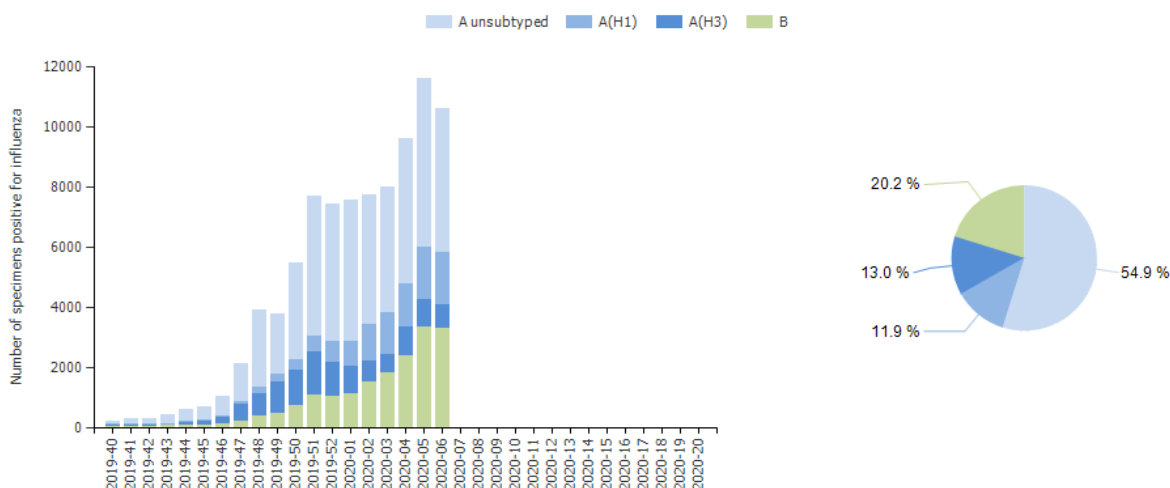
A. Percentage of sentinel ILI and ARI specimens testing positive for influenza by season and week, 2015-2020 (five seasons)



B. Sentinel ILI and ARI specimens testing positive for influenza by week, (sub)type and percentage positive, 2019–2020 season, weeks 40 through 06^a

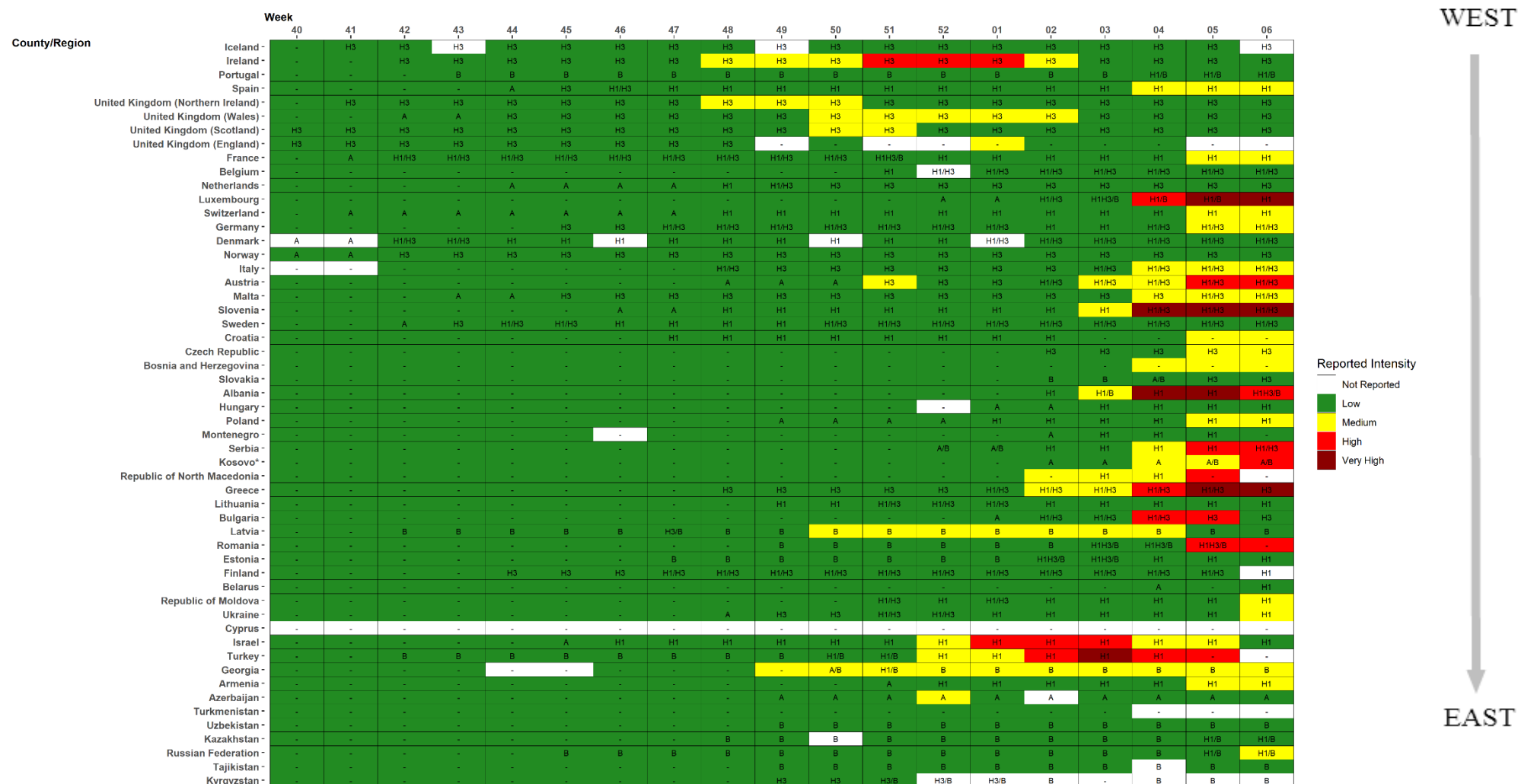


C. Non-sentinel ILI and ARI specimens testing positive for influenza by week and (sub)type, 2019–2020 season, weeks 40 through 06^a



^a Pie charts shows cumulative data for the period

Figure E2. Intensity of influenza activity and dominant virus (sub)type in the WHO European Region by week and Member State or area, 2019–2020 season (weeks 40 through 06)



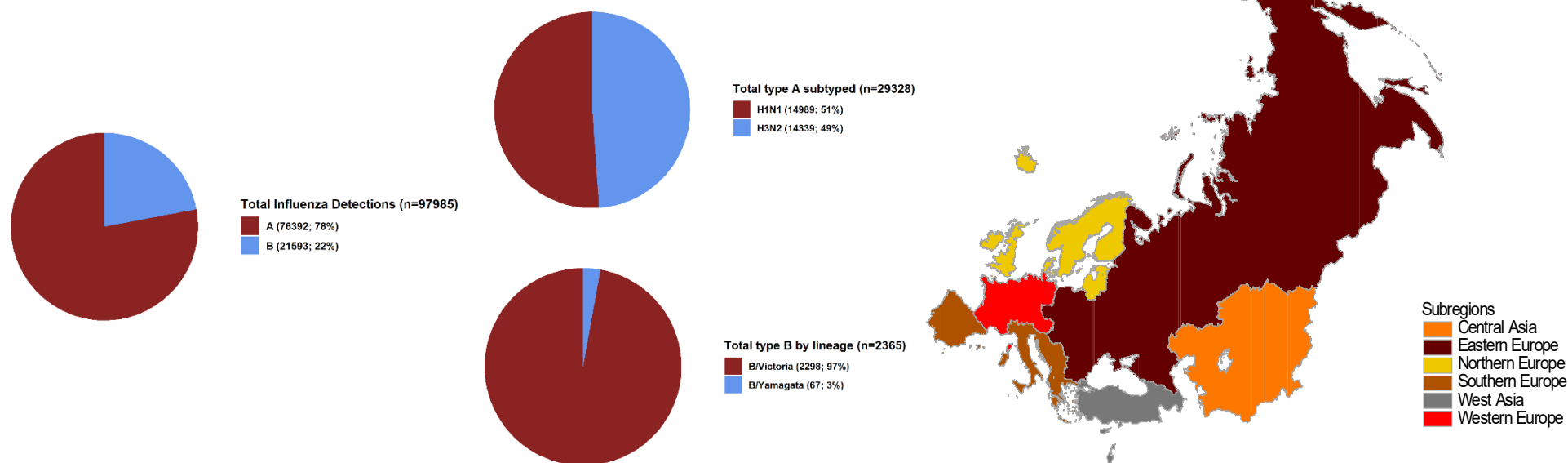
* In accordance with. United Nations Security Council. Resolution 1244 (1999).

Low intensity influenza activity is the level at which clinical influenza activity remains throughout the summer and most of the winter.

Dominant virus (sub)type is defined as at least 60% of detections.

Table E1. Distribution of Influenza Detections in the WHO European Region², 2019–2020 season (weeks 40/2019–06/2020)

Laboratory confirmed influenza from sentinel and non-sentinel sources by type and subtype/lineage



A. Laboratory confirmed influenza from sentinel and non-sentinel sources by geographic sub-region and influenza type and subtype/lineage

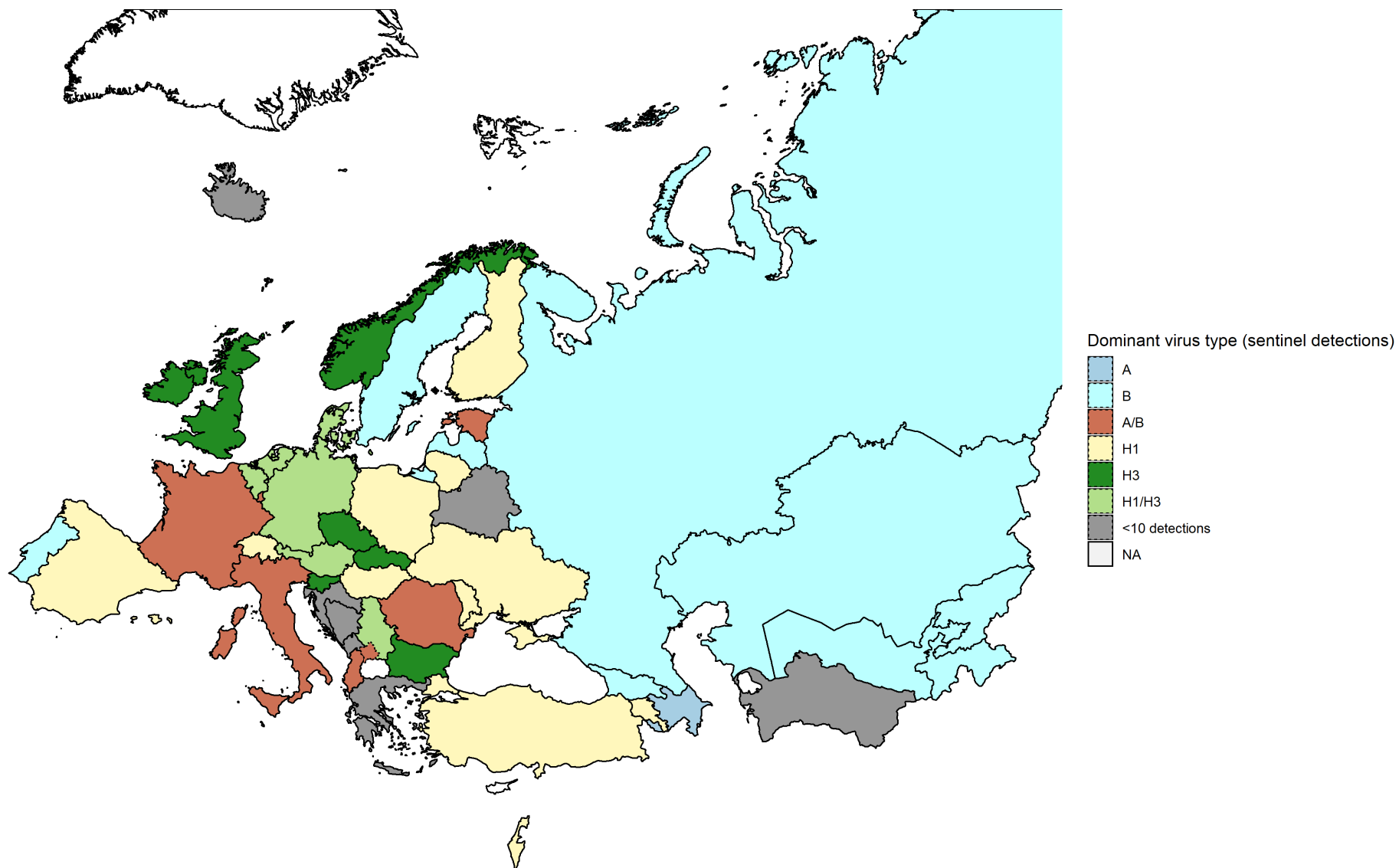
Influenza type and subtype/lineage	Southern Europe	Western Europe	Eastern Europe	Northern Europe	West Asia	Central Asia	WHO European Region
Influenza A	12,814 (73%)	14,937 (75%)	5,034 (58%)	40,368 (86%)	2,809 (70%)	430 (46%)	76,392 (78%)
Influenza A subtyped	6,521	3,082	4,085	12,549	2,671	420	29,328
A(H1N1)pdm09	4,474 (69%)	1,958 (64%)	3,039 (74%)	2,663 (21%)	2,625 (98%)	230 (55%)	14,989 (51%)
A(H3N2)	2,047 (31%)	1,124 (36%)	1,046 (26%)	9,886 (79%)	46 (2%)	190 (45%)	14,339 (49%)
Influenza B	4,666 (27%)	4,901 (25%)	3,637 (42%)	6,662 (14%)	1,213 (30%)	514 (54%)	21,593 (22%)
Influenza B lineage determined	702	468	95	995	105	0	2,365
B/Yamagata-lineage	3 (<1%)	10 (2%)	0 (0%)	54 (5%)	0 (0%)	-	67 (3%)
B/Victoria-lineage	699 (>99%)	458 (98%)	95 (100%)	941 (95%)	105 (100%)	-	2,298 (97%)
Total	17,480	19,838	8,671	47,030	4,022	944	97,985

² WHO European Region Member States grouped into sub-regions according to United Nations geographic composition. Available from: <http://unstats.un.org/UNSD/METHODS/M49/M49REGIN.HTM>

B. Laboratory confirmed influenza from sentinel site ILI and ARI specimens by geographic sub-region and influenza type and subtype/lineage

Influenza type and subtype/lineage	Southern Europe	Western Europe	Eastern Europe	Northern Europe	West Asia	Central Asia	WHO European Region
Influenza A	2,163 (62%)	1,608 (72%)	612 (81%)	1,659 (76%)	1,495 (65%)	175 (35%)	7,712 (67%)
Influenza A subtyped	1899	1,558	483	1,639	1,466	170	7,215
A(H1N1)pdm09	1,350 (71%)	927 (59%)	266 (55%)	373 (23%)	1,430 (98%)	31 (18%)	4,377 (61%)
A(H3N2)	549 (29%)	631 (41%)	217 (45%)	1,266 (77%)	36 (2%)	139 (82%)	2,838 (39%)
Influenza B	1,342 (38%)	634 (28%)	144 (19%)	513 (24%)	798 (35%)	318 (65%)	3,749 (33%)
Influenza B lineage determined	473	463	42	262	90	0	1,330
B/Yamagata-lineage	1 (<1%)	9 (2%)	0 (0%)	6 (2%)	0 (0%)	-	16 (1%)
B/Victoria-lineage	472 (>99%)	454 (98%)	42 (100%)	256 (98%)	90 (100%)	-	1,314 (99%)
Total	3,505	2,242	756	2,172	2,293	493	11,461

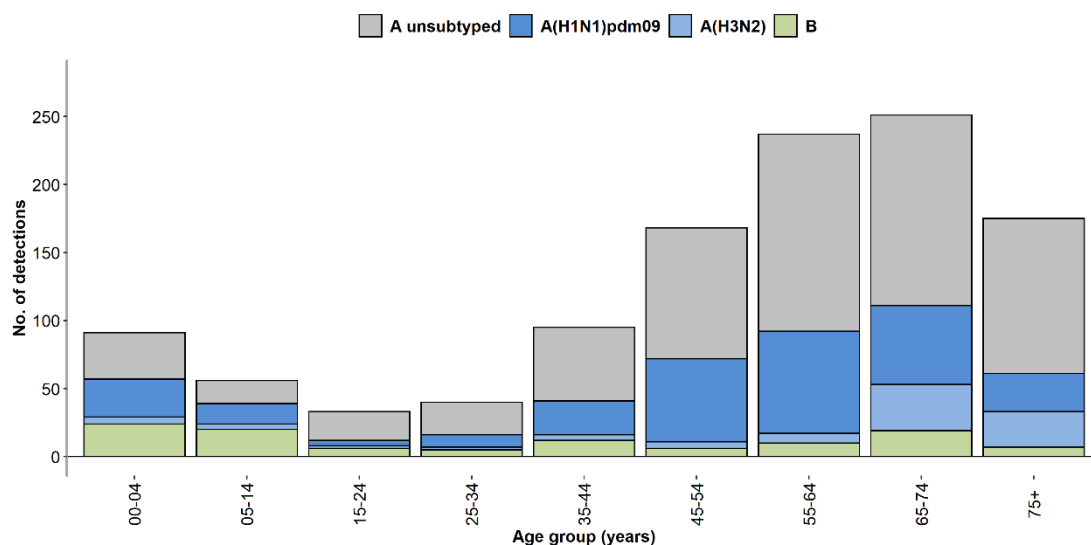
Figure E3. Influenza circulation³ in the WHO European Region by (sub)type and Member State or area, 2019–2020 season (weeks 40–06)



³ Dominance of (sub)type is based on at least 60% of detections and co-dominance is based on between 40% and 60% of detections. Calculations are only considered valid if at least 10 viruses were detected.

Figure E4. Hospitalised influenza cases in the WHO European Region, 2019–2020 season (weeks 40–06)

A. Distribution of virus (sub)type in laboratory-confirmed hospitalised cases in intensive care units^a by age group



^a Five reporting Member States account for 95% of the reported cases: Spain (47%); France (27%); Ireland (9%); Sweden (7%) and Czech Republic (6%)

B. Percentage distribution of severe acute respiratory infection (SARI) cases by age group and Member State or area, and influenza positives, 2019-2020 season (weeks 40-06)

Member State or area	SARI cases with age data	Percentage of total cases					All-age % positive
		0<5 years	5<15 years	15<30 years	30<65 years	≥65 years	
Albania	668	16	9	23	31	22	0.35
Armenia	2145	66	20	5	6	2	0.26
Azerbaijan	261	3	0	52	32	13	0
Belarus	259	22	7	20	41	10	0
Bosnia and Herzegovina	257	10	9	9	40	32	0.39
Georgia	1500	52	24	10	7	7	0.51
Kazakhstan	11142	58	16	14	10	3	0.38
Kyrgyzstan	627	88	6	3	3	1	0.18
Moldova, Republic of	1494	78	7	2	7	6	0.09
Montenegro	21	57	5	5	19	14	0.47
North Macedonia	259	50	9	8	22	11	0.36
Russian Federation	1545	31	14	25	21	9	0.12
Serbia	528	15	8	11	35	31	0.44
Tajikistan	303	59	39	0	0	3	0.23
Turkmenistan	133	70	25	1	0	5	0.02
Ukraine	730	19	16	40	22	4	0.30
Uzbekistan	229	56	24	14	7	0	0.18
Kosovo ^b	91	32	23	7	29	10	0.30

^b In accordance with: United Nations Security Council. Resolution 1244 (1999).

Influenza specimens received at the Francis Crick Institute.

Seasonal influenza virus samples with collection dates after 2019-08-31, totalling 1676, have been received from 53 countries in Europe, Africa, the Middle East, the Indian Ocean, Central Asia and the Far East. Influenza A viruses represented 68% of the samples, of which A(H1N1)pdm09 and A(H3N2) viruses were received in nearly equal numbers. Influenza B viruses represented 32% of the samples received with influenza B viruses of the B/Victoria lineage outnumbering those of the B/Yamagata lineage at a ratio of ~28:1 (**Table 1**).

Of the influenza-positive samples derived from individual patients (n = 1676) received, 505 (30%) were virus (V) isolates, 999 (60%) were clinical specimens (CS) and 172 (10%) were sent as CS/V pairs.

Table 2 shows the recovery/isolation rates from samples received with collection dates after 2019-08-31. Generally, viruses were recovered efficiently. For samples sent as clinical specimens, that were taken for culture, virus recoveries of between 22% and 83% were achieved, depending on virus subtype or lineage. Recoveries of samples sent as viruses were $\geq 97\%$. For A(H3N2) viruses, efficient recovery was achieved (258/278; 93%) as assessed by detection of virus neuraminidase activity using the MUNANA sialidase assay, but only 146/258 (57%) of these could be assessed by HI assay as the other viruses had low HA titres (<4 HA units in the presence of 20nM oseltamivir) or were unable to agglutinate red blood cells.

Table 3 summarises the reactivity of A(H1N1)pdm09, A(H3N2) and viruses of the B/Victoria and B/Yamagata lineages with post-infection ferret antisera raised against vaccine viruses or reference viruses in HI assays. For A(H3N2) viruses, the reactivity of test viruses in plaque reduction neutralisation assays (PRNA) is also summarised. Numbers and percentages of viruses showing ≤ 2 -, 4- and ≥ 8 -fold reductions in titres relative to the titres of the antisera with their respective homologous reference/vaccine viruses are shown. For A(H1N1)pdm09 test viruses the HI titres obtained with antisera raised against the vaccine viruses, egg-propagated A/Michigan/45/2015 and egg-propagated A/Brisbane/02/2018, were compared to the respective homologous titres. For A(H3N2) test viruses the HI titres of antisera raised against cell culture-propagated A/Hong Kong/5738/2014 (a 3C.2a virus), the vaccine virus egg-propagated A/Singapore/INFIMH-16-0019/2016 (a 3C.2a1 virus), the vaccine virus egg-propagated A/South Australia/34/2019 (a 3C.2a1b+T131K virus) and antisera raised against both cell culture- and egg-propagated A/Kansas/14/2017 (vaccine viruses of clade 3C.3a) were compared to the titres of the antisera with their homologous viruses. For the A(H3N2) test viruses assessed by PRNA the 50% neutralising titres of antisera raised against tissue culture-propagated cultivars of A/Norway/3275/2018 (3C.2a1b+T131K), A/La Rioja/2202/2018 (3C.2a1b+T135K-A) and A/Hong Kong/2671/2019 (3C.2a1b+T135K-B), egg-propagated A/South Australia/34/2019 (3C.2a1b+T131K) and both tissue culture- and egg-propagated cultivars of A/Kansas/14/2017 (clade 3C.3a) were compared to the titres of the antisera with their homologous viruses. For influenza B viruses of the B/Victoria lineage the HI titres of test viruses with antisera raised against both tissue culture- and egg-propagated cultivars of

vaccine viruses B/Colorado/06/2017 (subclade 1A(Δ 2)) and B/Washington/02/2019 (subclade 1A(Δ 3)B: with the additional HA1 amino acid substitution K136E) were compared to the titres of the antisera with their homologous viruses. For influenza B viruses of the B/Yamagata lineage the HI titres of test viruses with an antiserum raised against the egg-propagated vaccine virus, B/Phuket/3073/2013, were compared to the titre of the antiserum with the homologous virus.

HA and NA phylogenies for all viruses with collection dates after 2019-08-31 received at the Francis Crick Institute, for which genetic sequencing was performed successfully as of 2020-02-10, are presented in the report Annex (**Figures A-1 to A-8**). The following clade distribution of sequences originating from individual patient specimens was observed:

- A(H1N1)pdm09 – total 340 of which there were 315 (92.6%) 6B.1A5A, 16 (4.7%) 6B.1A5B, 8 (2.4%) 6B.1A7 and 1 (0.3%) viruses;
- A(H3N2) – total 403 of which there were 165 (40.9%) 3C.2a1b+T131K, 134 (33.3%) 3C.3a, 59 (14.6%) 3C.2a1b+T135K-B and 45 (11.2%) 3C.2a1b+T135K-A viruses;
- B/Victoria lineage – total 274 of which there were 265 (96.7%) 1A(Δ 3)B and 9 (3.3%) 1A(Δ 2) viruses;
- B/Yamagata lineage – total 10, all of which (100%) were clade 3 viruses.

Table 1a. Summary of clinical samples and virus isolates received, with collection dates after 2019-08-31, by country*

MONTH	TOTAL RECEIVED Seasonal	A		H1N1pdm09		H3N2		B		B Victoria lineage		B Yamagata lineage	
Country		Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ²	Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ¹
2019													
SEPTEMBER													
Czech Republic	1					1	1						
Finland	1					1	1						
France	6			1	in process	3	2	1		2	in process		
Ghana	1									1	in process		
Mauritius	1									1	1		
Norway	7			1	1	4	1	3		2	2		
Oman	12			7	6	2	1	1		3	3		
Palestine	1					1	in process						
Qatar	1									1	in process		
Romania	1					1	0	1					
Senegal	9					9	in process						
Sweden	3			2	2	1	1						
UK England	2			2	2								
UK Scotland	2					2	in process						
OCTOBER													
Denmark	3			2	2	1	1						
Finland	2			1	1	1	1						
France	5			3	3					2	2		
Germany	6			2	2	4	in process						
Ghana	5					1	in process			4	in process		
Greece	1					1	in process						
Hong Kong	1											1	1
Iceland	9					8	4	4		1	1		
Iran	10			6	6	4	in process						
Ireland	11			1	1	9	in process					1	0
Jordan	1									1	in process		
Latvia	3			1	1					2	2		
Lebanon	3					1	in process			2	in process		
Lithuania	1									1	in process		
Madagascar	7			4	in process			1	in process	2	in process		
Mauritius	3									3	in process		
Netherlands	3			1	1	2	in process						
Norway	28			5	4	19	3	15		3	2	1	0
Oman	67			28	28	24	6	18		15	13		
Poland	1	1	in process										
Portugal	5			2	in process	1	1	2	in process				
Portugal	2					1	in process	1	in process				
Qatar	20			1	in process	11	in process	1	in process	7	in process		
Spain	5			3	in process			1	in process	1	in process		
Sweden	3					2	2			1	1		
Switzerland	1									1	1		
Tajikistan	7			1	0			6	in process				
Ukraine	2			1	in process					1	1		
UK England	19			2	2	17	11	6					
UK Northern Ireland	1					1	in process						
UK Scotland	9			3	in process	3	in process			3	in process		
NOVEMBER													
Austria	4			2	2	1	0	1				1	1
Belgium	3			2	1	1	in process						
Bosnia & Herzegovina	1			1	in process								
Croatia	3			2	2					1	1		
Czech Republic	2					2	in process						
Denmark	16			7	7	6	3	3		3	3		
Finland	1			1	1								
France	16			8	in process	4	3	1		2	in process	2	2
Germany	8			5	in process	3	0	3					
Ghana	14					10	in process			4	4		
Greece	1					1	in process						
Hong Kong	12			3	3	5	5			4	4		
Iceland	3					2	0	2		1	1		
Iran	18			14	in process	3	in process			1	1		
Ireland	49			18	in process	22	in process	3	in process	6	in process		
Israel	5			5	in process								
Italy	7			2	in process	3	in process			2	in process		
Jordan	1			1	1								
Kazakhstan	5							5	in process				
Kyrgyzstan	7					5	in process			2	in process		
Latvia	10			2	2	3	3			5	5		
Lebanon	1			1	in process								
Lithuania	2			2	in process								
Madagascar	6			3	in process			1	in process	2	in process		
Mauritius	3			2	in process	1	in process						
Montenegro	1									1	in process		
Morocco	28			1	0					27	in process		
Netherlands	3			2	in process	1	in process						
Norway	22			6	5	9	3	4		4	4	3	1
Oman	21			10	10	2	1	1		9	8		
Palestine	5			1	in process	4	in process						
Poland	1	1	in process										
Portugal	83	1	0	13	in process	2	0	0	26	in process	41	in process	
Portugal	19			1	in process	1	in process	4	in process	13	in process		
Qatar	25			14	in process	10	in process			1	1		
Russia	24			8	in process	2	1	1		14	in process		
Saudi Arabia	1			1	1								
Slovenia	1			1	1								
Spain	6			2	in process	2	in process	2	in process				
Sweden	8			5	5	1	0	1		2	2		
Switzerland	6			4	4					2	1		
Tajikistan	3					1	0						
Tunisia	1							2	in process				
Ukraine	2			2	in process			1	in process				
UK England	20			4	in process	16	in process						
UK Northern Ireland	32			4	in process	28	in process						
UK Scotland	10			1	in process	8	in process			1	in process		

Table 1b. Summary of clinical samples and virus isolates received, with collection dates after 2019-08-31, by country*

MONTH	TOTAL RECEIVED	A		H1N1pdm09		H3N2		B		B Victoria lineage		B Yamagata lineage		
Country	Seasonal	Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ²	Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ¹	
DECEMBER														
Armenia	7			3	in process					4	in process			
Austria	19			5	in process	9	7	2		5	4			
Belgium	18			5	3	9	in process			4	3			
Bosnia & Herzegovina	11			11	in process									
Bulgaria	2			1	in process	1	in process							
Croatia	6			4	1	1	0	1		1	0			
Cyprus	2					1	in process		1	in process				
Czech Republic	2					2	in process							
Estonia	1			1	in process									
France	36			14	in process	7	3	4		15	in process			
Germany	13			6	in process	6	4	2		1	1			
Georgia	22			5	5	10	in process		2	in process	5	5		
Greece	5			4	in process	1	in process							
Greece	1					1	in process							
Hong Kong	17			10	10	6	6			1	1			
Iceland	5			2	2	2	0	1		1	1			
Israel	39			38	in process	1	1							
Israel	11									11	in process			
Italy	12			2	in process	6	in process			4	in process			
Jordan	8			7	in process					1	1			
Kazakhstan	23			4	in process	4	in process	15	in process					
Kyrgyzstan	33					21	in process			12	in process			
Latvia	1					1	0	1						
Lebanon	21	1	in process	5	in process	1	in process			14	in process			
Lithuania	20	1	in process	6	in process	12	in process			1	in process			
Madagascar	17			9	in process			4	in process	4	in process			
Mauritius	1			1	in process									
Montenegro	2					1	0	1		1	in process			
Morocco	2									2	2			
Netherlands	10			1	in process	9	in process							
Norway	15			8	in process	1	in process			1	in process	5	2	
Poland	5	2	in process	1	0	2	1	0						
Palestine	14			4	in process	5	in process	1	in process	4	in process			
Portugal	20			2	in process	3	2	1		15	in process			
Romania	8									8	in process			
Russia	32			12	in process	2	2			18	in process			
Serbia	4			4	in process									
Slovenia	9			5	5	3	3			1	1			
Spain	30			12	in process	6	in process	8	in process	4	4			
Switzerland	8			2	2	3	1	2		3	3			
Tajikistan	20			2	0	2	0	15	in process	1	in process			
Tunisia	9			1	in process			8	in process					
Ukraine	20			19	in process					1	1			
UK England	13			3	in process	9	in process			1	in process			
2020														
JANUARY														
Armenia	15			8	in process					7	in process			
Bulgaria	19			9	in process	8	in process			2	0			
Cyprus	20			4	in process	16	in process							
Czech Republic	5			2	in process	3	3							
Estonia	14			7	in process	3	in process			4	4			
France	1									1	in process			
Germany	7			2	in process	3	3			2	2			
Georgia	3					1	1			2	2			
Greece	44			21	in process	20	in process	2	in process	1	1			
Greece	1			1	in process									
Italy	3			1	in process	1	1			1	in process			
Kazakhstan	2			2	in process									
Lebanon	2									2	in process			
Lithuania	2					2	1	1						
Macedonia	20			8	in process	5	in process	1	in process	6	in process			
Mauritius	5			1	in process	2	in process			2	in process			
Montenegro	4			2	in process	1	0	1		1	in process			
Norway	17			1	in process	11	in process			5	in process			
Poland	4	1	in process	2	1	1	1							
Romania	15			4	in process	7	in process	1	in process	3	in process			
Serbia	21			14	in process	3	in process			4	in process			
Slovenia	4			2	2	1	1			1	1			
Spain	18			13	in process			5	in process					
Tunisia	4			2	in process			2	in process					
Ukraine	9			9	in process									
UK Northern Ireland	31			15	in process	10	in process	3	in process	3	in process			
53 Countries	1676	8	0	590	138	541	96	83	124	0	399	101	14	7
		0.48%		35.20%		32.28%			7.40%		23.81%		0.84%	
				68.0%							32.0%			

1. Propagated to sufficient titre to perform HI assay (the totalled number does not include any from batches that are in process)

2. Propagated to sufficient titre to perform HI assay in the presence of 20nM oseltamivir (the totalled number does not include any from batches that are in process)

Numbers in red indicate viruses recovered but with insufficient HA titre to permit HI assay

Includes clinical samples sent either in lysis-mix or as purified RNA, for which genetic characterisation only can be performed

* As of 2020-02-12

Table 2. Isolation rates for specimens collected after 2019-08-31

	Clinical sample			Virus isolate		
	Number processed	Number propagated ¹	Percent recovered	Number processed	Number propagated ¹	Percent recovered
H1	112	92	82%	133	130	98%
H3	118	41²	35%	160	105²	66%
		57³	83% (43%)⁴ 48%		55³	100% (66%)⁴ 34%
B Victoria	100	78	78%	72	70	97%
B Yamagata	9	2	22%	5	5	100%

1. Propagated to sufficient HA titre to perform HI assay

2. Propagated to sufficient HA titre in the presence of 20nM oseltamivir to allow HI assay

3. Viruses recovered (NA activity by MUNANA-based assay) but with insufficient HA titre to allow HI assay

4. The percentages for viruses recovered that could be assayed by HI are shown in parentheses

Table 3. Summary of isolates with collection dates after 2019-08-31 showing reduction in reactivity with antisera raised against reference viruses in HI & plaque-reduction neutralisation assays (PRNA)

Virus type/subtype	Number tested	≤2-fold ^a	4-fold ^a	≥8-fold ^a
H1 ^{1a}	222	161 (72.5%)	41 (18.5%)	20 (9.0%)
H1 ^{1b}	222	156 (70.3%)	49 (22.1%)	17 (7.7%)
H3 ^{2a}	146	77 (52.7%)	66 (45.2%)	3 (2.1%)
H3 ^{2b}	146	0	12 (8.2%)	134 (91.8%)
H3 ^{2c}	146	0	15 (10.3%)	131 (89.7%)
H3 ^{2d}	146	76 (52.1%)	23 (15.8%)	47 (32.2%)
H3 ^{3a}	120	65 (54.2%)	7 (5.8%)	48 (40.0%)
H3 ^{3b}	120	0	1 (0.8%)	119 (99.2%)
H3 ^{3c}	120	13 (10.8%)	6 (5.0%)	101 (84.2%)
H3 ^{3d}	120	0	0	120 (100.0%)
H3 ^{3e}	104	10 (9.6%)	38 (36.5%)	56 (53.8%)
H3 ^{3f}	120	24 (20.0%)	32 (26.7%)	64 (53.3%)
B/Victoria ^{4a}	148	28 (18.9%)	20 (13.5%)	100 (67.6%)
B/Victoria ^{4b}	148	11 (7.4%)	7 (4.7%)	130 (87.8%)
B/Victoria ^{4c}	148	109 (73.6%)	10 (6.8%)	29 (19.6%)
B/Victoria ^{4d}	148	113 (76.4%)	9 (6.1%)	26 (17.6%)
B/Yamagata ⁵	7	6 (85.7%)	0	1 (14.3%)

a. Reduction in titre, compared to the homologous titre with post-infection ferret antisera indicated below:

1a. Compared to homologous titres for ferret antiserum raised against vaccine virus A/Michigan/45/2015 (genetic clade 6B.1)

1b. Compared to homologous titres for ferret antiserum raised against vaccine virus A/Brisbane/02/2018 (genetic subgroup 6B.1A1)

2a. Compared to homologous titres for ferret antiserum raised against tissue culture-propagated A/Hong Kong/5738/2014 (genetic clade 3C.2a)

2b. Compared to homologous titres for ferret antiserum raised against egg-propagated A/South Australia/34/2019 (genetic subgroup 3C.2a1b+T131K)

2c. Compared to homologous titres for ferret antiserum raised against tissue culture-propagated A/Kansas/14/2017 (genetic clade 3C.3a)

2d. Compared to homologous titres for ferret antiserum raised against cell-propagated A/Kansas/14/2017 (genetic clade 3C.3a)

3a. Compared to homologous titres for ferret antiserum raised against tissue culture-propagated A/Norway/3275/2018 (genetic subgroup 3C.2a1b+T131K) in PRNA (2-fold)

3b. Compared to homologous titres for ferret antiserum raised against egg-propagated A/South Australia/34/2019 (genetic subgroup 3C.2a1b+T131K) in PRNA (2-fold)

3c. Compared to homologous titres for ferret antiserum raised against tissue culture-propagated A/Kansas/14/2017 (genetic clade 3C.3a) in PRNA (2-fold)

3d. Compared to homologous titres for ferret antiserum raised against egg-propagated A/Kansas/14/2017 (genetic clade 3C.3a) in PRNA (2-fold)

3e. Compared to homologous titres for ferret antiserum raised against tissue culture-propagated A/La Rioja/2202/2018 (genetic subgroup 3C.2a1b+T135K-A) in PRNA (2-fold)

3e. Compared to homologous titres for ferret antiserum raised against tissue culture-propagated A/Hong Kong/2671/2019 (genetic subgroup 3C.2a1b+T135K-B) in PRNA (2-fold)

4a. Compared to homologous titres for ferret antisera raised against tissue-culture propagated B/Colorado/06/2017 - Clade 1A(Δ2)

4b. Compared to homologous titres for ferret antisera raised against egg-propagated B/Colorado/06/2017 - Clade 1A(Δ2)

4c. Compared to homologous titres for ferret antisera raised against tissue-culture propagated B/Washington/02/2019 - Clade 1A(Δ3)B

4d. Compared to homologous titres for ferret antisera raised against egg-propagated B/Washington/02/2019 - Clade 1A(Δ3)B

5. Compared to homologous titres for ferret antiserum raised against egg-propagated vaccine virus B/Phuket/3073/2013

Influenza A(H1N1)pdm09 virus analyses.

Influenza A(H1N1)pdm09 viruses have been received from NICs in 50 countries or administrative regions in four WHO Regions: African (2: Madagascar and Mauritius), Eastern Mediterranean (9: Iran, Jordan, Lebanon, Morocco, Oman, Palestine, Qatar, Saudi Arabia, Tunisia,), European (38) and Western Pacific (1: Hong Kong SAR) (**Table 1**). **Table 4** summarises the antigenic (HI) results for viruses recovered with collection dates after 2019-08-31 with a summary of the recognition of viruses by antisera raised against the recommended vaccine virus for the Northern Hemisphere 2019/2020 and the Southern Hemisphere 2020 seasons, A/Brisbane/02/2018, and the former vaccine virus A/Michigan/45/2015.

HI analysis

Tables 5-1 to 5-12 show the results of HI assays of H1N1 viruses carried out since our September 2019 VCM report. Test viruses in each table are sorted by both date of collection and by genetic group and viruses with collection dates after 2019-08-31 are shown below red lines in **Table 5-4** and **Tables 5-7 to 5-12**. A summary of the HI results for viruses with collection dates after 2019-08-31 is shown in **Table 5-13** including a breakdown by the genetic group of the HA where known.

The vast majority (92%) of viruses collected after 2019-08-31 were antigenically similar to the egg-propagated vaccine virus recommended for use in the Northern Hemisphere 2019/2020 and the Southern Hemisphere 2020 seasons, A/Brisbane/02/2018, being recognised at titres within 4-fold of the titre of the antiserum with the homologous virus (**Table 4** and **Table 5-13**) with 70% being recognised at titres within 2-fold of the homologous titre of the antiserum. The majority (91%) were also recognised by the antiserum raised against the former vaccine virus, egg-propagated A/Michigan/45/2015, at titres within 4-fold of the homologous titre of the antiserum, with 72% being recognised at titres within 2-fold of the homologous titre.

Antisera raised against five reference viruses (A/Bayern/69/2009, A/Paris/1447/2017, A/Norway/3433/2018, A/Switzerland/3330/2017, and A/Ireland/84630/2018) recognised ≥90% of test viruses with collection dates after 2019-08-31 at titres within 4-fold of the titres of the antisera with their respective homologous viruses, with 70% or more being recognised at titres within 2-fold of the respective homologous titres (**Table 5-13**). The antisera raised against A/Slovenia/2903/2015 and A/Switzerland/2656/2017 recognised 74% and 70%, respectively, of test viruses at titres within 4-fold of the homologous titres of the antisera with the minority 48% and 35%, respectively, being recognised at titres within 2-fold of the homologous titres. The antiserum raised against cell culture-propagated A/Lviv/N6/2009 recognised only 60% of test viruses at titres within 4-fold of the homologous titres; this antiserum has an unusual virus/antiserum combination with A/Lviv/N6/2009 encoding HA1 amino acid substitutions of **G155G/E**, with E predominating, and **D222G**.

The vast majority of viruses analysed by HI had HA genes in subclade 6B.1A5A (90%) with a smaller number in subclade 6B.1A5B and very small numbers in subclades 6B.1A6 and 6B.1A7. The recognition of viruses in both these genetic groups by the

ferret antisera were similar. The majority of viruses that showed poorer recognition by the panel of antisera, HI titres highlighted in red in the tables, shared the amino acid substitution N156K in HA1.

Genetic analysis

All test viruses with collection dates after 2019-08-31, for which HA gene sequences were known, fell into subclade 6B.1A which is defined by the amino acid substitutions **S74R**, **S164T** (altering a glycosylation motif introduced at residues 162-164 in clade 6B.1 viruses) and **I295V** in **HA1**, compared with the clade 6B.1 prototype A/Michigan/45/2019 virus. **Figure 1** shows a phylogenetic tree for HA genes of representative A(H1N1)pdm09 viruses sequenced at the Francis Crick Institute and other representative viruses. A phylogenetic tree of the HA genes of all viruses with collection dates after 2019-08-31 analysed genetically at the Francis Crick Institute is shown in the Annex, **Figure A-1**.

The HA of most test viruses carry the **S183P** amino acid substitution in **HA1**, but this is not retained in all viruses. Within subclade 6B.1A genetic clusters of viruses encoding a range of **HA1** amino acid substitutions have emerged with the major groups being:

- 6B.1A5A, with HAs defined by **N260D** with **N129D** and **T185I** in **HA1**, many also have the substitutions **D187A** and **Q189E** in **HA1**;
- 6B.1A5B, with HAs defined by **E235D** and **N260D** with **K130N**, **K160M**, **T216K** and **H296N** in **HA1**, and **V193A** in **HA2** (**V520A**);
- 6B.1A7, with HAs defined by **E68D**, **S121N**, **T120A**, **L161L** and **K302T** in **HA1** and **I76M**, **N169S** and **E179D** in **HA2** (**I404M**, **N496S** and **E506D**).

Of the viruses with collection dates after 2019-08-31 analysed at the Francis Crick Institute: the large majority, 93%, had HA genes that fell into genetic group 6B.1A5A, 5% fell into genetic group 6B.1A5B, 2% into group 6B.1A7, and a single virus had an HA gene into group 6B.1A6. Recently collected viruses with **HA1 N156K** substitution, the majority of those viruses recognised less well by the panel of ferret antisera, had HA genes in group 6B.1A5A, but associated with alternative sets of substitutions. One set had the **N156K** substitution in association with **A195E** in **HA1** and another set had the **N156K** substitution associated with **K130N** and **L161I** in **HA1**, some of these with **V250A** in **HA1** and **E179D** in **HA2** (**E506D**) and some of these with the substitution **K209M** in **HA1**.

An amino acid sequence alignment of the HAs for representative H1N1pdm09 viruses is shown in **Figure 2**; the vaccine viruses are shown in red, reference viruses are coloured blue, glycosylation motifs are highlighted, the genetic group/subgroup is shown in green and the HA1/HA2 proteolytic-processing site is highlighted.

Figure 3 shows three H1pdm09-HA structures with the amino acid substitutions that differentiate subclade 6B.1A viruses (**S74R**, **S164T** and **I295V** in **HA1**) from clade 6B.1 viruses indicated in red. Those that define the main group of subclade 6B.1A5A viruses (**N129D**, **S183P**, **T185I** and **N260D** in **HA1**) indicated in yellow in **Figure 3a** and within

this genetic subgroup, viruses with the substitutions **D187A** and **Q189E** have been common and these are highlighted in green in **Figure 3a**. **Figure 3b** shows those substitutions that define the genetic group 6B.1A5B in green and **Figure 3c** shows those substitutions that define the genetic group 6B.1A7 highlighted in light blue.

Figure 4 shows a phylogenetic tree for the NA genes of representative H1N1pdm09 viruses sequenced at the Francis Crick Institute and other representative viruses. A phylogenetic tree of the NA genes of all viruses with collection dates after 2019-08-31 analysed genetically at the Francis Crick Institute is shown in the Annex, **Figure A-2**. The vast majority encoded the amino acid substitutions **G77R**, **V81A**, **I188T** and **N449D**, compared with the NA of A/Michigan/45/2015. Those with HA genes in genetic groups 6B.1A5A and 6B.1A6 also carry the NA substitution **D416N**, and viruses with HA genes in genetic group 6B.1A5A carry additional NA substitutions of **Q51K**, **F74S**, **I389K** and **T452I**, although some viruses with HA genes in 6B.1A5B viruses share these substitutions but others have NAs with **F74L** and **I436V**. Those with HA genes in group 6B.1A7 have the NA substitution **M314I** most with **F322L**.

Figure 5 shows a NA amino acid alignment for the same representative set of H1N1pdm09 viruses as used in **Figure 2**, annotated as **Figure 2** but lacking a proteolytic-processing site.

Antiviral resistance.

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on 283 viruses with collection dates after 2019-08-31. All but five viruses (>98%) showed normal inhibition (NI) by both neuraminidase inhibitors (**Table 13**). The exceptions were A/Sur/6980/2019 and A/Muscat/6997/2019, both showing highly reduced inhibition (HRI) by oseltamivir but normal inhibition (NI) by zanamivir, and A/Salalah/6780/2019 showing reduced inhibition (RI) by oseltamivir and NI by zanamivir (these three viruses had amino acid substitution H275Y or polymorphisms thereof in NA), and A/Denmark/3295/2019 and A/Denmark/3311/2019 showing HRI by zanamivir and NI by oseltamivir (these viruses shared the substitution Q136K in NA).

Conclusion.

All A(H1N1)pdm09 viruses received fell within subclade 6B.1A encoding the **HA1** amino acid substitutions **S74R**, **S164T** (which alters the glycosylation motif at residues **162 to 164**) and **I295V**. The majority of viruses in this subclade carry **HA1 S183P** amino acid substitution. The predominant genetic group was 6B.1A5A representing 93% of the viruses tested with collection dates after 2019-08-31. The majority of test viruses in each genetic group remained antigenically similar to the currently recommended vaccine virus, A/Brisbane/02/2018, as determined by assays using panels of post-infection ferret antisera. The vast majority of test viruses (>98%) retained susceptibility to oseltamivir and zanamivir.

Table 4a. Summary of influenza A(H1N1)pdm09 viruses collected after 2019-08-31, analysed by HI

MONTH	H1N1pdm09							
			Post-infection ferret antisera raised against					
			Clade 6B.1 virus			Clade 6B.1A1 virus		
Country	Number received ¹	Number propagated ²	≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴
2019								
SEPTEMBER								
France	1	0						
Norway	1	1	1			1		
Oman	7	6	2	3	1	5		1
Sweden	2	2	2			1	1	
UK England	2	2		1	1	1	1	
OCTOBER								
Denmark	2	2	2			2		
Finland	1	1	1			1		
France	3	3	3			3		
Germany	2	2	2			2		
Iran	6	6	6			6		
Ireland	1	1	1			1		
Latvia	1	1	1			1		
Madagascar	4	0						
Netherlands	1	1	1			1		
Norway	5	4		3	1	1	2	1
Oman	28	28	7	19	2	17	10	1
Portugal	2	1	1			1		
Qatar	1	0						
Spain	3	0						
Tajikistan	1	0						
Ukraine	1	0						
UK England	2	2		2		1	1	
NOVEMBER								
Austria	2	2	2			1		1
Belgium	2	1			1			1
Bosnia & Herzegovina	1	0						
Croatia	2	2	2			2		
Denmark	7	7	3	3	1	5	1	1
Finland	1	1	1			1		
France	8	4	4			4		
Germany	5	3	3			3		
Hong Kong	3	3	1		2	1		2
Iran	14	5	5			5		
Ireland	18	5	5			5		
Israel	5	0						
Italy	2	0						
Jordan	1	1	1				1	
Latvia	2	2	1	1		2		
Lebanon	1	0						
Lithuania	2	0						
Madagascar	3	0						
Mauritius	2	0						
Morocco	1	0						
Netherlands	2	1	1				1	
Norway	6	5	2	3		5		
Oman	10	10	1	5	4	4	6	
Palestine	1	0						
Portugal	13	4	4			4		
Qatar	14	5	5				5	
Russia	8	5	5			3	2	
Saudi Arabia	1	1	1			1		
Slovenia	1	1	1			1		
Spain	2	0						
Sweden	5	5	5			1	4	
Switzerland	4	4	4			2	2	
Ukraine	2	0						
UK England	4	0						

Table 4b. Summary of influenza A(H1N1)pdm09 viruses collected after 2019-08-31, analysed by HI

MONTH	H1N1pdm09							
			Post-infection ferret antisera raised against					
			Clade 6B.1 virus			Clade 6B.1A1 virus		
Country	Number received ¹	Number propagated ²	≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴
DECEMBER								
Armenia	3	0						
Austria	5	3	2		1		1	2
Belgium	5	3	2		1	1	1	1
Bosnia & Herzegovina	11	4	4			4		
Bulgaria	1	0						
Croatia	4	1	1			1		
Estonia	1	0						
France	14	10	8	1	1	7	1	2
Germany	6	0						
Georgia	5	5	5			5		
Greece	4	0						
Hong Kong	10	10	7		3	7		3
Iceland	2	2	2			1	1	
Israel	38	6	6			2	4	
Italy	2	0						
Jordan	7	3	3			1	2	
Kazakhstan	4	0						
Lebanon	5	0						
Lithuania	6	0						
Madagascar	9	0						
Mauritius	1	0						
Netherlands	1	0						
Norway	8	0						
Poland	1	0						
Palestine	4	0						
Portugal	2	0						
Russia	12	6	6			6		
Serbia	4	0						
Slovenia	5	5	5			5		
Spain	12	3	3			3		
Switzerland	2	2	2			2		
Tajikistan	2	0						
Tunisia	1	0						
Ukraine	19	11	10		1	10		1
UK England	3	0						
2020								
JANUARY								
Armenia	8	0						
Bulgaria	9	0						
Cyprus	4	0						
Czech Republic	2	0						
Estonia	7	0						
Germany	2	0						
Greece	21	3	3			2	1	
Italy	1	0						
Kazakhstan	2	0						
Macedonia	8	0						
Mauritius	1	0						
Montenegro	2	0						
Norway	1	0						
Poland	2	1	1			1		
Romania	4	0						
Serbia	14	0						
Slovenia	2	2	2			2		
Spain	13	1	1				1	
Tunisia	2	0						
Ukraine	9	1	1			1		
50 Countries	565	222	161	41	20	156	49	17
			72.5%	18.5%	9.0%	70.3%	22.1%	7.7%

¹ Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

² Propagated to sufficient titre to perform HI assay

³ Reduction in HI titre with post-infection ferret antiserum raised against A/Michigan/45/2015 (egg-propagated vaccine virus)

⁴ Reduction in HI titre with post-infection ferret antiserum raised against A/Brisbane/02/2018 (egg-propagated vaccine virus)

Table 5-1. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-09-17)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Mich 45/15 Egg	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK	A/HK 110/19 Egg
				F09/15 ¹	F13/18 ¹	F31/16 ¹	NIB F48/16 ¹	F03/18 ²	F20/18 ¹	F09/19 ¹	F04/19 ¹	F23/18 ¹	F08/19 ¹	F28/19 ¹
	Passage history													
	Ferret number													
	Genetic group													
REFERENCE VIRUSES														
A/Bayern/69/2009	G155E	2009-07-01	MDCK5/MDCK1	320	160	80	40	160	160	80	320	80	40	<
A/Lviv/N6/2009	G155E, D222G	2009-10-27	MDCK4/SIAT1/MDCK3	640	640	320	160	640	320	320	1280	320	160	40
A/Michigan/45/2015	6B.1	2015-09-07	E3/E4	320	320	1280	2560	2560	2560	1280	2560	1280	1280	40
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E2	320	320	1280	1280	1280	1280	640	2560	1280	1280	40
A/Paris/1447/2017	6B.1A	2017-10-20	MDCK1/MDCK3	160	80	640	640	1280	1280	640	1280	640	640	<
A/Switzerland/2656/2017	6B.1A	2017-12-21	E5/E2	640	640	1280	2560	2560	2560	1280	2560	1280	2560	40
A/Brisbane/02/2018	6B.1A1	2018-01-04	E3/E1	320	320	1280	1280	2560	1280	1280	2560	1280	1280	40
A/Norway/3433/2018	6B.1A5A	2018-10-30	MDCK4	80	40	320	320	320	320	320	1280	320	320	<
A/Switzerland/3330/2017	clone 35	2017-12-20	E6/E2	320	160	640	640	1280	1280	640	1280	1280	1280	<
A/Ireland/84630/2018	6B.1A6	2018-11-28	MDCK1/MDCK3	320	160	1280	1280	2560	1280	1280	2560	1280	1280	40
TEST VIRUSES														
A/Bosnia and Herzegovina/7368/2019	6B.1A5A	2019-01-16	MDCK1	160	80	320	640	640	320	320	1280	320	640	<
A/Bosnia and Herzegovina/14562/2019	6B.1A5A	2019-01-25	MDCK1	320	320	1280	1280	1280	1280	1280	2560	640	1280	40
A/Bosnia and Herzegovina/15261/2019	6B.1A5A	2019-01-25	MDCK1	320	160	1280	1280	1280	1280	640	2560	640	1280	40
A/Bosnia and Herzegovina/16488/2019	6B.1A5A	2019-01-28	MDCK1	320	160	640	640	640	640	640	2560	640	1280	40
A/Bosnia and Herzegovina/41866/2019	6B.1A5A	2019-02-22	MDCK1	160	160	640	640	640	640	640	1280	320	640	<
A/Bosnia and Herzegovina/55541/2019	6B.1A5A	2019-03-07	MDCK1	320	160	640	640	1280	640	640	2560	640	640	40
A/Norway/2133/2019	6B.1A5A	2019-06-03	MDCK1	320	320	1280	1280	1280	1280	640	2560	640	1280	40
A/Norway/2148/2019	6B.1A5A	2019-06-19	MDCK1	320	160	640	1280	1280	1280	640	2560	640	1280	40
A/Norway/2200/2019	6B.1A5A	2019-07-17	MDCK1/MDCK1	160	80	640	640	640	320	640	1280	320	640	<
A/Norway/2193/2019	6B.1A5A	2019-07-18	MDCK1/MDCK1	160	80	640	320	640	320	640	1280	640	640	<
A/Norway/2196/2019	6B.1A5A	2019-07-19	MDCK1/MDCK1	160	80	640	640	640	320	320	1280	320	640	<
A/Norway/2223/2019	6B.1A5A	2019-07-26	MDCK1/MDCK1	160	160	640	640	1280	640	640	2560	640	1280	<
A/Norway/2231/2019	6B.1A5B	2019-08-15	MDCK1	320	80	1280	1280	1280	1280	1280	2560	1280	2560	80
A/Norway/2228/2019	6B.1A5B	2019-08-15	MDCK1	640	160	2560	2560	2560	2560	1280	5120	1280	2560	160
A/Bosnia and Herzegovina/10482/2019	6B.1A7	2019-01-21	MDCK1	160	80	640	640	640	640	640	1280	640	1280	<
A/Bosnia and Herzegovina/11095/2019	6B.1A7	2019-01-22	MDCK1	320	160	1280	1280	2560	1280	1280	2560	1280	1280	40
A/Bosnia and Herzegovina/23151/2019	6B.1A7	2019-02-04	MDCK1	160	40	640	320	640	640	320	1280	320	640	<
A/Bosnia and Herzegovina/25764/2019	6B.1A7	2019-02-05	MDCK1	160	80	1280	1280	1280	1280	1280	2560	1280	1280	40
A/Bosnia and Herzegovina/32630/2019	6B.1A7	2019-02-13	MDCK1	160	80	640	640	640	640	640	1280	640	640	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Vaccine
SH 2019Vaccine
NH 2019-20
SH 2020

< 4-fold

4-fold

8-fold

> 8-fold

< not recognised by the antiserum

≥ 160 (when homologous titre ≥ 2560)

≥ 160 (no homologous titre)

Table 5-2. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-10-01)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Bayern 69/09	A/Lviv N6/09	A/Mich 45/15	A/Slov 2903/2015	A/Paris 1447/17	A/Swit 2656/17	A/Bris 02/18	A/Norway 3433/18	A/Swit 3330/17	A/Ire 84630/18
				MDCK	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK	Egg	MDCK
				F09/15 ^{†1}	F13/18 ^{†1}	F31/16 ^{†1}	NIB F48/16 ^{†1}	F03/18 ^{†2}	F20/18 ^{†1}	F09/19 ^{†1}	F04/19 ^{†1}	F23/18 ^{†1}	F08/19 ^{†1}
						6B.1	6B.1	6B.1A	6B.1A	6B.1A1	6B.1A5A	6B.1A5B	6B.1A6
REFERENCE VIRUSES													
A/Bayern/69/2009	G155E	2009-07-01	MDCK5/MDCK1	320	160	160	40	160	160	80	320	80	40
A/Lviv/N6/2009	G155E, D222G	2009-10-27	MDCK4/SIAT1/MDCK3	640	640	160	160	320	320	320	640	160	160
A/Michigan/45/2015	6B.1	2015-09-07	E3/E4	320	160	640	1280	1280	1280	640	1280	640	640
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E2	320	160	640	1280	1280	1280	640	1280	640	1280
A/Paris/1447/2017	6B.1A	2017-10-20	MDCK1/MDCK3	160	80	640	640	640	640	320	1280	320	640
A/Switzerland/2656/2017	6B.1A	2017-12-21	E5/E3	320	320	1280	1280	1280	1280	1280	2560	640	1280
A/Brisbane/02/2018	6B.1A1	2018-01-04	E3/E1	320	320	1280	1280	2560	1280	1280	2560	1280	1280
A/Norway/3433/2018	6B.1A5A	2018-10-30	MDCK4	80	<	160	320	320	320	160	640	160	320
A/Switzerland/3330/2017	clone 35	2017-12-20	E6/E2	160	160	640	640	640	640	640	1280	640	640
A/Ireland/84630/2018	6B.1A6	2018-11-28	MDCK1/MDCK3	160	80	640	1280	1280	1280	640	1280	640	1280
TEST VIRUSES													
A/Armenia/519/2019	6B.1A2	2019-02-08	MDCK1	320	160	1280	1280	1280	1280	1280	2560	640	1280
A/Armenia/688/2019	6B.1A2	2019-03-07	MDCK1	320	80	1280	1280	1280	1280	640	2560	640	1280
A/Armenia/872/2019	6B.1A5A	2019-04-17	MDCK1	160	160	640	640	1280	640	640	2560	640	640
A/Yaounde/19v-3503/2019	6B.1A5B	2019-05-03	MDCK2	640	160	1280	1280	1280	1280	640	2560	1280	1280
A/Norway/2227/2019	6B.1A5B	2019-08-15	MDCK2	640	160	1280	2560	2560	2560	1280	2560	1280	2560
A/Armenia/574/2019	6B.1A6	2019-02-14	MDCK1	320	80	1280	1280	1280	1280	640	2560	640	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80

Vaccine
SH 2019

Vaccine
NH 2019-20
SH 2020

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 5-3. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-10-30)

				Haemagglutination inhibition titre											
				Post-infection ferret antisera											
Viruses	Other information	Collection date	Passage history	NEW											
				A/Bayern	A/Lviv	A/Mich	A/Mich	A/Slov	A/Paris	A/Swit	A/Bris	A/Norway	A/Swit	A/Ire	
				69/09	N6/09	45/15	45/15	2903/2015	1447/17	2656/17	02/18	3433/18	3330/17	84630/18	
				Passage history	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg	MDCK	Egg	MDCK
				Ferret number	F09/15 ^{*1}	F13/18 ^{*1}	F31/16 ^{*1}	F40/19 ^{*1}	NIB F48/16 ^{*1}	F03/18 ^{*2}	F20/18 ^{*1}	F09/19 ^{*1}	F04/19 ^{*1}	F23/18 ^{*1}	F08/19 ^{*1}
Genetic group						6B.1	6B.1	6B.1	6B.1A	6B.1A	6B.1A1	6B.1A5A	6B.1A5B	6B.1A6	
REFERENCE VIRUSES															
A/Bayern/69/2009	G155E	2009-07-01	MDCK5/MDCK1	320	160	160	80	<	160	160	160	320	160	40	
A/Lviv/N6/2009	G155E, D222G	2009-10-27	MDCK4/SIAT1/MDCK3	320	640	160	80	80	320	320	160	640	160	80	
A/Michigan/45/2015		2015-09-07	E3/E4	320	320	1280	640	1280	1280	1280	1280	2560	640	1280	
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E2	320	320	1280	640	1280	2560	1280	1280	1280	1280	1280	
A/Paris/1447/2017		2017-10-20	MDCK1/MDCK3	160	40	320	320	640	640	640	320	1280	320	640	
A/Switzerland/2656/2017		2017-12-21	E5/E2	640	640	1280	1280	2560	2560	2560	1280	2560	1280	1280	
A/Brisbane/02/2018		2018-01-04	E3/E1	320	320	1280	1280	1280	2560	1280	1280	2560	1280	1280	
A/Norway/3433/2018		2018-10-30	MDCK4	80	<	160	160	320	320	320	160	640	320	320	
A/Switzerland/3330/2017	clone 35	2017-12-20	E6/E2	160	80	640	320	640	640	640	320	1280	640	640	
A/Ireland/84630/2018		2018-11-28	MDCK1/MDCK3	320	160	1280	1280	1280	2560	1280	1280	2560	1280	1280	
TEST VIRUSES															
A/Serbia/249/2019		2019-01-14	C1/MDCK1	320	160	1280	ND	640	1280	1280	640	2560	640	640	
A/Serbia/252/2019		2019-01-14	C1/MDCK1	320	160	1280	ND	640	1280	1280	640	2560	640	640	
A/Serbia/423/2019		2019-01-21	C1/MDCK1	320	320	640	ND	640	640	640	640	1280	640	640	
A/Serbia/469/2019		2019-01-22	C3/MDCK1	320	320	1280	ND	1280	1280	1280	640	2560	640	1280	
A/Serbia/815/2019		2019-01-30	C3/MDCK1	160	160	1280	ND	640	1280	1280	640	2560	640	1280	
A/Serbia/916/2019		2019-01-31	C3/MDCK1	160	160	640	ND	640	1280	1280	640	2560	640	1280	
A/Serbia/939/2019		2019-02-01	C3/MDCK1	160	80	640	ND	640	640	640	640	2560	640	640	
A/Serbia/2475/2019		2019-03-01	C2/MDCK1	320	320	1280	ND	640	1280	1280	640	2560	1280	1280	
A/Lebanon/325/2019		2019-03-15	MDCK1	320	160	1280	ND	640	1280	1280	640	2560	640	1280	
A/Serbia/657/2019		2019-01-26	C3/MDCK1	320	160	1280	ND	1280	2560	640	640	1280	640	640	
A/Serbia/900/2019		2019-01-31	C3/MDCK1	320	160	1280	ND	1280	2560	1280	1280	2560	1280	2560	
A/Lebanon/323/2019		2019-03-12	MDCK1	160	80	640	ND	640	640	640	640	1280	320	640	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80; ND =Not Done

Vaccine
SH 2019

Vaccine
NH 2019-20
SH 2020

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 5-4. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-11-05)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				A/Bayern 69/09	A/Lviv N6/09	A/Mich 45/15	A/Slov 2903/2015	A/Paris 1447/17	A/Swit 2656/17	A/Bris 02/18	A/Norway 3433/18	A/Swit 3330/17	A/Ire 84630/18	
				MDCK	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK	Egg	MDCK	
				F09/15 ^{*1}	F13/18 ^{*1}	F31/16 ^{*1}	NIB F48/16 ^{*1}	F03/18 ^{*2}	F20/18 ^{*1}	F09/19 ^{*1}	F04/19 ^{*1}	F23/18 ^{*1}	F08/19 ^{*1}	
Genetic group						6B.1	6B.1	6B.1A	6B.1A	6B.1A1	6B.1A5A	6B.1A5B	6B.1A6	
REFERENCE VIRUSES														
A/Bayern/69/2009	G155E	2009-07-01	MDCK5/MDCK1	160	160	80	<	160	80	80	160	80	40	
A/Lviv/N6/2009	G155E, D222G	2009-10-27	MDCK4/SIAT1/MDCK3	640	640	320	80	320	320	320	640	160	160	
A/Michigan/45/2015	6B.1	2015-09-07	E3/E4	320	320	640	640	1280	1280	640	1280	640	640	
A/Slovenia/2903/2015	clone 37	6B.1	E4/E2	320	320	1280	1280	1280	1280	1280	2560	640	1280	
A/Paris/1447/2017		6B.1A	MDCK1/MDCK3	160	40	640	640	1280	640	640	1280	320	640	
A/Switzerland/2656/2017		6B.1A	E5/E3	320	640	1280	1280	1280	1280	1280	2560	640	1280	
A/Brisbane/02/2018		6B.1A1	E3/E1	320	160	1280	1280	1280	1280	640	1280	640	1280	
A/Norway/3433/2018		6B.1A5A	MDCK4	80	40	320	160	320	320	160	640	320	320	
A/Switzerland/3330/2017	clone 35	6B.1A5B	E6/E2	160	160	640	640	640	640	640	1280	640	640	
A/Ireland/84630/2018		6B.1A6	MDCK1/MDCK3	160	160	640	640	1280	640	640	1280	640	640	
TEST VIRUSES														
A/Mauritius/I-345/2019		6B.1A5A	MDCKx/MDCK1	40	40	320	320	320	320	320	640	160	160	
A/Mauritius/I-609/2019		6B.1A5A	MDCKx/MDCK1	80	40	320	320	320	320	320	640	160	320	
A/Mauritius/I-337/2019		6B.1A6	MDCK1/MDCK1	80	80	320	320	320	320	320	640	160	320	
A/Mauritius/I-421/2019		6B.1A6	MDCK1/MDCK1	160	80	640	640	640	640	640	1280	320	640	
A/Norway/2276/2019		6B.1A5A	MDCK1	80	80	320	320	320	320	320	1280	320	320	
A/Norway/2288/2019		6B.1A5A	MDCK1	160	80	640	320	320	320	320	1280	320	320	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND =Not Done

Vaccine

SH 2019

Vaccine

NH 2019-20

SH 2020



< 4-fold

4-fold

8-fold

> 8-fold

< not recognised by the antiserum

≥ 160 (when homologous titre ≥ 2560)

≥ 160 (no homologous titre)

Table 5-5. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-11-12)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Mich 45/15 Egg	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK
				F09/15 ^{†1}	F13/18 ^{†1}	F31/16 ^{†1}	NIB F48/16 ^{†1}	F03/18 ^{†2}	F20/18 ^{†1}	F09/19 ^{†1}	F04/19 ^{†1}	F23/18 ^{†1}	F08/19 ^{†1}
	Passage history					6B.1	6B.1	6B.1A	6B.1A	6B.1A1	6B.1A5A	6B.1A5B	6B.1A6
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
A/Bayern/69/2009	G155E	2009-07-01	MDCK5/MDCK1	320	320	160	40	160	160	80	320	80	40
A/Lviv/N6/2009	G155E, D222G	2009-10-27	MDCK4/SIAT1/MDCK3	640	640	320	160	320	320	320	1280	160	160
A/Michigan/45/2015		2015-09-07	E3/E4	640	320	1280	1280	2560	2560	1280	2560	1280	1280
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E2	320	320	1280	1280	2560	2560	1280	2560	1280	1280
A/Paris/1447/2017		2017-10-20	MDCK1/MDCK3	160	80	640	1280	1280	1280	640	1280	640	1280
A/Switzerland/2656/2017		2017-12-21	E5/E3	640	320	1280	1280	2560	1280	1280	2560	1280	1280
A/Brisbane/02/2018		2018-01-04	E3/E1	320	160	1280	1280	1280	1280	1280	2560	640	1280
A/Norway/3433/2018		2018-10-30	MDCK4	80	40	320	320	640	320	320	1280	320	320
A/Switzerland/3330/2017	clone 35	2017-12-20	E6/E2	160	80	640	640	640	640	320	640	640	640
A/Ireland/84630/2018		2018-11-28	MDCK1/MDCK3	320	160	640	1280	1280	1280	640	1280	640	640
TEST VIRUSES													
A/Mauritius/I-273/2019		2019-02-06	MDCK2/MDCK2	320	320	1280	1280	640	640	640	2560	640	1280
A/Mauritius/I-456/2019	S190R	2019-02-26	MDCKx/MDCK2	40	<	80	80	80	80	80	320	80	80
A/Mauritius/I-323/2019		2019-02-08	MDCK1/MDCK2	160	80	640	640	320	640	640	1280	640	640
A/Mauritius/I-410/2019		2019-02-12	MDCKx/MDCK2	320	160	1280	1280	1280	1280	640	1280	1280	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND =Not Done

Vaccine
SH 2019

Vaccine
NH 2019-20
SH 2020

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 5-6. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-11-19)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Mich 45/15 Egg	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK
				F09/15 ¹	F13/18 ¹	F31/16 ¹	F48/16 ¹	F03/18 ²	F20/18 ¹	F09/19 ¹	F04/19 ¹	F23/18 ¹	F08/19 ¹
	Passage history												
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
A/Bayern/69/2009	G155E	2009-07-01	MDCK5/MDCK1	320	160	80	40	160	160	80	320	80	40
A/Lviv/N6/2009	G155E, D222G	2009-10-27	MDCK4/SIAT1/MDCK3	640	640	320	160	320	320	320	640	160	160
A/Michigan/45/2015		2015-09-07	E3/E4	640	320	2560	2560	2560	2560	1280	2560	1280	1280
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E2	320	320	1280	1280	1280	2560	640	1280	640	1280
A/Paris/1447/2017		2017-10-20	MDCK1/MDCK3	160	80	640	1280	1280	1280	640	1280	640	1280
A/Switzerland/2656/2017		2017-12-21	E5/E3	640	320	1280	2560	2560	2560	1280	2560	1280	2560
A/Brisbane/02/2018		2018-01-04	E3/E1	320	320	2560	2560	2560	5120	1280	2560	1280	2560
A/Norway/3433/2018		2018-10-30	MDCK4	80	<	320	160	320	320	160	640	160	320
A/Switzerland/3330/2017	clone 35	2017-12-20	E6/E2	160	80	640	640	640	640	640	1280	640	640
A/Ireland/84630/2018		2018-11-28	MDCK1/MDCK3	320	160	1280	1280	1280	1280	640	1280	640	1280
TEST VIRUSES													
A/Mauritius/I-106/2019			MDCKx/MDCK3	160	80	640	640	640	640	320	1280	320	640
A/Qatar/24-VI-19-0041928/2019		2019-07-08	MDCK1	320	160	1280	640	1280	1280	640	2560	640	640
A/Qatar/10-VI-19-0043385/2019		2019-07-17	MDCK1	160	80	640	640	1280	1280	320	1280	320	640
A/Qatar/16-VI-19-0044629/2019		2019-07-25	MDCK1	320	160	640	640	1280	1280	640	2560	640	640
A/Qatar/16-VI-19-0048544/2019		2019-08-23	MDCK1	320	160	640	640	1280	1280	640	2560	640	640
A/Qatar/47-VI-19-0048897/2019		2019-08-25	MDCK1	320	160	1280	640	1280	1280	640	2560	640	640
A/Qatar/16-VI-19-0048807/2019		2019-08-25	MDCK1	320	160	640	640	1280	1280	640	2560	640	640
A/Qatar/10-VI-19-0049421/2019		2019-08-27	MDCK1	80	40	320	320	320	320	320	640	160	320
A/Qatar/16-VI-19-0049631/2019		2019-08-28	MDCK1	160	40	640	640	640	640	640	1280	320	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND =Not Done

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20
SH 2020

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 5-7. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-12-10)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Mich 45/15 Egg	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK
					F09/15 [†]	F13/18 [†]	F31/16 [†]	F48/16 [†]	F03/18 [‡]	F20/18 [†]	F09/19 [†]	F04/19 [†]	F23/18 [†]	F08/19 [†]
							6B.1	6B.1	6B.1A	6B.1A	6B.1A1	6B.1A5A	6B.1A5B	6B.1A6
REFERENCE VIRUSES														
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	320	320	80	40	160	160	80	320	80	40
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	640	640	320	160	640	320	320	640	160	160
A/Michigan/45/2015		6B.1	2015-09-07	E3/E4	640	640	2560	2560	2560	2560	1280	2560	1280	2560
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	640	1280	2560	5120	5120	5120	2560	5120	1280	1280
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	320	160	640	1280	2560	2560	1280	2560	640	1280
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E3	640	640	2560	2560	2560	5120	2560	5120	2560	2560
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	320	320	1280	1280	2560	2560	1280	1280	1280	1280
A/Norway/3433/2018		6B.1A5A	2018-10-30	MDCK3	80	40	320	320	640	640	320	640	320	320
A/Switzerland/3330/2017	clone 35	6B.1A5B	2017-12-20	E6/E2	320	160	640	1280	1280	1280	640	640	1280	640
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	160	160	640	1280	1280	1280	640	640	640	1280
TEST VIRUSES														
A/Armenia/19/2019		6B.1A	2019-02-22	MDCK1/MDCK1	160	80	640	640	1280	1280	640	1280	640	640
A/Armenia/22/2019		6B.1A5A	2019-03-15	MDCK1/MDCK1	160	160	640	640	640	640	640	1280	320	640
A/Armenia/23/2019		6B.1A5A	2019-04-18	MDCK2/MDCK1	320	160	640	640	1280	1280	640	1280	640	640
A/Qatar/24-VI-19-0043262/2019		6B.1A5A	2019-07-06	MDCK2	320	320	1280	1280	1280	1280	640	2560	1280	1280
A/Armenia/14/2019		6B.1A6	2019-02-04	MDCK1/MDCK1	160	80	640	640	1280	1280	1280	1280	640	1280
A/England/292/2019		6B.1A5A	2019-09-06	SIAT1/MDCK1	160	80	320	320	640	320	320	1280	320	320
A/England/296/2019		6B.1A5A	2019-09-13	SIAT1/MDCK1	320	160	640	640	1280	640	640	1280	640	640
A/Sur/5609/2019		6B.1A5A	2019-09-26	MDCK1	160	160	640	640	1280	640	640	2560	640	640
A/Salalah/5678/2019		6B.1A5A	2019-09-27	MDCK1	320	160	1280	1280	1280	1280	640	2560	640	1280
A/Sohar/5731/2019		6B.1A5A	2019-09-27	MDCK1	160	160	640	640	640	640	640	1280	320	640
A/Nizwa/5713/2019		6B.1A5A	2019-09-29	MDCK1	80	40	320	160	320	160	160	640	160	320
A/Muscat/5728/2019		6B.1A5A	2019-09-29	MDCK1	160	160	640	640	640	640	640	2560	640	640
A/Salalah/5832/2019		6B.1A5A	2019-10-01	MDCK1	320	160	1280	1280	1280	1280	640	2560	640	1280
A/Salalah/6005/2019		6B.1A5A	2019-10-03	MDCK1	160	80	640	320	640	320	320	1280	320	640
A/I Buraymi/6054/2019		6B.1A5A	2019-10-07	MDCK1	320	160	1280	640	1280	640	640	2560	640	1280
A/Salalah/6044/2019		6B.1A5A	2019-10-07	MDCK1	160	160	640	320	640	640	320	1280	320	640
A/Muscat/6173/2019		6B.1A5A	2019-10-10	MDCK1	320	160	1280	1280	1280	1280	640	2560	640	1280
A/England/298/2019		6B.1A5A	2019-10-10	SIAT1/MDCK1	160	80	640	640	1280	640	320	1280	320	640
A/Sohar/6315/2019		6B.1A5A	2019-10-11	MDCK1	160	80	640	320	640	640	320	1280	320	640
A/I Buraymi/6204/2019		6B.1A5A	2019-10-11	MDCK1	160	80	320	320	320	320	320	1280	320	320
A/Ibra/6240/2019		6B.1A5A	2019-10-12	MDCK1	160	160	640	640	2560	640	640	2560	640	640
A/Muscat/6300/2019		6B.1A5A	2019-10-13	MDCK1	160	160	640	640	1280	640	640	1280	640	640
A/I Buraymi/6340/2019		6B.1A5A	2019-10-13	MDCK1	160	160	640	640	1280	640	640	1280	640	1280
A/Salalah/6453/2019		6B.1A5A	2019-10-15	MDCK1	320	160	1280	640	1280	640	640	2560	640	1280
A/Muqshin/6616/2019		6B.1A5A	2019-10-16	MDCK1	160	160	640	640	1280	640	640	2560	640	640
A/Norway/2316/2019		6B.1A5A	2019-10-16	MDCK1	160	160	640	640	1280	640	320	1280	640	640
A/Saham/6774/2019		6B.1A5A	2019-10-21	MDCK1	320	160	1280	1280	1280	1280	640	2560	640	1280
A/Salalah/6721/2019	K239N	6B.1A5A	2019-10-21	MDCK2	40	<	160	160	80	160	160	640	160	160
A/Saham/6772/2019		6B.1A5A	2019-10-22	MDCK1	160	160	640	640	640	640	320	1280	320	640
A/Norway/2347/2019		6B.1A5A	2019-10-23	MDCK1	80	40	320	160	320	160	160	640	160	160
A/Salalah/6780/2019		6B.1A5A	2019-10-23	MDCK1	320	160	1280	1280	1280	1280	640	2560	640	640
A/Muscat/6744/2019		6B.1A5A	2019-10-23	MDCK1	320	160	640	640	1280	640	640	1280	640	640
A/Muscat/6791/2019		6B.1A5A	2019-10-23	MDCK1	320	160	640	640	1280	640	640	2560	640	640
A/Salalah/6906/2019		6B.1A5A	2019-10-23	MDCK1	320	160	640	640	1280	640	640	1280	320	640
A/Ibra/6826/2019		6B.1A5A	2019-10-24	MDCK1	320	160	640	640	1280	640	640	2560	640	640
A/Ibra/6828/2019		6B.1A5A	2019-10-25	MDCK1	160	80	640	640	1280	640	320	1280	320	640
A/Salalah/6887/2019		6B.1A5A	2019-10-25	MDCK1	320	320	1280	640	1280	640	640	2560	640	640
A/I Buraymi/6870/2019		6B.1A5A	2019-10-26	MDCK1	160	80	640	320	640	320	320	1280	320	320
A/Sur/6980/2019		6B.1A5A	2019-10-27	MDCK1	160	160	640	640	640	640	640	1280	320	640
A/I Buraymi/7034/2019		6B.1A5A	2019-10-29	MDCK1	160	160	640	640	1280	640	320	1280	320	640
A/Muscat/6997/2019		6B.1A5A	2019-10-29	MDCK1	160	80	640	640	640	640	320	1280	320	640
A/I Buraymi/7023/2019		6B.1A5A	2019-10-29	MDCK1	160	80	640	640	640	160	320	1280	320	640
A/Ibra/7177/2019		6B.1A5A	2019-10-31	MDCK1	320	80	640	640	640	640	640	1280	640	640
A/Sohar/7230/2019		6B.1A5A	2019-11-02	MDCK1	320	160	640	1280	1280	1280	640	2560	640	1280
A/Sohar/7147/2019		6B.1A5A	2019-11-02	MDCK1	320	160	640	1280	1280	1280	640	2560	640	1280
A/Norway/2368/2019		6B.1A5A	2019-11-03	SIAT1/MDCK1	160	160	640	640	640	640	640	1280	640	640
A/Muscat/7221/2019		6B.1A5A	2019-11-03	MDCK1	160	160	320	640	640	640	320	1280	640	640
A/Sohar/7260/2019		6B.1A5A	2019-11-03	MDCK1	320	160	640	640	1280	640	640	2560	640	640
A/Ar Rustaq/7243/2019		6B.1A5A	2019-11-04	MDCK1	160	160	640	640	640	640	320	2560	640	640
A/Hayma/7385/2019		6B.1A5A	2019-11-05	MDCK1	160	160	320	640	640	320	320	1280	640	640
A/Ibra/7362/2019		6B.1A5A	2019-11-06	MDCK1	160	80	320	640	640	320	320	1280	640	640
A/Muscat/7459/2019		6B.1A5A	2019-11-07	MDCK1	320	160	1280	1280	1280	1280	640	2560	1280	1280
A/Sohar/7409/2019		6B.1A5A	2019-11-09	MDCK1	160	80	320	640	640	320	320	1280	320	640
A/Norway/2317/2019		6B.1A5B	2019-10-21	MDCK1	80	40	640	640	640	320	320	640	320	640
A/Norway/2412/2019		6B.1A5B	2019-11-06	MDCK1	320	160	1280	1280	1280	1280	640	2560	640	1280
A/Norway/2406/2019		6B.1A5B	2019-11-08	MDCK1	160	80	640	640	640	640	640	1280	640	640
A/England/297/2019		6B.1A6	2019-10-08	SIAT1/MDCK1	320	160	640	640	1280	640	640	1280	640	640
A/Norway/2471/2019		6B.1A7	2019-11-14	MDCK1	320	160	640	1280	2560	1280	1280	2560	1280	1280
A/Norway/2475/2019		6B.1A7	2019-11-16	MDCK1	320	160	1280	1280	1280	1280	1280	2560	640	1280
A/Sohar/7367/2019			2019-11-04	MDCK1	160	80	640	640	640	640	320	1280	640	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80; ND =Not Done

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20
SH 2020

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 5-8. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-12-18)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Mich 45/15 Egg	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK
					F09/15 ^{*1}	F13/18 ^{*1}	F31/16 ^{*1}	NIB F48/16 ^{*1}	F03/18 ^{*2}	F20/18 ^{*1}	F09/19 ^{*1}	F04/19 ^{*1}	F23/18 ^{*1}	F08/19 ^{*1}
							6B.1	6B.1	6B.1A	6B.1A	6B.1A1	6B.1A5A	6B.1A5B	6B.1A6
REFERENCE VIRUSES														
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	320	320	160	40	320	160	160	640	160	40
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	640	640	320	80	320	320	160	640	160	80
A/Michigan/45/2015		6B.1	2015-09-07	E3/E4	320	320	1280	1280	2560	2560	1280	2560	1280	1280
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	640	640	2560	2560	2560	2560	2560	2560	1280	2560
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	320	80	1280	1280	1280	1280	640	1280	640	1280
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E3	640	1280	5120	5120	5120	5120	2560	5120	5120	5120
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	320	320	1280	1280	1280	1280	640	1280	640	1280
A/Norway/3433/2018		6B.1A5A	2018-10-30	MDCK3	160	40	640	640	640	640	320	1280	640	640
A/Switzerland/3330/2017	clone 35	6B.1A5B	2017-12-20	E6/E2	320	160	1280	640	1280	1280	640	1280	1280	1280
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	320	160	1280	1280	1280	1280	640	1280	1280	1280
TEST VIRUSES														
A/Ar Rustaq/5758/2019		6B.1A5A	2019-09-30	MDCK2	160	160	640	640	1280	640	640	1280	640	640
A/Iran/184643/2019		6B.1A5A	2019-10-16	SIAT1/SIAT1	640	320	1280	1280	1280	1280	1280	2560	640	1280
A/Azores/9/2019		6B.1A5A	2019-10-31	MDCK2	320	320	1280	640	1280	1280	640	1280	320	640
A/Azores/10/2019		6B.1A5A	2019-11-07	MDCK1	320	160	1280	640	1280	1280	640	2560	640	640
A/Azores/11/2019		6B.1A5A	2019-11-11	MDCK2	320	320	1280	1280	1280	1280	640	2560	640	1280
A/Azores/12/2019		6B.1A5A	2019-11-19	MDCK2	320	320	1280	1280	1280	1280	1280	1280	640	1280
A/Azores/13/2019		6B.1A5A	2019-11-21	MDCK1	320	160	640	640	1280	640	640	1280	320	640
A/Iran/187528/2019		6B.1A5A	2019-11-26	SIAT1	320	160	1280	640	1280	1280	640	2560	640	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80; ND =Not Done

Vaccine
NH 2018-19
SH 2019

Vaccine
NH 2019-20
SH 2020

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 5-9. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2020-01-14)

Haemagglutination inhibition titre														
Viruses	Other information	Passage history	Collection date	Passage history	Post-infection ferret antisera									
					A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Mich 45/15 Egg	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK
					F09/15 ¹	F13/18 ¹	F31/16 ¹	F48/16 ¹	F03/18 ²	F20/18 ¹	F09/19 ¹	F04/19 ¹	F23/18 ¹	F08/19 ¹
					Genetic group									
REFERENCE VIRUSES														
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	160	160	80	40	160	80	80	320	80	<
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	640	640	320	80	320	320	160	640	160	80
A/Michigan/45/2015		6B.1	2015-09-07	E3/E4	320	320	1280	1280	1280	2560	640	2560	640	1280
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	320	320	1280	2560	1280	1280	1280	1280	640	1280
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	160	80	640	1280	1280	1280	1280	2560	640	1280
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E3	640	1280	2560	5120	2560	2560	1280	2560	2560	2560
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	320	160	1280	1280	1280	1280	640	2560	640	1280
A/Norway/3433/2018		6B.1A5A	2018-10-30	MDCK3	80	<	320	640	320	320	160	640	320	320
A/Switzerland/3330/2017	clone 35	6B.1A5B	2017-12-20	E6/E2	320	80	640	640	640	1280	320	1280	1280	640
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	160	80	640	1280	1280	640	640	1280	640	1280
TEST VIRUSES														
A/Czech Republic/1428/2019		6B.1A5A	2019-08-20	MDCK2/MDCK1	320	320	1280	640	1280	640	640	2560	640	1280
A/Denmark/3254/2019		6B.1A5A	2019-10-06	MDCK3/MDCK1	160	80	640	640	1280	640	640	1280	320	640
A/Mougins/1891/2019		6B.1A5A	2019-10-07	MDCK3/MDCK1	160	80	640	640	640	320	320	1280	640	640
A/Latvia/10-025804/2019		6B.1A5A	2019-10-09	MDCK2/MDCK1	320	160	640	1280	1280	640	640	2560	640	1280
A/Le Cannel/1893/2019		6B.1A5A	2019-10-10	MDCK3/MDCK1	320	160	640	640	1280	640	640	1280	640	1280
A/Le Cannel/1892/2019		6B.1A5A	2019-10-11	MDCK3/MDCK1	160	160	640	640	1280	640	640	1280	320	1280
A/Hessen/70/2019		6B.1A5A	2019-10-15	C2/MDCK1	320	160	640	640	640	640	640	1280	320	1280
A/Denmark/3256/2019		6B.1A5A	2019-10-15	MDCK3/MDCK1	160	80	640	640	640	640	640	1280	640	640
A/Hessen/71/2019		6B.1A5A	2019-10-16	C2/MDCK1	160	160	640	640	640	640	640	1280	320	640
A/Iran/187169/2019		6B.1A5A	2019-10-23	SIAT1/MDCK1	320	160	640	640	1280	640	640	2560	640	640
A/Iran/186980/2019		6B.1A5A	2019-10-23	SIAT1/MDCK1	160	80	640	640	640	640	320	1280	320	640
A/Iran/187210/2019		6B.1A5A	2019-10-25	SIAT1/MDCK1	160	160	640	640	1280	640	320	1280	640	640
A/Iran/187375/2019		6B.1A5A	2019-10-26	SIAT1/MDCK1	320	320	1280	1280	2560	1280	640	2560	640	1280
A/Iran/187215/2019		6B.1A5A	2019-10-26	SIAT1/MDCK1	320	160	1280	1280	1280	640	640	2560	640	1280
A/Finland/133/2019		6B.1A5A	2019-10-30	MDCK1/MDCK1	320	160	1280	640	1280	640	640	2560	640	1280
A/Latvia/11-003941/2019		6B.1A5A	2019-11-02	MDCK2/MDCK1	320	160	640	640	1280	640	320	2560	640	640
A/Latvia/11-003858/2019		6B.1A5A	2019-11-02	MDCK2/MDCK1	160	80	320	640	640	640	320	1280	320	640
A/Finland/132/2019		6B.1A5A	2019-11-03	MDCK1/MDCK1	320	160	640	640	1280	640	640	1280	640	640
A/Denmark/3288/2019		6B.1A5A	2019-11-08	MDCK4/MDCK1	80	80	320	320	640	1280	160	1280	320	320
A/Denmark/3280/2019	N156K, K130N, L161I, V250A	6B.1A5A	2019-11-10	MDCK4/MDCK1	80	40	80	40	40	80	40	320	80	40
A/Berlin/57/2019		6B.1A5A	2019-11-11	C1/MDCK1	160	160	640	640	1280	640	640	1280	640	640
A/Iran/192588/2019		6B.1A5A	2019-11-12	SIAT1/MDCK1	320	320	640	1280	1280	1280	640	2560	640	640
A/Iran/192506/2019		6B.1A5A	2019-11-12	SIAT1/MDCK1	320	160	1280	1280	1280	640	640	1280	640	1280
A/Lyon/2003/2019		6B.1A5A	2019-11-12	MDCK2/MDCK1	320	160	1280	640	1280	640	640	2560	640	1280
A/Iran/193134/2019		6B.1A5A	2019-11-13	SIAT1/MDCK1	320	160	640	640	1280	640	640	1280	640	640
A/Iran/184509/2019		6B.1A5A	2019-11-14	SIAT2/MDCK1	160	160	640	640	1280	640	640	1280	640	640
A/Lyon/2037/2019		6B.1A5A	2019-11-18	MDCK2/MDCK1	320	320	1280	1280	1280	1280	640	1280	320	1280
A/Nordrhein-Westfalen/136/2019		6B.1A5A	2019-11-20	C1/MDCK1	320	160	640	640	1280	640	640	1280	640	640
A/Lyon/CHU/R19.132.13/2019		6B.1A5A	2019-11-23	MDCK2/MDCK1	160	160	640	640	1280	640	640	1280	640	640
A/Niedersachsen/195/2019		6B.1A5A	2019-11-25	C1/MDCK1	320	160	640	640	1280	640	640	2560	640	1280
A/Hong Kong/4380/2019	N156K, K130N, L161I, V250A	6B.1A5A	2019-11-27	MDCK1/MDCK2	40	<	40	<	<	<	<	80	40	<
A/Hong Kong/4372/2019		6B.1A5A	2019-11-28	MDCK1/MDCK1	160	80	320	320	640	320	160	640	160	320
A/Hong Kong/4394/2019	N156K, K130N, L161I, V250A, K209M	6B.1A5A	2019-11-30	MDCK1/MDCK1	40	<	<	<	<	<	<	40	<	<
A/Hong Kong/4395/2019		6B.1A5A	2019-12-01	MDCK1/MDCK1	320	320	1280	1280	1280	1280	640	2560	1280	1280
A/Hong Kong/4388/2019	N156K, K130N, L161I, V250A	6B.1A5A	2019-12-01	MDCK1/MDCK1	80	40	40	<	<	80	40	160	40	<
A/Hong Kong/4387/2019		6B.1A5A	2019-12-01	MDCK1/MDCK1	160	160	640	1280	640	1280	640	1280	640	640
A/Hong Kong/4414/2019		6B.1A5A	2019-12-02	MDCK1/MDCK1	160	160	1280	1280	1280	1280	640	2560	640	1280
A/Hong Kong/4413/2019		6B.1A5A	2019-12-02	MDCK1/MDCK1	160	80	640	640	640	640	320	1280	640	640
A/Hong Kong/4416/2019	N156K, K130N, L161I, V250A	6B.1A5A	2019-12-03	MDCK1/MDCK1	40	<	40	<	<	40	<	80	40	<
A/Hong Kong/4432/2019	N156K, A195E	6B.1A5A	2019-12-04	MDCK1/MDCK1	80	<	80	<	80	80	80	640	80	<
A/Hong Kong/4431/2019		6B.1A5A	2019-12-04	MDCK1/MDCK1	320	320	1280	1280	1280	1280	640	2560	640	1280
A/Hong Kong/4430/2019		6B.1A5A	2019-12-04	MDCK1/MDCK1	160	80	640	640	640	640	320	1280	320	640
A/Hong Kong/4429/2019		6B.1A5A	2019-12-04	MDCK1/MDCK1	320	160	640	640	1280	640	640	2560	640	1280
A/Denmark/3284/2019		6B.1A5B	2019-11-08	MDCK4/MDCK1	80	<	320	320	640	<	320	640	320	640
A/Denmark/3283/2019		6B.1A5B	2019-11-08	MDCK3/MDCK1	320	80	1280	1280	1280	1280	640	2560	640	1280
A/Denmark/3279/2019		6B.1A5B	2019-11-10	MDCK4/MDCK1	80	40	320	640	640	640	320	640	320	640
A/Denmark/3295/2019		6B.1A5B	2019-11-15	MDCK3/MDCK1	320	80	1280	1280	1280	<	1280	2560	640	1280
A/Denmark/3311/2019		6B.1A5B	2019-11-16	MDCK3/MDCK1	160	80	640	640	1280	640	640	1280	640	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND =Not Done

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 5-10. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2020-01-21)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Mich 45/15 Egg	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK
				F09/15 ¹	F13/18 ¹	F31/16 ¹	F48/16 ¹	F03/18 ²	F20/18 ¹	F09/19 ¹	F04/19 ¹	F23/18 ¹	F08/19 ¹
	Passage history					6B.1	6B.1	6B.1A	6B.1A	6B.1A1	6B.1A5A	6B.1A5B	6B.1A6
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
A/Bayern/69/2009	G155E	2009-07-01	MDCK5/MDCK1	320	320	80	40	160	160	80	320	80	40
A/Lviv/N6/2009	G155E, D222G	2009-10-27	MDCK4/SIAT1/MDCK3	640	640	320	160	320	320	160	640	320	160
A/Michigan/45/2015	clone 37	2015-09-07	E3/E3	160	160	640	640	640	1280	640	1280	320	640
A/Slovenia/2903/2015		2015-10-26	E4/E2	320	160	1280	1280	1280	1280	640	1280	640	640
A/Paris/1447/2017		2017-10-20	MDCK1/MDCK3	320	80	640	1280	1280	1280	640	1280	640	1280
A/Switzerland/2656/2017		2017-12-21	E5/E3	640	640	2560	1280	1280	2560	1280	2560	1280	1280
A/Brisbane/02/2018		2018-01-04	E3/E1	320	320	1280	1280	2560	2560	1280	2560	1280	1280
A/Norway/3433/2018		2018-10-30	MDCK3	160	80	640	640	640	640	320	1280	640	640
A/Switzerland/3330/2017	clone 35	2017-12-20	E6/E2	160	80	640	640	640	640	320	1280	640	640
A/Ireland/84630/2018		2018-11-28	MDCK1/MDCK3	320	160	1280	1280	1280	1280	640	2560	640	1280
TEST VIRUSES													
A/Orebro/3/2019	6B.1A5A	2019-09-15	MDCK0/MDCK1	320	160	640	640	640	640	640	2560	640	640
A/Falun/4/2019	6B.1A5A	2019-09-17	MDCK0/MDCK1	160	80	640	640	320	640	320	1280	640	640
A/Switzerland/9887/2019	6B.1A5A	2019-11-06	SIAT1/MDCK1	160	80	320	640	640	640	320	640	320	320
A/Umea/2/2019	6B.1A5A	2019-11-10	MDCK0/MDCK1	160	80	640	640	320	640	320	1280	320	640
A/Umea/4/2019	6B.1A5A	2019-11-11	MDCK0/MDCK1	160	80	640	640	320	640	320	1280	320	640
A/Umea/3/2019	6B.1A5A	2019-11-11	MDCK0/MDCK1	160	160	640	640	320	640	320	1280	640	640
A/Switzerland/9795/2019	6B.1A5A	2019-11-14	SIAT1/MDCK1	320	160	640	640	640	640	320	1280	320	640
A/Lulea/1/2019	6B.1A5A	2019-11-18	MDCK0/MDCK1	160	80	640	640	320	640	320	1280	320	640
A/S.Petersburg/RII-326/2019	6B.1A5A	2019-11-20	MDCK1/MDCK1	320	160	1280	1280	1280	640	640	1280	640	640
A/S.Petersburg/RII-251S/2019	6B.1A5A	2019-11-22	MDCK1	320	160	640	640	1280	640	320	1280	320	640
A/S.Petersburg/RII-329/2019	6B.1A5A	2019-11-23	MDCK1/MDCK1	320	80	640	640	320	640	320	1280	320	640
A/S.Petersburg/RII-252S/2019	6B.1A5A	2019-11-23	MDCK1	320	160	640	1280	640	1280	640	1280	640	640
A/S.Petersburg/RII-327/2019	6B.1A5A	2019-11-25	MDCK1/MDCK1	320	160	640	640	320	640	640	1280	640	640
A/Croatia/8025/2019	6B.1A5A	2019-12-02	MDCKx/MDCK1	320	160	640	1280	1280	1280	640	1280	640	640
A/Switzerland/5139/2019	6B.1A5A	2019-12-02	SIAT1/MDCK1	320	160	640	1280	1280	1280	640	2560	1280	1280
A/Israel/487/2019	6B.1A5A	2019-12-05	C1/MDCK1	160	80	320	320	320	320	320	640	160	320
A/Iceland/86/2019	6B.1A5A	2019-12-08	MDCK1/MDCK1	160	80	640	640	320	320	320	1280	320	320
A/Israel/463/2019	6B.1A5A	2019-12-08	C1/MDCK1	160	80	320	320	320	320	320	640	160	320
A/Israel/483/2019	6B.1A5A	2019-12-09	C1/MDCK1	320	160	640	640	1280	640	640	1280	640	640
A/Israel/482/2019	6B.1A5A	2019-12-09	C1/MDCK1	320	160	640	640	1280	640	640	2560	640	640
A/Israel/477/2019	6B.1A5A	2019-12-09	C1/MDCK1	160	160	640	640	640	640	320	1280	320	640
A/Switzerland/4068/2019	6B.1A5A	2019-12-11	SIAT1/MDCK1	320	160	640	1280	1280	1280	640	2560	1280	1280
A/Israel/10539/2019	6B.1A5A	2019-12-19	C1/MDCK1	160	80	320	640	640	320	320	640	320	320
A/Karlstad/3/2019	6B.1A5B	2019-11-15	MDCK0/MDCK1	320	80	1280	1280	640	1280	640	1280	640	1280
A/Switzerland/3565/2019	6B.1A5B	2019-11-26	SIAT1/MDCK1	320	80	1280	1280	1280	1280	640	1280	640	1280
A/Iceland/85/2019	6B.1A5B	2019-12-01	MDCK1/MDCK1	160	80	640	640	640	640	640	1280	640	640
A/Croatia/7847/2019	6B.1A7	2019-11-20	MDCKx/MDCK1	640	160	1280	1280	2560	1280	1280	2560	1280	1280
A/Croatia/7846/2019	6B.1A7	2019-11-20	MDCKx/MDCK1	320	160	1280	1280	1280	1280	1280	2560	1280	1280
A/Switzerland/4501/2019	6B.1A7	2019-11-26	SIAT1/MDCK1	320	80	1280	1280	1280	1280	640	1280	640	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80; ND =Not Done

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20
SH 2020

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 5-11. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2020-01-28)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Mich 45/15 Egg	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Bris 02/18 Egg	A/Norway 3433/18 MDCK	A/Swit 3330/17 Egg	A/Ire 84630/18 MDCK
				F09/15 ¹	F13/18 ¹	F31/16 ¹	F48/16 ¹	F03/18 ²	F20/18 ¹	F09/19 ¹	F04/19 ¹	F23/18 ¹	F08/19 ¹
	Passage history												
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
A/Bayern/69/2009	G155E	2009-07-01	MDCK5/MDCK1	640	320	160	40	160	160	160	320	160	40
A/Lviv/N6/2009	G155E, D222G	2009-10-27	MDCK4/SIAT1/MDCK3	640	640	160	160	320	320	320	1280	160	80
A/Michigan/45/2015	6B.1	2015-09-07	E3/E3	160	160	640	640	1280	1280	640	1280	640	640
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E2	320	160	640	640	1280	640	640	1280	640	640
A/Paris/1447/2017	6B.1A	2017-10-20	MDCK1/MDCK3	320	80	640	640	1280	1280	640	1280	640	640
A/Switzerland/2656/2017	6B.1A	2017-12-21	E5/E3	640	640	1280	1280	1280	1280	1280	2560	1280	1280
A/Brisbane/02/2018	6B.1A1	2018-01-04	E3/E1	320	320	1280	1280	2560	1280	640	2560	1280	1280
A/Norway/3433/2018	6B.1A5A	2018-10-30	MDCK3	160	40	320	320	320	320	320	640	320	320
A/Switzerland/3330/2017	clone 35	2017-12-20	E6/E2	320	160	640	1280	1280	1280	640	1280	1280	640
A/Ireland/84630/2018	6B.1A6	2018-11-28	MDCK1/MDCK3	320	160	1280	1280	1280	1280	640	2560	640	1280
TEST VIRUSES													
A/Ireland/87362/2019	6B.1A5A	2019-10-30	MDCK1	160	160	640	640	640	640	640	1280	640	640
A/Ireland/87733/2019	6B.1A5A	2019-11-03	MDCK2	160	80	320	320	640	320	320	640	160	320
A/Slovenia/2924/2019	6B.1A5A	2019-11-18	MDCK2/MDCK1	160	160	640	640	640	640	320	1280	320	320
A/Saudi Arabia/2086/2019	6B.1A5A	2019-11-19	SIAT1	80	80	320	320	320	320	320	640	320	320
A/Ireland/95067/2019	6B.1A5A	2019-11-26	MDCK1	160	80	640	640	640	640	320	1280	320	640
A/Ireland/95264/2019	6B.1A5A	2019-11-27	MDCK2	160	80	320	640	640	640	320	1280	320	640
A/Ireland/95337/2019	6B.1A5A	2019-11-28	MDCK1	160	160	640	640	640	640	320	1280	320	640
A/Ireland/95766/2019	6B.1A5A	2019-11-29	MDCK2	160	160	640	640	1280	640	640	1280	640	640
A/Alsace/2631/2019	6B.1A5A	2019-12-02	MDCK2/MDCK1	320	80	640	640	640	640	640	1280	320	640
A/Slovenia/3106/2019	6B.1A5A	2019-12-04	MDCK1/MDCK1	320	160	640	640	640	640	640	1280	320	640
A/Yakutsk/5782/2019	6B.1A5A	2019-12-05	MDCK3/MDCK1	160	160	640	640	640	640	320	1280	320	640
A/Georgia/1098/2019	6B.1A5A	2019-12-06	MDCK1	160	160	640	640	640	640	320	1280	320	640
A/Slovenia/3139/2019	6B.1A5A	2019-12-06	MDCKx/MDCK1	160	80	640	640	640	640	320	1280	320	640
A/Kyiv/311/2019	6B.1A5A	2019-12-07	SIAT1/MDCK1	160	160	640	640	640	640	640	1280	320	640
A/Georgia/1112/2019	6B.1A5A	2019-12-09	MDCK1	160	80	640	320	640	640	320	1280	320	640
A/Franche Comte/2784/2019	6B.1A5A	2019-12-09	MDCK1/MDCK1	160	160	640	320	640	640	320	1280	320	320
A/Bretagne/2725/2019	6B.1A5A	2019-12-09	MDCK1/MDCK1	320	320	1280	1280	1280	1280	640	2560	640	1280
A/Astrakhan/13/2019	6B.1A5A	2019-12-10	MDCK3/MDCK1	160	160	320	320	640	320	320	1280	640	640
A/Vladivostok/131/2019	6B.1A5A	2019-12-10	MDCK3/MDCK1	320	160	640	640	1280	640	640	1280	320	640
A/Yakutsk/5942/2019	6B.1A5A	2019-12-10	MDCK3/MDCK1	160	80	640	640	640	640	320	1280	320	640
A/Khmelnitskyi/323/2019	6B.1A5A	2019-12-11	SIAT1/MDCK1	320	160	640	640	1280	640	320	1280	320	640
A/Slovenia/3222/2019	6B.1A5A	2019-12-12	SIAT2/MDCK1	160	160	640	640	640	640	320	1280	320	640
A/Slovenia/3218/2019	6B.1A5A	2019-12-12	SIAT1/MDCK1	160	80	320	640	640	640	320	1280	320	320
A/Yakutsk/5992/2019	6B.1A5A	2019-12-12	MDCK3/MDCK1	160	160	640	640	640	640	320	1280	320	640
A/Yakutsk/5976/2019	6B.1A5A	2019-12-12	MDCK3/MDCK1	320	160	640	640	640	640	640	1280	640	640
A/Slovenia/3247/2019	6B.1A5A	2019-12-13	MDCK3/MDCK1	80	80	640	640	640	640	320	1280	320	640
A/Georgia/1231/2019	6B.1A5A	2019-12-18	MDCK1	160	80	640	640	640	640	320	1280	320	640
A/Ukraine/8375/2019	6B.1A5A	2019-12-18	MDCK3/MDCK1	320	160	1280	640	1280	640	640	2560	640	640
A/Ukraine/8374/2019	6B.1A5A	2019-12-18	MDCK1/MDCK1	320	160	640	640	1280	640	640	1280	640	640
A/Kyiv/334/2019	6B.1A5A	2019-12-19	SIAT2/MDCK1	160	160	640	640	640	640	320	1280	320	320
A/Georgia/1178/2019	6B.1A5A	2019-12-20	MDCK1	160	80	320	640	640	640	320	1280	320	640
A/Paris/2834/2019	6B.1A5A	2019-12-23	MDCK2/MDCK1	160	160	640	640	640	640	640	1280	320	640
A/Bosnia and Herzegovia/77/2019	6B.1A5A	2019-12-23	MDCK1	320	160	640	640	1280	1280	640	2560	640	640
A/Ukraine/8378/2019	6B.1A5A	2019-12-23	MDCK2/MDCK1	160	160	640	640	640	640	320	1280	320	320
A/Ukraine/8377/2019	6B.1A5A	2019-12-23	MDCK1/MDCK1	160	80	640	640	640	640	320	1280	320	320
A/Ukraine/8376/2019	6B.1A5A	2019-12-23	MDCK3/MDCK1	320	320	1280	640	1280	1280	640	2560	640	640
A/Pays de Loire/2930/2019	6B.1A5A	2019-12-26	MDCK1/MDCK1	160	80	640	320	80	320	320	1280	320	320
A/Bosnia and Herzegovia/90/2019	6B.1A5A	2019-12-27	MDCK1	320	160	640	1280	1280	1280	640	2560	640	6340
A/Kyiv/354/2019	6B.1A5A	2019-12-27	SIATx/MDCK1	160	160	640	640	640	640	640	1280	640	640
A/Georgia/1252/2019	6B.1A5A	2019-12-30	MDCK1	160	160	640	640	640	640	320	1280	320	640
A/Bosnia and Herzegovia/96/2019	6B.1A5A	2019-12-30	MDCK1	160	80	640	320	320	320	320	1280	320	320
A/Bosnia and Herzegovia/95/2019	6B.1A5A	2019-12-30	MDCK1	320	320	640	640	1280	640	640	2560	640	640
A/Kyiv/355/2019	6B.1A5A	2019-12-30	SIATx/MDCK1	160	80	640	640	640	640	320	1280	320	640
A/Kyiv/1/2020	6B.1A5A	2020-01-03	SIATx/MDCK1	320	160	640	640	1280	640	640	1280	320	640
A/Slovenia/161/2020	6B.1A5A	2020-01-08	SIATx/MDCK1	160	160	640	640	640	640	320	1280	320	640
A/Slovenia/148/2020	6B.1A5A	2020-01-08	SIATx/MDCK1	160	80	320	320	320	320	320	640	160	320
A/Centre/2599/2019	6B.1A5B	2019-11-27	MDCK1/MDCK1	160	40	640	1280	1280	640	640	1280	640	1280
A/Alsace/2798/2019	6B.1A5B	2019-12-18	MDCK2/MDCK1	160	80	640	640	1280	640	640	1280	640	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND =Not Done

Vaccine
NH 2018-19
SH 2019Vaccine
NH 2019-20
SH 2020

< 4-fold
4-fold
8-fold
> 8-fold
not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 5-12. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2020-02-04)

Viruses	Other information	Passage history	Collection date	Passage history	A/Bayern	A/Lviv	A/Mich	A/Slov	A/Paris	A/Swit	A/Bris	A/Norway	A/Swit	A/Ire	A/HK	
					69/09	N6/09	45/15	2903/2015	1447/17	2656/17	02/18	3433/18	3330/17	84630/18	110/19	
					MDCK	MDCK	Egg	Egg	MDCK	Egg	Egg	MDCK	Egg	MDCK	MDCK	
					F09/15 ¹	F13/18 ¹	F31/16 ¹	F48/16 ¹	F03/18 ²	F20/18 ¹	F09/19 ¹	F04/19 ¹	F23/18 ¹	F08/19 ¹	F28/19	
							6B.1	6B.1	6B.1A	6B.1A	6B.1A1	6B.1A5	6B.1A5	6B.1A6	6B.1A2	
REFERENCE VIRUSES																
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	640	160	80	<	160	160	80	160	80	40	ND	
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	640	640	320	160	640	320	320	640	320	160	ND	
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	320	160	640	640	1280	1280	640	1280	640	640	ND	
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	320	320	1280	1280	1280	1280	640	1280	640	1280	ND	
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	320	80	1280	1280	1280	1280	1280	2560	640	1280	ND	
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E3	640	640	1280	1280	2560	1280	1280	2560	1280	1280	ND	
A/Brisbane/02/2018		6B.1A1	2018-01-04	E3/E1	320	320	1280	1280	2560	1280	1280	2560	1280	1280	40	
A/Norway/3433/2018		6B.1A5A	2018-10-30	MDCK3	160	80	640	640	640	640	640	1280	640	640	ND	
A/Switzerland/3330/2017	clone 35	6B.1A5B	2017-12-20	E6/E2	320	160	640	1280	1280	1280	640	2560	1280	1280	ND	
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK1/MDCK3	320	160	1280	1280	1280	1280	640	1280	640	1280	ND	
A/Hong Kong/110/2019	N156K	6B.1A2	2019-01-01	MDCK1/MDCK2	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	2560	
TEST VIRUSES																
A/Netherlands/10258/2019		6B.1A5A	2019-10-31	MDCK-MIX2/MDCK1	160	160	640	640	1280	640	640	2560	640	640	ND	
A/Belgium/G0460/2019	N156K, K54E, P137S, A195E	6B.1A5A	2019-11-05	MDCK1	80	80	80	40	80	80	80	640	80	<	160	
A/Austria/1200911/2019		6B.1A5A	2019-11-13	SIAT1/MDCK1	320	160	640	640	1280	640	640	1280	640	640	ND	
A/Qatar/34-VI-19-0071996/2019		6B.1A5A	2019-11-14	MDCK1	80	80	320	320	320	320	320	1280	160	320	ND	
A/Austria/1200546/2019		6B.1A5A	2019-11-14	SIAT1/MDCK1	160	80	320	320	320	320	160	640	160	320	ND	
A/Qatar/34-VI-19-0072438/2019		6B.1A5A	2019-11-16	MDCK1	160	80	640	320	640	320	320	1280	320	640	ND	
A/Qatar/10-VI-19-0072485/2019		6B.1A5A	2019-11-16	MDCK1	160	80	320	320	640	320	320	1280	320	320	ND	
A/Qatar/10-VI-19-0072693/2019		6B.1A5A	2019-11-17	MDCK1	80	80	320	320	640	320	320	1280	160	320	ND	
A/Qatar/21-VI-19-0072564/2019		6B.1A5A	2019-11-17	MDCK1	160	80	320	320	320	320	320	640	320	640	ND	
A/Jordan/30261/2019		6B.1A5A	2019-11-18	MDCK1	160	160	640	640	640	640	320	1280	320	640	ND	
A/Netherlands/10263/2019		6B.1A5A	2019-11-25	MDCK-MIX2/MDCK1	160	80	640	640	640	640	320	1280	320	640	ND	
A/Jordan/12191/2019		6B.1A5A	2019-12-05	MDCK1	160	80	640	320	640	320	320	1280	320	640	ND	
A/Austria/1206292/2019		6B.1A5A	2019-12-08	SIATx/MDCK1	160	160	640	640	1280	640	320	1280	320	640	ND	
A/Austria/1207286/2019	N156K, K130N, L161I, V250A, K209M	6B.1A5A	2019-12-11	SIATx/MDCK2	40	<	<	<	<	<	<	80	<	<	320	
A/Catalonia/3512178NS/2019		6B.1A5A	2019-12-11	P0/MDCK1	160	160	640	640	1280	640	640	1280	320	640	ND	
A/Catalonia/3512137NS/2019		6B.1A5A	2019-12-11	P0/MDCK1	320	160	1280	1280	1280	1280	640	2560	640	1280	ND	
A/Saint Etienne/2287/2019		6B.1A5A	2019-12-12	MDCK2/MDCK1	160	160	640	640	640	640	640	1280	640	640	ND	
A/Kyiv/327/2019	N156K, K130N, L161I, V250A	6B.1A5A	2019-12-14	SIAT3/MDCK2	80	<	40	<	<	40	40	160	40	<	1280	
A/La Rochelle/2309/2019		6B.1A5A	2019-12-14	MDCK2/MDCK1	160	160	640	320	640	320	320	1280	320	640	ND	
A/Jordan/12270/2019		6B.1A5A	2019-12-15	MDCK1	320	160	640	320	640	640	320	1280	320	640	ND	
A/Austria/1205297/2019		6B.1A5A	2019-12-16	SIAT1/MDCK1	80	80	320	320	640	320	160	640	160	320	ND	
A/Jordan/12344/2019		6B.1A5A	2019-12-17	MDCK1	320	160	640	640	1280	640	640	1280	640	640	ND	
A/Belgium/G0526/2019		6B.1A5A	2019-12-17	MDCK1	160	160	640	640	640	640	320	1280	320	640	ND	
A/Belgium/G0537/2019	N156K, K130N, L161I, V250A, K209M	6B.1A5A	2019-12-18	MDCK1	80	40	40	<	<	40	<	80	40	<	640	
A/Belgium/G0001/2020		6B.1A5A	2019-12-19	MDCK1	320	160	640	640	1280	640	640	2560	640	640	ND	
A/Lyon/2343/2019	N156K, K130N, L161I, V250A, K209M	6B.1A5A	2019-12-19	MDCK2/MDCK1	<	<	<	<	<	<	<	40	<	<	160	
A/Catalonia/3512868NS/2019		6B.1A5A	2019-12-26	P0/MDCK1	320	320	1280	1280	1280	1280	640	2560	640	1280	ND	
A/Catalonia/2174495NS/2020		6B.1A5A	2020-01-09	P0/MDCK1	160	80	320	320	640	320	320	1280	320	320	ND	
A/Poland/1535/2020		6B.1A5A	2020-01-12	MDCK1/MDCK1	160	160	640	640	1280	640	640	1280	640	640	ND	
A/Greece/59/2020		6B.1A5A	2020-01-13	MDCK1	160	160	640	640	640	640	320	1280	320	640	ND	
A/Athens.GR/93/2020		6B.1A5A	2020-01-14	MDCK1/MDCK1	160	160	640	640	1280	640	640	1280	640	640	ND	
A/Athens.GR/120/2020		6B.1A5A	2020-01-16	Cx/MDCK1	320	160	640	640	640	640	640	1280	640	640	ND	
A/Saint Etienne/2362/2019			2019-12-16	MDCK2/MDCK1	80	80	160	320	320	320	160	640	160	320	ND	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND =Not Done

< 4-fold
4-fold
8-fold
> 8-fold
not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 5-13. Antigenic analyses of influenza A(H1N1)pdm09 viruses - Summary

Viruses	Other information	A/Bayern 69/09 MDCK F09/15 ^{*1} Genetic group	A/Lviv N6/09 MDCK F13/18 ^{*1}	A/Mich 45/15 Egg F31/16 ^{*1} 6B.1	A/Slov 2903/2015 Egg NIB F48/16 ^{*1} 6B.1	A/Paris 1447/17 MDCK F03/18 ^{*2} 6B.1A	A/Swit 2656/17 Egg F20/18 ^{*1} 6B.1A	A/Bris 02/18 Egg F09/19 ^{*1} 6B.1A1	A/Norway 3433/18 MDCK F04/19 ^{*1} 6B.1A5A	A/Swit 3330/17 Egg F23/18 ^{*1} 6B.1A5B	A/Ire 84630/18 MDCK F08/19 ^{*1} 6B.1A6	
REFERENCE VIRUSES												
A/Bayern/69/2009	G155E	160	160	80	40	160	80	80	320	80	<	
A/Lviv/N6/2009	G155E, D222G	640	640	320	80	320	320	160	640	160	80	
A/Michigan/45/2015	6B.1	320	320	1280	1280	1280	2560	640	2560	640	1280	
A/Slovenia/2903/2015	clone 37	320	320	1280	2560	1280	1280	1280	1280	640	1280	
A/Paris/1447/2017	6B.1A	160	80	640	1280	1280	1280	1280	2560	640	1280	
A/Switzerland/2656/2017	6B.1A	640	1280	2560	5120	2560	2560	1280	2560	2560	2560	
A/Brisbane/02/2018	6B.1A1	320	160	1280	1280	1280	1280	640	2560	640	1280	
A/Norway/3433/2018	6B.1A5A	80	<	320	640	320	320	160	640	320	320	
A/Switzerland/3330/2017	clone 35	320	80	640	640	640	1280	320	1280	1280	640	
A/Ireland/84630/2018	6B.1A6	160	80	640	1280	1280	640	640	1280	640	1280	
TEST VIRUSES												
Genetic group	No tested	No with										
All viruses	222	Titre reduced ≤2-fold	154	41	161	106	166	77	155	214	123	180
		%	69.4	18.5	72.5	47.8	74.8	34.7	69.8	96.4	55.5	81.1
		Titre reduced =4-fold	56	92	41	58	40	80	50	1	77	29
		%	25.2	41.4	18.5	26.1	18.0	36.0	22.5	0.4	34.7	13.1
		Titre reduced ≥8-fold	12	89	20	58	16	65	17	7	22	13
		%	5.4	40.1	9.0	26.1	7.2	29.3	7.7	3.2	9.8	5.8
Unknown	2	Titre reduced ≤2-fold	1						2	1	1	
		Titre reduced =4-fold		2	1	2	1	1			1	
		Titre reduced ≥8-fold	1	2	1	1	1	1		1		
6B.1A5A	201	Titre reduced ≤2-fold	137	40	148	96	149	69	137	193	106	160
		%	68.2	19.9	73.6	47.9	74.1	34.3	68.2	96.0	52.7	79.6
		Titre reduced =4-fold	53	87	33	52	36	73	48	1	74	28
		%	26.4	43.3	16.4	26.0	17.9	36.3	23.9	0.5	36.8	13.9
		Titre reduced ≥8-fold	11	74	20	53	16	59	16	7	21	13
		%	5.4	36.8	10.0	26.1	8.0	29.4	7.9	3.5	10.5	6.5
6B.1A5B	13	Titre reduced ≤2-fold	10		9	7	11	5	12	13	10	13
		%	76.9		69.2	53.8	84.6	38.4	92.3	100.0	76.9	100.0
		Titre reduced =4-fold	3	1	4	3	2	4	1		3	
		%	23.1	7.7	30.8	23.1	15.4	30.8	7.7		23.1	
		Titre reduced ≥8-fold		12		3		4				
		%		92.3		23.1		30.8				
6B.1A6	1	Titre reduced ≤2-fold	1	1			1		1	1	1	1
		Titre reduced =4-fold			1							
		Titre reduced ≥8-fold				1		1				
6B.1A7	5	Titre reduced ≤2-fold	5		4	3	5	3	5	5	5	5
		Titre reduced =4-fold		4	1	2		2				
		Titre reduced ≥8-fold		1								
				Vaccine NH 2018-19 SH 2019					Vaccine NH 2019-20 SH 2020			

Reference virus results are taken from an individual table as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed

Figure 1. Phylogenetic comparison of influenza A(H1N1)pdm09 HA genes

Vaccine virus
Reference viruses

Collection date

Sep 2019
Oct 2019
Nov 2019
Dec 2019
Jan 2020

HA2 numbering

Substitutions at HA1 position 156

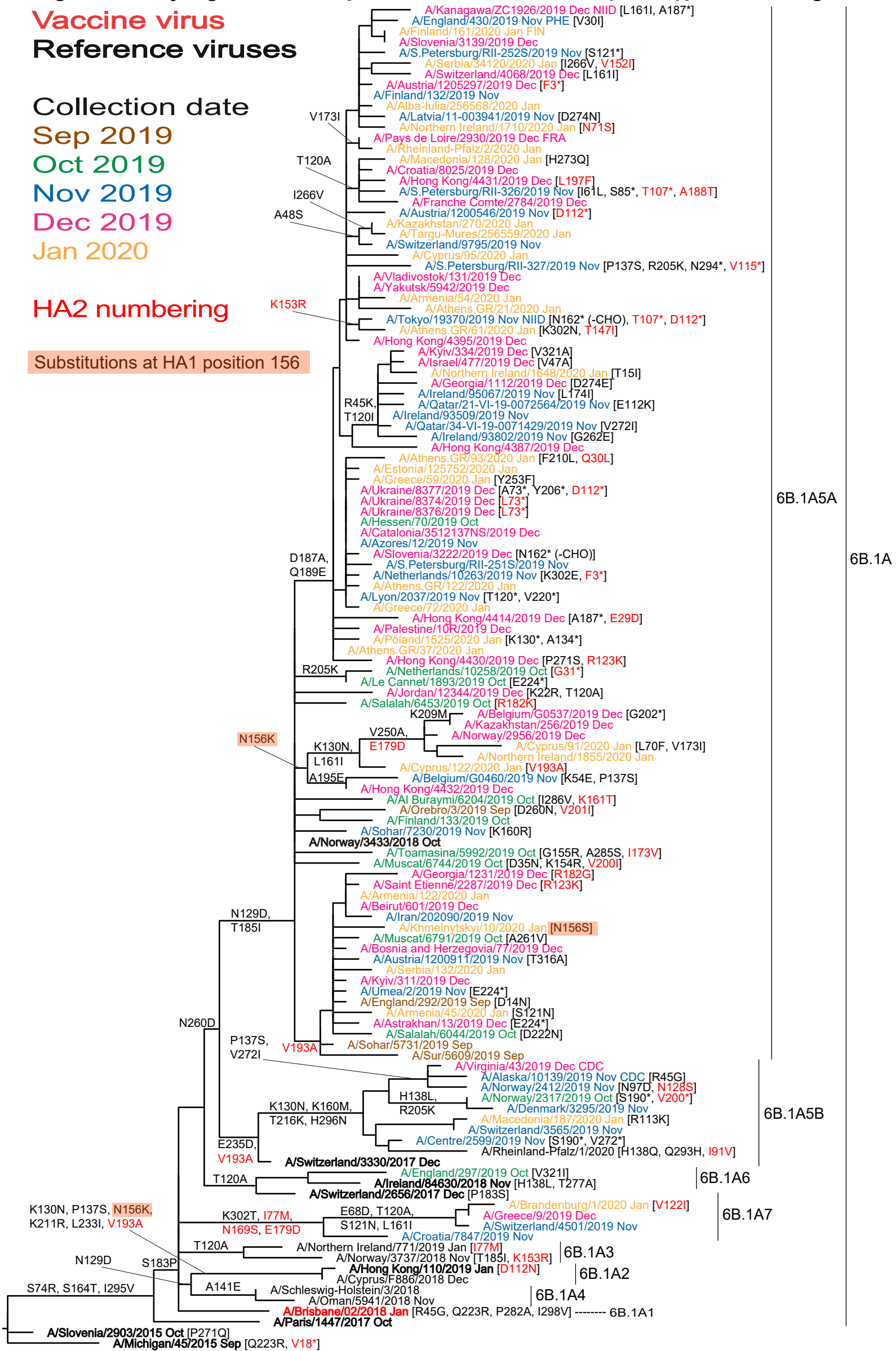


Figure 2. HA protein alignment for A(H1N1)pdm09 viruses

Vaccine virus: Reference virus: 6B Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100
A/Michigan/45/2015 Sep	DTLCIGYHANNST	DTVDVTVLEKNVT	VTHSVNLLEDKHKGLCKLRGVAPLHLGKCN	IAGWILGNPECESL	STASSWSYIVETSNSD	NGT	CYPGDFIN	YEE	1	
A/Paris/1447/2017 Oct							R.		1A	
A/Brisbane/02/2018 Jan				G.			R.		1A1	
A/Hong Kong/110/2019 Jan							R.		1A2	
A/Croatia/7847/2019 Nov							R.		1A7	
A/Brandenburg/1/2020 Jan						D.	R.		1A7	
A/Switzerland/2656/2017 Dec							R.		1A	
A/Ireland/84630/2018 Nov							R.		1A6	
A/Switzerland/3330/2017 Dec							R.		1A5B	
A/Macedonia/172/2020 Jan							R.		1A5B	
A/Denmark/3283/2019 Nov							R.		1A5B	
A/Salalah/6906/2019 Oct							R.		1A5A	
A/Kazakhstan/278/2020 Jan							K.		1A5A	
A/Jordan/30261/2019 Nov							R.		1A5A	
A/Bosnia&Herzegovia/96/2019 De							R.		1A5A	
A/Georgia/1231/2019 Dec							R.		1A5A	
A/Jordan/12191/2019 Dec							R.		1A5A	
A/Norway/3433/2018 Oct							R.		1A5A	
A/Mougins/1891/2019 Oct							R.		1A5A	
A/Cyprus/122/2020 Jan							R.		1A5A	
A/Hong Kong/4388/2019 Dec							R.		1A5A	
A/Athens.GR/145/2020 Jan							R.		1A5A	
A/Hong Kong/4432/2019 Dec							R.		1A5A	
A/Belgium/G0460/2019 Nov					E.		R.		1A5A	
A/Nizwa/5713/2019 Sep							R.		1A5A	
A/Ireland/87362/2019 Oct							R.		1A5A	
A/Greece/59/2020 Jan							R.		1A5A	
A/Ireland/95337/2019 Nov					K.		R.		1A5A	
A/Qatar/21VI190072564/2019 Nov					K.		R.		1A5A	
A/Croatia/8025/2019 Dec							R.		1A5A	
A/Macedonia/183/2020 Jan							R.		1A5A	
A/Kazakhstan/270/2020 Jan					S.		R.		1A5A	
	110	120	130	140	150	160	170	180	190	200
A/Michigan/45/2015 Sep	IREQLSSVSSFERFEIFPKTSSWP	NHDSNKGVTAA	CPHAGAKSFYK	NLIWL	VKKGNSYPK	LN	QSYINDK	KGKVLVLWG	IHHPTTAD	QOSLYQNADAYVF
A/Paris/1447/2017 Oct							T.			
A/Brisbane/02/2018 Jan							T.		P.	
A/Hong Kong/110/2019 Jan			DN.	S.		K.	T.		P.	
A/Croatia/7847/2019 Nov							T.		P.	
A/Brandenburg/1/2020 Jan							I.		P.	
A/Switzerland/2656/2017 Dec		AN.					T.			
A/Ireland/84630/2018 Nov		A.					T.		P.	
A/Switzerland/3330/2017 Dec				L.			T.		P.	
A/Macedonia/172/2020 Jan			K.		N.		M.		P.	
A/Denmark/3283/2019 Nov				N.		L.	M.		P.	R.
A/Salalah/6906/2019 Oct				D.			T.		P.	I.
A/Kazakhstan/278/2020 Jan				D.			T.		P.	I.
A/Jordan/30261/2019 Nov			N.	D.			T.		P.	I.
A/Bosnia&Herzegovia/96/2019 De				D.			T.		P.	I.
A/Georgia/1231/2019 Dec				D.			T.		P.	I.
A/Jordan/12191/2019 Dec						I.	T.		P.	I.
A/Norway/3433/2018 Oct				D.			T.		P.	I.
A/Mougins/1891/2019 Oct				D.			T.		P.	I.
A/Cyprus/122/2020 Jan				DN.			K.	I.	T.	P.
A/Hong Kong/4388/2019 Dec				DN.			K.	I.	T.	P.
A/Athens.GR/145/2020 Jan				DN.			K.	I.	T.	P.
A/Hong Kong/4432/2019 Dec				D.			K.		T.	P.
A/Belgium/G0460/2019 Nov				D.		S.		K.		T.
A/Nizwa/5713/2019 Sep				D.					T.	P.
A/Ireland/87362/2019 Oct				D.					T.	P.
A/Greece/59/2020 Jan				D.					T.	P.
A/Ireland/95337/2019 Nov				I.	D.				T.	P.
A/Qatar/21VI190072564/2019 Nov		K.		I.	D.				T.	P.
A/Croatia/8025/2019 Dec			A.	D.					T.	P.
A/Macedonia/183/2020 Jan			A.	D.					T.	P.
A/Kazakhstan/270/2020 Jan				D.					T.	P.
	210	220	230	240	250	260	270	280	290	300
A/Michigan/45/2015 Sep	VGTSRYSKKFKPEIATR	PKVRDREG	RNMYYWTL	VEPGDKIT	FEATGNL	VVP	RYAFTMERNAGSGII	ISDTPVHDC	NTTCQT	PEGAIN
A/Paris/1447/2017 Oct		Q								V
A/Brisbane/02/2018 Jan									A.	V
A/Hong Kong/110/2019 Jan		R.	Q	I.						V
A/Croatia/7847/2019 Nov										V
A/Brandenburg/1/2020 Jan										V
A/Switzerland/2656/2017 Dec										V
A/Ireland/84630/2018 Nov								A.		V
A/Switzerland/3330/2017 Dec										V
A/Macedonia/172/2020 Jan		K.	Q	D.		D.				V
A/Denmark/3283/2019 Nov			K.	Q	D.					V
A/Salalah/6906/2019 Oct										V
A/Kazakhstan/278/2020 Jan										V
A/Jordan/30261/2019 Nov										V
A/Bosnia&Herzegovia/96/2019 De			NQ							V
A/Georgia/1231/2019 Dec										V
A/Jordan/12191/2019 Dec										V
A/Norway/3433/2018 Oct										V
A/Mougins/1891/2019 Oct		K.	Q							V
A/Cyprus/122/2020 Jan										V
A/Hong Kong/4388/2019 Dec						A.				V
A/Athens.GR/145/2020 Jan			M.	Q		A.				V
A/Hong Kong/4432/2019 Dec										V
A/Belgium/G0460/2019 Nov										V
A/Nizwa/5713/2019 Sep										V
A/Ireland/87362/2019 Oct				N.						V
A/Greece/59/2020 Jan							F.			V
A/Ireland/95337/2019 Nov								D.	E.	V
A/Qatar/21VI190072564/2019 Nov										V
A/Croatia/8025/2019 Dec										V
A/Macedonia/183/2020 Jan								Q.		V
A/Kazakhstan/270/2020 Jan										V

	310	320	330	340	350	360	370	380	390	400
A/Michigan/45/2015 Sep	GKCPKYVKSTKLR	LATGLRNVPSIQ	SRGLFGA	IAGFIEGGWT	GMXDGWYGYHHQ	NEQSGYAADLK	STQNAIDKITN	KNVSVIEKMNT	QFTAVGKEFNHL	
A/Paris/1447/2017 Oct					V					
A/Brisbane/02/2018 Jan					V					
A/Hong Kong/110/2019 Jan					V					
A/Croatia/7847/2019 Nov	T				V					
A/Brandenburg/1/2020 Jan	T				V					
A/Switzerland/2656/2017 Dec					V					
A/Ireland/84630/2018 Nov			X		V					
A/Switzerland/3330/2017 Dec					V					
A/Macedonia/172/2020 Jan					V					
A/Denmark/3283/2019 Nov					V					
A/Salalah/6906/2019 Oct					V					
A/Kazakhstan/278/2020 Jan					V					
A/Jordan/30261/2019 Nov		I			V					
A/Bosnia&Herzegovia/96/2019 De					V					
A/Georgia/1231/2019 Dec					V					
A/Jordan/12191/2019 Dec					V					
A/Norway/3433/2018 Oct					V					
A/Mougins/1891/2019 Oct					V					
A/Cyprus/122/2020 Jan					V					
A/Hong Kong/4388/2019 Dec					V					
A/Athens.GR/145/2020 Jan					V					
A/Hong Kong/4432/2019 Dec					V					
A/Belgium/G0460/2019 Nov					V					
A/Nizwa/5713/2019 Sep					V					
A/Ireland/87362/2019 Oct					V					
A/Greece/59/2020 Jan					V					
A/Ireland/95337/2019 Nov					V					
A/Qatar/21VI190072564/2019 Nov					V					
A/Croatia/8025/2019 Dec					V					
A/Macedonia/183/2020 Jan					V					
A/Kazakhstan/270/2020 Jan					V					

	410	420	430	440	450	460	470	480	490	500
A/Michigan/45/2015 Sep	EKRIENLNKKVDD	GFLDIWTYNAB	LLVLLNERTLD	YHDSNVKNLYE	KVRNQLKNNAKE	IGNGCFEFYHK	DNTCMESVKN	NGTYDPKYSEE	AKLNREKI	
A/Paris/1447/2017 Oct										
A/Brisbane/02/2018 Jan										
A/Hong Kong/110/2019 Jan				N						
A/Croatia/7847/2019 Nov	M								S	
A/Brandenburg/1/2020 Jan	M				I				S	
A/Switzerland/2656/2017 Dec										
A/Ireland/84630/2018 Nov										
A/Switzerland/3330/2017 Dec										
A/Macedonia/172/2020 Jan										
A/Denmark/3283/2019 Nov										
A/Salalah/6906/2019 Oct										
A/Kazakhstan/278/2020 Jan						N				
A/Jordan/30261/2019 Nov										
A/Bosnia&Herzegovia/96/2019 De										
A/Georgia/1231/2019 Dec										
A/Jordan/12191/2019 Dec										
A/Norway/3433/2018 Oct										
A/Mougins/1891/2019 Oct										
A/Cyprus/122/2020 Jan										
A/Hong Kong/4388/2019 Dec										
A/Athens.GR/145/2020 Jan										
A/Hong Kong/4432/2019 Dec										
A/Belgium/G0460/2019 Nov										
A/Nizwa/5713/2019 Sep										
A/Ireland/87362/2019 Oct										
A/Greece/59/2020 Jan										
A/Ireland/95337/2019 Nov										
A/Qatar/21VI190072564/2019 Nov										
A/Croatia/8025/2019 Dec										
A/Macedonia/183/2020 Jan										
A/Kazakhstan/270/2020 Jan										

	510	520	530	540
A/Michigan/45/2015 Sep	DGVKLESTRIYQ	ILAIYSTVASS	LVLVVSIGAI	SFWMCSNGSLQCRICI
A/Paris/1447/2017 Oct				
A/Brisbane/02/2018 Jan				
A/Hong Kong/110/2019 Jan		A		
A/Croatia/7847/2019 Nov	D			
A/Brandenburg/1/2020 Jan	D			
A/Switzerland/2656/2017 Dec				
A/Ireland/84630/2018 Nov				
A/Switzerland/3330/2017 Dec		A		
A/Macedonia/172/2020 Jan		A		
A/Denmark/3283/2019 Nov		A	N	
A/Salalah/6906/2019 Oct		A		
A/Kazakhstan/278/2020 Jan		A		
A/Jordan/30261/2019 Nov		A		
A/Bosnia&Herzegovia/96/2019 De		A		
A/Georgia/1231/2019 Dec	G	A		
A/Jordan/12191/2019 Dec		A		
A/Norway/3433/2018 Oct				
A/Mougins/1891/2019 Oct				
A/Cyprus/122/2020 Jan		A		
A/Hong Kong/4388/2019 Dec	D			
A/Athens.GR/145/2020 Jan	D			
A/Hong Kong/4432/2019 Dec				
A/Belgium/G0460/2019 Nov				
A/Nizwa/5713/2019 Sep				
A/Ireland/87362/2019 Oct				
A/Greece/59/2020 Jan				
A/Ireland/95337/2019 Nov				
A/Qatar/21VI190072564/2019 Nov				
A/Croatia/8025/2019 Dec				
A/Macedonia/183/2020 Jan				
A/Kazakhstan/270/2020 Jan				

Vaccine virus: Reference virus: 6B Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 3a. HA amino acid substitutions defining H1N1pdm09 subgroup 6B.1A5A viruses

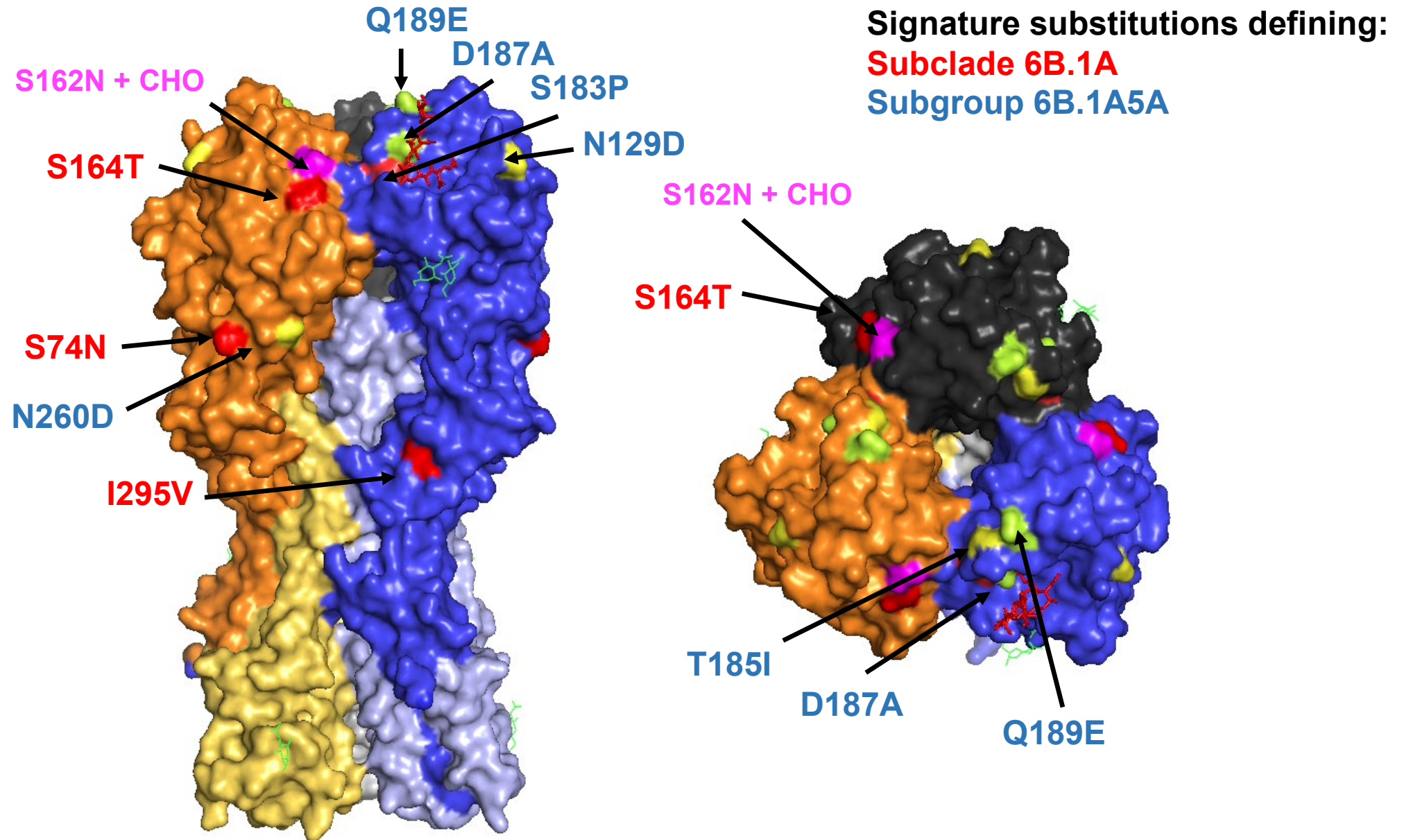


Figure 3b. HA amino acid substitutions defining H1N1pdm09 subgroup 6B.1A5B viruses

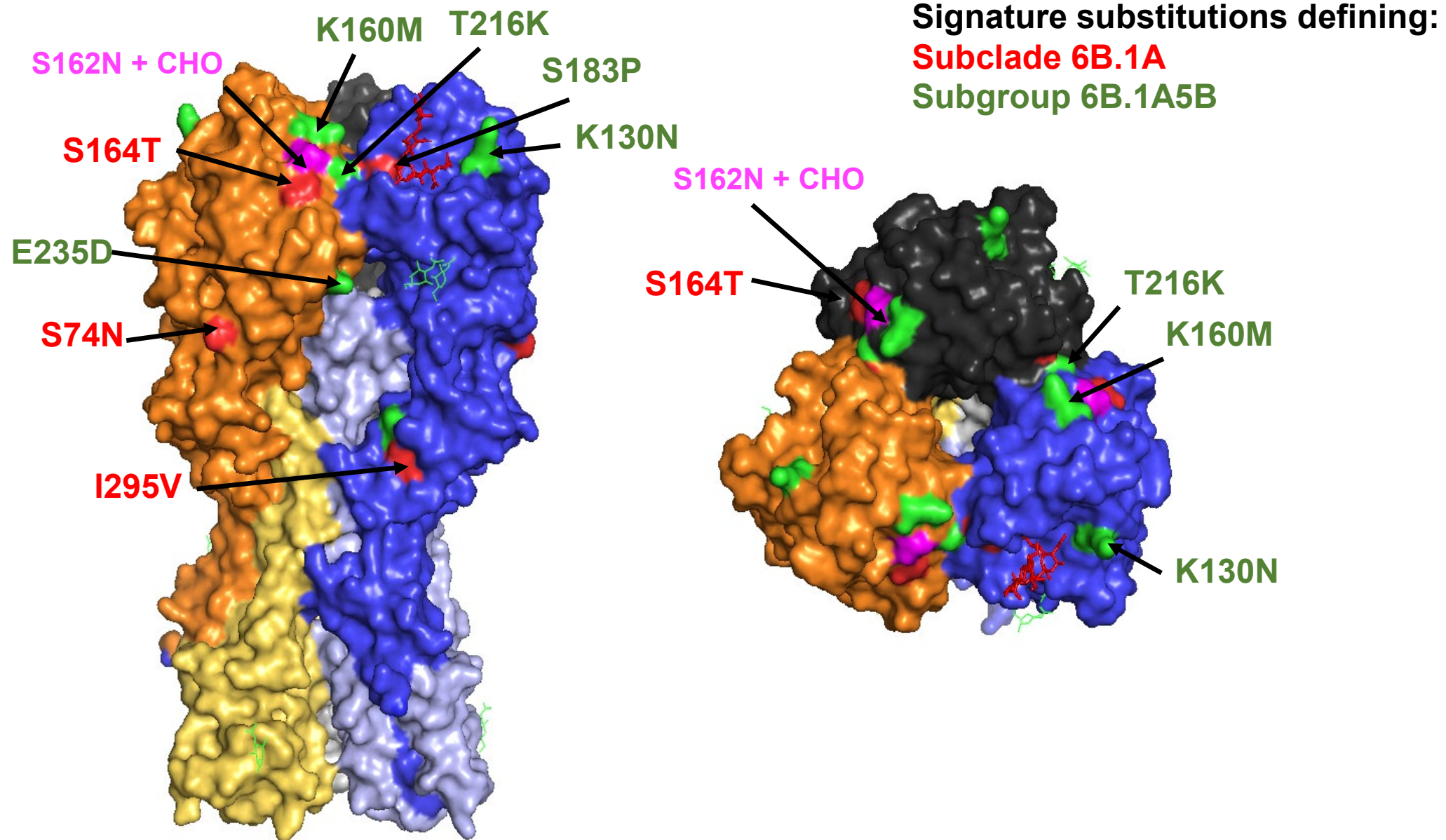
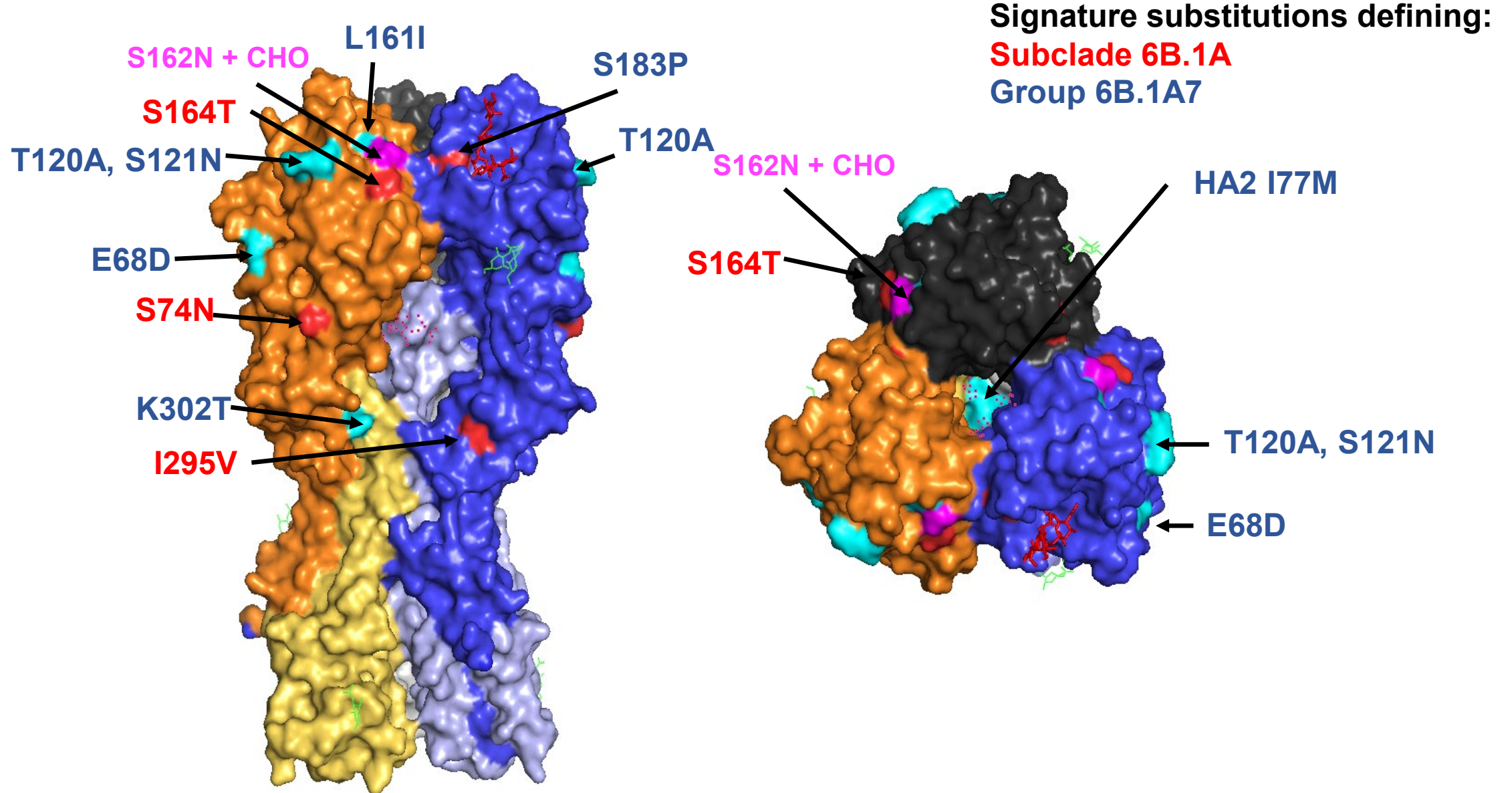


Figure 3c. HA amino acid substitutions defining H1N1pdm09 group 6B.1A7 viruses



Vaccine virus: **Reference virus:** 6B **Genetic group:** Potential N-linked glycosylation site



	310	320	330	340	350	360	370	380	390	400
A/Michigan/45/2015 Sep	RPWVSF	NQ	LEYQ	MGYIC	SGVFG	DNPR	PNDKT	GSCGP	VSSNG	ANGV
A/Paris/1447/2017 Oct										
A/Brisbane/02/2018 Jan										
A/Hong Kong/110/2019 Jan										
A/Croatia/7847/2019 Nov			I							
A/Brandenburg/1/2020 Jan			I		L					
A/Switzerland/2656/2017 Dec										
A/Ireland/84630/2018 Nov								T		
A/Switzerland/3330/2017 Dec										
A/Macedonia/172/2020 Jan										K
A/Denmark/3283/2019 Nov										
A/Salalah/6906/2019 Oct										K
A/Kazakhstan/278/2020 Jan										K
A/Jordan/30261/2019 Nov										K
A/Bosnia&Herzegovia/96/2019 De										K
A/Georgia/1231/2019 Dec										K
A/Jordan/12191/2019 Dec										K
A/Norway/3433/2018 Oct										
A/Mougins/1891/2019 Oct										K
A/Cyprus/122/2020 Jan										K
A/Hong Kong/4388/2019 Dec										K
A/Athens.GR/145/2020 Jan										K
A/Hong Kong/4432/2019 Dec										K
A/Belgium/G0460/2019 Nov										K
A/Nizwa/5713/2019 Sep										K
A/Ireland/87362/2019 Oct										K
A/Greece/59/2020 Jan										K
A/Ireland/95337/2019 Nov						S				K
A/Qatar/21VII90072564/2019 Nov										K
A/Croatia/8025/2019 Dec										K
A/Macedonia/183/2020 Jan										K
A/Kazakhstan/270/2020 Jan										K
	410	420	430	440	450	460				
A/Michigan/45/2015 Sep	GYSGSF	VQHP	ELTGL	DCIRP	CFWV	ELIR	GRPE	ENTIT	WTS	SGSS
A/Paris/1447/2017 Oct										D
A/Brisbane/02/2018 Jan										D
A/Hong Kong/110/2019 Jan										D
A/Croatia/7847/2019 Nov										D
A/Brandenburg/1/2020 Jan										D
A/Switzerland/2656/2017 Dec				N						D
A/Ireland/84630/2018 Nov				N						D
A/Switzerland/3330/2017 Dec				N						D
A/Macedonia/172/2020 Jan				N						D
A/Denmark/3283/2019 Nov										D
A/Salalah/6906/2019 Oct				N						D
A/Kazakhstan/278/2020 Jan				N						D
A/Jordan/30261/2019 Nov				N						D
A/Bosnia&Herzegovia/96/2019 De				N						D
A/Georgia/1231/2019 Dec				N						D
A/Jordan/12191/2019 Dec				N						D
A/Norway/3433/2018 Oct				N						D
A/Mougins/1891/2019 Oct				N						D
A/Cyprus/122/2020 Jan										D
A/Hong Kong/4388/2019 Dec				N						D
A/Athens.GR/145/2020 Jan				N						D
A/Hong Kong/4432/2019 Dec				N						D
A/Belgium/G0460/2019 Nov				N						D
A/Nizwa/5713/2019 Sep				N						D
A/Ireland/87362/2019 Oct				N						D
A/Greece/59/2020 Jan				N						D
A/Ireland/95337/2019 Nov				N						D
A/Qatar/21VII90072564/2019 Nov				N						D
A/Croatia/8025/2019 Dec				N						D
A/Macedonia/183/2020 Jan				N						D
A/Kazakhstan/270/2020 Jan				N						D

Vaccine virus: Reference virus: 6B Genetic group: Potential N-linked glycosylation site

Influenza A(H3N2) virus analyses.

Influenza A(H3N2) viruses collected since 2019-08-31 have been received from NICs in 45 countries or administrative regions in four WHO Regions: African (3: Ghana, Mauritius and Senegal), Eastern Mediterranean (5: Iran, Lebanon, Oman, Palestine and Qatar), Western Pacific (1: Hong Kong SAR) and European (36 countries) (**Table 1**).

Influenza A(H3N2) viruses have become increasingly difficult to characterise antigenically by HI assay due to either the variable agglutination of red blood cells (RBCs) from guinea pigs, turkeys and humans or the loss of the ability of viruses to agglutinate any of these RBCs. HI analyses performed since our September 2019 VCM report for viruses able to agglutinate RBCs are shown in **Tables 7-1 to 7-12** and virus neutralisation by plaque reduction neutralisation assay (PRNA) was done on a selected subset of 120 test viruses with collection dates after 2019-08-31 (**Tables 8-1 to 8-6**).

HI analyses.

A total of 146 viruses with collection dates after 2019-08-31 could be analysed by HI assay and summaries of the HI results are shown in **Tables 3** and **6**. Individual HI results are shown in **Tables 7-1 to 7-12**. Test viruses in each table are sorted by date of collection and genetic clade/subclade for those viruses that have been sequenced and those below the red horizontal lines in the tables are viruses with collection dates after 2019-08-31 (**Tables 7-4** and **7-6 to 7-12**). The viruses sequenced and analysed by HI with collection dates after 2019-08-31 fell into clade 3C.3a (44%) or subclades 3C.2a1b+131K (16%), 3C.2a1b+135K-A (16%) and 3C.2a1b+135K-B (viruses in the 135K subclade with the HA1 substitutions S137F, A138S and F193S) (24%) of clade 3C.2a.

A summary of the HI results for viruses collected since 2019-08-31 with antisera raised against the egg-propagated cultivars of A/South Australia/34/2019 and A/Kansas/14/2017 (NYMC X-327) (the most recently recommended vaccine viruses) and cell culture-propagated A/Hong Kong/5738/2014 (a 3C.2a clade virus that is A/Hong Kong/4801/2014-like) and A/Kansas/14/2017 (a clade 3C.3a virus) is shown in **Table 6**. A more detailed summary for the whole panel of antisera is shown in **Table 7-13**.

Viruses in clade 3C.3a:

- antisera raised against clade 3C.3a viruses, cell culture-propagated A/Kansas/14/2017 and A/England/538/2018, recognised viruses in clade 3C.3a very well at titres within 2-fold of the homologous titres, and they were recognised at similar titres by the antiserum raised against NYMC X-327 (derived from egg-propagated A/Kansas/14/2017) although the titre of the

- antiserum for the homologous virus was higher and recognised only 23% of these test viruses at titres within 4-fold of the high homologous titre;
- the antiserum raised against cell culture-propagated A/Hong Kong/5738/2014 recognised 98% of the 3C.3a viruses at titres within 4-fold of the homologous titre of the antisera, 23% within 2-fold, while antisera raised against other cell culture-propagated reference viruses in clade 3C.2a did so less well:
 - the antisera raised against A/Bretagne/1413/2017 (subclade 3C.2a2) and A/Hong Kong/2669/2019 (subclade 3C.2a1b+135K-B) recognised 15% and none, respectively, of the clade 3C.3a test viruses at titres within 4-fold of the homologous titre;
 - antisera raised against cell culture-propagated A/La Rioja/2202/2018 and A/Norway/3275/2018, with HA genes in 3C.2a1b+T135K-A and 3C.2a1b+T131K subgroups, respectively, had no homologous titres since the viruses are unable to agglutinate red blood cells but recognised all the clade 3C.3a viruses at titres of <160;
 - the antiserum raised against egg-propagated A/Singapore/INFIMH-16-0019/2016 (subclade 3C.2a1) recognised 84% of the clade 3C.3a viruses at titres within 4-fold of the homologous titre of the antiserum and 28% within 2-fold;
 - the antiserum raised against egg-propagated A/Norway/2279/2019 (subclade 3C.2a1b+131K) recognised only 18% of the clade 3C.3a viruses at titres within 4-fold of the homologous titre of the antiserum and none within 2-fold;
 - the antisera raised against the egg-propagated cultivars of A/South Australia/34/2019 (subclade 3C.2a1+131K) and A/Hong Kong/2671/2019 (subclade 3C.2a1b+135K-B) recognised none of the test viruses at titres within 4-fold of the homologous titres of the antisera.

Viruses in subclade 3C.2a1b with the HA1 T131K substitution:

- antisera raised against the cell culture-propagated clade 3C.3a viruses A/Kansas/14/2017 and A/England/538/2018 recognised 44% and 30%, respectively, of viruses in subclade 3C.2a1b+T131K at titres within 4-fold of the homologous titres, and 9% and none, respectively, at titres within 2-fold. All were recognised poorly by the antiserum raised against NYMC X-327 (derived from egg-propagated A/Kansas/14/2017);
- antiserum raised against A/Bretagne/1413/2017 (subclade 3C.2a2) recognised 56% of the test viruses at titres within 4-fold of the homologous titre, 27% within 2-fold. Only two subclade 3C.2a1b+131K viruses were tested with the antiserum raised against A/Hong Kong/2669/2019 (subclade 3C.2a1b+135K-B) of which one only was recognised well;
- all but one of the 23 test viruses in subclade 3C.2a1b+T131K were recognised at titres of ≥ 160 by the antiserum raised against cell culture-propagated A/Norway/3275/2018, in the same group, but only 13 of 23 (56%) were recognised at titres of ≥ 160 by the antiserum raised against A/La Rioja/2202/2018;

- all 23 test viruses in subclade 3C.2a1b+T131K were recognised at titres within 4-fold, and 87% within 2-fold, of the homologous titre by the antiserum raised against egg-propagated A/Singapore/INFIMH-16-0019/2016;
- the antiserum raised against the vaccine virus egg-propagated A/South Australia/34/2019 (subclade 3C.2a1b+131K) recognised only 48% of these test viruses at titres within 4-fold of the higher homologous titre, none within 2-fold, and the antiserum raised against egg-propagated A/Norway/2279/2019 (also subclade 3C.2a1b+131K) recognised the single test virus with which it was tested at a titre 4-fold lower than the homologous titre;
- the antiserum raised against egg-propagated A/Hong Kong/2671/2019, subclade 3C.2a1b+T135K-B, poorly recognised the test viruses in subclade 3C.2a1b+T131K.

Viruses in subclade 3C.2a1b with the T135K substitution (3C.2a1b+T135K-A):

- antisera raised against the cell culture-propagated clade 3C.3a viruses A/Kansas/14/2017 and A/England/538/2018 recognised 66% and 52%, respectively, of viruses of subclade 3C.2a1b+T135K-A at titres within 4-fold of the homologous titres but all were recognised less well by the antiserum raised against NYMC X-327 (derived from egg-propagated A/Kansas/14/2017);
- the antiserum raised against A/Bretagne/1413/2017 (subclade 3C.2a2) recognised only a single test virus of the seven tested at a titre within 4-fold of the homologous titre;
- the antiserum raised against A/Hong Kong/2669/2019 (subclade 3C.2a1b+135K-B) recognised all seven viruses with which it was tested at titres within 2-fold of the homologous titre;
- 21 of 23 (91%) test viruses in subclade 3C.2a1b+T135K-A were recognised at titres of ≥ 160 by the antiserum raised against cell culture-propagated A/La Rioja/2202/2018, in the same genetic group;
- the antiserum raised against cell culture-propagated A/Norway/3275/2018 (subclade 3C.2a1b+131K) also recognised the majority (83%) of test viruses at titres of ≥ 160 ;
- all 23 test viruses were recognised at titres within 4-fold of the homologous titre by the antiserum raised against egg-propagated A/Singapore/INFIMH-16-0019/2016, 61% within 2-fold;
- the antisera raised against the vaccine virus egg-propagated A/South Australia/34/2019 (subclade 3C.2a1b+T131K), and egg-propagated A/Hong Kong/2671/2019 (subclade 3C.2a1b+T135K-B) recognised none of these test viruses at titres within 4-fold of the homologous titres of the antisera. However, the antiserum raised against egg-propagated A/Norway/2279/2019 (subclade 3C.2a1b+131K) recognised five of the six test viruses (83%) at titres within 4-fold of the titre of the antiserum for the homologous virus.

Viruses in subclade 3C.2a1b+T135K with the HA1 substitutions S137F, A138S and F193S (3C.2a1b+T135K-B):

- antisera raised against the cell culture-propagated clade 3C.3a viruses A/Kansas/14/2017 and A/England/538/2018 recognised 18% and 12%, respectively, of viruses in subclade 3C.2a1b+T135K-B at titres within 4-fold of the homologous titres but all were recognised poorly by the antiserum raised against NYMC X-327 (derived from egg-propagated A/Kansas/14/2017);
- the antiserum raised against A/Bretagne/1413/2017 (subclade 3C.2a2) recognised none of the test viruses at a titre within 4-fold of the homologous titre;
- the antiserum raised against A/Hong Kong/2669/2019 (subclade 3C.2a1b+135K-B) recognised all seven viruses with which it was tested at titres within 2-fold of the homologous titre;
- the antisera raised against cell culture-propagated A/Norway/3275/2018 (subclade 3C.2a1b+T131K) and against A/La Rioja/2202/2018 (subclade 3C.2a1b+T135K-A) recognised 65% and 38%, respectively, at titres of ≥ 160 ;
- the antiserum raised against egg-propagated A/Singapore/INFIMH-16-0019/2016 recognised 85% of test viruses in this group at titres within 4-fold of the homologous titre, 38% within 2-fold;
- the antiserum raised against egg-propagated A/Hong Kong/2671/2019 (subclade 3C.2a1b+T135K-B) recognised 31% of the test viruses at titres within 4-fold of the homologous titre, but only 6% within 2-fold;
- the antiserum raised against egg-propagated A/Norway/2279/2019 (subclade 3C.2a1b+131K) recognised two of the seven test viruses (29%) at titres within 4-fold of the titre of the antiserum for the homologous virus, but none within 2-fold;
- the antiserum raised against the vaccine virus egg-propagated A/South Australia/34/2019 recognised none of these test viruses at a titre within 4-fold of the homologous titre of this antiserum.

Virus Neutralisation results

Viruses neutralisation results by PRNA are shown in **Tables 8-1 to 8-6** and a summary is presented in **Table 8-7**. Of the viruses analysed 14% were in clade 3C.3a, 62% were in subclade 3C.2a1b+T131K, 9.2% were in subclade 3C.2a1b+T135K-A and 15% were in subclade 3C.2a1b+T135K-B. A phylogenetic tree for the HA genes of the viruses analysed by PNRA is shown in **Figure 6a**.

Viruses in clade 3C.3a:

- the antiserum raised against the cell culture-propagated cultivar of A/Kansas/14/2017 recognised all test viruses within 4-fold of the homologous titre, and 76% within 2-fold, while the antiserum raised against NYMC X-327 (derived from egg-propagated A/Kansas/14/2017) recognised the test viruses at similar titres although the titre of the antiserum for the homologous virus was higher and so recognised none of the test viruses at a titre within 4-fold of that for the homologous virus;
- all antisera raised against clade 3C.2a viruses recognised these test viruses poorly.

Viruses in subclade 3C.2a1b with the T131K substitution:

- antisera raised against egg-propagated and cell culture-propagated A/Kansas/14/2007 recognised all but a single virus in this subclade poorly;
- the antiserum raised against cell culture-propagated A/Norway/3275/2018, in the same genetic group, recognised the large majority (95%) of test viruses in subclade 3C.2a1b+T131K at titres within 4-fold of the homologous titre of the antiserum, 88% within 2-fold;
- the antiserum raised against cell culture-propagated A/La Rioja/2202/2018 (subclade 3C.2a1b+T135K-A) recognised 63% of the 3C.2a1b+T131K test viruses at titres within 4-fold of the homologous titre but only 9% at titres within 2-fold;
- the antiserum raised against cell culture-propagated A/Hong Kong/2671/2019 (subclade 3C.2a1b+T135K-B) recognised only 41% of test viruses with the T131K substitution at titres within 4-fold of the homologous titre and only 4% at titres within 2-fold. An antiserum raised against the egg-propagated cultivar of A/Hong Kong/2671/2019, used in **Table 8-6**, recognised none of the viruses tested;
- antisera raised against egg-propagated viruses in the 3C.2a1b+T131K group, A/South Australia/34/2019 and A/Christchurch/515/2019, recognised test viruses in this genetic group less well compared to the high homologous titres of the antisera;
 - although the antiserum raised against A/Christchurch/515/2019 recognised, 51% within 4-fold of the homologous titre (but only 4% within 2-fold), the antiserum raised against A/South Australia/34/2019 recognised only a single virus (1.4%) at a titre within 4-fold of the homologous titre of the antiserum;
 - however, the antiserum raised against A/Christchurch/515/2019 recognised a substantial proportion of the test viruses (98% of tested viruses) at titres ≥ 120 but the antiserum raised against A/South Australia/34/2019 recognised only 57% at titres ≥ 120 .

Viruses in subclade 3C.2a1b with the T135K substitution (3C.2a1b+T135K-A):

- antisera raised against egg-propagated and cell culture-propagated A/Kansas/14/2007 recognised all viruses in this subclade poorly;
- the antiserum raised against cell culture-propagated A/Norway/3275/2018 (subclade 3C.2a1b+T131K) recognised only 2 of 11 (18%) 3C.2a1b+T135K-A test viruses at titres within 4-fold of the homologous titre of the antiserum, none within 2-fold;
- the antiserum raised against cell culture-propagated A/La Rioja/2202/2018, (subclade 3C.2a1b+T135K-A) recognised 4 of 5 (80%) 3C.2a1b+T135K-A test viruses at titres within 2-fold of the homologous titre;
- the antiserum raised against cell culture-propagated A/Hong Kong/2671/2019 (subgroup 3C.2a1b+T135-B) recognised 8 of 11 (73%) 3C.2a1b+T135K-A test viruses at titres within 4-fold of the homologous titre and 3 of 11 (27%) at titres within 2-fold. An antiserum raised against the egg-propagated cultivar of

A/Hong Kong/2671/2019, used in **Table 8-6**, recognised none of the viruses tested;

- antisera raised against egg-propagated viruses in subclade 3C.2a1b+T131K, A/South Australia/34/2019 and A/Christchurch/515/2019, recognised only a single test virus (1 of 11) at a titre within 4-fold of the homologous titres but the antiserum raised against A/Christchurch/515/2019 recognised all the test viruses at titres ≥ 120 .

Viruses in subclade 3C.2a1b with the T135K substitution with substitutions S137F, A138S and F193S (subclade 3C.2a1b+T135K-B):

- antisera raised against egg-propagated and cell culture-propagated A/Kansas/14/2007 recognised all viruses in this subclade poorly with only one of 18 being recognised at a titre within 4-fold of the homologous titre;
- the antiserum raised against cell culture-propagated A/Norway/3275/2018 (subclade 3C.2a1b+T131K) recognised none of the 18 3C.2a1b+T135K-B test viruses at titres within 4-fold of the homologous titre of the antiserum;
- the antiserum raised against cell culture-propagated A/La Rioja/2202/2018, (subclade 3C.2a1b + T135K-A) also recognised only one of 18 (5.6%) 3C.2a1b+T135K-B test viruses at titres within 4-fold of the homologous titre;
- the antiserum raised against cell culture-propagated A/Hong Kong/2671/2019 (subclade 3C.2a1b + T135-B) recognised all 18 test viruses at titres within 2-fold of the homologous titre of the antiserum;
- antisera raised against egg-propagated viruses in subclade 3C.2a1b+T131K, A/South Australia/34/2019 and A/Christchurch/515/2019, recognised none of the test viruses in this genetic group at a titre within 4-fold of the homologous titre; although A/Christchurch/515/2019 recognised all but one of the test viruses (95%) at titres ≥ 120 .

Genetic analyses.

Phylogenetic analysis of the HA genes of representative recently circulating A(H3N2) viruses analysed at the Francis Crick Institute and other representative viruses is shown in **Figure 6b**. As described above Figure 6a is a phylogenetic tree of the HA genes of those viruses analysed by PRNA. A phylogenetic tree of the HA genes of all viruses with collection dates after 2019-08-31 analysed genetically at the Francis Crick Institute is shown in the Annex, **Figure A-3**. The HA genes fell within clade 3C.2a, primarily subclade 3C.2a1b, and in clade 3C.3a. Of those with collection dates after 2019-08-31 41% had HA genes in subclade 3C.2a1b+T131K, 33% were in clade 3C.3a, 11% were in subclade 3C.2a1b+T131K-A and 14% were in subclade 3C.2a1b+T131K-B (with the A137T, S138A, and F193S HA1 substitutions).

Major clades/subclades

- Clade **3C.2a** viruses differ from clade **3C.3a** viruses with substitutions **N128T** (resulting in the gain of a potential glycosylation site), **S138A**, **G142R**, **N144S** (resulting in the loss of a potential glycosylation site), **S159Y**, **K160T** (resulting in the gain of a potential glycosylation site at residue **158**), **Q311H** and **R326K** in **HA1**;
- Subclade **3C.2a1b** viruses, with the substitutions **E62G**, **K92R**, **N121K**, a reversion **R142G**, **N171K** and **H311Q** in **HA1** and **I77V** and **G155E** in **HA2**, have predominated. Within this subclade there are two main genetic groups defined by:
 - **T128A** and **T135K** in **HA1** both resulting in the loss of glycosylation motifs (e.g. A/La Rioja/2202/2018), group **3C.2a1b+135K-A**
 - Many viruses in this group have the additional **HA1** substitutions **S137F**, **A138S** and **F193S** (e.g. A/Hong Kong/2671/2019), group **3C.2a1b+135K-B**
 - Another group has the substitution **S198P** in **HA1**.
 - **T131K** in **HA1** and **V200I** in **HA2** (e.g. A/Norway/3275/2018). Within this subclade viruses fall into two subgroups:
 - The HA of viruses in one subgroup within this subclade have the substitutions **K83E** and **Y94N** in **HA1** and **I193M** in **HA2**
 - The HA of another group has the substitutions **Q197R** and **S219F** in **HA1**, many with **K207R**, and **V18M** and **E155G** in **HA2**.
- Clade **3C.3a** viruses have acquired the substitutions **S91N**, **N144K** (resulting in the loss of a glycosylation site) **P193S** and **K326R** in **HA1** (e.g. A/Kansas/14/2017) with most sharing **M17L**, **I149M** and **A201V** in **HA2**.

No viruses with collection dates after 2019-08-31 with HA in other subclades of 3C.2a have been received.

Figure 7 shows an HA amino acid sequence alignment for a representative selection of the viruses used to generate the phylogeny. The vaccine viruses A/Kansas/14/2017 (3C.3a) and A/South Australia/34/2019 (subclade 3C.2a1b+T131K) are shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the HA1/HA2 proteolytic cleavage site is highlighted. The clade/subclade to which each of the HA sequences belongs is also shown.

Figure 8a shows the location on a 2005 H3 HA structure of amino acid substitutions that define recent subclades of viruses compared with the vaccine virus A/Singapore/INFIMH-16-0019/2016. **Figure 8b** shows the location of HA amino acid substitutions found in recently circulating clade 3C.3a viruses compared with A/Switzerland/9715293/2013.

Figure 9 shows a NA phylogenetic tree for representative viruses with their HA genetic groups indicated. A phylogenetic tree of the HA genes of all viruses with collection dates after 2019-08-31 analysed genetically at the Francis Crick Institute is shown in the Annex, **Figure A-4**. **Figure 10** shows a NA amino acid sequence alignment for a representative selection of these viruses. As above, the vaccine viruses are shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the clade/subclade to which the corresponding HA gene sequences belong is shown.

The NA gene sequences of recent viruses, with HA genes in genetic clusters within clade 3C.2a and its subclades, show generally similar subclade clustering. There are several subgroups defined by amino acid substitutions in NA:

- Many recently collected viruses with HA genes in subclade 3C.2a1b share the amino acid substitutions **P126L**, **K220N**, **V303I** and **N329S** (losing a potential glycosylation motif) and **E344K** in the NA and many also have substitution **S315R**.
- Recent HA clade **3C.3a** viruses have NA genes evolved from clade 3C.2a viruses and have NA substitutions **K75R**, **L140I**, **V149A**, **Y155H**, **N329T** (resulting in the loss of a potential glycosylation site) and **E344K**. Many share the additional substitutions **I57M**, **N161S** and **P386S**.

Antiviral resistance

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on 237 viruses with collection dates after 2019-08-31. All but one showed normal inhibition by oseltamivir and zanamivir (**Table 13**). A/Latvia/2326/2019 showed reduced inhibition (RI) by zanamivir and the NA carried the amino acid substitution T148I resulting in the loss of a glycosylation motif.

Conclusion.

Recently circulating viruses received have fallen in clade 3C.3a and subclade 3C.2a1b. Viruses in clade 3C.3a were recognised well in HI assays and PRNA by antisera raised against A/Kansas/14/2017-like viruses but not well by antisera raised against reference viruses in subclade 3C.2a1b. Viruses in subclade 3C.2a1b fell into three main genetic groups (3C.2a1b+T131K, 3C.2a1b+T135K-A and 3C.2a1b+T135KB) and all were recognised poorly by antisera raised against clade 3C.3a viruses. Those in subclade 3C.2a1b+T131K were recognised best by antisera raised against reference viruses in the same group; likewise, those in subclade 3C.2a1b+T1315-A were recognised best by antisera raised against reference viruses in the same group, and those in subclade 3C.2a1b+T1315-B were recognised best by antisera raised against viruses in the same group. Most H3N2 viruses retained susceptibility to the neuraminidase inhibitors.

Table 6a. Summary of influenza A(H3N2) viruses collected after 2019-08-31, analysed by HI

MONTH	H3N2														
	Country	Number received ¹	Number propagated ²	3C.2a: Cell-propagated A/Hong Kong/5738/2014			3C.2a1b+T131K: Egg-propagated A/South Australia/34/2019			3C.3a:Egg-propagated NYMC X-327 A/Kansas/14/2017			3C.3a:Cell-propagated A/Kansas/14/2017		
				≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴	≤2 fold ⁵	4 fold ⁵	≥8 fold ⁵	≤2 fold ⁶	4 fold ⁶	≥8 fold ⁶
2019															
SEPTEMBER															
Czech Republic	1	1	1				1				1			1	
Finland	1	1		1				1			1	1			
France	3	2	2					2			2			2	
Norway	4	1		1				1		1		1			
Oman	2	1	1					1			1			1	
Palestine	1	0													
Romania	1	0													
Senegal	9	5	5				1	4			5	2	1	2	
Sweden	1	1	1				1				1			1	
OCTOBER															
Denmark	1	1	1					1			1		1		
Finland	1	1		1				1			1	1			
Germany	4	2	2				1	1			2		2		
Ghana	1	0													
Greece	1	0													
Iceland	8	4	2	2			1	3			4		1	3	
Iran	4	0													
Ireland	9	0													
Lebanon	1	0													
Netherlands	2	0													
Norway	19	3	1	2				3		1	2	1	1	1	
Oman	24	6	6					6			6			6	
Portugal	1	1	1					1			1			1	
Qatar	11	0													
Sweden	2	2	2				2				2		1	1	
UK England	17	11	4	7				11		5	6	10		1	
NOVEMBER															
Austria	1	0													
Belgium	1	0													
Czech Republic	2	0													
Denmark	6	3	1	2				3			3			3	
France	4	3	2	1				3			3	1	2		
Germany	3	0													
Ghana	10	3	3					3			3	1	1	1	
Greece	1	0													
Hong Kong	5	5	4	1				5			5	1	4		
Iceland	2	0													
Iran	3	0													
Ireland	22	3	2	1				3		1	2	3			
Italy	3	0													
Kyrgyzstan	5	0													
Latvia	3	3	3				1	2			3		1	2	
Mauritius	1	0													
Netherlands	1	0													
Norway	9	3	1	2				3		1	2	1		2	
Oman	2	1	1					1			1			1	
Palestine	4	0													
Portugal	2	0													
Qatar	10	1		1				1			1			1	
Russia	2	1	1					1			1		1		
Spain	2	0													
Sweden	1	0													
Tajikistan	1	0													
UK England	16	1	1					1			1	1			

Table 6b. Summary of influenza A(H3N2) viruses collected after 2019-08-31, analysed by HI

MONTH	H3N2															
Country	Number received ¹	Number propagated ²	3C.2a: Cell-propagated A/Hong Kong/5738/2014			3C.2a1b+T131K: Egg-propagated A/South Australia/34/2019			3C.3a:Egg-propagated NYMC X-327 A/Kansas/14/2017			3C.3a:Cell-propagated A/Kansas/14/2017				
			≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴	≤2 fold ⁵	4 fold ⁵	≥8 fold ⁵	≤2 fold ⁶	4 fold ⁶	≥8 fold ⁶		
DECEMBER																
Austria	9	7	3	3	1			7		3	4	7				
Belgium	9	5	2	3				5		1	4	5				
Bulgaria	1	0														
Croatia	1	0														
Cyprus	1	0														
Czech Republi	2	1			1			1			1	1				
France	7	3	2	1				3		2	1	3				
Germany	6	4	2	2				4			4	2	1	1		
Georgia	10	1	1					1			1	1				
Greece	1	0														
Hong Kong	6	6	3	3				6			6		1	5		
Iceland	2	0														
Israel	1	1	1					1			1			1		
Italy	6	2	1	1				2			2	1	1			
Kazakhstan	4	0														
Kyrgyzstan	21	6	6					6			6		1	5		
Latvia	1	0														
Lebanon	1	0														
Lithuania	12	2		2				2			2	2				
Montenegro	1	0														
Netherlands	9	2		2				2			2	2				
Norway	1	0														
Poland	2	1			1			1			1	1				
Palestine	5	0														
Portugal	3	2	1	1				2			2	1		1		
Russia	2	2	1	1			1	1			2	1	1			
Slovenia	3	3	2	1			1	2			3	2	1			
Spain	6	0														
Switzerland	3	1	1					1			1		1			
Tajikistan	2	0														
UK England	9	4	1	3				4			4	2		2		
2020																
JANUARY																
Bulgaria	8	4		4				4			4	4				
Cyprus	16	0														
Czech Republi	3	3		3				3			3	3				
Estonia	3	1		1				1			1	1				
Germany	3	3		3				3			3	3				
Georgia	1	1	1				1				1	1				
Greece	20	4		4				4			4	3		1		
Italy	1	1		1			1				1			1		
Lithuania	2	1		1				1			1	1				
Macedonia	5	0														
Mauritius	2	0														
Montenegro	1	0														
Norway	11	2		2				2			2	2				
Poland	1	1		1				1			1	1				
Romania	7	1		1				1			1	1				
Serbia	3	0														
Slovenia	1	1	1					1			1	1				
	486	146	77	66	3		0	12	134		0	15	131	76	23	47
45 Countries			52.7%	45.2%	2.1%		0.0%	8.2%	91.8%		0.0%	10.3%	89.7%	52.1%	15.8%	32.2%

- Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared
- Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir
- Compared to homologous titres for ferret antisera raised against tissue culture-propagated A/Hong Kong/5738/2014 (3C.2a)
- Compared to homologous titres for ferret antisera raised against egg-propagated A/South Australia/34/2019 (3C.2a1b+T131K)
- Compared to homologous titres for ferret antisera raised against egg-propagated NYMC X-327 A/Kansas/14/2017 (3C.3a)
- Compared to homologous titres for ferret antisera raised against tissue culture-propagated A/Kansas/14/2017 (3C.3a)

Table 7-1. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-09-27

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				NEW									
				Post-infection ferret antisera									
				A/HK 5738/14	A/Bretagne 1413/17	A/Switz 8060/17	A/Singapore 0019/16	A/Norway 3275/18	A/La Rioja 2202/18	A/HK 2671/19	A/Eng 538/18	NYMC X-327 A/Kansas/14	A/Kansas 14/17
	Passage history			MDCK	SIAT	Egg	Egg 10 ⁻⁴	SIAT	SIAT	SIAT	SIAT	Egg	SIAT
	Ferret number			St Jude's F60/17 ^{*1}	F01/18 ^{*1}	F27/18 ^{*1}	F46/17 ^{*1}	F03/19 ^{*1}	F26/18 ^{*1}	F33/19 ^{*1}	F31/18 ^{*1}	F16/19 ^{*1}	F17/19 ^{*1}
	Genetic group			3C.2a	3C.2a2	3C.2a2	3C.2a1	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.2a1b+T135K-B	3C.3a	3C.3a	3C.3a
REFERENCE VIRUSES													
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	160	160	160	320	160	160	40	160	160
A/Bretagne/1413/2017		3C.2a2	2017-10-09	MDCK1/SIAT5	160	640	640	320	160	160	40	160	160
A/Switzerland/8060/2017	clone 57	3C.2a2	2017-12-12	E7/E1	80	640	640	320	40	80	40	80	<
A/Singapore/INFIMH-16-0019/2016		3C.2a1	2016-04-14	E5/E2	80	40	80	320	<	160	40	<	<
A/England/538/2018		3C.3a	2018-02-26	MDCK1/SIAT3	<	<	<	40	<	<	<	320	160
NYMC X-327 (A/Kansas/14/17)		3C.3a	2017-12-14	Ex/E1	<	<	<	40	<	<	<	160	640
A/Kansas/14/2017		3C.3a	2017-12-14	SIAT3/SIAT2	<	40	<	40	<	<	<	320	160
TEST VIRUSES													
A/Seychelles/537/2019			2019-05-31	SIAT1	80	80	80	160	160	160	80	40	40

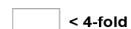
* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40Vaccine
SH 2019Vaccine
NH 2018-19Vaccine
NH 2019-20 < 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table 7-2. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-10-11

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/HK 5738/14	A/Bretagne 1413/17	A/Switz 8060/17	A/Singapore 0019/16	A/Norway 3275/18	A/La Rioja 2202/18	A/Eng 538/18	NYMC X-327 A/Kansas/14	A/Kansas 14/17
				MDCK	SIAT	Egg	Egg 10 ⁻⁴	SIAT	SIAT	SIAT	Egg	SIAT
	Passage history			St Jude's F60/17 ⁻¹	F01/18 ⁻¹	F27/18 ⁻¹	F46/17 ⁻¹	F03/19 ⁻¹	F26/18 ⁻¹	F31/18 ⁻¹	F16/19 ⁻¹	F17/19 ⁻¹
	Genetic group			3C.2a	3C.2a2	3C.2a2	3C.2a1	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.3a	3C.3a	3C.3a
REFERENCE VIRUSES												
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3/SIAT2	160	80	160	160	80	80	160	160	160
A/Bretagne/1413/2017	3C.2a2	2017-10-09	MDCK1/SIAT5	160	640	640	320	160	160	160	160	160
A/Switzerland/8060/2017	clone 57 3C.2a2	2017-12-12	E7/E1	160	1280	640	640	80	160	160	40	<
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	80	40	160	640	40	160	80	40	<
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT3	<	40	40	40	40	<	640	160	320
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	<	<	<	40	<	<	320	1280	320
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	40	40	80	40	40	640	160	320
TEST VIRUSES												
A/Cote d'Ivoire/752/2019	3C.2a1b+T135K-A	2019-05-03	C1/SIAT1	<	<	<	40	80	160	80	<	<
A/Cote d'Ivoire/830/2019	3C.2a1b+T135K-A	2019-05-13	C2/SIAT1	<	<	<	<	40	80	80	<	<
A/Cote d'Ivoire/1024/2019	3C.2a1b+T135K-A	2019-06-14	C1/SIAT1	40	40	<	160	80	160	80	40	<
A/Cote d'Ivoire/1021/2019	3C.2a1b+T135K-A	2019-06-14	C1/SIAT1	40	40	<	80	80	160	80	40	<
A/Cote d'Ivoire/1038/2019	3C.2a1b+T135K-A	2019-06-16	C1/SIAT1	40	40	40	160	80	160	40	<	<
A/Cote d'Ivoire/1116/2019	3C.2a1b+T135K-A	2019-06-26	C0/SIAT1	40	<	<	80	80	160	40	40	<
A/Cote d'Ivoire/1093/2019	3C.2a1b+T135K-A	2019-06-26	C1/SIAT1	40	<	40	80	80	160	40	40	<
A/Cote d'Ivoire/1132/2019	3C.2a1b+T135K-A	2019-06-28	C1/SIAT1	80	<	80	80	160	160	80	40	40
A/Cote d'Ivoire/1126/2019	3C.2a1b+T135K-A	2019-06-28	C0/SIAT1	80	<	80	80	160	160	80	40	40
A/Cote d'Ivoire/1155/2019	3C.2a1b+T135K-A	2019-06-30	C1/SIAT1	80	<	40	160	160	160	80	40	40
A/Cote d'Ivoire/1171/2019	3C.2a1b+T135K-A	2019-07-03	C0/SIAT1	80	40	160	160	160	320	80	80	40
A/Cote d'Ivoire/1442/2019	3C.2a1b+T135K-A	2019-07-08	C2/SIAT1	40	40	40	80	160	160	80	40	<
A/Cote d'Ivoire/1278/2019	3C.2a1b+T135K-A	2019-07-14	C0/SIAT1	80	<	40	80	160	160	160	80	40
A/Cote d'Ivoire/1279/2019	3C.2a1b+T135K-A	2019-07-15	C0/SIAT1	40	<	40	80	160	160	80	80	40
A/Cote d'Ivoire/1299/2019	3C.2a1b+T135K-A	2019-07-17	C2/SIAT1	80	40	80	160	160	160	80	80	80
A/Cote d'Ivoire/1339/2019	3C.2a1b+T135K-A	2019-07-18	C2/SIAT1	80	40	40	160	160	160	160	80	80
A/Cote d'Ivoire/1301/2019	3C.2a1b+T135K-A	2019-07-18	C2/SIAT1	40	<	80	160	160	160	80	80	40
A/Cote d'Ivoire/1417/2019	3C.2a1b+T135K-A	2019-07-23	C2/SIAT1	40	<	<	40	80	160	80	80	40
A/Cote d'Ivoire/1422/2019	3C.2a1b+T135K-A	2019-07-25	C1/SIAT1	40	<	40	40	80	160	40	40	40
A/Cote d'Ivoire/1438/2019	3C.2a1b+T135K-A	2019-07-27	C2/SIAT1	40	40	40	80	80	160	40	<	40
A/Cote d'Ivoire/1521/2019	3C.2a1b+T135K-A	2019-08-10	C1/SIAT1	40	40	40	160	160	160	40	80	40
A/Cote d'Ivoire/1458/2019	3C.2a3	2019-07-30	C2/SIAT1	40	40	40	40	40	40	40	40	<
A/Cote d'Ivoire/1439/2019		2019-07-27	C2/SIAT1	40	40	40	80	160	160	40	40	40

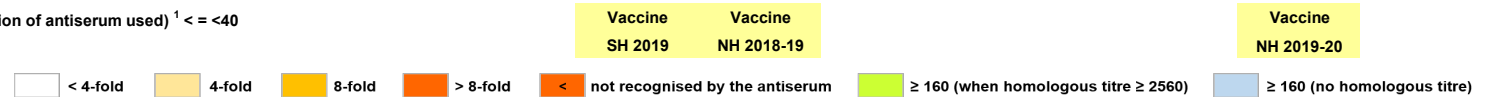
* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Table 7-3. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-10-25

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/HK 5738/14	A/Bretagne 1413/17	A/Switz 8060/17	A/Singapore 0019/16	A/Norway 3275/18	IVR-197 A/Sth Aus/34/19	A/La Rioja 2202/18	A/Eng 538/18	NYMC X-327 A/Kansas/14	A/Kansas 14/17
				MDCK	SIAT	Egg	Egg 10 ⁻⁴	SIAT	Egg	SIAT	SIAT	Egg	SIAT
				St Jude's F60/17 [†]	F01/18 [†]	F27/18 [†]	F46/17 [†]	F03/19 [†]	F39/19 [†]	F26/18 [†]	F31/18 [†]	F16/19 [†]	F17/19 [†]
	Passage history			3C.2a	3C.2a2	3C.2a2	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.3a	3C.3a	3C.3a
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK3/SIAT2	160	160	160	320	160	80	80	160	160	80
A/Bretagne/1413/2017		2017-10-09	MDCK1/SIAT5	160	320	640	160	160	160	80	160	160	80
A/Switzerland/8060/2017	clone 57	2017-12-12	E7/E1	80	640	640	320	40	80	80	80	40	<
A/Singapore/INFIMH-16-0019/2016		2016-04-14	E5/E2	80	40	160	640	40	40	160	80	40	<
IVR-197 (A/South Australia/34/2019)		2019-02-06	E5/D8/E1	80	160	160	80	320	320	40	40	<	<
A/England/538/2018		2018-02-26	MDCK1/SIAT3	<	<	<	<	<	<	<	320	80	160
NYMC X-327 (A/Kansas/14/17)		2017-12-14	Ex/E1	<	<	<	<	<	<	<	160	640	160
A/Kansas/14/2017		2017-12-14	SIAT3/SIAT2	40	40	40	80	40	<	<	320	160	320
TEST VIRUSES													
A/Serbia/883/2019		2019-01-31	C4/SIAT1	80	40	40	80	80	40	160	40	40	40
A/Serbia/1985/2019		2019-02-18	C4/SIAT1	40	<	40	80	80	40	160	40	<	<
A/Serbia/1868/2019		2019-02-18	C4/SIAT1	80	40	40	160	160	80	160	80	40	40
A/Serbia/584/2019		2019-02-23	C4/SIAT1	40	<	<	80	80	40	80	40	<	<
A/Serbia/3031/2019		2019-03-18	C4/SIAT1	40	40	40	80	80	40	160	40	<	<
A/Lebanon/I 471/2019		2019-03-15	SIAT6/SIAT1	40	<	<	<	160	<	40	40	<	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) [†] < = <40

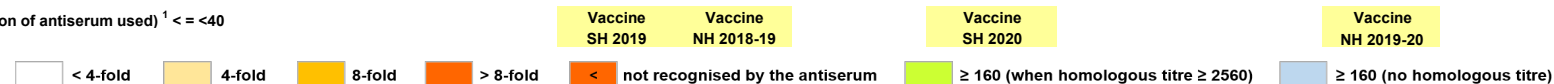


Table 7-4. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-11-08

				Haemagglutination inhibition titre										
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera										
				A/HK 5738/14	A/Bretagne 1413/17	A/Switz 8060/17	A/Singapore 0019/16	A/Norway 3275/18	IVR-197 A/Sth Aus/34/19	A/La Rioja 2202/18	A/Eng 538/18	NYMC X-327 A/Kansas/14	A/Kansas 14/17	
				Passage history	MDCK	SIAT	Egg	Egg 10 ⁻⁴	SIAT	Egg	SIAT	SIAT	Egg	SIAT
				Ferret number	St Jude's F60/17 ⁻¹	F01/18 ⁻¹	F27/18 ⁻¹	F13/19 ⁻¹	F03/19 ⁻¹	F39/19 ⁻¹	F26/18 ⁻¹	F31/18 ⁻¹	F16/19 ⁻¹	F17/19 ⁻¹
	Genetic group			3C.2a	3C.2a2	3C.2a2	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.3a	3C.3a	3C.3a	
REFERENCE VIRUSES														
A/Hong Kong/5738/2014	clone 57	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	160	160	160	160	160	80	160	320	160	160
A/Bretagne/1413/2017		3C.2a2	2017-10-09	MDCK1/SIAT4	160	640	320	160	160	160	80	160	160	80
A/Switzerland/8060/2017		3C.2a2	2017-12-12	E7/E1	160	1280	640	160	80	80	80	80	40	<
A/Singapore/INFIMH-16-0019/2016		3C.2a1	2016-04-14	E5/E2	80	40	80	160	40	40	160	40	40	<
IVR-197 (A/South Australia/34/2019)		3C.2a1b+T131K	2019-02-06	E5/D8/E1	160	320	320	160	640	640	40	80	40	40
A/England/538/2018		3C.3a	2018-02-26	MDCK1/SIAT4	<	40	<	40	<	<	<	640	160	320
NYMC X-327 (A/Kansas/14/17)		3C.3a	2017-12-14	Ex/E1	<	<	<	<	<	<	<	160	640	160
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	40	40	80	40	<	<	640	160	320	
TEST VIRUSES														
A/Mauritius/SF 225/2019	3C.2a1b+T131K	2019-07-05	MDCK1/SIAT2	80	80	40	80	320	160	160	80	80	80	
A/Mauritius/I-1921/2019	3C.2a1b+T131K	2019-07-12	MDCK1/SIAT1	160	160	80	160	320	160	160	80	80	80	
A/Mauritius/I-330/2019	3C.2a1b+T135K-A	2019-02-10	MDCK2/SIAT1	160	80	80	320	160	160	320	80	80	80	
A/Lebanon/307/2019	3C.2a1b+T135K-A	2019-02-27	SIAT1	80	80	80	320	160	160	320	80	80	40	
A/Lebanon/397/2019	3C.3a	2019-04-09	SIAT1	<	40	40	80	40	<	40	640	160	320	
A/Norway/2282/2019	3C.3a	2019-09-25	SIAT1	40	40	40	40	<	<	<	320	160	160	
A/Norway/2298/2019	3C.3a	2019-10-10	SIAT1	40	40	40	40	<	<	<	320	160	320	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine SH 2019	Vaccine NH 2018-19	Vaccine SH 2020	Vaccine NH 2019-20
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< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 7-5. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-11-20

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/HK 5738/14 MDCK	A/Bretagne 1413/17 SIAT	A/Switz 8060/17 Egg	A/Singapore 0019/16 Egg 10 ⁻⁴	A/Norway 3275/18 SIAT	IVR-197 A/Sth Aus/34/19 Egg	A/La Rioja 2202/18 SIAT	A/Eng 538/18 SIAT	NYMC X-327 A/Kansas/14 Egg	A/Kansas 14/17 SIAT
				St Jude's F60/17 ¹	F01/18 ¹	F27/18 ¹	F13/19 ¹	F03/19 ¹	F39/19 ¹	F26/18 ¹	F31/18 ¹	F16/19 ¹	F17/19 ¹
	Passage history			3C.2a	3C.2a2	3C.2a2	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.3a	3C.3a	3C.3a
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT1	160	160	80	160	160	40	80	160	160	80
A/Bretagne/1413/2017	clone 57	2017-10-09	MDCK1/SIAT4	160	640	320	320	160	160	160	160	160	80
A/Switzerland/8060/2017		2017-12-12	E7/E1	160	1280	1280	320	80	80	160	80	40	40
A/Singapore/INFIMH-16-0019/2016		2016-04-14	E5/E2	80	80	80	320	40	80	160	40	40	<
IVR-197 (A/South Australia/34/2019)		2019-02-06	E5/D8/E1	160	640	160	160	640	1280	80	80	40	40
A/England/538/2018		2018-02-26	MDCK1/SIAT4	<	<	<	40	<	<	<	320	160	320
NYMC X-327 (A/Kansas/14/17)		2017-12-14	Ex/E1	<	<	<	40	<	<	<	320	1280	320
A/Kansas/14/2017		2017-12-14	SIAT3/SIAT2	40	40	40	40	<	<	<	640	160	320
TEST VIRUSES													
A/Newcastle/82/2018		2018-12-23	E4/E1	80	160	80	80	320	160	40	160	80	<
A/South Australia/34/2019		2019-02-06	E6/E1	160	640	160	320	640	640	40	40	40	<
A/Christchurch/516/2019		2019-04-03	E3/E1	80	320	160	160	320	640	40	40	80	<
A/Qatar/13-VI-19-0046447/2019		2019-08-08	SIAT1	40	80	40	80	80	40	40	40	40	40
A/Qatar/10-VI-19-0048255/2019		2019-08-21	SIAT1	80	80	40	80	160	40	40	40	40	40
A/Qatar/10-VI-19-0048252/2019		2019-08-21	SIAT1	40	40	<	40	80	40	40	<	40	<
A/Qatar/16-VI-19-0048499/2019		2019-08-22	SIAT1	80	80	80	80	160	40	80	40	40	<
A/Qatar/10-VI-19-0048332/2019		2019-08-22	SIAT1	40	40	40	80	160	40	40	40	40	<
A/Qatar/10-VI-19-0048317/2019		2019-08-22	SIAT1	80	40	40	80	160	40	40	40	40	<
A/Qatar/10-VI-19-0048393/2019		2019-08-22	SIAT1	40	40	40	80	160	40	80	80	40	<
A/Qatar/16-VI-19-0048823/2019		2019-08-25	SIAT1	80	80	80	80	160	40	80	80	40	80
A/Qatar/16-VI-19-0049409/2019		2019-08-27	SIAT1	80	80	40	40	160	<	80	80	80	80
A/Qatar/10-VI-19-0049653/2019		2019-08-29	SIAT1	80	80	80	160	320	80	160	80	80	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

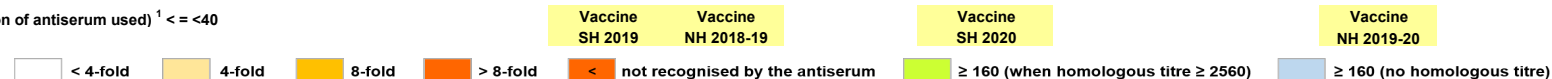


Table 7-6. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-12-06

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/HK 5738/14 MDCK	A/Bretagne 1413/17 SIAT	A/Switz 8060/17 Egg	A/Singapore 0019/16 Egg 10 ⁻⁴	A/Norway 3275/18 SIAT	IVR-197 A/Sth Aus/34/19 Egg	A/La Rioja 2202/18 SIAT	A/Eng 538/18 SIAT	NYMC X-327 A/Kansas/14 Egg	A/Kansas 14/17 SIAT
				St Jude's F60/17 ¹	F01/18 ¹	F27/18 ¹	F13/19 ¹	F03/19 ¹	F39/19 ¹	F26/18 ¹	F31/18 ¹	F16/19 ¹	F17/19 ¹
	Passage history			3C.2a	3C.2a2	3C.2a2	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.3a	3C.3a	3C.3a
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK3/SIAT2	160	160	160	160	160	80	160	320	160	160
A/Bretagne/1413/2017		2017-10-09	MDCK1/SIAT4	160	640	640	160	320	320	160	320	160	160
A/Switzerland/8060/2017	clone 57	2017-12-12	E7/E1	160	1280	640	160	80	160	80	80	80	<
A/Singapore/INFIMH-16-0019/2016		2016-04-14	E5/E2	80	40	80	160	40	80	320	80	40	<
IVR-197 (A/South Australia/34/2019)		2019-02-06	E5/D8/E1	1280	1280	320	320	2560	2560	160	160	40	80
A/England/538/2018		2018-02-26	MDCK1/SIAT4	<	40	<	40	<	<	40	640	160	640
NYMC X-327 (A/Kansas/14/17)		2017-12-14	Ex/E1	<	<	<	<	<	<	<	320	1280	320
A/Kansas/14/2017		2017-12-14	SIAT3/SIAT2	80	40	40	80	40	<	<	640	320	640
TEST VIRUSES													
A/Armenia/13/2019	3C.2a1b+T135K-A	2019-01-29	MDCK1/SIAT1	80	<	<	160	160	40	160	<	<	<
A/Armenia/15/2019	3C.2a1b+T135K-A	2019-02-05	MDCK1/SIAT1	40	<	<	80	80	40	80	<	<	<
A/Salalah/5662/2019	3C.2a1b+T135K-B	2019-09-27	SIAT1	80	40	<	160	160	<	80	40	<	80
A/Salalah/6016/2019	3C.2a1b+T135K-B	2019-10-03	SIAT1	80	<	<	80	80	<	80	40	<	40
A/Salalah/5980/2019	3C.2a1b+T135K-B	2019-10-03	SIAT1	80	<	<	80	80	<	80	40	<	40
A/Salalah/6611/2019	3C.2a1b+T135K-B	2019-10-16	SIAT1	80	<	<	80	160	<	80	40	<	40
A/Saham/6943/2019	3C.2a1b+T135K-B	2019-10-23	SIAT1	80	<	<	40	160	<	80	80	<	40
A/Salalah/6905/2019	3C.2a1b+T135K-B	2019-10-23	SIAT1	80	<	<	80	320	<	160	80	<	80
A/Salalah/7183/2019	3C.2a1b+T135K-B	2019-10-31	SIAT1	80	<	<	80	320	<	160	80	<	80
A/Sur/7325/2019	3C.2a1b+T135K-B	2019-11-06	SIAT1	80	<	<	80	160	<	80	80	<	40
A/England/307/2019	3C.3a	2019-10-05	SIAT1/SIAT1	40	40	<	80	<	<	40	640	160	320
A/England/320/2019	3C.3a	2019-10-07	MDCK1/SIAT1	80	80	40	160	<	<	40	640	320	320
A/England/323/2019	3C.3a	2019-10-14	SIAT1/SIAT1	40	80	40	80	<	<	40	640	320	320
A/England/309/2019	3C.3a	2019-10-15	SIAT1/SIAT1	40	80	40	80	<	<	40	640	160	320
A/England/325/2019	3C.3a	2019-10-16	SIAT1/SIAT1	80	80	40	80	<	<	40	640	320	640
A/England/344/2019	3C.3a	2019-10-21	SIAT1/SIAT1	40	80	40	80	<	<	40	640	160	640
A/England/330/2019	3C.3a	2019-10-24	SIAT1/SIAT1	40	80	40	80	<	<	40	640	160	320
A/England/331/2019	3C.3a	2019-10-25	MDCK1/SIAT1	80	80	40	160	40	<	40	640	320	640
A/England/335/2019	3C.3a	2019-10-28	SIAT1/SIAT1	40	40	40	40	<	<	<	640	160	320
A/England/347/2019	3C.3a	2019-10-30	SIAT1/SIAT1	80	80	<	80	40	40	40	640	320	640
A/England/350/2019	3C.3a	2019-11-11	SIAT1/SIAT1	80	40	<	40	<	<	<	640	160	320

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

					Vaccine SH 2019	Vaccine NH 2018-19		Vaccine SH 2020		Vaccine NH 2019-20	
< 4-fold	4-fold	8-fold	> 8-fold	<	not recognised by the antiserum	≥ 160 (when homologous titre ≥ 2560)	≥ 160 (no homologous titre)				

Table 7-7. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-12-13

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				NEW										
				Post-infection ferret antisera										
				A/HK 5738/14 MDCK St Jude F60/17 ¹	A/Bretagne 1413/17 SIAT F01/18 ¹	A/Switz 8060/17 Egg F27/18 ¹	A/Singapore 0019/16 Egg F13/19 ¹	A/Norway 3275/18 SIAT F03/19 ¹	IVR-197 A/Sth Aus/34/19 Egg F39/19 ¹	A/Sth Aus 34/19 Egg F45/19	A/La Rioja 2202/18 SIAT F26/18 ¹	A/Eng 538/18 SIAT F31/18 ¹	NYMC X-327 A/Kansas/14 Egg F16/19 ¹	A/Kansas 14/17 SIAT F17/19 ¹
	Passage history			3C.2a	3C.2a2	3C.2a2	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.3a	3C.3a	3C.3a
	Ferret number													
	Genetic group													
REFERENCE VIRUSES														
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK3/SIAT2	160	160	160	320	160	80	160	160	320	160
A/Bretagne/1413/2017		3C.2a2	2017-10-09	MDCK1/SIAT4	160	640	640	320	320	160	320	160	320	160
A/Switzerland/8060/2017	clone 57	3C.2a2	2017-12-12	E7/E1	160	1280	1280	320	80	80	160	160	40	40
A/Singapore/INFIMH-16-0019/2016		3C.2a1	2016-04-14	E5/E2	160	80	160	320	40	40	80	160	80	<
IVR-197 (A/South Australia/34/2019)		3C.2a1b+T131K	2019-02-06	E5/D8/E1	320	640	320	640	1280	1280	1280	80	80	40
A/South Australia/34/2019		3C.2a1b+T131K	2019-02-06	E6/E1	160	640	320	640	1280	1280	1280	80	80	40
A/England/538/2018		3C.3a	2018-02-26	MDCK1/SIAT3	80	40	40	80	<	<	40	40	640	320
NYMC X-327 (A/Kansas/14/17)		3C.3a	2017-12-14	Ex/E1	40	40	<	80	<	40	80	40	320	1280
A/Kansas/14/2017		3C.3a	2017-12-14	SIAT3/SIAT2	80	40	40	80	40	<	40	40	640	320
TEST VIRUSES														
A/Dakar/17/2019		3C.2a1b+T135K-A	2019-09-04	C2/SIAT2	80	40	80	80	80	80	80	160	80	80
A/Dakar/21/2019		3C.2a1b+T135K-A	2019-09-11	C2/SIAT1	80	<	80	160	160	160	160	320	160	80
A/Dakar/18/2019		3C.2a1b+T135K-A	2019-09-13	C2/SIAT2	80	40	40	80	80	40	40	80	80	40
A/Dakar/19/2019		3C.2a1b+T135K-A	2019-09-13	C2/SIAT2	80	40	40	80	80	80	80	160	80	80
A/Dakar/24/2019		3C.2a1b+T135K-A	2019-09-18	C2/SIAT1	160	40	80	320	160	160	320	320	160	80
A/Norway/2323/2019		3C.2a1b+T135K-B	2019-10-06	SIAT1/SIAT1	160	40	80	160	160	160	160	160	160	80
A/England/94540758/2019		3C.2a1b+T135K-B	2019-10-25	SIAT1/SIAT1	40	<	<	40	80	<	ND	40	40	<
A/Norway/2356/2019		3C.2a1b+T135K-B	2019-10-28	SIAT1/SIAT1	40	<	<	40	80	<	<	40	40	<
A/Norway/2425/2019		3C.2a1b+T135K-B	2019-11-05	SIAT1	40	<	<	40	80	<	<	40	40	<
A/Norway/2438/2019		3C.2a1b+T135K-B	2019-11-12	SIAT1	40	<	<	80	80	<	<	80	40	<
A/Norway/2379/2019		3C.3a	2019-11-06	SIAT1/SIAT1	80	80	80	160	40	40	80	40	640	320

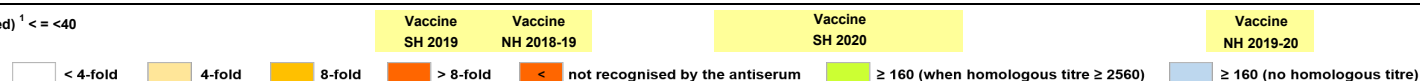
* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ <= <40

Table 7-8. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2020-01-10

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/HK 5738/14	A/Bretagne 1413/17	A/Singapore 0019/16	A/Norway 3275/18	A/Sth Aus 34/19	A/La Rioja 2202/18	A/Eng 538/18	NYMC X-327 A/Kansas/14	A/Kansas 14/17
				MDCK	SIAT	Egg 10 ⁻⁴	SIAT	Egg	SIAT	SIAT	Egg	SIAT
				St Jude's F60/17 ¹	F01/18 ¹	F13/19 ¹	F03/19 ¹	F45/19 ¹	F26/18 ¹	F31/18 ¹	F16/19 ¹	F17/19 ¹
	Passage history			3C.2a	3C.2a2	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.3a	3C.3a	3C.3a
	Ferret number											
	Genetic group											
REFERENCE VIRUSES												
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	160	160	320	160	160	160	160	160	160
A/Bretagne/1413/2017	3C.2a2	2017-10-09	MDCK1/SIAT4	320	640	320	320	320	160	160	160	160
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	160	40	320	40	40	160	80	40	40
A/South Australia/34/2019	3C.2a1b+T131K	2019-02-06	E6/E1	160	640	320	640	1280	80	80	40	40
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT4	40	<	80	40	<	<	640	160	320
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	40	<	80	<	<	<	320	1280	320
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	<	40	<	<	<	320	160	320
TEST VIRUSES												
A/Stockholm/22/2019	3C.2a1b+T131K	2019-08-23	MDCK1/SIAT1	80	40	80	160	160	80	80	40	40
A/Iceland/78/2019	3C.2a1b+T131K	2019-10-08	MDCK1/SIAT1	80	40	160	160	160	80	80	80	40
A/Iceland/81/2019	3C.2a1b+T131K	2019-10-15	MDCK1/SIAT1	160	80	160	160	320	80	160	80	80
A/Moscow/15/2019	3C.2a1b+T131K	2019-11-05	MDCK1/SIAT1	80	80	160	160	160	160	160	80	80
A/Latvia/11-028714/2019	3C.2a1b+T131K	2019-11-12	MDCK1/SIAT1	80	80	160	160	320	160	80	80	40
A/Latvia/11-068832/2019	3C.2a1b+T131K	2019-11-26	MDCK1/SIAT1	80	40	160	160	160	80	80	40	40
A/Hong Kong/4434/2019	3C.2a1b+T131K	2019-11-29	MDCK1/SIAT1	80	80	160	160	160	160	80	80	80
A/Hong Kong/4399/2019	3C.2a1b+T131K	2019-12-02	MDCK1/SIAT1	80	40	80	160	160	80	80	80	40
A/Bremen/24/2019	3C.2a1b+T135K-A	2019-10-02	C1/SIAT1	80	40	160	160	160	160	160	80	80
A/Denmark/3264/2019	3C.2a1b+T135K-A	2019-10-25	SIAT3/SIAT1	160	40	160	160	80	160	160	80	80
A/Denmark/3275/2019	3C.2a1b+T135K-B	2019-11-06	SIAT3/SIAT1	80	<	80	160	<	80	40	<	40
A/Denmark/3292/2019	3C.2a1b+T135K-B	2019-11-13	SIAT2/SIAT1	40	<	80	80	<	40	40	<	40
A/Denmark/3331/2019	3C.2a1b+T135K-B	2019-11-21	SIAT2/SIAT1	40	<	80	80	<	40	40	<	40
A/Hong Kong/4435/2019	3C.2a1b+T135K-B	2019-11-29	MDCK1/SIAT1	160	80	160	160	80	160	80	80	80
A/Hong Kong/4398/2019	3C.2a1b+T135K-B	2019-12-01	MDCK1/SIAT1	80	40	80	160	160	80	80	80	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

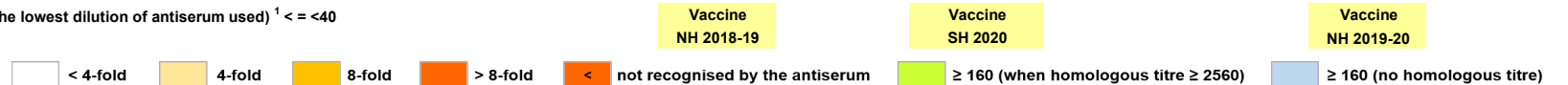


Table 7-9. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2020-01-17

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/HK 5738/14	A/Bretagne 1413/17	A/Singapore 0019/16	A/Norway 3275/18	A/Sth Aus 34/19	A/La Rioja 2202/18	A/HK 2671/19	A/Eng 538/18	NYMC X-327 A/Kansas/14	A/Kansas 14/17
				MDCK	SIAT	Egg 10 ⁻⁴	SIAT	Egg	SIAT	Egg	SIAT	Egg	SIAT
				St Jude's F60/17 ¹	F01/18 ¹	F13/19 ¹	F03/19 ¹	F45/19 ¹	F26/18 ¹	F44/19 ¹	F31/18 ¹	F16/19 ¹	F17/19 ¹
	Passage history			3C.2a	3C.2a2	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.2a1b+T135K-B	3C.3a	3C.3a	3C.3a
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3/SIAT2	160	160	320	160	160	80	<	160	160	80
A/Bretagne/1413/2017	3C.2a2	2017-10-09	MDCK1/SIAT4	160	640	320	160	320	80	<	160	80	80
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	160	80	320	40	80	160	40	80	40	<
A/South Australia/34/2019	3C.2a1b+T131K	2019-02-06	E6/E1	160	640	320	1280	1280	80	80	80	40	40
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8	40	<	80	<	80	40	320	160	320	80
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT4	40	40	40	<	40	<	<	640	160	320
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	40	40	80	40	80	<	160	320	1280	320
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	40	80	40	40	<	<	320	160	320
TEST VIRUSES													
A/Cagnes sur Mer/1850/2019	3C.2a1b+T131K	2019-09-11	MDCK3/SIAT1	80	80	160	160	160	80	<	40	<	40
A/Cannes/1846/2019	3C.2a1b+T131K	2019-09-12	MDCK3/SIAT1	80	80	160	160	160	80	<	80	40	40
A/Czech Republic/1454/2019	3C.2a1b+T131K	2019-09-18	MDCK4/SIAT1	80	160	160	320	320	160	40	80	80	40
A/Skovde/3/2019	3C.2a1b+T131K	2019-09-30	MDCK1/SIAT1	80	160	160	320	320	160	40	80	40	40
A/Iceland/77/2019	3C.2a1b+T131K	2019-10-08	MDCK1/SIAT1	40	80	160	160	160	160	<	80	40	<
A/Iceland/79/2019	3C.2a1b+T131K	2019-10-09	MDCK1/SIAT2	40	80	160	160	160	80	<	40	40	<
A/Bayern/113/2019	3C.2a1b+T131K	2019-10-10	C2/SIAT1	80	160	160	640	320	160	<	160	80	80
A/Stockholm/32/2019	3C.2a1b+T131K	2019-10-14	MDCK0/SIAT1	80	80	160	160	320	80	40	80	<	40
A/Stockholm/30/2019	3C.2a1b+T131K	2019-10-14	MDCK0/SIAT1	80	160	160	320	320	160	40	160	80	80
A/Latvia/11-032248/2019	3C.2a1b+T131K	2019-11-12	MDCK2/SIAT2	80	160	160	320	160	160	<	80	80	80
A/Switzerland/7204/2019	3C.2a1b+T131K	2019-12-06	SIAT1/SIAT1	160	160	160	640	160	160	<	160	80	80
A/Switzerland/9947/2019	3C.2a1b+T135K-A	2019-11-06	SIAT1/SIAT1	<	<	<	<	40	<	<	<	<	<
A/Lyon/2017/2019	3C.2a1b+T135K-A	2019-11-07	MDCK2/SIAT1	160	40	160	320	80	160	<	160	80	80
A/Israel/338/2019	3C.2a1b+T135K-A	2019-12-04	C1/SIAT1	80	<	160	160	80	160	40	160	80	40
A/Finland/136/2019	3C.2a1b+T135K-B	2019-10-25	SIAT1/SIAT1	40	<	80	160	<	80	40	40	<	40
A/Hong Kong/4325/2019	3C.2a1b+T135K-B	2019-11-20	MDCK1/SIAT1	40	<	160	160	160	80	160	40	80	160
A/Hong Kong/4357/2019	3C.2a1b+T135K-B	2019-11-28	MDCK1/SIAT1	160	40	160	320	160	160	80	160	80	80
A/Hong Kong/4377/2019	3C.2a1b+T135K-B	2019-11-29	MDCK1/SIAT1	80	40	160	160	80	160	80	160	80	80
A/Hong Kong/4436/2019	3C.2a1b+T135K-B	2019-12-01	MDCK1/SIAT1	80	40	80	160	80	160	80	160	80	80
A/Hong Kong/4440/2019	3C.2a1b+T135K-B	2019-12-03	MDCK1/SIAT1	40	<	80	160	40	40	40	40	<	<
A/Hong Kong/4439/2019	3C.2a1b+T135K-B	2019-12-03	MDCK1/SIAT1	40	<	80	80	<	80	40	40	<	<
A/Hong Kong/4437/2019	3C.2a1b+T135K-B	2019-12-03	MDCK1/SIAT1	40	<	80	80	<	80	80	80	40	40
A/Finland/130/2019	3C.3a	2019-09-13	SIAT1/SIAT1	40	40	80	<	40	<	<	320	160	160
A/Croatia/8436/2019		2019-12-12	MDCKx/SIAT3	<	40	40	<	80	<	<	<	<	<

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40Vaccine
NH 2018-19Vaccine
SH 2020Vaccine
NH 2019-20

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 7-10. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2020-01-24

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/HK 5738/14	A/Bretagne 1413/17	A/Singapore 0019/16	A/Norway 3275/18	A/Sth Aus 34/19	A/La Rioja 2202/18	A/HK 2671/19	A/Eng 538/18	NYMC X-327 A/Kansas/14	A/Kansas 14/17
				MDCK	SIAT	Egg 10 ⁻⁴	SIAT	Egg	SIAT	Egg	SIAT	Egg	SIAT
	Passage history			St Jude's F60/17 ¹	F01/18 ¹	F13/19 ¹	F03/19 ¹	F45/19 ¹	F26/18 ¹	F44/19 ¹	F31/18 ¹	F16/19 ¹	F17/19 ¹
	Ferret number			3C.2a	3C.2a2	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.2a1b+T135K-B	3C.3a	3C.3a	3C.3a
	Genetic group												
REFERENCE VIRUSES													
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3/SIAT2	160	80	160	160	160	80	<	160	160	80
A/Bretagne/1413/2017	3C.2a2	2017-10-09	MDCK1/SIAT4	160	640	320	160	160	80	<	160	160	80
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	160	40	320	40	40	160	40	40	40	<
A/South Australia/34/2019	3C.2a1b+T131K	2019-02-06	E6/E1	160	640	320	640	1280	80	80	80	40	40
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E1	<	<	80	<	40	40	320	160	320	80
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT4	40	40	40	<	<	<	<	640	160	320
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	40	<	40	<	40	<	160	320	1280	320
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	40	40	40	<	<	<	320	160	160
TEST VIRUSES													
A/Slovenia/3126/2019	3C.2a1b+T131K	2019-12-05	SIAT2/SIAT1	80	80	160	320	320	160	<	160	80	80
A/S.Petersburg/RIL-338/2019	3C.2a1b+T131K	2019-12-14	MDCK2/SIAT1	80	80	160	160	320	160	<	80	80	40
A/Georgia/49/2020	3C.2a1b+T131K	2020-01-03	SIAT1	80	160	160	320	320	160	<	160	80	80
A/Ghana/722/2019	3C.2a1b+T135K-A	2019-11-11	SIAT1	80	<	80	160	80	160	40	80	40	40
A/Ghana/3793/2019	3C.2a1b+T135K-A	2019-11-12	SIAT1	80	<	80	160	<	80	40	40	<	<
A/Ghana/4021/2019	3C.2a1b+T135K-A	2019-11-16	SIAT1	80	40	160	160	80	160	40	160	80	80
A/Centre/2559/2019	3C.2a1b+T135K-A	2019-11-25	MDCK2/SIAT2	80	40	160	80	160	160	<	80	80	40
A/Slovenia/3251/2019	3C.2a1b+T135K-A	2019-12-13	SIAT1/SIAT1	80	<	80	160	40	160	40	80	80	40
A/Belgium/G0003/2020	3C.2a1b+T135K-A	2019-12-26	SIAT1	160	80	160	320	160	160	40	160	80	80
A/Georgia/1246/2019	3C.2a1b+T135K-A	2019-12-27	SIAT1	80	40	80	160	160	160	40	160	80	80
A/Qatar/16-VI-19-0069037/2019	3C.2a1b+T135K-B	2019-11-12	SIAT1	40	<	40	80	<	40	40	<	<	<
A/Paris/2560/2019	3C.3a	2019-11-25	SIAT1	40	40	80	<	<	<	<	320	160	320
A/Pays de Loire/2718/2019	3C.3a	2019-12-10	MDCK1/SIAT1	80	80	160	40	40	40	<	640	320	320
A/Lorraine/2777/2019	3C.3a	2019-12-16	SIAT1	80	80	160	40	40	40	40	640	320	320
A/S.Petersburg/RIL-337/2019	3C.3a	2019-12-21	MDCK1/SIAT1	40	40	80	40	40	<	<	640	160	320
A/Belgium/G0539/2019	3C.3a	2019-12-23	SIAT1	160	80	160	80	80	80	<	640	320	320
A/Belgium/G0540/2019	3C.3a	2019-12-24	SIAT1	40	80	80	40	40	40	<	640	160	320
A/Belgium/G0538/2019	3C.3a	2019-12-24	SIAT1	40	80	40	40	40	<	<	640	160	320
A/Belgium/G0004/2020	3C.3a	2019-12-30	SIAT1	40	80	80	40	40	40	<	640	160	320
A/Slovenia/3636/2019	3C.3a	2019-12-30	SIATx/SIAT1	40	80	80	40	40	<	40	640	160	320
A/Slovenia/23/2020	3C.3a	2020-01-01	SIATx/SIAT1	160	160	160	40	40	40	<	640	160	320

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40Vaccine
NH 2018-19Vaccine
SH 2020Vaccine
NH 2019-20

 < 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 7-11. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2020-01-31

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				NEW											
				A/HK 5738/14	A/Bretagne 1413/17	A/Singapore 0019/16	A/Norway 3275/18	A/Sth Aus 34/19	A/Norway 2279/19	A/La Rioja 2202/18	A/HK 2671/19	A/HK 2669/19	A/Eng 538/18	NYMC X-327 A/Kansas/14	A/Kansas 14/17
				MDCK	SIAT	Egg 10 ⁻⁴	SIAT	Egg	Egg	SIAT	Egg	SIAT	SIAT	Egg	SIAT
Passage history	Ferret number	Genetic group		St Jude's F60/17 ¹	F01/18 ¹	F13/19 ¹	F03/19 ¹	F45/19 ¹	F03/20 ¹	F26/18 ¹	F44/19 ¹	F04/20 ¹	F31/18 ¹	F16/19 ¹	F17/19 ¹
				3C.2a	3C.2a2	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.3a	3C.3a	3C.3a
REFERENCE VIRUSES															
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3/SIAT2	160	80	320	160	160	80	160	40	40	160	160	160
A/Bretagne/1413/2017	3C.2a2	2017-10-09	MDCK1/SIAT4	160	320	320	160	160	80	160	40	40	160	160	80
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	160	80	320	80	40	80	320	80	160	80	40	<
A/South Australia/34/2019	3C.2a1b+T131K	2019-02-06	E6/E1	160	640	320	640	1280	160	80	160	<	80	40	40
A/Norway/2279/2019	3C.2a1b+T131K	2019-09-25	E4	ND	ND	640	ND	320	160	ND	160	160	ND	80	ND
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E1	<	<	80	<	40	<	40	640	160	160	320	80
A/Hong Kong/2669/2019	3C.2a1b+T135K-B	2019-06-18	MDCK1/SIAT5	160	40	160	160	80	40	320	160	320	80	160	80
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT4	40	40	40	<	<	<	40	40	<	640	160	320
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	40	<	40	<	40	40	40	160	40	320	1280	320
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	40	80	40	40	40	40	40	<	640	320	320
TEST VIRUSES															
A/Portugal/6/2019	3C.2a1b+T135K-A	2019-10-22	SIAT1/SIAT1	80	40	160	160	80	40	160	40	160	80	40	40
A/Ireland/95061/2019	3C.2a1b+T135K-A	2019-11-25	SIAT1	160	80	160	320	160	40	320	80	320	160	160	160
A/England/195160658/2019	3C.2a1b+T135K-A	2019-12-10	SIAT1/SIAT1	40	40	80	160	40	<	160	40	160	40	40	40
A/Bayern/115/2019	3C.2a1b+T135K-A	2019-12-16	C1/SIAT1	80	40	160	160	160	40	160	40	160	160	160	80
A/Portugal/16/2019	3C.2a1b+T135K-A	2019-12-16	SIAT1/SIAT1	80	40	160	160	80	40	160	80	160	80	80	40
A/England/195200780/2019	3C.2a1b+T135K-A	2019-12-23	SIAT1/SIAT1	80	40	160	160	80	40	160	40	160	80	80	40
A/Athens.GR/25/2020	3C.2a1b+T131K	2020-01-07	MDCK1/SIAT1	40	40	80	80	160	40	80	<	40	<	<	<
A/Saarland/25/2019	3C.2a1b+T135K-B	2019-12-16	C2/SIAT1	40	<	40	80	<	<	80	<	40	160	40	<
A/Kyrgyzstan/292/2019	3C.2a1b+T135K-B	2019-12-18	SIAT1	80	<	80	160	40	<	160	80	320	80	40	40
A/Kyrgyzstan/291/2019	3C.2a1b+T135K-B	2019-12-18	SIAT1	80	<	160	160	40	<	160	80	320	80	40	40
A/Kyrgyzstan/205/2019	3C.2a1b+T135K-B	2019-12-18	SIAT1	80	<	80	160	40	<	160	80	320	80	40	40
A/Kyrgyzstan/177/2019	3C.2a1b+T135K-B	2019-12-18	SIAT1	80	<	80	160	80	<	160	80	320	80	40	40
A/Kyrgyzstan/308/2019	3C.2a1b+T135K-B	2019-12-19	SIAT1	80	<	80	160	80	40	160	40	160	80	40	40
A/Kyrgyzstan/118/2019	3C.2a1b+T135K-B	2019-12-19	SIAT1	80	<	80	160	80	40	160	80	160	80	40	80
A/Ireland/93664/2019	3C.3a	2019-11-22	SIAT3	40	40	80	40	40	<	40	40	<	320	160	320
A/Ireland/95068/2019	3C.3a	2019-11-25	SIAT1	80	80	160	40	40	40	40	<	<	640	320	320
A/England/670/2019	3C.3a	2019-12-09	SIAT1/SIAT1	40	40	40	<	<	<	40	40	<	320	160	320
A/England/652/2019	3C.3a	2019-12-09	SIAT1/SIAT1	40	40	80	40	40	<	40	40	<	320	160	320
A/Hessen/72/2019	3C.3a	2019-12-10	C1/SIAT1	80	40	80	<	40	<	40	40	<	640	160	320
A/Austria/1207674/2019	3C.3a	2019-12-12	SIATx/SIAT1	40	40	80	<	<	40	40	<	<	640	160	320
A/Austria/1208858/2019	3C.3a	2019-12-16	SIATx/SIAT1	80	80	80	40	40	<	80	40	<	640	320	320
A/Austria/1208486/2019	3C.3a	2019-12-16	SIATx/SIAT1	40	40	80	40	<	<	40	40	<	320	160	320
A/Berlin/62/2019	3C.3a	2019-12-16	C1/SIAT1	40	40	80	<	40	<	<	<	<	320	160	160
A/Austria/1209023/2019	3C.3a	2019-12-17	SIATx/SIAT1	<	<	40	<	<	<	<	<	<	640	160	160
A/Austria/1208875/2019	3C.3a	2019-12-17	SIATx/SIAT1	40	80	160	40	40	<	40	40	<	640	160	320
A/Austria/1209271/2019	3C.3a	2019-12-18	SIATx/SIAT1	80	80	160	40	80	40	80	40	<	640	320	320
A/Austria/1209022/2019	3C.3a	2019-12-18	SIATx/SIAT1	80	80	160	40	40	40	40	40	<	640	320	320
A/Lyon/2358/2019	3C.3a	2019-12-20	MDCK2/SIAT1	40	40	80	<	<	<	40	<	<	320	160	320
A/Netherlands/10282/2019	3C.3a	2019-12-24	MDCK-MIX2/SIAT1	40	40	40	<	<	<	<	<	<	320	80	160
A/Lisboa/118/2019	3C.3a	2019-12-25	SIAT1/SIAT1	40	40	80	<	<	<	40	40	<	320	160	320
A/Netherlands/10286/2019	3C.3a	2019-12-30	MDCK-MIX2/SIAT1	40	40	80	40	40	40	40	40	<	640	160	320
A/Berlin/1/2020	3C.3a	2020-01-03	C1/SIAT1	40	40	80	<	<	<	40	40	<	320	160	160
A/Bulgaria/021/2020	3C.3a	2020-01-06	SIAT1	40	40	80	<	<	<	40	40	<	320	160	320
A/Thuringen/3/2020	3C.3a	2020-01-06	C1/SIAT1	40	40	80	<	<	<	40	40	<	320	160	320
A/Berlin/2/2020	3C.3a	2020-01-09	C1/SIAT1	40	40	80	<	<	<	40	40	<	320	160	320
A/Bulgaria/062/2020	3C.3a	2020-01-10	SIAT1	40	80	80	<	<	<	40	40	<	320	160	320
A/Bulgaria/061/2020	3C.3a	2020-01-10	SIAT1	40	40	80	<	<	<	<	40	<	320	160	320
A/Bulgaria/090/2020	3C.3a	2020-01-13	SIAT1	40	80	80	40	80	<	40	40	<	640	160	320
A/Greece/60/2020	3C.3a	2020-01-13	SIAT1	40	40	40	<	<	<	<	<	<	320	160	320
A/Estonia/125785/2020	3C.3a	2020-01-13	SIAT1/SIAT1	40	40	40	<	<	<	<	<	<	320	160	320
A/Athens.GR/79/2020	3C.3a	2020-01-14	MDCK1/SIAT1	40	40	80	<	<	<	40	40	<	320	160	320
A/Athens.GR/128/2020	3C.3a	2020-01-16	Cx/SIAT1	40	40	40	<	<	<	<	<	<	320	160	160

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40Vaccine
NH 2018-19Vaccine
SH 2020Vaccine
NH 2019-20

< 4-fold

4-fold

8-fold

> 8-fold

< not recognised by the antiserum

≥ 160 (when homologous titre ≥ 2560)

≥ 160 (no homologous titre)

Table 7-12. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2020-02-07

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/HK 5738/14	A/Singapore 0019/16	A/Norway 3275/18	A/Sth Aus 34/19	A/La Rioja 2202/18	A/HK 2671/19	A/HK 2669/19	A/Eng 538/18	NYMC X-327 A/Kansas/14	A/Kansas 14/17
				MDCK	Egg 10 ⁻⁴	SIAT	Egg	SIAT	Egg	SIAT	SIAT	Egg	SIAT
				St Jude's F60/17 ⁻¹	F13/19 ⁻¹	F03/19 ⁻¹	F45/19 ⁻¹	F26/18 ⁻¹	F44/19 ⁻¹	F04/20 ⁻¹	F31/18 ⁻¹	F16/19 ⁻¹	F17/19 ⁻¹
	Passage history			3C.2a	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.3a	3C.3a	3C.3a
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3/SIAT2	160	320	160	160	160	40	40	320	160	160
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E2	80	320	80	40	160	40	80	80	40	<
A/South Australia/34/2019	3C.2a1b+T131K	2019-02-06	E6/E1	160	320	640	640	80	80	<	80	40	40
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	2019-06-17	E8/E1	<	80	<	40	40	640	160	160	320	80
A/Hong Kong/2669/2019	3C.2a1b+T135K-B	2019-06-18	MDCK1/SIAT5	80	80	80	40	160	80	320	80	160	40
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT4	40	40	40	<	<	40	<	640	160	320
NYMC X-327 (A/Kansas/14/17)	3C.3a	2017-12-14	Ex/E1	40	80	<	<	<	160	40	320	1280	320
A/Kansas/14/2017	3C.3a	2017-12-14	SIAT3/SIAT2	40	80	<	<	<	40	<	320	160	320
TEST VIRUSES													
A/Delaware/41/2019	3C.2a1b+T131K	2019-03-06	SIAT2/SIAT1	80	160	160	160	80	<	160	80	80	40
A/Aichi/218/2019	3C.3a	2019-04-02	MDCK2+SIAT2/SIAT1	40	80	<	40	40	40	<	640	160	320
A/Shanghai-Jiading/1977/2019	3C.2a1b+T131K	2019-12-03	C1/SIAT1/SIAT1	80	160	320	320	160	40	160	80	80	40
A/Tianjin-Nankai/3210/2019	3C.2a1b+T135K-A	2019-11-24	C2/SIAT1/SIAT1	40	<	40	<	40	<	160	<	<	<
A/Palermo/36/2019	3C.2a1b+T135K-A	2019-12-11	SIAT2/SIAT1	80	80	160	80	160	40	320	160	80	80
A/Palermo/02/2020	3C.2a1b+T131K	2020-01-02	SIAT2/SIAT1	40	80	160	160	80	<	160	80	40	40
A/Kanagawa/ZC1907/2019	3C.2a1b+T135K-B	2019-11-01	SIAT2/SIAT1	80	80	160	80	160	80	320	80	40	40
A/Jilin-Dongchang/1574/2019	3C.2a1b+T135K-B	2019-12-09	C1/SIAT1/SIAT1	40	40	80	<	40	40	160	40	<	<
A/Lithuania/MB42384/2019	3C.3a	2019-12-24	SIAT2/SIAT1	40	80	40	<	<	<	<	320	160	160
A/Parma/391/2019	3C.3a	2019-12-27	SIAT1/SIAT1	40	80	40	40	<	<	<	640	160	320
A/Lithuania/MB170/2019	3C.3a	2019-12-31	SIAT1/SIAT1	40	40	<	<	<	<	<	320	160	160
A/Norway/47/2020	3C.3a	2020-01-03	SIAT1/SIAT1	40	80	40	40	40	<	<	320	160	320
A/Poland/1521/2020	3C.3a	2020-01-05	SIAT1	40	80	<	<	<	<	<	640	40	320
A/Lithuania/MB398/2020	3C.3a	2020-01-05	SIAT1/SIAT1	40	40	<	<	<	<	<	320	160	160
A/Norway/108/2020	3C.3a	2020-01-06	SIAT1/SIAT1	40	80	<	<	40	<	<	640	160	320
A/Czech Republic/38/2020	3C.3a	2020-01-07	P2/SIAT1	40	80	<	<	<	<	<	320	160	320
A/Czech Republic/87/2020	3C.3a	2020-01-10	P1/SIAT1	40	80	<	<	<	<	<	320	160	320
A/Czech Republic/101/2020	3C.3a	2020-01-13	P1/SIAT1	40	80	<	<	<	<	<	320	160	320
A/Bucuresti/256299/2020	3C.3a	2020-01-13	MDCK1/SIAT1	40	40	<	<	<	<	<	320	160	160
A/Czech Republic/1691/2019		2019-12-05	P3/SIAT1	<	40	<	<	<	<	<	320	80	160
A/Poland/605/2019		2019-12-10	SIAT1	<	<	<	<	<	<	<	320	80	160

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40Vaccine
NH 2018-19Vaccine
SH 2020Vaccine
NH 2019-20

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Haemagglutination inhibition titre													
Post-infection ferret antisera													
Viruses	A/HK 5738/14	A/Bretagne 1413/17	A/Singapore 0019/16	A/Norway 3275/18	A/Sth Aus 34/19	A/Norway 2279/19	A/La Rioja 2202/18	A/HK 2671/19	A/HK 2669/19	A/Eng 538/18	NYMC X-327 A/Kansas/14	A/Kansas 14/17	
Passage history	MDCK	SIAT	Eqq 10 ⁻⁴	SIAT	Egg	Egg	SIAT	Egg	SIAT	SIAT	Egg	SIAT	
Ferret number	F60/17 ⁻¹	F01/18 ⁻¹	F13/19 ⁻¹	F03/19 ⁻¹	F45/19 ⁻¹	F03/20 ⁻¹	F26/18 ⁻¹	F44/19 ⁻¹	F04/20 ⁻¹	F31/18 ⁻¹	F16/19 ⁻¹	F17/19 ⁻¹	
Genetic group	3C.2a	3C.2a2	3C.2a1	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.2a1b+T135K-B	3C.2a1b+T135K-B	3C.3a	3C.3a	3C.3a	
REFERENCE VIRUSES													
A/Hong Kong/5738/2014	3C.2a	160	80	320	160	160	80	160	40	40	160	160	
A/Bretagne/1413/2017	3C.2a2	160	320	320	160	160	80	160	40	40	160	160	
A/Singapore/INFIMH-16-0019/2016	3C.2a1	160	80	320	80	40	80	320	80	160	80	40	
A/South Australia/34/2019	3C.2a1b+T131K	160	640	320	640	1280	160	80	160	<	80	40	
A/Norway/2279/2019	3C.2a1b+T131K	ND	ND	640	ND	320	160	ND	160	ND	80	ND	
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	<	<	80	<	40	<	40	640	160	320	80	
A/Hong Kong/2669/2019	3C.2a1b+T135K-B	160	40	160	160	80	40	320	160	320	80	80	
A/England/538/2018	3C.3a	40	40	40	<	<	40	40	40	<	640	160	
NYMC X-327 (A/Kansas/14/17)	3C.3a	40	<	40	<	40	40	40	160	40	320	1280	
A/Kansas/14/2017	3C.3a	40	40	80	40	40	40	40	40	<	640	320	
TEST VIRUSES													
Number of viruses tested	146	133	146	146*	146	42	146*	100	57	146	146	146	
No with titre reduction ≤2-fold	77	6	65	63	65	47	1	15	66	75	75	75	
%	52.7	4.5	44.5	43.2	44.5	32.2	1.0	26.3	45.2	51.4	51.4	51.4	
No with titre reduction =4-fold	66	17	64	11	13	4	4	23	15	23	23	23	
%	45.2	12.8	43.8	7.5	31.0	4.0	4.0	15.8	10.3	15.7	15.7	15.7	
No with titre reduction ≥8-fold	3	110	17	135	29	95	42	57	131	48	48	48	
%	2.8	82.7	11.7	92.5	69.0	95.0	95.0	73.7	39.0	89.7	32.9	32.9	
Subgroup 3C.2a1b+T131K viruses													
Number of viruses tested	23	22	23	23*	23	1	23*	16	2	23	23	23	
No with titre reduction ≤2-fold	19	6	20	22	20	13	1	50.0	7	8	8	8	
%	82.6	27.3	87.0	95.7	87.0	56.5	5.0	30.4	16	23	13	13	
No with titre reduction =4-fold	4	6	3	11	47.8	100.0	16	100.0	50.0	69.6	100.0	56.5	
%	17.4	27.3	13.0	47.8	100.0	100.0	100.0	100.0	100.0	100.0	100.0	56.5	
No with titre reduction ≥8-fold	10	45.4	52.2	52.2	52.2	52.2	52.2	52.2	52.2	52.2	52.2	52.2	
%	43.5	20.5	36.8	36.8	36.8	36.8	36.8	36.8	36.8	36.8	36.8	36.8	
Subgroup 3C.2a1b+T135K-A viruses													
Number of viruses tested	23	23	23	23*	23	6	23*	16	7				

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Table 8-1. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2020-01-09

Viruses	Passage history Ferret number Genetic group	Passage History	Neutralisation Titre ¹														HA1 substitutions for 3C.2a/3a viruses compared to cell culture-propagated A/Switzerland/9715293/2013. Egg adaptation HA substitutions compared to the corresponding cell culture-propagated virus. Virus specific substitutions.
			Post-infection ferret antisera														
			A/Norway 3275/2018 SIAT F03/19		A/Hong Kong 2671/19 SIAT F31/19		A/SouthAustralia 34/2019 Egg F43/19		A/Christchurch 515/19 Egg F32/19		A/La Rioja 2202/18 SIAT F26/18		A/Kansas 14/2017 SIAT F17/19		NYMC X-327 A/Kansas/14 Egg F16/19		
			3C.2a1b+T131K 2-fold Read		3C.2a1b+T135K-B 2-fold Read		3C.2a1b+T131K 2-fold Read		3C.2a1b+T131K 2-fold Read		3C.2a1b+T135K-A 2-fold Read		3C.3a 2-fold Read		3C.3a 2-fold Read		
REFERENCE VIRUSES																	
A/Norway/3275/2018	3C.2a1b+T131K	SIAT2 10 ⁻³	640	712	80	101	80	78	320	302	80	68	<	0	<	5	L3I, E62G, K92R, A106V, N121K, A128T (+CHO), T131K, S138A, N144S (-CHO), S159Y, K160T (+CHO), N171K, R326K
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	SIAT1/SIAT1 10 ⁻¹	80	78	640	638	<	0	320	282	160	126	40	59	40	40	L3I, E62G, K92R, N121K, T135K (-CHO), S137F, N144S (-CHO), S159Y, K160T (+CHO), N171K, F193S, R326K
A/SouthAustralia/34/2019	3C.2a1b+T131K	E5/D8 10 ⁻⁸	640	509	<	0	1280	1844	80	76	<	20	<	0	<	0	L3I, E62G, K92R, N121K, A128T (+CHO), T131K, S138A, N144S (-CHO), S159Y, T160K (-CHO), N171K, G186I, S219F, R326K
A/Christchurch/515/2019	3C.2a1b+T131K	E3/E1	640	600	<	0	40	46	1280	2007	640	564	40	20	<	10	L3I, E62G, K92R, N121K, A128T (+CHO), T131K, S138A, N144S (-CHO), S159Y, K160T (-CHO), N171K, L194P, Q197R, S219F, R326K
A/La Rioja/2202/2018	3C.2a1b+T135K-A	SIAT1/SIAT4 10 ⁻³	160	210	320	255	40	44	320	469	320	411	<	0	<	0	L3I, E62G, K92R, N121K, T135K (-CHO), S138A, N144S (-CHO), S159Y, K160T (+CHO), N171K, R326K
A/Kansas/14/2017	3C.3a	SIAT3/SIAT2 10 ⁻³	<	0	<	0	<	0	<	0	<	0	160	235	320	251	L3I, S91N, N144K (-CHO), F193S
NYMC X-327 (A/Kansas/14/17)	3C.3a	Ex/E1	<	0	<	0	<	0	<	0	<	20	320	250	2560	2368	L3I, S91N, N144K (-CHO), G186V, D190N, F193S, S219Y
TEST VIRUSES																	
A/Norway/2280/2019	3C.2a1b+T131K	SIAT1	1280	1100	160	206	160	168	640	617	160	150	<	0	<	0	K83E, Y94N
A/Norway/2279/2019	3C.2a1b+T131K	SIAT1	1280	1034	160	134	160	152	320	464	160	148	<	0	<	0	K83E, Y94N
A/Norway/2278/2019	3C.2a1b+T131K	SIAT1	640	742	160	173	160	124	640	603	80	115	<	0	<	0	K83E, Y94N
A/England/94120567/2019	3C.2a1b+T131K	SIAT1/SIAT1	640	949	320	334	160	142	640	613	80	111	<	0	<	20	K83E, Y94N
A/England/94460567/2019	3C.2a1b+T131K	SIAT1/SIAT1	640	950	80	74	160	149	640	519	160	144	<	0	<	0	K83E, Y94N
A/Muscat/6972/2019	3C.2a1b+T131K	SIAT1	1280	1382	160	169	160	210	640	629	160	170	<	0	<	0	K83E, Y94N, A106T
A/Muscat/7070/2019	3C.2a1b+T131K	SIAT1	640	892	320	308	160	120	640	587	80	77	<	20	<	10	T10M (-CHO), K83E, Y94N
A/England/94240408/2019	3C.2a1b+T131K	SIAT1/SIAT1	640	628	320	259	160	127	1280	1024	80	107	<	0	<	3	Q197R, K207R, S219F
A/Norway/2296/2019	3C.2a1b+T135K-A	SIAT1	320	287	320	260	80	77	320	444	640	524	<	20	40	48	S198P, R261Q
A/Salalah/5980/2019	3C.2a1b+T135K-B	SIAT1	80	72	1280	1040	<	0	160	200	40	49	<	5	<	0	E50K, N122D (-CHO)
A/Salalah/6016/2019	3C.2a1b+T135K-B	SIAT1	80	74	640	612	<	0	320	256	40	58	<	10	<	1	E50K, N122D (-CHO)
A/Salalah/6611/2019	3C.2a1b+T135K-B	SIAT1	80	111	640	541	<	0	160	177	80	77	<	10	<	10	E50K, N122D (-CHO), S124N
A/Salalah/6404/2019	3C.2a1b+T135K-B	SIAT1	160	120	640	522	<	3	160	152	80	115	80	67	<	10	E50K, N122D (-CHO), P239S
A/Norway/2282/2019	3C.3a	SIAT1	<	0	<	0	<	0	<	0	<	0	320	247	320	252	none
A/England/307/2019	3C.3a	SIAT1/SIAT1	<	1	<	0	<	0	<	0	<	0	320	272	160	125	none
A/England/331/2019	3C.3a	MDCK1/SIAT1	<	0	<	0	<	0	<	0	<	0	320	274	320	270	none
A/England/320/2019	3C.3a	MDCK1/SIAT1	<	0	<	0	<	0	<	0	<	0	160	234	320	262	K82R
A/England/330/2019	3C.3a	SIAT1/SIAT1	<	0	<	0	<	0	<	0	<	3	160	232	160	122	K82R
Vaccine SH 2020							Vaccine NH 2019-20										

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 8-2. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2020-01-13

Viruses	Passage history Ferret number Genetic group	Passage History	Neutralisation Titre ¹														HA1 substitutions for 3C.2a/3a viruses compared to cell culture-propagated A/Switzerland/9715293/2013. Egg adaptation HA substitutions compared to the corresponding cell culture-propagated virus. Virus specific substitutions.
			Post-infection ferret antisera														
			A/Norway 3275/2018 SIAT F03/19		A/Hong Kong 2671/19 SIAT F31/19		A/SouthAustralia 34/2019 Egg F43/19		A/Christchurch 515/19 Egg F32/19		A/La Rioja 2202/18 SIAT F26/18		A/Kansas 14/2017 SIAT F17/19		NYMC X-327 A/Kansas/14 Egg F16/19		
			3C.2a1b+T131K		3C.2a1b+T135K-B		3C.2a1b+T131K		3C.2a1b+T131K		3C.2a1b+T135K-A		3C.3a		3C.3a		
			2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																	
A/Norway/3275/2018	3C.2a1b+T131K	SIAT2 10 ⁻³	640	712	80	101	80	78	320	302	80	68	<	0	<	5	L3I, E62G, K92R, A106V, N121K, A128T (+CHO), T131K, S138A, N144S (-CHO), S159Y, K160T (+CHO), N171K, R326K
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	SIAT1/SIAT1 10 ⁻¹	80	78	640	638	<	0	320	282	160	126	40	59	40	40	L3I, E62G, K92R, N121K, T135K (-CHO), S137F, N144S (-CHO), S159Y, K160T (+CHO), N171K, F193S, R326K
A/SouthAustralia/34/2019	3C.2a1b+T131K	E5/D8 10 ⁻⁶	640	509	<	0	1280	1844	80	76	<	20	<	0	<	0	L3I, E62G, K92R, N121K, A128T (+CHO), T131K, S138A, N144S (-CHO), S159Y, T160K (-CHO), N171K, G186I, S219F, R326K
A/Christchurch/515/2019	3C.2a1b+T131K	E3/E1	640	600	<	0	40	46	1280	2007	640	564	40	20	<	10	L3I, E62G, K92R, N121K, A128T (+CHO), T131K, S138A, N144S (-CHO), S159Y, K160T (+CHO), N171K, L194P, Q197R, S219F, R326K
A/La Rioja/2202/2018	3C.2a1b+T135K-A	SIAT1/SIAT4 10 ⁻³	160	210	320	255	40	44	320	469	320	411	<	0	<	0	L3I, E62G, K92R, N121K, T135K (-CHO), S138A, N144S (-CHO), S159Y, K160T (+CHO), N171K, R326K
A/Kansas/14/2017	3C.3a	SIAT3/SIAT2 10 ⁻³	<	0	<	0	<	0	<	0	<	0	160	235	320	251	L3I, S91N, N144K (-CHO), F193S
NYMC X-327 (A/Kansas/14/17)	3C.3a	Ex/E1	<	0	<	0	<	0	<	0	<	20	320	250	2560	2368	L3I, S91N, N144K (-CHO), G186V, D190N, F193S, S219Y
TEST VIRUSES																	
A/Ibra/6555/2019	3C.2a1b+T131K	SIAT1	640	847	160	154	160	156	640	547	160	183	<	0	<	0	K83E, Y94N
A/Norway/2421/2019	3C.2a1b+T131K	SIAT1	640	862	80	77	160	147	640	823	160	156	<	0	<	0	K83E, Y94N
A/Iran/189177/2019	3C.2a1b+T131K	SIAT1	640	934	160	127	160	137	320	380	160	129	<	10	<	0	K83E, Y94N
A/Iran/189156/2019	3C.2a1b+T131K	SIAT1/SIAT1	80	67	160	138	<	0	640	604	80	77	<	0	<	0	K83E, Y94N
A/Iran/M191919/2019	3C.2a1b+T131K	SIAT1/SIAT1	640	819	80	113	80	93	320	396	160	125	<	10	<	0	K83E, Y94N
A/England/94580411/2019	3C.2a1b+T131K	SIAT1/SIAT1	640	911	40	40	160	147	640	560	160	160	<	0	<	0	K83E, Y94N
A/England/94540759/2019	3C.2a1b+T131K	SIAT1/SIAT1	640	826	80	117	160	183	640	569	160	142	<	0	<	0	E50K, K83E, Y94N
A/Ibra/6688/2019	3C.2a1b+T131K	SIAT1	320	240	640	668	<	20	320	301	120	102	80	109	80	67	K83E, Y94N, F193S
A/Norway/2477/2019	3C.2a1b+T131K	SIAT1	320	307	640	691	60	54	320	346	160	156	<	0	160	128	K83E, N94S, F193S
A/England/94220533/2019	3C.2a1b+T131K	SIAT1/SIAT1	640	601	160	234	160	124	640	573	160	142	<	0	<	0	G78S, S219F, P289S
A/Iran/182342/2019	3C.2a1b+T131K	SIAT2/SIAT1	640	592	320	255	160	153	640	947	160	178	<	0	<	10	Q197R, K207R, S219F
A/Ibri/6486/2019	3C.2a1b+T131K	SIAT1	640	791	160	231	160	160	640	739	320	259	<	0	<	0	Q197R, S219F, P273T
A/Norway/2438/2019	3C.2a1b+T135K-B	SIAT1	80	67	640	672	<	0	320	261	80	76	<	0	<	0	E50K
A/Saham/6943/2019	3C.2a1b+T135K-B	SIAT1	40	56	640	604	<	1	160	198	40	43	<	20	<	3	E50K, N122D (-CHO)
A/England/323/2019	3C.3a	SIAT1/SIAT1	<	10	<	0	<	0	<	0	40	41	160	201	160	233	none
A/England/309/2019	3C.3a	SIAT1/SIAT1	<	0	<	0	<	0	<	0	<	0	160	205	320	249	none
A/England/344/2019	3C.3a	SIAT1/SIAT1	<	20	<	0	<	5	<	0	<	20	160	206	320	243	none
A/England/335/2019	3C.3a	SIAT1/SIAT1	<	0	<	0	<	0	<	0	<	5	160	184	160	157	K82R
A/England/347/2019	3C.3a	SIAT1/SIAT1	<	0	<	0	<	0	<	0	<	0	160	175	160	125	K82R
A/Norway/2379/2019	3C.3a	SIAT1/SIAT1	<	0	<	0	<	0	<	0	<	0	160	210	320	265	H56Y, K82R, N121D
Vaccine SH 2020							Vaccine NH 2019-20										

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Viruses	Passage History	Neutralisation Titre ¹														HA1 substitutions for 3C.2a/3a viruses compared to cell culture-propagated A/Switzerland/9715293/2013. Egg adaptation HA substitutions compared to the corresponding cell culture-propagated virus. Virus specific substitutions.		
		Post-infection ferret antisera																
		A/Norway 3275/2018 SIAT F03/19	A/Hong Kong 2671/19 SIAT F31/19	A/SouthAustralia 34/2019 Egg F43/19	A/Christchurch 515/19 Egg F32/19	A/La Rioja 2202/18 SIAT F26/18	A/Kansas 14/2017 SIAT F17/19	NYMC X-327 A/Kansas/14 Egg F16/19										
		3C.2a1b+T131K	3C.2a1b+T135K-B	3C.2a1b+T131K	3C.2a1b+T131K	3C.2a1b+T135K-A	3C.3a	3C.3a										
Passage history Ferret number Genetic group		2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read			
REFERENCE VIRUSES																		
A/Norway/3275/2018	3C.2a1b+T131K	SIAT2	10 ⁻³	640	712	80	101	80	78	320	302	80	68	<	0	<	5	L3I, E62G, K92R, A106V, N121K, A128T (+CHO), T131K, S138A, N144S (-CHO), S159Y, K160T (+CHO), N171K, R326K
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	SIAT1/SIAT1	10 ⁻¹	80	78	640	638	<	0	320	282	160	126	40	59	40	40	L3I, E62G, K92R, N121K, T135K (-CHO), S137F, N144S (-CHO), S159Y, K160T (+CHO), N171K, F193S, R326K
A/SouthAustralia/34/2019	3C.2a1b+T131K	E5/D8	10 ⁻⁸	640	509	<	0	1280	1844	80	76	<	20	<	0	<	0	L3I, E62G, K92R, N121K, A128T (+CHO), T131K, S138A, N144S (-CHO), S159Y, T160K (-CHO), N171K, G186I, S219F, R326K
A/Christchurch/515/2019	3C.2a1b+T131K	E3/E1		640	600	<	0	40	46	1280	2007	640	564	40	20	<	10	L3I, E62G, K92R, N121K, A128T (+CHO), T131K, S138A, N144S (-CHO), S159Y, K160T (+CHO), N171K, L194P, Q197R, S219F, R326K
A/La Rioja/2202/2018	3C.2a1b+T135K-A	SIAT1/SIAT4	10 ⁻³	160	210	320	255	40	44	320	469	320	411	<	0	<	0	L3I, E62G, K92R, N121K, T135K (-CHO), S138A, N144S (-CHO), S159Y, K160T (+CHO), N171K, R326K
A/Kansas/14/2017	3C.3a	SIAT3/SIAT2	10 ⁻³	<	0	<	0	<	0	<	0	<	0	160	235	320	251	L3I, S91N, N144K (-CHO), F193S
NYMC X-327 (A/Kansas/14/17)	3C.3a	Ex/E1		<	0	<	0	<	0	<	0	<	20	320	250	2560	2368	L3I, S91N, N144K (-CHO), G186V, D190N, F193S, S219Y
TEST VIRUSES																		
A/Norway/2294/2019	3C.2a1b+T131K			640	795	320	296	160	137	320	386	80	109	<	0	<	0	K83E, Y94N
A/Muscat/6630/2019	3C.2a1b+T131K			640	612	160	147	80	83	320	415	80	68	<	0	<	0	K83E, Y94N
A/Jaalan BBA/6445/2019	3C.2a1b+T131K			640	917	160	130	160	132	320	311	80	95	<	0	<	0	K83E, Y94N
A/Norway/2339/2019	3C.2a1b+T131K			640	731	80	116	80	101	320	290	80	78	<	0	<	0	K83E, Y94N
A/Norway/2334/2019	3C.2a1b+T131K			640	762	160	166	160	120	640	533	80	92	<	0	<	0	K83E, Y94N
A/lbra/5856/2019	3C.2a1b+T131K			1280	1061	160	139	160	128	640	547	80	79	<	0	<	0	167T, K83E, Y94N
A/Norway/2418/2019	3C.2a1b+T131K			2560	2504	160	213	320	346	1280	1074	160	235	<	0	<	0	D73G, K83E, Y94N
A/Norway/2297/2019	3C.2a1b+T131K			1280	1197	80	114	160	158	320	345	160	148	<	0	<	0	T10M (-CHO), I25L, K83E, Y94N
A/lbra/5866/2019	3C.2a1b+T131K			1280	1143	160	132	160	166	320	431	160	137	<	0	<	0	T10M (-CHO), K83E, Y94N
A/Norway/2428/2019	3C.2a1b+T131K			640	709	160	127	80	79	320	376	80	70	<	0	<	0	T10M (-CHO), K83E, Y94N, T128A (-CHO)
A/Hayma/7383/2019	3C.2a1b+T131K			640	693	160	123	80	92	320	320	80	93	<	0	<	0	T10M (-CHO), K83E, Y94N, P169T
A/Sur/6859/2019	3C.2a1b+T131K																	

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 8-4. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2020-01-27

Viruses	Passage history Ferret number Genetic group	Passage History	Neutralisation Titre ¹ Post-infection ferret antisera														HA1 substitutions for 3C.2a/3a viruses compared to cell culture-propagated A/Switzerland/9715293/2013. Egg adaptation HA substitutions compared to the corresponding cell culture-propagated virus. Virus specific substitutions.
			A/Norway 3275/2018 SIAT F03/19		A/Hong Kong 2671/19 SIAT F31/19		A/SouthAustralia 34/2019 Egg F43/19		A/Christchurch 515/19 Egg F32/19		A/La Rioja 2202/18 SIAT F26/18		A/Kansas 14/2017 SIAT F17/19		NYMC X-327 A/Kansas/14 Egg F16/19		
			3C.2a1b+T131K		3C.2a1b+T135K-B		3C.2a1b+T131K		3C.2a1b+T131K		3C.2a1b+T135K-A		3C.3a		3C.3a		
			2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																	
A/Norway/3275/2018	3C.2a1b+T131K	SIAT2 10 ⁻³	640	712	80	101	80	78	320	302	80	68	<	0	<	5	L3I, E62G, K92R, A106V, N121K, A128T (+CHO), T131K, S138A, N144S (-CHO), S159Y, K160T (+CHO), N171K, R326K
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	SIAT1/SIAT1 10 ⁻¹	80	78	640	638	<	0	320	282	160	126	40	59	40	40	L3I, E62G, K92R, N121K, T135K (-CHO), S137F, N144S (-CHO), S159Y, K160T (+CHO), N171K, F193S, R326K
A/SouthAustralia/34/2019	3C.2a1b+T131K	E5/D8 10 ⁻⁸	640	509	<	0	1280	1844	80	76	<	20	<	0	<	0	L3I, E62G, K92R, N121K, A128T (+CHO), T131K, S138A, N144S (-CHO), S159Y, T160K (-CHO), N171K, G186I, S219F, R326K
A/Christchurch/515/2019	3C.2a1b+T131K	E3/E1	640	600	<	0	40	46	1280	2007	640	564	40	20	<	10	L3I, E62G, K92R, N121K, A128T (+CHO), T131K, S138A, N144S (-CHO), S159Y, K160T (+CHO), N171K, L194P, Q197R, S219F, R326K
A/La Rioja/2202/2018	3C.2a1b+T135K-A	SIAT1/SIAT4 10 ⁻³	160	210	320	255	40	44	320	469	320	411	<	0	<	0	L3I, E62G, K92R, N121K, T135K (-CHO), S138A, N144S (-CHO), S159Y, K160T (+CHO), N171K, R326K
A/Kansas/14/2017	3C.3a	SIAT3/SIAT2 10 ⁻³	<	0	<	0	<	0	<	0	<	0	160	235	320	251	L3I, S91N, N144K (-CHO), F193S
NYMC X-327 (A/Kansas/14/17)	3C.3a	Ex/E1	<	0	<	0	<	0	<	0	<	20	320	250	2560	2368	L3I, S91N, N144K (-CHO), G186V, D190N, F193S, S219Y
TEST VIRUSES																	
A/Norway/2330/2019	3C.2a1b+T131K	SIAT1/SIAT1	640	668	320	282	160	134	320	466	80	80	<	0	<	0	D53N
A/Baden-Wuerttemberg/312/2019	3C.2a1b+T131K	C2/SIAT1	640	762	320	293	160	168	320	473	160	152	<	0	<	0	D53N, K83E, Y94N
A/Bayern/114/2019	3C.2a1b+T131K	C2/SIAT1	640	701	160	177	160	188	320	409	160	142	40	20	40	40	K83E, Y94N
A/Denmark/3312/2019	3C.2a1b+T131K	SIAT2/SIAT1	320	472	320	280	80	72	320	384	80	74	<	0	<	0	K83E, Y94N
A/Iceland/82/2019	3C.2a1b+T131K	MDCK1/SIAT1	640	601	160	151	160	138	320	296	80	108	<	0	<	0	K83E, Y94N
A/Stockholm/37/2019	3C.2a1b+T131K	MDCK0/SIAT1	1280	1371	160	194	160	198	640	537	160	146	<	0	<	0	H56Y, K83E, Y94N, S199P
A/Iceland/81/2019	3C.2a1b+T131K	MDCK1/SIAT1	320	263	80	99	80	60	160	173	40	43	<	0	<	0	K83E, Y94N, N158* (-CHO)
A/Denmark/3339/2019	3C.2a1b+T131K	SIAT2/SIAT1	320	347	80	107	80	65	320	453	80	73	<	0	<	0	K83E, Y94N, S198P
A/Hong Kong/4399/2019	3C.2a1b+T131K	MDCK1/SIAT1	640	533	160	129	80	96	320	347	80	73	<	0	<	0	K83E, Y94N, I260M
A/Hong Kong/4434/2019	3C.2a1b+T131K	MDCK1/SIAT1	320	427	80	72	80	78	160	207	40	58	<	0	<	0	K83E, Y94N, I260M
A/Denmark/3310/2019	3C.2a1b+T131K	SIAT2/SIAT1	320	407	160	127	80	74	320	415	80	73	<	0	<	0	S124G, I214T, S219F
A/Norway/2312/2019	3C.2a1b+T131K	SIAT1/SIAT1	640	949	160	200	160	217	640	670	160	224	<	0	<	0	Q197R, S219F
A/Iceland/84/2019	3C.2a1b+T131K	MDCK1/SIAT1	160	126	80	83	40	48	160	147	40	51	<	0	<	0	Q197R, K207R, S219F
A/Latvia/11-032248/2019	3C.2a1b+T131K	MDCK2/SIAT1	320	392	160	166	160	121	640	490	160	135	<	0	<	0	I48M, V112I, Q197R, K207R, I214T, S219F
A/Latvia/12-006525/2019	3C.2a1b+T131K	MDCKx/SIAT1	320	384	160	216	80	118	320	314	80	117	<	0	<	0	Q1*, I48M, V112I, Q197R, K207R, I214T, S219F
A/Latvia/11-068832/2019	3C.2a1b+T131K	MDCK1/SIAT1	320	413	160	189	160	155	320	384	160	175	<	0	<	0	I48M, V112I, T160* (-CHO), Q197R, K207R, I214T, S219F
A/Latvia/11-028714/2019	3C.2a1b+T131K	MDCK1/SIAT1	160	160	80	64	80	60	160	150	40	55	<	0	<	0	I48M, V112I, T160* (-CHO), Q197R, K207R, I214T, S219F
A/Baden-Wuerttemberg/310/2019	3C.2a1b+T135K-A	C1/SIAT1	80	92	160	180	40	40	320	452	320	371	<	0	<	0	S198P, R261Q
A/Bremen/24/2019	3C.2a1b+T135K-A	C1/SIAT1	160	132	320	372	80	67	320	366	320	441	<	0	40	20	S198P, R261Q
A/Denmark/3264/2019	3C.2a1b+T135K-A	SIAT3/SIAT1	40	42	640	600	<	0	160	192	80	71	<	0	<	0	A138S, G186D, D190N, F193S, S198P
A/Hong Kong/4398/2019	3C.2a1b+T135K-B	MDCK1/SIAT1	<	0	320	403	<	0	160	143	40	40	<	0	<	0	none
A/Hong Kong/4435/2019	3C.2a1b+T135K-B	MDCK1/SIAT1	<	0	640	499	<	0	160	160	40	43	<	0	<	0	none
A/Denmark/3331/2019	3C.2a1b+T135K-B	SIAT2/SIAT1	<	0	640	584	<	0	160	203	40	40	<	0	<	0	E50K
A/Denmark/3292/2019	3C.2a1b+T135K-B	SIAT2/SIAT1	<	0	320	351	<	0	80	110	<	0	<	0	<	0	E50K
A/Denmark/3275/2019	3C.2a1b+T135K-B	SIAT3/SIAT1	<	0	640	496	<	0	160	218	40	40	<	0	<	0	E50K
Vaccine SH 2020							Vaccine NH 2019-20										

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Viruses	Passage history Ferret number Genetic group	Passage History	Neutralisation Titre ¹														HA1 substitutions for 3C.2a/3a viruses compared to cell culture-propagated A/Switzerland/9715293/2013. Egg adaptation HA substitutions compared to the corresponding cell culture-propagated virus. Virus specific substitutions.	
			Post-infection ferret antisera															
			A/Norway 3275/2018 SIAT F03/19		A/Hong Kong 2671/19 SIAT F31/19		A/SouthAustralia 34/2019 Egg F43/19		A/Christchurch 515/19 Egg F32/19		A/La Rioja 2202/18 SIAT F26/18		A/Kansas 14/2017 SIAT F17/19		NYMC X-327 A/Kansas/14 Egg F16/19			
			3C.2a1b+T131K		3C.2a1b+T135K-B		3C.2a1b+T131K		3C.2a1b+T131K		3C.2a1b+T135K-A		3C.3a		3C.3a			
			2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES																		
A/Norway/3275/2018	3C.2a1b+T131K	SIAT2	10 ⁻³	640	712	80	101	80	78	320	302	80	68	<	0	<	5	L3I, E62G, K92R, A106V , N121K, A128T (+CHO), T131K, S138A, N144S (-CHO), S159Y, K160T (+CHO), N171K, R326K
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	SIAT1/SIAT1	10 ⁻¹	80	78	640	638	<	0	320	282	160	126	40	59	40	40	L3I, E62G, K92R, N121K, T135K (-CHO), S137F, N144S (-CHO), S159Y, K160T (+CHO), N171K, F193S, R326K
A/SouthAustralia/34/2019	3C.2a1b+T131K	Es/D8	10 ⁻⁸	640	509	<	0	1280	1844	80	76	<	20	<	0	<	0	L3I, E62G, K92R, N121K, A128T (+CHO), T131K, S138A, N144S (-CHO), S159Y, T160K (-CHO), N171K, G186I , S219F , R326K
A/Christchurch/515/2019	3C.2a1b+T131K	E3/E1		320	391	<	0	40	46	1280	2007	640	711	40	20	<	10	L3I, E62G, K92R, N121K, A128T (+CHO), T131K, S138A, N144S (-CHO), S159Y, K160T (+CHO), N171K, L194P , Q197R , S219F , R326K
A/La Rioja/2202/2018	3C.2a1b+T135K-A	SIAT1/SIAT4	10 ⁻³	160	210	320	255	40	44	320	469	320	411	<	0	<	0	L3I, E62G, K92R, N121K, T135K (-CHO), S138A, N144S (-CHO), S159Y, K160T (+CHO), N171K, R326K
A/Kansas/14/2017	3C.3a	SIAT3/SIAT2	10 ⁻³	<	0	<	0	<	0	<	0	<	0	160	235	320	251	L3I, S91N, N144K (-CHO), F193S
NYMC X-327 (A/Kansas/14/17)	3C.3a	Ex/E1		<	0	<	0	<	0	<	0	<	20	320	250	2560	2368	L3I, S91N, N144K (-CHO), G186V , D190N , F193S, S219Y
TEST VIRUSES																		
A/Iceland/80/2019	3C.2a1b+T131K	MDCK1/SIAT1		640	804	160	143	80	112	640	640	80	104	<	0	<	0	K83E, Y94N
A/Iceland/24506/2019	3C.2a1b+T131K	SIAT1		640	583	<	<	80	97	320	378	80	115	<	0	<	0	K83E, Y94N
A/Qatar/10-VI-19-0072992/2019	3C.2a1b+T131K	SIAT1		640	789	80	113	80	103	640	566	80	109	<	0	<	0	K83E, Y94N
A/Qatar/16-VI-19-0070720/2019	3C.2a1b+T131K	SIAT1		640	823	<	14	80	117	640	587	160	140	<	0	<	0	K83E, Y94N, Y105H
A/Iceland/79/2019	3C.2a1b+T131K	MDCK1/SIAT1		320	415	<	<	80	72	320	400	40	56	<	0	<	0	K83E, Y94N, N158* (-CHO)
A/Iceland/77/2019	3C.2a1b+T131K	MDCK1/SIAT1		320	306	<	<	80	109	80	112	40	46	<	0	<	0	K83E, Y94N, T160I (-CHO)
A/Iceland/78/2019	3C.2a1b+T131K	MDCK1/SIAT1		640	640	80	117	80	77	640	498	80	80	<	0	<	0	K83E, Y94N, P221*
A/Qatar/39-VI-19-0070534/2019	3C.2a1b+T131K	SIAT1		640	821	320	251	80	107	640	629	80	98	<	0	<	0	T10M (-CHO), K83E, Y94N
A/Iceland/22847/2019	3C.2a1b+T131K	SIAT1		640	800	160	170	160	222	640	640	160	216	<	0	<	0	Q197R , K207R , S219F
A/Skovde/3/2019	3C.2a1b+T131K	MDCK1/SIAT1		160	181	<	0	80	101	160	143	40	53	<	0	<	0	V43I , Q197R , K207R , T160* (-CHO), S219F</

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 8-6. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2020-02-10

Viruses	Passage history Ferret number Genetic group	Passage History	Neutralisation Titre ¹														HA1 substitutions for 3C.2a/3a viruses compared to cell culture-propagated A/Switzerland/9715293/2013. Egg adaptation HA substitutions compared to the corresponding cell culture-propagated virus. Virus specific substitutions.
			Post-infection ferret antisera														
			A/Norway 3275/2018 SIAT F03/19		A/Hong Kong 2671/19 SIAT F31/19		A/Hong Kong 2671/19 Egg F44/19		A/SouthAustralia 34/2019 Egg F43/19		A/Christchurch 515/19 Egg F32/19		A/Kansas 14/2017 SIAT F17/19		NYMC X-327 A/Kansas/14 Egg F16/19		
			3C.2a1b+T131K		3C.2a1b+T135K-B		3C.2a1b+T135K-B		3C.2a1b+T131K		3C.2a1b+T131K		3C.3a		3C.3a		
			2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																	
A/Norway/3275/2018	3C.2a1b+T131K	SIAT2 10 ⁻³	640	712	80	101	<	0	80	78	320	302	<	0	<	5	L3I, E62G, K92R, A106V, N121K, A128T (+CHO), T131K, S138A, N144S (-CHO), S159Y, K160T (+CHO), N171K, R326K
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	SIAT1/SIAT1 10 ⁻¹	80	78	640	638	160	171	<	0	320	282	40	59	40	40	L3I, E62G, K92R, N121K, T135K (-CHO), S137F, N144S (-CHO), S159Y, K160T (+CHO), N171K, F193S, R326K
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	E8/E1	<	0	80	92	1280	1823	160	160	<	0	40	42	160	235	L3I, E62G, K92R, N121K, T135K (-CHO), S137F, N144S (-CHO), S159Y, T160I (-CHO), N171K, G186V, F193S, S219F, D225N, R326K
A/SouthAustralia/34/2019	3C.2a1b+T131K	E5/D8 10 ⁻⁸	640	509	<	0	80	61	1280	1844	80	76	<	0	<	0	L3I, E62G, K92R, N121K, A128T (+CHO), T131K, S138A, N144S (-CHO), S159Y, T160K (-CHO), N171K, G186I, S219F, R326K
A/Christchurch/515/2019	3C.2a1b+T131K	E3/E1	320	391	<	0	40	20	40	46	1280	2007	40	20	<	10	L3I, E62G, K92R, N121K, A128T (+CHO), T131K, S138A, N144S (-CHO), S159Y, K160T (+CHO), N171K, L194P, Q197R, S219F, R326K
A/Kansas/14/2017	3C.3a	SIAT3/SIAT2 10 ⁻³	<	0	<	0	ND	ND	<	0	<	0	160	235	320	251	L3I, S91N, N144K (-CHO), F193S
NYMC X-327 (A/Kansas/14/17)	3C.3a	Ex/E1	<	0	<	0	ND	ND	<	0	<	0	320	250	2560	2368	L3I, S91N, N144K (-CHO), G186V, D190N, F193S, S219Y
TEST VIRUSES																	
A/Slovenia/3126/2019	3C.2a1b+T131K	SIAT2/SIAT1	640	533	80	83	<	0	80	89	640	549	<	0	<	0	K83E, Y94N
A/Austria/1204784/2019	3C.2a1b+T131K	SIAT1/SIAT1	640	574	80	62	<	0	80	65	640	640	<	0	<	0	K83E, Y94N
A/Austria/1202557/2019	3C.2a1b+T131K	SIAT1/SIAT1	640	768	80	95	<	0	160	160	640	594	<	0	<	0	K83E, Y94N
A/Berlin/63/2019	3C.2a1b+T131K	C2/SIAT1	640	800	160	125	<	0	80	117	640	947	<	0	<	0	Q197R, S219F
A/Nordrhein-Westfalen/137/2019	3C.2a1b+T131K	C1/SIAT1	640	496	320	305	<	0	160	96	1280	974	<	0	<	0	Q197R, S219F
A/Bretagne/2709/2019	3C.2a1b+T131K	SIAT1	640	557	160	143	<	0	80	77	1280	960	<	0	<	0	P55H, Q197R, K207R, S219F
A/Centre/2608/2019	3C.2a1b+T135K-A	SIAT1	80	65	160	152	<	0	<	0	320	368	<	0	<	0	S198P, R261Q
A/Austria/1204623/2019	3C.2a1b+T135K-A	SIAT1/SIAT1	80	117	160	160	<	0	<	0	640	489	<	0	<	0	S198P, R261Q
A/Ghana/4021/2019	3C.2a1b+T135K-A	SIAT1	<	0	160	148	<	0	<	0	320	251	<	0	<	0	A138S, G186D, D190N, F193S, S198P
A/Ghana/722/2019	3C.2a1b+T135K-A	SIAT1	<	0	160	160	40	40	<	0	320	247	<	0	<	0	A138S, G186D, D190N, F193S, S198P
A/Belgium/G0003/2020	3C.2a1b+T135K-A	SIAT1	<	0	160	201	40	54	<	0	320	291	<	0	<	0	A138S, G186D, D190N, F193S, S198P
A/Ghana/3793/2019	3C.2a1b+T135K-A	SIAT1	40	40	160	151	80	94	<	0	320	269	<	0	<	0	E50K, A138S, G186D, D190N, F193S, S198P
A/Belgium/G0540/2019	3C.3a	SIAT1	<	0	<	0	<	0	<	0	<	0	80	101	160	160	none
A/Belgium/G0538/2019	3C.3a	SIAT1	<	0	<	0	<	0	<	0	<	0	80	112	160	155	none
A/Slovenia/23/2020	3C.3a	SIATx/SIAT1	<	0	<	0	<	0	<	0	<	0	80	87	320	240	none
A/Paris/2560/2019	3C.3a	SIAT1	<	0	<	0	<	0	<	0	<	0	80	76	80	116	none
									Vaccine SH 2020								
													Vaccine NH 2019-20				

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 8-7. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) - Summary

Viruses		Neutralisation Titre																	
		Post-infection ferret antisera																	
		A/Norway 3275/2018 SIAT F03/19		A/SouthAustralia 34/2019 Egg F43/19		A/Christchurch 515/19 Egg F32/19		A/La Rioja 2202/18 SIAT F26/18		A/Hong Kong 2671/19 SIAT F31/19		A/Hong Kong 2671/19 Egg F44/19		A/Kansas 14/2017 SIAT F17/19		NYMC X-327 A/Kansas/14 Egg F16/19			
		3C.2a1b+T131K		3C.2a1b+T131K		3C.2a1b+T131K		3C.2a1b+T135K-A		3C.2a1b+T135K-B		3C.2a1b+T135K-B		3C.3a		3C.3a			
Passage history Ferret number Genetic group		2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES																			
A/Norway/3275/2018	3C.2a1b+T131K	640	712	80	78	320	302	80	68	80	101	<	0	<	0	<	5		
A/SouthAustralia/34/2019	3C.2a1b+T131K	640	509	1280	1844	80	76	<	20	<	0	80	61	<	0	<	0		
A/Christchurch/515/2019	3C.2a1b+T131K	320	391	40	46	1280	2007	640	711	<	0	40	20	40	20	<	10		
A/La Rioja/2202/2018	3C.2a1b+T135K-A	160	210	40	44	320	469	320	411	320	255	<	0	<	0	<	0		
A/Hong Kong/2671/2019	3C.2a1b+T135K-B	80	78	<	0	320	282	160	126	640	638	80	102	40	59	40	40		
A/Hong Kong/2671/2019	3C.2a1b+T135K-B							1280	1823										
A/Kansas/14/2017	3C.3a	<	0	<	0	<	0	<	0	<	0	ND	ND	160	235	320	251		
NYMC X-327 (A/Kansas/14/17)	3C.3a	<	0	<	0	<	0	<	20	<	0	ND	ND	320	250	2560	2368		
TEST VIRUSES																			
Number of viruses tested		120		120		120		104		120		16		120		120			
No with titre reduction ≤2-fold		65				3		10		24				13					
%		54.2				2.5		9.6		20.0				10.8					
No with titre reduction =4-fold		7		1		36		38		32				6					
%		5.8		0.8		30.0		36.5		26.7				5.0					
No with titre reduction ≥8-fold		48		119		81		56		64		16		101		120			
%		40.0		99.2		67.5		53.8		53.3		100.0		84.2		100.0			
Subgroup 3C.2a1b viruses + T131K																			
Number of viruses tested		74		74		74		68		74		6		74		74			
No with titre reduction ≤2-fold		65				3		6		3									
%		87.8				4.1		8.8		4.1									
No with titre reduction =4-fold		5		1		35		37		27				1					
%		6.8		1.4		47.3		54.4		36.5				1.4					
No with titre reduction ≥8-fold		4		73		36		25		44		6		73		74			
%		5.4		98.6		48.6		36.8		59.5		100.0		98.6		100.0			
Subgroup 3C.2a1b viruses + T135K-A																			
Number of viruses tested		11		11		11		5		11		6		11		11			
No with titre reduction ≤2-fold								4		3									
%								80.0		27.3									
No with titre reduction =4-fold		2				1				5									
%		18.2				9.1				45.5									
No with titre reduction ≥8-fold		9		11		10		1		3		6		11		11			
%		81.8		100.0		90.9		20.0		27.3		100.0		100.0		100.0			
Subgroup 3C.2a1b viruses + T135K-B																			
Number of viruses tested		18		18		18		18		18		0		18		18			
No with titre reduction ≤2-fold										18									
%										100.0									
No with titre reduction =4-fold								1						1					
%								5.6						5.6					
No with titre reduction ≥8-fold		18		18		18		17						17		18			
%		100.0		100.0		100.0		94.4						94.4		100.0			
Clade 3C.3a viruses																			
Number of viruses tested		17		17		17		13		17		4		17		17	17		
No with titre reduction ≤2-fold														13			15*		
%														76.5					
No with titre reduction =4-fold														4					
%														23.5					
No with titre reduction ≥8-fold		17		17		17		13		17		4				17			
%		100.0		100.0		100.0		100.0		100.0		100.0				100.0			
				Vaccine SH 2019												Vaccine NH 2019-20			

* Number of clade 3C.3a viruses giving a titre of at least 160 with this antiserum

Figure 6a. HA gene phylogeny of A(H3N2) viruses used in VN assay

Vaccine viruses

Reference viruses

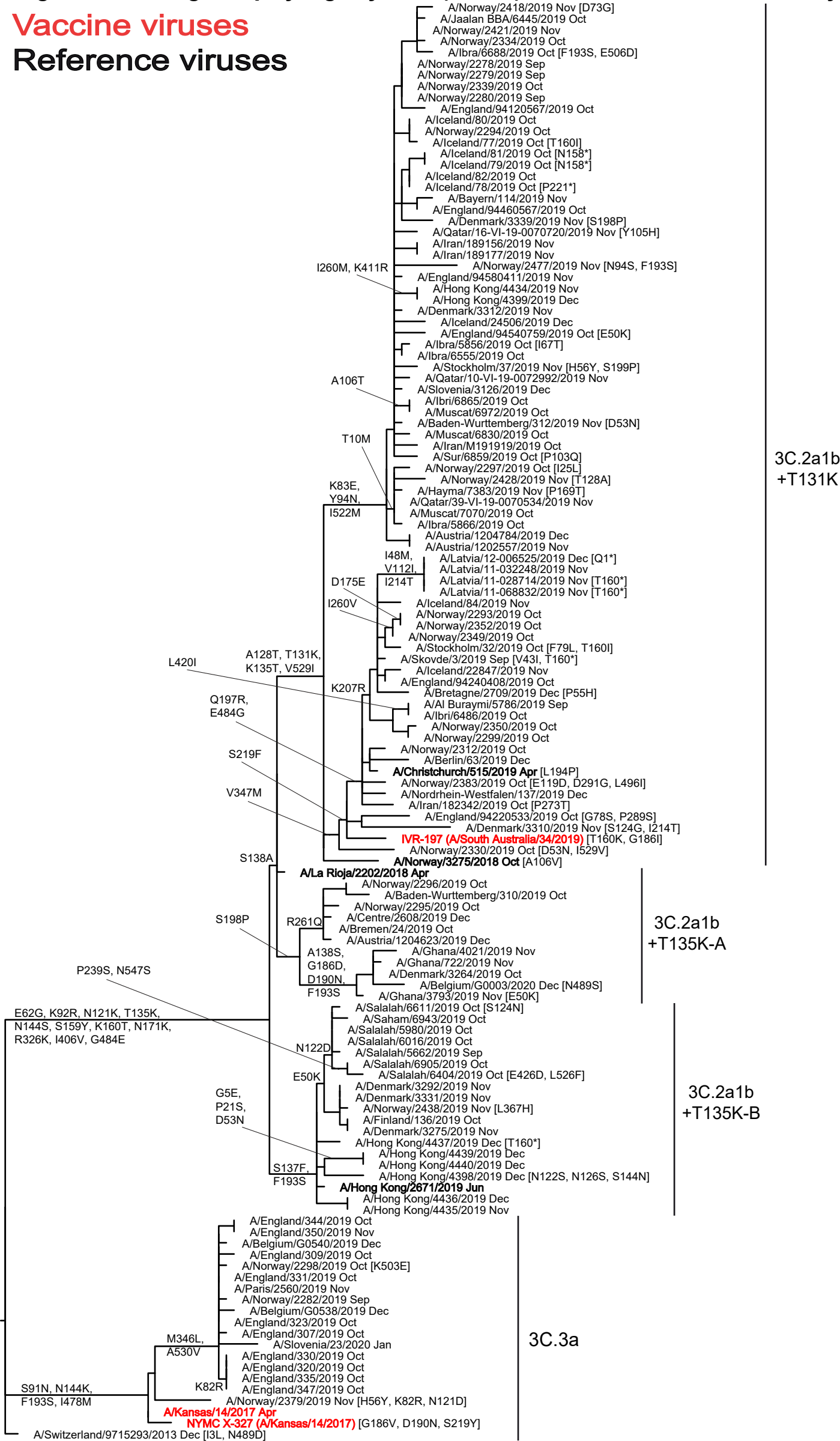


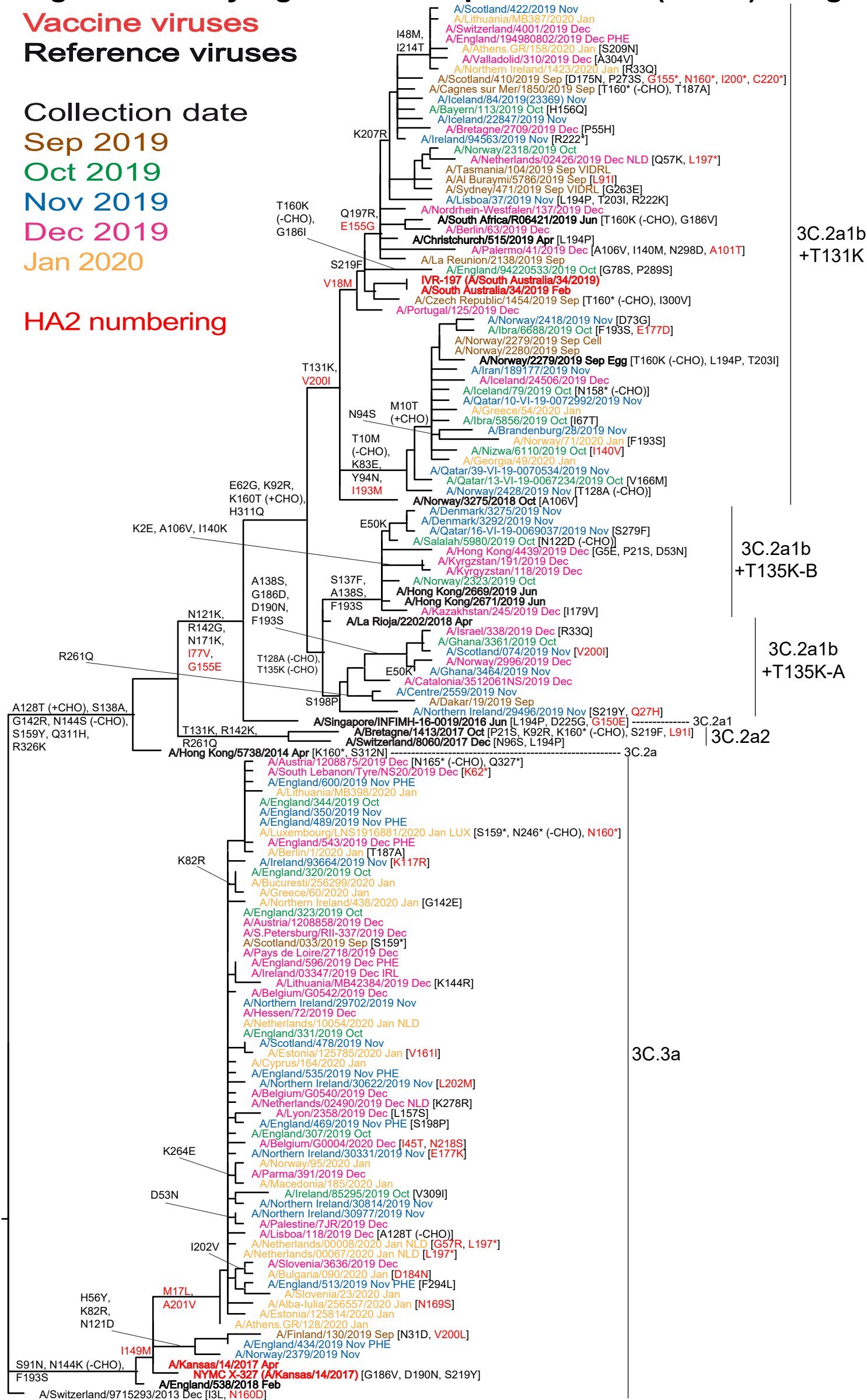
Figure 6b. Phylogenetic comparison of A(H3N2) HA genes

Vaccine viruses
Reference viruses

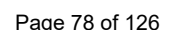
Collection date

Sep 2019
Oct 2019
Nov 2019
Dec 2019
Jan 2020

HA2 numbering



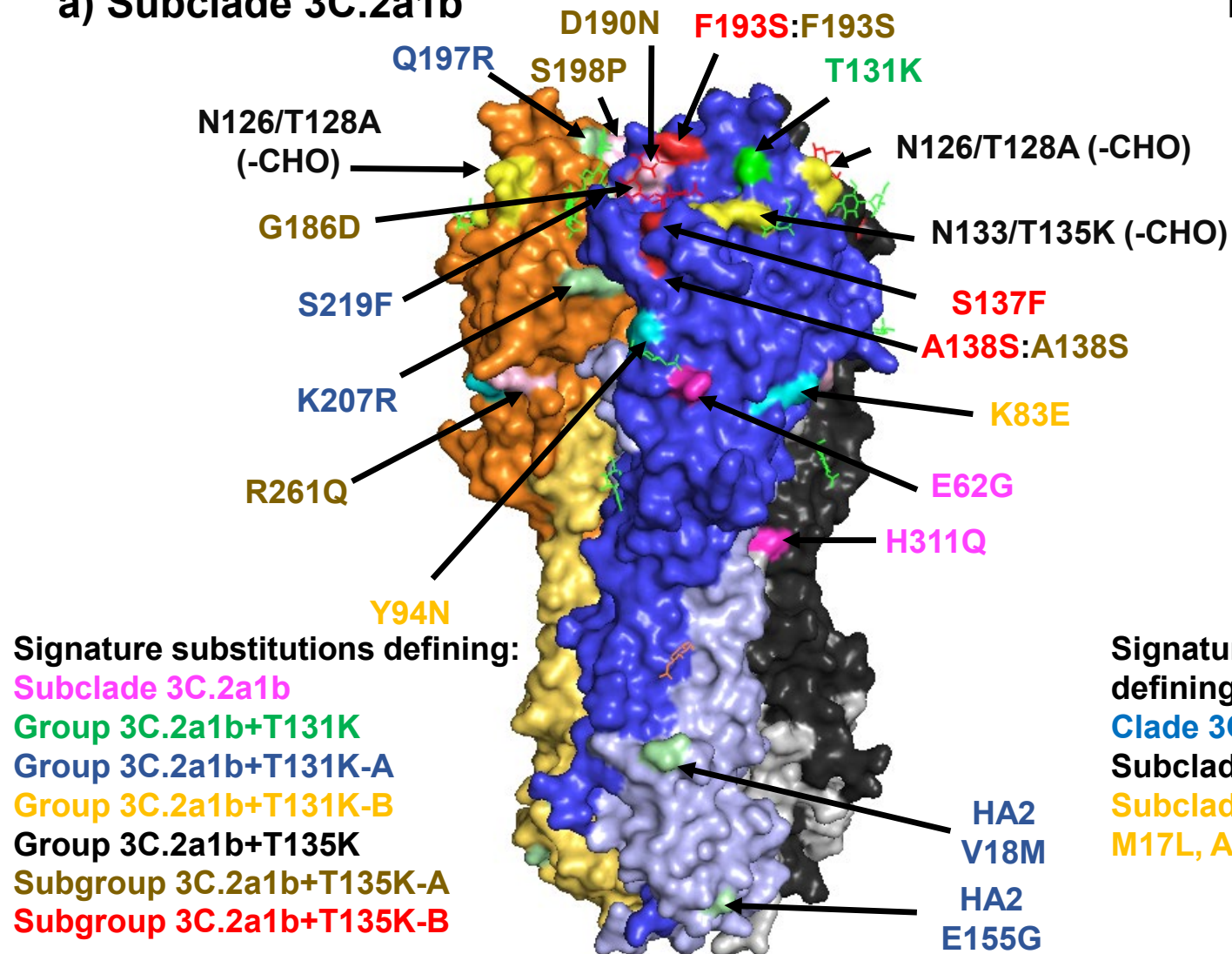
Vaccine virus: **Reference virus:** **Genetic group:** **HA1/HA2 boundary:** **Potential N-linked glycosylation site**



	310	320	330	340	350	360	370	380	390	400
A/Switzerland/9715293/2013 Dec	TYGACPRYVKQSTLKLATGM	RNVPERQTRGIFGAIAGFI	ENGWEGMVDGWYGFRRH	QNSEGRGQAADLKSTQAA	IDQINGKLNRLIGKTNEK	FHQIEKFS				
A/England/538/2018 Feb										
A/Kansas/14/2017 Apr										
A/Finland/130/2019 Sep										
A/Athens.GR/84/2020 Jan					L					
A/Northern Ireland/29790/2019					L					
A/Parma/391/2019 Dec					L					
A/Palestine/7JR/2019 Dec					L					
A/Bucuresti/256299/2020 Jan					L					
A/Bulgaria/061/2020 Jan					L					
A/Slovenia/3636/2019 Dec					L					
A/Palestine/20N/2019 Dec					L					
A/Hong Kong/5738/2014 Apr		HN	K							
A/Singapore/INFIMH1619/2016 Ap		H	K							
A/La Rioja/2202/2018 Apr			K							
A/Switzerland/9947/2019 Nov	X		K							
A/Ghana/3793/2019 Nov			K							
A/Palermo/36/2019 Dec			K							
A/Hong Kong/2669/2019 Jun			K							
A/Salalah/6611/2019 Oct			K							
A/Kyrgyzstan/291/2019 Dec			K							
A/Norway/3275/2018 Oct			K							
A/South Australia/34/2019 Feb			K		M					
A/Berlin/63/2019 Dec			K		M					
A/Catalonia/120508/2019 Dec			K		M					
A/Stockholm/32/2019 Oct			K		M					
A/Hayma/7383/2019 Nov			K							
A/Georgia/1237/2019 Dec			K							
A/Georgia/1269/2019 Dec			K							
A/Norway/2279/2019 Sep			K							
A/Norway/100/2020 Jan			K							
A/Brandenburg/28/2019 Nov			K							
	410	420	430	440	450	460	470	480	490	500
A/Switzerland/9715293/2013 Dec	EVEGRIQDLEKYVEDTKID	LWSYNAELLVALENQHT	IDLTDS	EMNKLFEKTKKQLRENA	EDMGNGCFRIYHKCDNAC	IGSIRNGTYD	HDVVRDEALNNRF			
A/England/538/2018 Feb									N	
A/Kansas/14/2017 Apr								M	N	
A/Finland/130/2019 Sep								M	N	
A/Athens.GR/84/2020 Jan								M	N	S
A/Northern Ireland/29790/2019								M	N	
A/Parma/391/2019 Dec								M	N	
A/Palestine/7JR/2019 Dec								M	N	
A/Bucuresti/256299/2020 Jan								M	N	
A/Bulgaria/061/2020 Jan								M	N	
A/Slovenia/3636/2019 Dec								M	N	
A/Palestine/20N/2019 Dec								M	N	
A/Hong Kong/5738/2014 Apr									N	
A/Singapore/INFIMH1619/2016 Ap	V							E	N	
A/La Rioja/2202/2018 Apr		V						E	N	
A/Switzerland/9947/2019 Nov		V						E	N	
A/Ghana/3793/2019 Nov		V						E	N	
A/Palermo/36/2019 Dec		V X						E	N	
A/Hong Kong/2669/2019 Jun		V						E	N	
A/Salalah/6611/2019 Oct		V						E	N	
A/Kyrgyzstan/291/2019 Dec		V						E	N	
A/Norway/3275/2018 Oct		V						E	N	
A/South Australia/34/2019 Feb		V						E	N	
A/Berlin/63/2019 Dec		V							N	
A/Catalonia/120508/2019 Dec		V							N	
A/Stockholm/32/2019 Oct		V							N	
A/Hayma/7383/2019 Nov		V						E	N	
A/Georgia/1237/2019 Dec		V						E	N	
A/Georgia/1269/2019 Dec		V						E	N	
A/Norway/2279/2019 Sep		V						E	N	
A/Norway/100/2020 Jan		V						E	N	
A/Brandenburg/28/2019 Nov		V						E	N	
	510	520	530	540	550					
A/Switzerland/9715293/2013 Dec	QIKGV	ELKSGYKDWILWISFAIS	CFLLCVALLGFI	MMWACQKGNIR	CNICI					
A/England/538/2018 Feb										
A/Kansas/14/2017 Apr										
A/Finland/130/2019 Sep			L							
A/Athens.GR/84/2020 Jan			V							
A/Northern Ireland/29790/2019			VM							
A/Parma/391/2019 Dec			V							
A/Palestine/7JR/2019 Dec			V							
A/Bucuresti/256299/2020 Jan			V							
A/Bulgaria/061/2020 Jan			V							
A/Slovenia/3636/2019 Dec			V							
A/Palestine/20N/2019 Dec			V							
A/Hong Kong/5738/2014 Apr										
A/Singapore/INFIMH1619/2016 Ap										
A/La Rioja/2202/2018 Apr										
A/Switzerland/9947/2019 Nov										
A/Ghana/3793/2019 Nov										
A/Palermo/36/2019 Dec										
A/Hong Kong/2669/2019 Jun										
A/Salalah/6611/2019 Oct										
A/Kyrgyzstan/291/2019 Dec										
A/Norway/3275/2018 Oct			I							
A/South Australia/34/2019 Feb			I							
A/Berlin/63/2019 Dec			I							
A/Catalonia/120508/2019 Dec			I							
A/Stockholm/32/2019 Oct			I							
A/Hayma/7383/2019 Nov		M	I							
A/Georgia/1237/2019 Dec		M	I							
A/Georgia/1269/2019 Dec		M	I							
A/Norway/2279/2019 Sep		M	I							
A/Norway/100/2020 Jan		M	I							
A/Brandenburg/28/2019 Nov		M	I							

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

a) Subclade 3C.2a1b



b) Clade 3C.3a

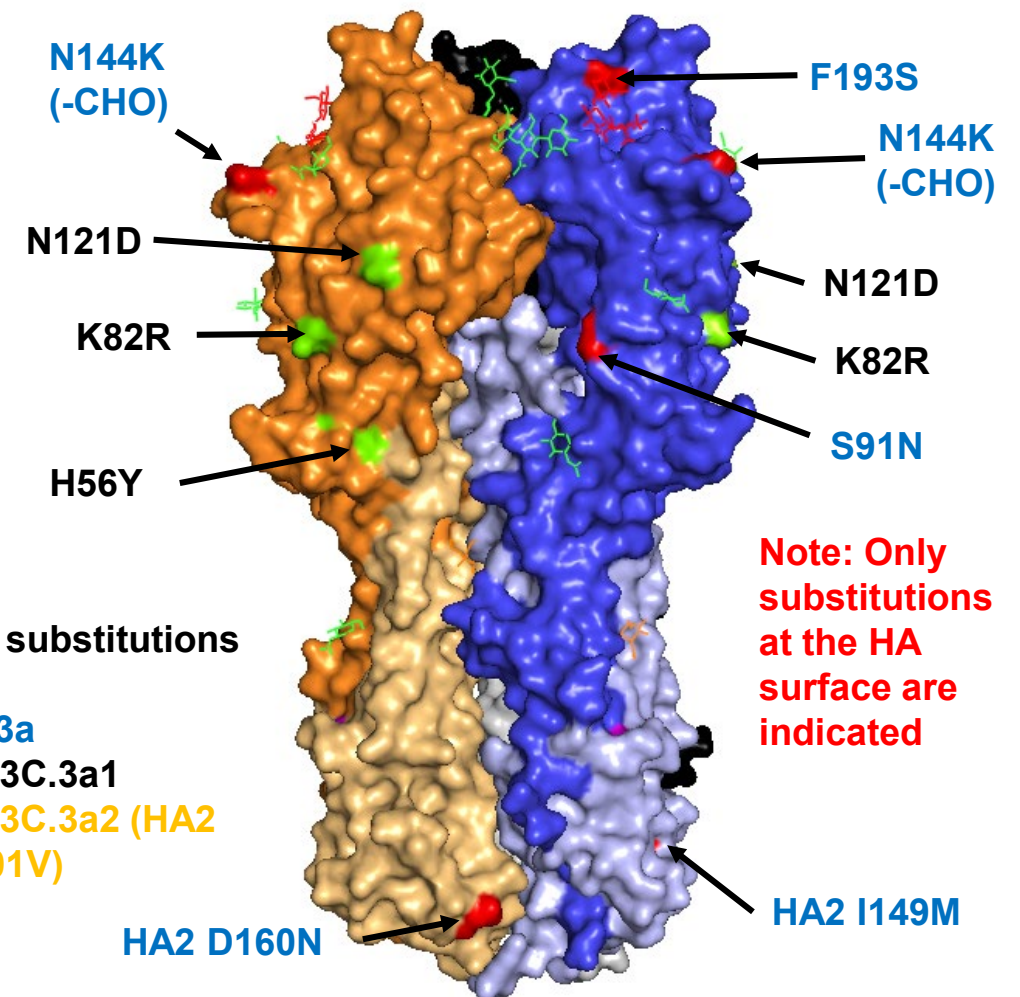
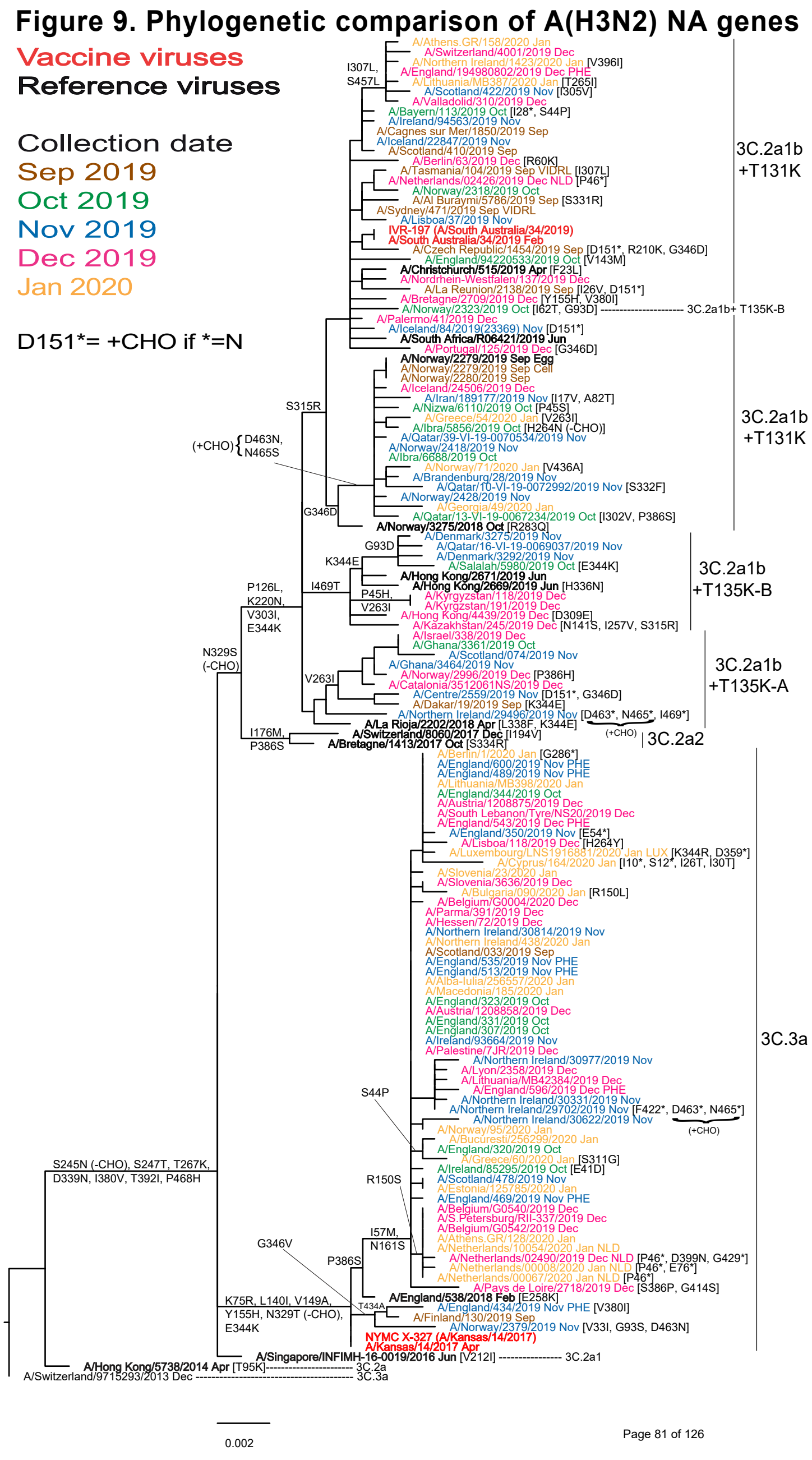
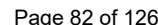


Figure 8. Amino acid substitutions in the HA of (a) recent group 3C.2a1b H3N2 viruses compared with A/Singapore/INFIMH-16-0019/2016 and (b) clade 3C.3a viruses compared with A/Switzerland/9715293/2013



Vaccine virus: **Reference virus:** **Genetic group:** **Potential N-linked glycosylation site**



Vaccine virus: **Reference virus:** **Genetic group:** **Potential N-linked glycosylation site**

Influenza B virus analyses.

Influenza B viruses represented 32% of the samples received with collection dates after 2019-08-31 and were received from NICs in 42 countries or administrative regions in four WHO Regions: African (3: Ghana, Madagascar and Mauritius), Eastern Mediterranean (7: Iran, Jordan, Lebanon, Morocco, Oman, Palestine and Qatar), European (31) and Western Pacific (1: Hong Kong SAR) (**Table 1**). B/Victoria lineage viruses outnumbered B/Yamagata lineage viruses at a ratio of ~28:1.

B/Victoria lineage.

HI analyses

The collection details and summary results for viruses that were recovered and assayed by HI are shown in **Table 9**.

HI results for the propagated viruses analysed since our September 2019 VCM report are shown in **Tables 10-1 to 10-8**, test viruses in each table are sorted by date of collection and those below the red horizontal lines in **Tables 10-2** and **10-4 to 10-8** are those with collection dates after 2019-08-31. The HA genetic group is indicated for those viruses that have been sequenced. A summary of the recognition of the test viruses collected after 2019-08-31 is shown in **Table 10-9**. Two clearly distinct patterns of antigenicity were detected.

A total of 149 viruses with collection dates after 2019-08-31 were analysed; gene sequencing has been completed for all but one of the viruses analysed antigenically. The large majority of viruses (141 of 149, 95%) were identified as having HA genes encoding a triple deletion $\Delta 162-164$ in HA1. The majority of these viruses were recognised well by antisera raised against viruses with the three amino acid deletion in HA1 ($\Delta 162-164$) – cell culture-propagated B/Cote d'Ivoire/1662/2018, and the cell culture-propagated and egg-propagated cultivars of B/Washington/02/2019, the vaccine virus recommended for use in the Southern Hemisphere 2020 season. These antisera recognised 100%, 84% and 83%, respectively, of the tested viruses with the ($\Delta 162-164$) deletion at titres within 4-fold of the titres of the antisera for the homologous viruses, and 100%, 78% and 81% within 2-fold. These viruses were recognised poorly by antisera raised against egg-propagated or cell culture-propagated viruses with no deletion in the HA (B/Brisbane/60/2008, B/South Australia/81/2012, B/Ireland/3154/2016, and B/Nordrhein-Westfalen/1/2016). The majority of $\Delta 162-164$ viruses were recognised at titres >4-fold lower than the homologous titres by antisera raised against viruses having HA genes with a two amino acid deletion $\Delta 162-163$ in HA1: the antisera raised against cell culture-propagated B/Norway/2409/2017 and B/Colorado/06/2017 and the egg-propagated cultivar of B/Colorado/06/2017 recognising 100%, 73% and 91%, respectively of test viruses collected after 2019-08-31 at titres >4-fold lower than the homologous titres.

Several viruses with the triple deletion $\Delta 162-164$ in HA1 were recognised poorly by one or both of the antisera raised against cell culture-propagated and egg-propagated cultivars of B/Washington/02/2019 each recognising approximately 15% of the test viruses poorly. The poorly recognised viruses had HA genes that encoded amino acid

substitutions in HA1: N126K; E128K some with T155A; D129N; N150K with G184E, N197D and R279K; or T155K; some had additional substitutions.

Seven viruses had HA genes with the two amino acid deletion in HA1 (Δ 162-163). These were recognised best by antisera raised against cell culture-propagated B/Norway/2409/2017 and B/Colorado/06/2017 and the egg-propagated cultivar of B/Colorado/06/2017 which recognised six, all and six, respectively, of the seven tested viruses at titres within 4-fold of the titres of the antisera for the homologous viruses. These viruses were all poorly recognised by antisera raised against viruses with no deletion in the HA and those with the Δ 162-164 deletion in HA1.

Genetic analysis

Phylogenetic analysis of the HA genes of B/Victoria lineage viruses sequenced at the Crick, and other representatives, is shown in **Figure 11**. A phylogenetic tree of the HA genes of all viruses in the B/Victoria lineage with collection dates after 2019-08-31 analysed genetically at the Francis Crick Institute is shown in the Annex, **Figure A-5**. An alignment of the HA glycoproteins of a selection of these viruses is shown in **Figure 12**. All recently circulating viruses analysed carried HA genes that fell into genetic clade 1A, the B/Brisbane/60/2008 clade, and all fell in a genetic group defined by three **HA1** amino acid substitutions compared with B/Brisbane/60/2008: **I117V**, **N129D** and **V146I**. There are three clusters within clade 1A with deletions in HA1. Test viruses with HA genes encoding the two amino acid deletion in **HA1** (Δ 162-163) also encoded amino acid substitutions **D129G** and **I180V** in **HA1** and **R151K** in **HA2** and represented 3% of the samples received since 2019-08-31, (**genetic group 1A(Δ 2)**). The HA genes of viruses collected since 2019-08-31 with a three amino acid deletion in **HA1** (Δ 162-164) fell into a group encoding **K136E** substitution in **HA1** (**genetic group 1A(Δ 3)B**) and represented the vast majority of viruses (97%) received with collection dates after 2019-08-31. Most viruses in this group also have HA genes encoding **G133R** substitution in **HA1**. One minor subgroup within **genetic group 1A(Δ 3)B** without the **G133R** substitution had **HA1** substitutions **N150K**, **G184E**, **N197D** (resulting in the loss of the glycosylation motif at **197-199** in **HA1** and **R275K**; another minor subgroup had the substitution **N126K**. No viruses with the Δ 162-164 deletion with the substitutions **I180T** and **K209N** in **HA1** (**genetic group 1A(Δ 3)A**) with collection dates after 2019-08-31 have been received. **Figure 13** shows the locations of amino acid substitutions that define the three groups of deletion variants on a B/Brisbane/60/2008 HA structure.

Figure 14 shows a phylogenetic tree for the NA gene of representative viruses and a phylogenetic tree for the NA genes of all viruses in the B/Victoria lineage with collection dates after 2019-08-31 analysed genetically at the Francis Crick Institute is shown in the Annex, **Figure A-6**. **Figure 15** shows a corresponding amino acid sequence alignment for a selection of NA glycoproteins. The NA gene phylogeny is largely congruent with the HA phylogeny and most viruses in the main recently emerging Δ 162-164 genetic group (**genetic group 1A(Δ 3)B**) encode a **A395T** substitution in the NA, whilst the Δ 162-163 group of viruses (**genetic group 1A(Δ 2)**) share an **NA K371Q** amino acid substitution. Minor genetic subgroups have been seen within the NA and are associated with those observed in the HA.

B/Yamagata lineage.

The collection details and summary results for viruses that were recovered and assayed by HI are shown in **Table 12**. B/Yamagata lineage viruses were received only from five countries/administrative regions: Austria, France, Ireland, Hong Kong SAR and Norway.

HI analyses

The results of HI analyses for propagated viruses of the B/Yamagata lineage are shown in **Tables 12-1** to **12-4**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines in **Tables 12-3** and **12-4** are viruses with collection dates after 2019-08-31. All viruses fell into clade 3, the B/Wisconsin/1/2010 – B/Phuket/3073/2013 clade.

Post-infection ferret antiserum raised against the egg-propagated vaccine virus B/Phuket/3073/2013, recommended for use in quadrivalent vaccines for the Northern Hemisphere 2019-2020 and the Southern Hemisphere 2020 seasons, recognised all but one of the seven test viruses collected after 2019-08-31 at titres within 2-fold of the titre with the homologous virus. A post-infection ferret antiserum raised against a cell culture-propagated cultivar of B/Phuket/3073/2013 recognised six of the seven recently collected test viruses at titres within 4-fold of its titre with the homologous virus. but only one (14%) within 2-fold. The test virus recognised less well was B/Norway/2465/2019.

Antisera raised against four other viruses from the B/Wisconsin/1/2010-B/Phuket/3073/2013 clade (clade 3), egg-propagated B/Wisconsin/1/2010, B/Stockholm/12/2011 and cell culture-propagated B/Mauritius/1791/2017 and B/Mauritius/I-762/2018, recognised well the test viruses collected after 2019-08-31, with the exception of B/Norway/2465/2019, with each recognising 6 of the 7 tested viruses at titres within 4-fold of the homologous titres. However, the antiserum raised against B/Norway/2134/2019 recognised only B/Norway/2465/2019 of the recently collected viruses at titres within 4-fold of the titre of the antiserum for the homologous virus.

An antiserum raised against the former vaccine virus egg-propagated B/Massachusetts/02/2012 (clade 2) recognised six of the seven recently collected viruses (85%) at titres within 4-fold of the titre of the antiserum with the homologous virus, but only one at a titre within 2-fold of the homologous titre. However, an antiserum raised against the cell culture-propagated cultivar of B/Massachusetts/02/2012 recognised none of the seven recently collected test viruses at titres within 4-fold of its homologous titre and an antiserum raised against another cell culture-propagated clade 2 virus, B/Estonia/55669/2011, also recognised none of the test viruses at titres within 4-fold of the titre of the antiserum with the homologous virus.

Genetic analyses

Figure 16 shows a phylogenetic analysis of the HA genes of B/Yamagata lineage viruses sequenced at the Crick and other representative viruses. An amino acid sequence alignment of a selection of the HA glycoproteins is shown in **Figure 17**. All ten viruses analysed genetically with collection dates after 2019-08-31 had HA gene sequences falling in clade 3, the B/Phuket/3073/2013 clade, and, like the vast majority of recently circulating viruses, fall into a genetic group defined by two **HA1** amino acid substitutions compared with B/Phuket/3073/2013: **L172Q** and **M251V**.

Figure 18 shows a phylogenetic tree for NA genes of the same viruses represented in the HA phylogeny and **Figure 19** shows a corresponding amino acid sequence alignment for selected NA glycoproteins. As for the HA gene, the vast majority of NA genes of recently collected viruses fell into the B/Phuket/3073/2013 clade (clade 3) with the great majority of viruses falling in a group defined by **NA I49M, I171M, D342K** and **K373Q** amino acid substitutions compared to the B/Phuket/3073/2013 NA sequence. A number of subgroups have emerged within this group, notably those defined by either **R65H** and **S402P** substitutions, or an **A395S** amino acid substitution, or an **M50V** substitution.

The reference virus B/Norway/2134/2019 has seven substitutions in **HA1, R69G, G141R, K149E, I150N, G183E, N196I** (resulting in the loss of the glycosylation motif at residues **196-198** in **HA1**) and **P240Q**, and six, **K69E, R188K, I248V, G346R, G378E** and **K435R**, in NA.

Antiviral Analysis.

Susceptibility to the neuraminidase inhibitors (NAI) zanamivir and oseltamivir was assessed phenotypically and by full NA gene sequencing, for 185 B/Victoria-lineage and seven B/Yamagata-lineage viruses with collection dates after 2019-08-31 (**Table 14**). All but two (99%) showed normal inhibition (NI: no reduced susceptibility) by both antivirals. The exceptions were that B/Beirut/570/2019 and B/Estonia/125782/2019 both showed reduced inhibition by zanamivir but normal inhibition by oseltamivir. The NA gene of B/Estonia/125782/2019 encoded the substitutions I384V and V403I in the NA. NA gene sequencing of B/Beirut/570/2019 is incomplete.

Conclusion.

Viruses of the B/Victoria and the B/Yamagata lineages were received in small numbers with those of the B/Victoria lineage predominating over those of the B/Yamagata lineage at a ratio of ~28:1. Viruses of the B/Victoria lineage fell into two genetic groups. The predominant group, representing 97% of the viruses sequenced, had HAs with a three amino acid deletion in **HA1 (Δ162-164)** with the additional **HA1** amino acid substitution **K136E**. Viruses with HA genes encoding a deletion of two amino acids in **HA1 (Δ162-163)** represented 3% of those sequenced, but no viruses lacking amino acid deletions were received with collection dates after 2019-08-31. Viruses with either deletion are antigenically distinct from the other viruses in clade 1A and the **Δ162-163** and **Δ162-164** variants are antigenically distinct from each other.

Viruses of the B/Yamagata lineage fell within genetic clade 3, the B/Phuket/3073/2013 clade, and the majority (6 of 7) of recently collected viruses were recognised well by antisera raised against egg-propagated clade 3 vaccine/reference viruses.

All but two influenza B viruses (99%) showed normal inhibition by the NA inhibitors oseltamivir and zanamivir.

Table 9a. Summary of influenza B (Victoria-lineage) viruses analysed, collected after 2019-08-31a

MONTH	B Victoria lineage														
	Country	Number received ¹	Number propagated ²	Post infection ferret antisera raised against											
				Clade 1A(Δ2) virus			Clade 1A(Δ2) virus			Clade 1A(Δ3)B virus			Clade 1A(Δ3)B virus		
				≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴	≤2 fold ⁵	4 fold ⁵	≥8 fold ⁵	≤2 fold ⁶	4 fold ⁶	≥8 fold ⁶
2019															
SEPTEMBER															
France	2	0													
Ghana	1	0													
Mauritius	1	1		1				1	1			1			
Norway	2	2		2				2	1	1		1		1	
Oman	3	3		3				3	3			3			
Qatar	1	0													
OCTOBER															
France	2	2	1		1	1		1	1		1	1		1	
Ghana	4	1			1			1	1					1	
Iceland	1	1			1			1	1			1			
Jordan	1	0													
Latvia	2	2			2			2	2			2			
Lebanon	2	0													
Lithuania	1	0													
Madagascar	2	0													
Mauritius	3	0													
Norway	3	2		1	1			2	1	1		1		1	
Oman	15	13	2	5	6			13	7		6	9	4		
Qatar	7	5	5			5				4	1	5			
Spain	1	0													
Sweden	1	1			1			1	1			1			
Switzerland	1	1			1			1	1					1	
Ukraine	1	1			1			1	1			1			
NOVEMBER															
Croatia	1	1	1			1					1	1			
Denmark	3	3			3			3	3			3			
France	2	0													
Ghana	4	4			4		1	3	3		1		2	2	
Hong Kong	4	4	1		3	1		3	3	1		4			
Iceland	1	1	1				1				1			1	
Iran	1	1			1			1			1			1	
Ireland	6	3			3			3	3			3			
Italy	2	0													
Kyrgyzstan	2	0													
Latvia	5	5			5			5	5			5			
Madagascar	2	0													
Montenegro	1	0													
Morocco	27	3			3		1	2	2		1	1		2	
Norway	4	4		1	3			4	3	1		3	1		
Oman	9	8	6	2				8	8			8			
Portugal	41	11	2	4	5		1	10	11			11			
Qatar	1	1	1			1					1	1			
Russia	14	6			6	1		5	6			6			
Sweden	2	2			2			2	2			2			
Switzerland	2	1			1			1			1	1			

Table 9b. Summary of influenza B (Victoria-lineage) viruses analysed, collected after 2019-08-31

MONTH	B Victoria lineage														
	Country	Number received ¹	Number propagated ²	Post infection ferret antisera raised against											
				Clade 1A(Δ2) virus			Clade 1A(Δ2) virus			Clade 1A(Δ3)B virus			Clade 1A(Δ3)B virus		
				≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴	≤2 fold ⁵	4 fold ⁵	≥8 fold ⁵	≤2 fold ⁶	4 fold ⁶	≥8 fold ⁶
DECEMBER															
Armenia	4	0													
Austria	5	4	3		1		2	2	2		2	2		2	
Belgium	4	3			3			3	3				3		
Croatia	1	0													
France	15	11	1	1	9			11	6	2	3	5	1	5	
Germany	1	1			1			1	1			1			
Georgia	5	5			5			5	5			5			
Hong Kong	1	1			1			1	1			1			
Iceland	1	1			1			1	1			1			
Italy	4	0													
Jordan	1	1			1			1			1			1	
Kyrgyzstan	12	4			4			4	3		1	3	1		
Lebanon	14	0													
Lithuania	1	0													
Madagascar	4	0													
Montenegro	1	0													
Morocco	2	2			2			2			2			2	
Norway	1	0													
Palestine	4	0													
Portugal	15	0													
Romania	8	0													
Russia	18	3			3			3	2		1	2		1	
Slovenia	1	1	1					1	1			1			
Spain	4	4	2		2	1	1	2	1		3	1		3	
Switzerland	3	3			3			3	3			3			
Tajikistan	1	0													
Ukraine	1	1			1			1	1			1			
UK England	1	0													
2020															
JANUARY															
Armenia	7	0													
Bulgaria	2	0													
Estonia	4	4			4			4	3		1	3		1	
France	1	0													
Germany	2	2			2			2	2			2			
Georgia	2	2			2			2	2			2			
Greece	1	1			1			1	1			1			
Italy	1	0													
Lebanon	2	0													
Macedonia	6	0													
Mauritius	2	0													
Montenegro	1	0													
Norway	5	0													
Romania	3	0													
Serbia	4	0													
Slovenia	1	1	1					1	1			1			
42 Countries	368	148	28	20	100	11	7	130	109	10	29	113	9	26	
			18.9%	13.5%	67.6%	7.4%	4.7%	87.8%	73.6%	6.8%	19.6%	76.4%	6.1%	17.6%	

1 Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2 Propagated to sufficient titre to perform HI assay

3 Compared to homologous titres for ferret antisera raised against tissue culture-propagated B/Colorado/06/2017 - Clade 1A(Δ2)

4 Compared to homologous titres for ferret antisera raised against egg-propagated B/Colorado/06/2017 - Clade 1A(Δ2)

5 Compared to homologous titres for ferret antisera raised against tissue culture-propagated B/Washington/02/2019 - Clade 1A(Δ3)B

6 Compared to homologous titres for ferret antisera raised against egg-propagated B/Washington/02/2019 - Clade 1A(Δ3)B

Table 10-1. Antigenic analyses of influenza B viruses (Victoria lineage) 2019-09-16

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Bris 60/08	B/Bris 60/08	B/Malta 636714/11	B/Sth Aus 81/12	B/Ireland 3154/16	B/Nord-West 1/16	B/HK 514/09	B/Norway 2409/17	B/Colorado 06/17	B/Colorado 06/17	B/CIV 1662/18
				Egg	Egg	Egg	Egg	MDCK	MDCK	MDCK	MDCK	MDCK	Egg	MDCK
	Passage history			Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F29/13 ²	F25/16 ⁴	F15/16 ²	F16/16 ²	NIB F47/16 ²	F40/17 ²	F09/18 ⁴	F11/18 ²	F37/18 ⁴
	Ferret number													
	Genetic group			1A	1A	1A	1A	1A	1A	1B	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)B
REFERENCE VIRUSES														
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	640	320	640	40	40	160	<	40	160	<
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	640	320	320	40	20	160	<	20	80	<
B/South Australia/81/2012	1A	2012-11-28	E4/E2	2560	1280	320	640	80	40	320	<	40	160	<
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	40	80	40	160	80	80	<	<	<	<
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	1280	40	40	40	80	80	40	<	<	<	<
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	2560	320	160	80	80	80	160	<	20	<	<
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	<	<	<	<	<	<	80	80	160	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	<	<	<	<	<	<	80	80	320	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	1280	320	80	80	<	<	20	40	160	640	<
B/Cote D'Ivoire/1662/2018	1A(Δ3)B	2018-07-25	P0/MDCK3	80	20	10	20	<	<	<	<	20	40	40
TEST VIRUSES														
B/Norway/2115/2019	1A	2019-06-12	MDCK1/MDCK1	320	40	40	80	80	40	80	<	20	<	<
B/Norway/2225/2019	1A(Δ2)	2019-07-25	MDCK1/MDCK1	40	<	40	<	<	<	<	80	320	320	<
B/Norway/1410/2019	1A(Δ3)B	2019-03-18	MDCK1/MDCK1	40	<	40	<	<	<	<	<	40	80	40
B/Norway/1717/2019	1A(Δ3)B	2019-04-12	MDCK1/MDCK1	20	<	40	<	<	<	<	<	40	<	40
B/Norway/2092/2019	1A(Δ3)B	2019-05-24	MDCK1/MDCK1	20	<	40	<	<	<	<	<	40	40	40
B/Norway/2116/2019	1A(Δ3)B	2019-06-06	MDCK1/MDCK1	80	<	40	40	<	<	<	<	20	160	80
B/Oman/4412/2019	1A(Δ3)B	2019-07-08	MDCK2	80	<	20	<	<	<	<	<	10	<	20
B/Oman/4504/2019	1A(Δ3)B	2019-07-17	MDCK2	160	80	10	80	<	<	<	<	10	40	40
B/Norway/2211/2019	1A(Δ3)B	2019-07-28	MDCK1/MDCK1	40	<	40	<	<	<	<	20	20	<	40
B/Norway/2210/2019	1A(Δ3)B	2019-07-31	MDCK1/MDCK1	320	<	80	<	<	<	<	20	40	80	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not DoneVaccine
NH 2018-19
SH 2019
NH 2019-20

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 10-2. Antigenic analyses of influenza B viruses (Victoria lineage) 2019-11-05

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Sth Aus 81/12 Egg	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/CIV 1662/18 MDCK	B/Wash'ton 02/19 MDCK	B/Wash'ton 02/19 Egg
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F25/16 ⁴	F15/16 ²	F16/16 ²	F40/17 ²	F09/18 ⁴	F11/18 ²	F37/18 ⁴	F37/19 ⁴	F38/19 ⁴
	Passage history													
	Ferret number													
	Genetic group													
REFERENCE VIRUSES														
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	640	320	40	40	<	20	80	10	<	160
B/South Australia/81/2012	1A	2012-11-28	E4/E2	1280	640	320	40	20	<	20	40	<	<	320
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	80	20	80	80	<	<	<	<	<	<
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	1280	80	20	40	40	<	<	<	<	<	<
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	40	10	<	<	<	20	20	40	<	<	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	10	<	<	<	20	20	80	<	<	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	1280	320	80	<	<	20	80	320	<	<	160
B/Cote D'Ivoire/1662/2018	1A(Δ3)B	2018-07-25	P0/MDCK3	80	40	20	<	<	<	<	10	20	<	20
B/Washington/02/2019	1A(Δ3)B	2019-01-19	C2/MDCK2	320	80	20	<	<	<	<	20	20	20	160
B/Washington/02/2019	1A(Δ3)B	2019-01-19	E3/E2	640	160	80	<	<	<	10	160	40	20	320
TEST VIRUSES														
B/Mauritius/I-1862/2019			MDCKx/MDCK2	320	80	<	<	<	<	<	20	10	<	80
B/Mauritius/I-1940/2019	1A(Δ3)B		MDCKx/MDCK1	<	20	<	<	<	<	<	<	20	20	160
B/Mauritius/I-1953/2019	1A(Δ3)B	2019-07-21	MDCKx/MDCK1	<	40	<	<	<	<	<	<	20	20	160
B/Mauritius/I-2003/2019	1A(Δ3)B		MDCKx/MDCK1	160	40	<	<	<	<	<	<	20	20	160
B/Norway/2286/2019	E128K, T155A	2019-09-27	MDCK1	320	40	<	<	<	<	<	<	10	<	20
B/Norway/2284/2019		2019-09-29	MDCK1	160	40	<	<	<	<	<	<	10	20	160
B/Mauritius/I-2602/2019		2019-09-09	MDCKx/MDCK1	<	20	<	<	<	<	<	<	20	20	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

Vaccine
NH 2018-19
SH 2019
NH 2019-20

Vaccine
SH 2020

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 10-3. Antigenic analyses of influenza B viruses (Victoria lineage) 2019-11-22

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Sth Aus 81/12 Egg	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/CIV 1662/18 MDCK	B/Wash'ton 02/19 MDCK	B/Wash'ton 02/19 Egg	
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F25/16 ⁴	F15/16 ²	F16/16 ²	F40/17 ²	F09/18 ⁴	F11/18 ²	F37/18 ⁴	F37/19 ⁴	F38/19 ⁴	
				Genetic group	1A	1A	1A	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)B	1A(Δ3)B	1A(Δ3)B
REFERENCE VIRUSES															
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	2560	640	320	40	40	<	40	80	<	<	160	
B/South Australia/81/2012	1A	2012-11-28	E4/E2	2560	640	640	40	40	<	20	80	<	<	160	
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	80	20	80	80	<	<	<	<	<	<	
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	2560	80	20	80	80	<	<	<	<	<	<	
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	40	20	<	<	<	40	40	80	<	<	<	
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	40	<	<	<	20	40	80	<	<	<	
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	1280	160	40	<	<	20	80	160	<	<	160	
B/Cote D'Ivoire/1662/2018	1A(Δ3)B	2018-07-25	P0/MDCK3	320	80	40	<	<	<	10	40	40	20	40	
B/Washington/02/2019	1A(Δ3)B	2019-01-19	C2/MDCK2	320	160	80	<	<	<	20	160	80	80	320	
B/Washington/02/2019	1A(Δ3)B	2019-01-19	E3/E2	640	160	80	<	<	<	10	160	20	20	320	
TEST VIRUSES															
B/Qatar/24-VI-19-0045756/2019	1A(Δ3)B	2019-08-04	MDCK1	320	40	<	<	<	<	10	40	20	20	160	
B/Qatar/10-VI-19-0045830/2019	1A(Δ3)B	2019-08-04	MDCK1	640	320	160	<	<	<	40	160	<	10	160	
B/Qatar/16-VI-19-0045885/2019	1A(Δ3)B	2019-08-04	MDCK2	160	20	<	<	<	<	<	<	<	20	80	
B/Qatar/10-VI-19-0045921/2019	1A(Δ3)B	2019-08-05	MDCK1	320	40	<	<	<	<	10	40	40	40	160	
B/Qatar/10-VI-19-0046191/2019	1A(Δ3)B	2019-08-06	MDCK1	320	40	<	<	<	<	20	<	10	80	160	
B/Qatar/10-VI-19-0046169/2019	1A(Δ3)B	2019-08-06	MDCK1	160	40	<	<	<	<	10	<	20	20	160	
B/Qatar/16-VI-19-0047470/2019	1A(Δ3)B	2019-08-18	MDCK1	320	20	<	<	<	<	20	<	40	40	160	
B/Qatar/10-VI-19-0048096/2019	1A(Δ3)B	2019-08-20	MDCK1	640	320	320	<	<	<	20	160	10	10	320	
B/Qatar/10-VI-19-0049104/2019	1A(Δ3)B	2019-08-26	MDCK1	1280	320	320	<	<	<	40	160	10	20	320	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not DoneVaccine
NH 2018-19
SH 2019
NH 2019-20Vaccine
SH 2020

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 10-4. Antigenic analyses of influenza B viruses (Victoria lineage) 2019-12-17

Viruses	Other information	Passage history	Collection date	Passage history	Post-infection ferret antisera											
					B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Sth Aus 81/12 Egg	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/CIV 1662/18 MDCK	B/Wash'ton 02/19 MDCK	B/Wash'ton 02/19 Egg	
					Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F25/16 ⁴	F15/16 ²	F16/16 ²	F40/17 ²	F09/18 ⁴	F11/18 ²	F37/18 ⁴	F37/19 ⁴	F38/19 ⁴	
					1A	1A	1A	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)B	1A(Δ3)B	1A(Δ3)B	
REFERENCE VIRUSES																
B/Brisbane/60/2008		1A	2008-08-04	E4/E4	2560	640	640	80	40	<	20	80	<	<	320	
B/South Australia/81/2012		1A	2012-11-28	E4/E2	2560	640	640	40	20	<	20	80	<	<	160	
B/Ireland/3154/2016		1A	2016-01-14	MDCK1/MDCK4	2560	160	40	160	80	<	<	<	<	<	<	
B/Nordrhein-Westfalen/1/2016		1A	2016-01-04	C2/MDCK2	2560	160	40	160	80	<	<	<	<	<	<	
B/Norway/2409/2017		1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	20	10	<	<	40	40	80	<	<	<	
B/Colorado/06/2017		1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	20	20	<	<	40	40	80	<	<	20	
B/Colorado/06/2017		1A(Δ2)	2017-02-05	E5/E2	1280	320	80	<	<	40	80	320	<	<	160	
B/Cote D'Ivoire/1662/2018		1A(Δ3)B	2018-07-25	P0/MDCK3	320	40	40	<	<	<	<	20	20	80	80	
B/Washington/02/2019		1A(Δ3)B	2019-01-19	C2/MDCK2	640	160	80	<	<	<	40	160	80	80	320	
B/Washington/02/2019		1A(Δ3)B	2019-01-19	E3/E2	640	160	40	<	<	<	20	160	40	<	320	
TEST VIRUSES																
B/Khasab/5820/2019		1A(Δ3)B	2019-09-26	MDCK1	80	40	10	<	<	<	10	10	40	40	320	
B/Salah/5670/2019		1A(Δ3)B	2019-09-26	MDCK1	160	40	20	<	<	<	10	20	80	80	320	
B/Muscat/5733/2019		1A(Δ3)B	2019-09-29	MDCK1	320	40	10	<	<	<	10	10	20	40	320	
B/Muscat/5812/2019		1A(Δ3)B	2019-10-02	MDCK1	40	20	10	<	<	<	10	40	40	40	320	
B/Salah/5825/2019		1A(Δ3)B	2019-10-02	MDCK1	320	40	10	<	<	<	10	<	10	40	320	
B/Khasab/6396/2019		1A(Δ3)B	2019-10-15	MDCK2	160	40	10	<	<	<	10	20	40	40	320	
B/Nizwa/6350/2019	N150K, G184E, N197D (-CHO), R279K	1A(Δ3)B	2019-10-15	MDCK1	320	80	40	<	<	<	<	20	10	<	80	
B/libra/6682/2019		1A(Δ3)B	2019-10-20	MDCK1	320	40	20	<	<	<	10	40	40	40	320	
B/AI Buraymi/6667/2019	N150K, G184E, N197D (-CHO), R279K	1A(Δ3)B	2019-10-20	MDCK1	320	80	40	<	<	<	<	40	20	<	160	
B/AI Buraymi/6660/2019		1A(Δ3)B	2019-10-20	MDCK1	320	80	10	<	<	<	10	40	40	80	320	
B/Hayma/6659/2019	N150K, G184E, N197D (-CHO), R279K	1A(Δ3)B	2019-10-20	MDCK1	320	80	40	<	<	<	<	20	20	<	80	
B/Salah/6730/2019		1A(Δ3)B	2019-10-21	MDCK1	160	40	<	<	<	<	20	20	40	40	320	
B/Muscat/6666/2019	N150K, G184E, N197D (-CHO), R279K	1A(Δ3)B	2019-10-21	MDCK1	320	80	40	<	<	<	<	20	20	<	160	
B/Nizwa/6760/2019	N150K, G184E, N197D (-CHO), R279K	1A(Δ3)B	2019-10-22	MDCK1	320	80	40	<	<	<	<	20	20	<	80	
B/Muscat/6935/2019		1A(Δ3)B	2019-10-24	MDCK1	320	80	20	<	<	<	20	40	40	80	320	
B/Norway/2386/2019	N126K	1A(Δ3)B	2019-10-29	MDCK1/MDCK1	80	20	10	<	<	<	<	<	20	20	40	
B/Sur/7107/2019	N150K, G184E, N197D (-CHO), R279K	1A(Δ3)B	2019-10-31	MDCK2	320	80	40	<	<	<	<	20	20	<	80	
B/Norway/2375/2019		1A(Δ3)B	2019-10-31	MDCK1/MDCK1	160	40	10	<	<	<	10	<	20	80	320	
B/Sohar/7135/2019		1A(Δ3)B	2019-11-01	MDCK1	320	80	40	<	<	<	20	40	40	80	320	
B/libra/7356/2019		1A(Δ3)B	2019-11-06	MDCK1	160	40	20	<	<	<	20	40	20	80	320	
B/Muscat/7319/2019		1A(Δ3)B	2019-11-06	MDCK2	320	40	20	<	<	<	20	40	40	40	320	
B/Muscat/7322/2019		1A(Δ3)B	2019-11-06	MDCK1	320	40	20	<	<	<	20	40	40	40	320	
B/Sohar/7369/2019		1A(Δ3)B	2019-11-06	MDCK1	160	40	10	<	<	<	10	40	40	40	320	
B/Muscat/7373/2019		1A(Δ3)B	2019-11-07	MDCK2	160	80	40	<	<	<	20	40	40	80	640	
B/Norway/2426/2019		1A(Δ3)B	2019-11-07	MDCK1	320	40	40	<	<	<	10	10	40	80	320	
B/Sur/7472/2019		1A(Δ3)B	2019-11-08	MDCK1	160	40	10	<	<	<	10	20	20	80	320	
B/Norway/2429/2019		1A(Δ3)B	2019-11-08	MDCK1	160	40	10	<	<	<	<	<	40	20	80	
B/Norway/2446/2019	D129N	1A(Δ3)B	2019-11-12	MDCK2	320	40	10	<	<	<	<	<	10	80	320	
B/Norway/2464/2019		1A(Δ3)B	2019-11-14	MDCK1	80	40	10	<	<	<	<	<	20	40	320	
B/libri/7701/2019		1A(Δ3)B	2019-11-16	MDCK1	160	80	20	<	<	<	20	40	40	80	320	
B/Madeira/67/2019		1A(Δ3)B	2019-11-22	MDCK1	80	40	10	<	<	<	<	<	20	40	160	
B/Madeira/68/2019		1A(Δ3)B	2019-11-23	MDCK2	640	40	40	<	<	<	20	80	40	80	320	
B/Lisboa/RL036/2019		1A(Δ3)B	2019-11-23	MDCK2	640	160	80	<	<	<	<	40	40	80	160	
B/Guarda/171/2019		1A(Δ3)B	2019-11-24	MDCK1	320	40	20	<	<	<	10	<	40	40	320	
B/Guarda/170/2019		1A(Δ3)B	2019-11-24	MDCK2	640	80	40	<	<	<	40	40	40	80	320	
B/Madeira/69/2019		1A(Δ3)B	2019-11-24	MDCK1	320	40	20	<	<	<	10	<	40	40	160	
B/Madeira/70/2019		1A(Δ3)B	2019-11-26	MDCK1	320	40	20	<	<	<	10	<	40	80	320	
B/Braga/135/2019		1A(Δ3)B	2019-11-27	MDCK1	160	40	10	<	<	<	10	<	40	80	320	

* Superscripts refer to antisera properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Vaccine
NH 2018-19
SH 2019
NH 2019-20

Vaccine
SH 2020

Table 10-5. Antigenic analyses of influenza B viruses (Victoria lineage) 2020-01-15

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Sth Aus 81/12 Egg	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/Wash'ton 02/19 MDCK	B/Wash'ton 02/19 Egg	
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F25/16 ⁴	F15/16 ²	F38/17 ²	F40/17 ²	F09/18 ⁴	F11/18 ²	F37/19 ⁴	F38/19 ⁴	
				Genetic group	1A	1A	1A	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)B	1A(Δ3)B
REFERENCE VIRUSES														
B/Brisbane/60/2008		1A	2008-08-04	E4/E4	2560	640	640	40	160	<	20	80	<	320
B/South Australia/81/2012		1A	2012-11-28	E4/E2	2560	640	640	80	160	<	40	80	<	320
B/Ireland/3154/2016		1A	2016-01-14	MDCK1/MDCK4	2560	80	<	80	160	<	<	<	<	<
B/Nordrhein-Westfalen/1/2016		1A	2016-01-04	C2/MDCK2	2560	80	<	80	160	<	<	<	<	<
B/Norway/2409/2017		1A(Δ2)	2017-04-27	MDCK1/MDCK3	160	20	<	<	<	40	40	80	<	<
B/Colorado/06/2017		1A(Δ2)	2017-02-05	MDCK1/MDCK2	160	20	<	<	<	40	40	80	<	<
B/Colorado/06/2017		1A(Δ2)	2017-02-05	E5/E2	1280	160	40	<	<	20	80	320	<	160
B/Cote D'Ivoire/1662/2018		1A(Δ3)B	2018-07-25	P0/MDCK3	640	80	<	<	<	<	<	<	10	80
B/Washington/02/2019		1A(Δ3)B	2019-01-19	C2/MDCK2	640	160	<	<	<	<	<	160	80	640
B/Washington/02/2019		1A(Δ3)B	2019-01-19	E3/E2	640	80	40	<	<	<	10	160	20	320
TEST VIRUSES														
B/Lulea/1/2019	E128K, T155A	1A(Δ3)B	2019-08-01	MDCK0/MDCK1	80	20	<	<	<	<	<	<	10	10
B/Finland/124/2019		1A(Δ3)B	2019-08-14	MDCK1/MDCK1	320	80	<	<	<	<	<	<	40	640
B/Lyon/1861/2019		1A(Δ2)	2019-10-04	MDCK3/MDCK1	320	40	<	<	<	<	20	160	<	<
B/Iceland/83/2019		1A(Δ2)	2019-11-13	MDCK1/MDCK1	320	20	<	<	<	40	20	80	<	<
B/Latvia/10-010670/2019		1A(Δ3)B	2019-10-03	MDCK2/MDCK1	640	80	<	<	<	<	<	<	80	640
B/Latvia/10-010619/2019		1A(Δ3)B	2019-10-03	MDCK1/MDCK1	320	40	<	<	<	<	<	<	40	320
B/Stockholm/5/2019		1A(Δ3)B	2019-10-05	MDCK0/MDCK1	160	40	<	<	<	<	<	<	80	320
B/Lyon/CHU/ R19.116.25/2019		1A(Δ3)B	2019-10-08	MDCK2/MDCK1	640	40	<	<	<	<	<	<	80	640
B/Denmark/14/2019		1A(Δ3)B	2019-11-02	SIAT2/MDCK1	640	80	<	<	<	<	<	<	80	320
B/Iran/189332/2019	D129N	1A(Δ3)B	2019-11-03	SIAT1/MDCK1	640	40	<	<	<	<	<	<	10	40
B/Hong Kong/991/2019		1A(Δ3)B	2019-11-10	MDCK1/MDCK1	320	40	<	<	<	<	<	<	40	320
B/Croatia/7789/2019	N150K, G184E, N197D (-CHO), R279K	1A(Δ3)B	2019-11-11	MDCKx/MDCK1	1280	320	160	<	<	<	20	160	10	320
B/Latvia/11-034083/2019		1A(Δ3)B	2019-11-12	MDCK1/MDCK1	320	40	<	<	<	<	<	<	80	640
B/Latvia/11-034080/2019		1A(Δ3)B	2019-11-12	MDCK1/MDCK1	320	40	<	<	<	<	<	<	80	320
B/Latvia/11-034075/2019		1A(Δ3)B	2019-11-12	MDCKx/MDCK1	160	40	<	<	<	<	<	<	40	320
B/Latvia/11-034069/2019		1A(Δ3)B	2019-11-12	MDCK2/MDCK1	320	40	<	<	<	<	<	<	80	320
B/Hong Kong/992/2019		1A(Δ3)B	2019-11-12	MDCK1/MDCK1	80	20	<	<	<	<	<	<	40	320
B/Latvia/11-031992/2019		1A(Δ3)B	2019-11-13	MDCKx/MDCK1	320	80	<	<	<	<	<	<	80	640
B/S.Petersburg/R11-325/2019		1A(Δ3)B	2019-11-14	MDCK3/MDCK1	1280	80	<	<	<	<	<	40	40	320
B/Lulea/2/2019		1A(Δ3)B	2019-11-21	MDCK0/MDCK1	160	40	<	<	<	<	<	<	40	320
B/Stockholm/9/2019		1A(Δ3)B	2019-11-21	MDCK0/MDCK1	160	20	<	<	<	<	<	<	40	320
B/Denmark/17/2019		1A(Δ3)B	2019-11-23	SIAT2/MDCK1	640	160	<	<	80	<	<	<	80	640
B/Denmark/18/2019		1A(Δ3)B	2019-11-25	SIAT2/MDCK1	160	40	<	<	<	<	<	<	40	320
B/Hong Kong/997/2019		1A(Δ3)B	2019-11-28	MDCK1/MDCK1	320	40	<	<	<	<	<	<	40	640
B/Hong Kong/996/2019	N150K, G184E, N197D (-CHO), R279K	1A(Δ3)B	2019-11-30	MDCK1/MDCK1	1280	640	<	320	<	<	40	320	20	640
B/Hong Kong/998/2019		1A(Δ3)B	2019-12-02	MDCK1/MDCK1	640	80	<	<	<	<	<	<	40	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

Vaccine	Vaccine
NH 2018-19	SH 2020
SH 2019	
NH 2019-20	

< 4-fold	4-fold	8-fold	> 8-fold	< not recognised by the antiserum	≥ 160 (when homologous titre ≥ 2560)	≥ 160 (no homologous titre)
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Table 10-6. Antigenic analyses of influenza B viruses (Victoria lineage) 2020-01-22

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre										
					B/Bris 60/08 Egg Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3} 1A	Post-infection ferret antisera									
						B/Bris 60/08 Egg F44/17 ² 1A	B/Sth Aus 81/12 Egg F25/16 ² 1A	B/Ireland 3154/16 MDCK F15/16 ² 1A	B/Nord-West 1/16 MDCK F38/17 ² 1A	B/Norway 2409/17 MDCK F40/17 ² 1A(Δ2)	B/Colorado 06/17 MDCK F09/18 ⁴ 1A(Δ2)	B/Colorado 06/17 Egg F11/18 ⁴ 1A(Δ2)	B/Wash'ton 02/19 MDCK F37/19 ⁴ 1A(Δ3)B	B/Wash'ton 02/19 Egg F38/19 ⁴ 1A(Δ3)B	
REFERENCE VIRUSES															
B/Brisbane/60/2008		1A	2008-08-04	E4/E4	1280	640	640	40	160	<	20	40	<	160	
B/South Australia/81/2012		1A	2012-11-28	E4/E2	2560	640	640	40	160	<	20	40	<	160	
B/Ireland/3154/2016		1A	2016-01-14	MDCK1/MDCK4	1280	80	40	80	320	<	<	<	<	<	
B/Nordrhein-Westfalen/1/2016		1A	2016-01-04	C2/MDCK2	1280	80	40	40	160	<	<	<	<	<	
B/Norway/2409/2017		1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	10	10	<	<	80	20	40	<	<	
B/Colorado/06/2017		1A(Δ2)	2017-02-05	MDCK1/MDCK2	160	40	20	<	<	80	40	80	<	<	
B/Colorado/06/2017		1A(Δ2)	2017-02-05	E5/E2	640	160	80	<	<	20	40	160	<	80	
B/Washington/02/2019		1A(Δ3)B	2019-01-19	C2/MDCK2	320	80	40	<	<	<	<	40	40	320	
B/Washington/02/2019		1A(Δ3)B	2019-01-19	E3/E2	640	80	80	<	<	<	<	80	20	320	
TEST VIRUSES															
B/Switzerland/6569/2019	E128K	1A(Δ3)B	2019-10-03	SIAT1/MDCK1	160	40	40	<	<	<	<	<	80	40	
B/Qatar/13-VI-19-0063403/2019	N150K, G184E, N197D (-CHO), R279K	1A(Δ3)B	2019-10-20	MDCK1	640	320	320	<	<	<	20	160	10	320	
B/Qatar/47-VI-19-0063810/2019	N150K, G184E, N197D (-CHO), R279K	1A(Δ3)B	2019-10-21	MDCK1	640	320	320	<	<	<	20	160	<	320	
B/Iceland/21277/2019		1A(Δ3)B	2019-10-26	MDCK1	320	40	40	<	<	<	<	<	40	320	
B/Qatar/47-VI-19-0066815/2019	N150K, G184E, N197D (-CHO), R279K	1A(Δ3)B	2019-10-30	MDCK1	640	160	320	<	<	<	20	80	10	320	
B/Qatar/47-VI-19-0066811/2019	N150K, G184E, N197D (-CHO), R279K	1A(Δ3)B	2019-10-30	MDCK1	640	320	320	<	<	<	40	160	10	320	
B/Qatar/47-VI-19-0066813/2019	N150K, G184E, N197D (-CHO), R279K	1A(Δ3)B	2019-10-30	MDCK1	1280	320	320	<	<	<	40	160	10	320	
B/Ghana/3443/2019	D129N, N233S (-CHO)	1A(Δ3)B	2019-11-02	MDCK1	160	20	20	<	<	<	<	<	<	40	
B/Qatar/13-VI-19-0068458/2019	N150K, G184E, N197D (-CHO), R279K	1A(Δ3)B	2019-11-04	MDCK1	640	160	320	<	<	<	20	80	<	160	
B/Ghana/3613/2019	D129N, N233S (-CHO)	1A(Δ3)B	2019-11-06	MDCK1	160	40	40	<	<	<	<	<	20	40	
B/Ghana/3801/2019	D129N, N233S (-CHO)	1A(Δ3)B	2019-11-10	MDCK1	640	80	80	<	<	<	<	40	20	80	
B/Switzerland/9761/2019	N150K, G184E, N197D (-CHO), R279K	1A(Δ3)B	2019-11-15	MDCK1/MDCK1	640	80	20	<	<	<	<	<	<	320	
B/Coimbra/142/2019		1A(Δ3)B	2019-11-21	MDCK1	160	40	40	<	<	<	<	<	40	320	
B/Madeira/66/2019		1A(Δ3)B	2019-11-21	MDCK1	160	40	40	<	<	<	<	<	40	320	
B/Lisboa/RL050/2019		1A(Δ3)B	2019-11-21	MDCK1	160	40	20	<	<	<	<	<	40	320	
B/Ghana/3931/2019	D129N, N233S (-CHO)	1A(Δ3)B	2019-11-22	MDCK1	160	40	40	<	<	<	<	<	20	80	
B/Ireland/95062/2019		1A(Δ3)B	2019-11-26	MDCK1	160	40	40	<	<	<	<	<	40	320	
B/Ireland/95248/2019		1A(Δ3)B	2019-11-27	MDCK1	160	40	40	<	<	<	<	<	40	320	
B/Rabat/173/2019	N126K	1A(Δ3)B	2019-11-28	MDCK1	640	80	40	<	<	<	<	40	20	40	
B/Fes/206/2019	N126K, E136K	1A(Δ3)B	2019-11-29	MDCK1	160	20	10	<	<	<	<	<	<	10	
B/Fes/207/2019		1A(Δ3)B	2019-11-30	MDCK1	160	40	40	<	<	<	<	<	40	320	
B/Switzerland/5198/2019		1A(Δ3)B	2019-12-02	SIAT1/MDCK2	160	20	20	<	<	<	<	<	40	320	
B/Rabat/228/2019	N126K, R117W	1A(Δ3)B	2019-12-04	MDCK1	160	20	10	<	<	<	<	<	<	20	
B/Rabat/233/2019	N126K	1A(Δ3)B	2019-12-06	MDCK1	160	40	20	<	<	<	<	<	<	40	
B/Switzerland/7299/2019		1A(Δ3)B	2019-12-06	SIAT1/MDCK1	160	40	40	<	<	<	<	<	40	320	
B/Switzerland/7929/2019		1A(Δ3)B	2019-12-09	SIAT1/MDCK1	160	20	20	<	<	<	<	<	40	320	
B/Iceland/24875/2019		1A(Δ3)B	2019-12-12	MDCK1	160	40	20	<	<	<	<	<	20	160	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

Vaccine	Vaccine
NH 2018-19	SH 2020
SH 2019	

< 4-fold	4-fold	8-fold	> 8-fold	< not recognised by the antiserum	≥ 160 (when homologous titre ≥ 2560)	≥ 160 (no homologous titre)
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Table 10-7. Antigenic analyses of influenza B viruses (Victoria lineage) 2020-01-29

					Haemagglutination inhibition titre										
Viruses	Other information	Passage history	Collection date	Passage history	Post-infection ferret antisera										
					B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Sth Aus 81/12 Egg	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/Wash'ton 02/19 MDCK	B/Wash'ton 02/19 Egg	
					Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F25/16 ²	F15/16 ²	F38/17 ²	F40/17 ²	F21/18 ⁴	F11/18 ⁴	F37/19 ⁴	F38/19 ⁴	
					Genetic group	1A	1A	1A	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)B	1A(Δ3)B
REFERENCE VIRUSES															
B/Brisbane/60/2008		1A	2008-08-04	E4/E4	2560	640	320	40	80	<	<	40	<	160	
B/South Australia/81/2012		1A	2012-11-28	E4/E2	2560	640	640	80	320	<	<	80	<	320	
B/Ireland/3154/2016		1A	2016-01-14	MDCK1/MDCK4	2560	160	80	160	320	<	<	<	<	<	
B/Nordrhein-Westfalen/1/2016		1A	2016-01-04	C2/MDCK2	2560	160	80	40	320	<	<	<	<	<	
B/Norway/2409/2017		1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	20	10	<	<	80	20	80	<	<	
B/Colorado/06/2017		1A(Δ2)	2017-02-05	MDCK1/MDCK2	320	80	40	<	<	80	40	80	<	20	
B/Colorado/06/2017		1A(Δ2)	2017-02-05	E5/E2	1280	320	80	<	<	40	20	160	<	160	
B/Washington/02/2019		1A(Δ3)B	2019-01-19	C2/MDCK2	640	160	80	<	<	<	<	80	40	320	
B/Washington/02/2019		1A(Δ3)B	2019-01-19	E3/E2	640	160	80	<	<	<	<	160	40	320	
TEST VIRUSES															
B/Austria/1209211/2019		1A(Δ2)	2019-12-18	SIATx/MDCK1	160	20	20	<	<	40	40	40	<	<	
B/Austria/1209210/2019		1A(Δ2)	2019-12-18	SIATx/MDCK1	160	10	10	<	<	40	20	40	<	<	
B/Paris/2931/2019		1A(Δ2)	2019-12-26	MDCK1/MDCK1	320	<	<	<	<	20	10	20	<	<	
B/Kyiv/277/2019		1A(Δ3)B	2019-10-01	SIATx/MDCK1	320	40	20	<	<	<	<	<	40	320	
B/Ghana/699/2019	D129N, N233S (-CHO)	1A(Δ3)B	2019-10-31	MDCK2	320	80	20	<	<	<	<	<	20	40	
B/Vladivostok/01/2019		1A(Δ3)B	2019-11-17	MDCK2/MDCK1	320	40	20	<	<	<	<	<	20	160	
B/Vladivostok/03/2019		1A(Δ3)B	2019-11-22	MDCK2/MDCK1	640	80	40	<	<	<	<	80	20	160	
B/Ireland/93804/2019		1A(Δ3)B	2019-11-25	MDCK3	640	80	20	<	<	<	<	<	40	320	
B/Vladivostok/05/2019		1A(Δ3)B	2019-11-26	MDCK2/MDCK1	1280	160	160	<	<	<	<	<	80	320	
B/Vladivostok/07/2019		1A(Δ3)B	2019-11-27	MDCK2/MDCK1	1280	160	80	<	<	<	<	<	80	320	
B/Vladivostok/09/2019		1A(Δ3)B	2019-11-28	MDCK2/MDCK1	160	40	20	<	<	<	<	<	40	160	
B/Paris/2643/2019		1A(Δ3)B	2019-12-03	MDCK1/MDCK2	80	20	10	<	<	<	<	<	20	160	
B/Georgia/1101/2019		1A(Δ3)B	2019-12-04	MDCK1	320	80	20	<	<	<	<	<	40	320	
B/Austria/1206293/2019		1A(Δ3)B	2019-12-07	SIATx/MDCK1	640	80	20	<	<	<	40	<	40	320	
B/S. Petersburg/RIL-336/2019		1A(Δ3)B	2019-12-09	MDCK1/MDCK1	320	80	40	<	<	<	<	<	40	320	
B/Baden-Wuerttemberg/2/2019		1A(Δ3)B	2019-12-09	C2/MDCK1	160	20	10	<	<	<	<	<	40	160	
B/Pays de Loire/2775/2019		1A(Δ3)B	2019-12-11	MDCK1/MDCK1	160	40	20	<	<	<	<	<	40	320	
B/Alsace/2705/2019	N126K, D207N, R279K, K345R	1A(Δ3)B	2019-12-11	MDCK1/MDCK1	640	80	40	<	<	<	<	<	20	40	
B/Vladivostok/13/2019	T168A (-CHO), A202T	1A(Δ3)B	2019-12-13	MDCK2/MDCK1	160	20	10	<	<	<	<	<	<	40	
B/Vladivostok/11/2019		1A(Δ3)B	2019-12-13	MDCK2/MDCK1	640	80	40	<	<	<	<	<	40	320	
B/Nord Pas de Calais/2774/2019	N126K, K345R	1A(Δ3)B	2019-12-16	MDCK1/MDCK1	640	40	40	<	<	<	<	<	10	40	
B/Alsace/2766/2019		1A(Δ3)B	2019-12-16	MDCK1/MDCK1	320	40	40	<	<	<	<	<	40	320	
B/Slovenia/3418/2019		1A(Δ3)B	2019-12-19	SIAT1/MDCK1	1280	160	40	<	<	<	80	<	80	640	
B/Georgia/1179/2019		1A(Δ3)B	2019-12-20	MDCK1	320	80	20	<	<	<	<	<	40	320	
B/Austria/1210344/2019		1A(Δ3)B	2019-12-20	SIATx/MDCK1	160	40	40	<	<	<	<	<	20	320	
B/Georgia/1204/2019		1A(Δ3)B	2019-12-21	MDCK1	160	40	10	<	<	<	<	<	40	160	
B/Georgia/1240/2019		1A(Δ3)B	2019-12-26	MDCK1	320	80	20	<	<	<	<	<	40	320	
B/Georgia/1229/2019		1A(Δ3)B	2019-12-26	MDCK1	320	80	20	<	<	<	<	<	40	320	
B/Centre/2940/2019		1A(Δ3)B	2019-12-27	MDCK1/MDCK1	1280	160	40	<	<	<	20	<	40	320	
B/Bayern/1/2020		1A(Δ3)B	2020-01-03	C1/MDCK1	80	40	10	<	<	<	<	<	40	160	
B/Georgia/25/2020		1A(Δ3)B	2020-01-08	MDCK1	640	80	20	<	<	<	<	<	40	320	
B/Slovenia/149/2020		1A(Δ3)B	2020-01-08	SIATx/MDCK2	320	80	40	<	<	<	40	<	40	320	
B/Georgia/42/2020		1A(Δ3)B	2020-01-09	MDCK1	320	80	20	<	<	<	<	<	40	320	
B/Berlin/1/2020		1A(Δ3)B	2020-01-10	C1/MDCK1	160	40	10	<	<	<	<	<	40	160	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

Vaccine NH 2018-19 SH 2019	Vaccine SH 2020
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< 4-fold	4-fold	8-fold	> 8-fold	< not recognised by the antiserum	≥ 160 (when homologous titre ≥ 2560)	≥ 160 (no homologous titre)
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Table 10-8. Antigenic analyses of influenza B viruses (Victoria lineage) 2020-02-05

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Sth Aus 81/12 Egg	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/Wash'ton 02/19 MDCK	B/Wash'ton 02/19 Egg
					Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ²	F25/16 ²	F15/16 ²	F38/17 ²	F40/17 ²	F21/18 ⁴	F11/18 ⁴	F37/19 ⁴	F38/19 ⁴
		Passage history												
		Ferret number												
		Genetic group			1A	1A	1A	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)B	1A(Δ3)B
REFERENCE VIRUSES														
B/Brisbane/60/2008		1A	2008-08-04	E4/E4	1280	640	320	40	80	<	<	40	<	80
B/South Australia/81/2012		1A	2012-11-28	E4/E2	2560	1280	640	80	160	<	<	80	<	160
B/Ireland/3154/2016		1A	2016-01-14	MDCK1/MDCK4	2560	160	80	80	320	<	<	<	<	<
B/Nordrhein-Westfalen/1/2016		1A	2016-01-04	C2/MDCK2	2560	160	80	80	160	<	<	<	<	<
B/Norway/2409/2017		1A(Δ2)	2017-04-27	MDCK1/MDCK3	160	20	20	<	<	80	20	80	<	<
B/Colorado/06/2017		1A(Δ2)	2017-02-05	MDCK1/MDCK2	160	40	20	<	<	80	40	80	<	<
B/Colorado/06/2017		1A(Δ2)	2017-02-05	E5/E2	640	160	40	<	<	40	20	160	<	80
B/Washington/02/2019		1A(Δ3)B	2019-01-19	C2/MDCK2	640	80	40	<	<	<	<	80	40	320
B/Washington/02/2019		1A(Δ3)B	2019-01-19	E3/E2	1280	160	80	<	<	<	<	160	40	320
TEST VIRUSES														
B/Catalonia/12041S/2020		1A(Δ2)	2019-12-17	P0/MDCK1	80	40	20	<	<	80	40	80	<	10
B/Catalonia/12049S/2020		1A(Δ2)	2019-12-17	P0/MDCK1	40	40	20	<	<	40	20	40	<	<
B/Jordan/30272/2019	E128K, V87A	1A(Δ3)B	2019-12-10	MDCK1	80	40	10	<	<	<	<	<	<	20
B/Belgium/G0496/2019		1A(Δ3)B	2019-12-05	MDCK1	160	40	20	<	<	<	<	<	20	320
B/Belgium/G0513/2019		1A(Δ3)B	2019-12-09	MDCK1	160	80	20	<	<	<	<	<	80	320
B/Poitiers/2311/2019	T155A	1A(Δ3)B	2019-12-16	MDCK2/MDCK1	20	20	20	<	<	<	<	<	10	40
B/Catalonia/12038S/2020	N126K, V248A	1A(Δ3)B	2019-12-16	P0/MDCK1	80	40	40	<	<	<	<	<	<	40
B/Kyrgyzstan/250/2019	N150K, G184E, N197D (-CHO), R279K, V177I, V220M, P241Q	1A(Δ3)B	2019-12-17	MDCK1	320	80	40	<	<	<	<	<	<	80
B/Catalonia/12044S/2020		1A(Δ3)B	2019-12-17	P0/MDCK1	40	40	20	<	<	<	<	<	20	320
B/Kyrgyzstan/293/2019		1A(Δ3)B	2019-12-18	MDCK1	160	40	40	<	<	<	<	<	40	320
B/Kyrgyzstan/208/2019		1A(Δ3)B	2019-12-19	MDCK1	80	40	20	<	<	<	<	<	40	320
B/Saint Etienne/2361/2019	N126K, R80K, R279K, K345R	1A(Δ3)B	2019-12-19	MDCK2/MDCK1	80	40	40	<	<	<	<	<	<	40
B/Belgium/G0533/2019		1A(Δ3)B	2019-12-20	MDCK1	80	40	10	<	<	<	<	<	40	320
B/Kyrgyzstan/264/2019		1A(Δ3)B	2019-12-23	MDCK1	160	80	40	<	<	<	<	<	40	320
B/Limoges/2376/2019		1A(Δ3)B	2019-12-23	MDCK2/MDCK1	160	40	40	<	<	<	<	<	40	320
B/Ukraine/8369/2019		1A(Δ3)B	2019-12-30	MDCK2	160	40	20	<	<	<	<	<	40	320
B/Estonia/125661/2020		1A(Δ3)B	2020-01-10	SIAT1/MDCK1	160	40	20	<	10	<	<	<	40	320
B/Estonia/125784/2020		1A(Δ3)B	2020-01-13	SIAT1/MDCK1	160	40	20	<	<	<	<	<	40	320
B/Estonia/125783/2020		1A(Δ3)B	2020-01-13	SIAT1/MDCK1	160	40	20	<	10	<	<	<	40	320
B/Estonia/125782/2020	N126K, E198G	1A(Δ3)B	2020-01-13	SIAT1/MDCK1	80	40	40	<	<	<	<	<	<	20
B/Athens.GR/112/2020		1A(Δ3)B	2020-01-16	MDCK1/MDCK1	80	40	20	<	<	<	<	<	80	320
B/Athens.GR/130/2020		1A(Δ3)B		MDCK1/MDCK1	80	20	20	<	<	<	<	<	40	320
B/Lyon/2271/2019			2019-12-12	MDCK2/MDCK1	80	80	40	<	<	<	<	<	<	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not DoneVaccine
NH 2018-19
SH 2019Vaccine
SH 2020

 < 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 10-9. Antigenic analyses of influenza B viruses (Victoria lineage) - Summary

Viruses	Other information	Haemagglutination inhibition titre									
		Post-infection ferret antisera									
		B/Bris 60/08 Egg	B/Sth Aus 81/12 Egg	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Norway 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/CIV 1662/18 MDCK	B/Wash'ton 02/19 MDCK	B/Wash'ton 02/19 Egg
		F44/17 ¹²	F25/16 ¹⁴	F38/17 ¹²	F16/16 ¹²	F40/17 ¹²	F09/18 ¹⁴	F11/18 ¹²	F37/18 ¹⁴	F37/19 ¹⁴	F38/19 ¹⁴
	Passage history										
	Ferret number										
	Genetic group	1A	1A	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)B	1A(Δ3)B	1A(Δ3)B
REFERENCE VIRUSES											
B/Brisbane/60/2008	1A	640	640	40	160	<	20	80	<	<	320
B/South Australia/81/2012	1A	640	640	80	160	<	40	80	<	<	320
B/Ireland/3154/2016	1A	80	<	80	160	<	<	<	<	<	<
B/Nordrhein-Westfalen/1/2016	1A	80	<	80	160	<	<	<	<	<	<
B/Norway/2409/2017	1A(Δ2)	20	<	<	<	40	40	80	<	<	<
B/Colorado/06/2017	1A(Δ2)	20	<	<	<	40	40	80	<	<	<
B/Colorado/06/2017	1A(Δ2)	160	40	<	<	20	80	320	<	<	160
B/Cote D'Ivoire/1662/2018	1A(Δ3)B	80	<	<	<	<	<	<	20	10	80
B/Washington/02/2019	1A(Δ3)B	160	<	<	<	<	<	160	80	80	640
B/Washington/02/2019	1A(Δ3)B	80	40	<	<	<	10	160	40	20	320
TEST VIRUSES											
Genetic group	No tested	No with									
All viruses			149	149	149	149	149	149	41	149	149
		Titre reduced ≤2-fold	6	6	1	1	5	28	11	41	110
		%	4.0	4.0	0.7	0.7	3.3	18.8	7.4	100.0	73.8
		Titre reduced =4-fold	8	2			1	17	7		9
		%	5.4	1.4			0.7	11.4	4.7		6.1
		Titre reduced ≥8-fold	135	141	148	148	143	104	131		30
		%	90.6	94.6	99.3	99.3	96.0	69.8	87.9		20.1
											17.4
1A(Δ3)B	No tested	No with	141	141	141	141	141	141	41	141	141
		Titre reduced ≤2-fold	6	6	1	1		22	9	41	110
		%	4.3	4.3	0.7	0.7		15.6	6.4	100.0	78.0
		Titre reduced =4-fold	8	2				16	3		9
		%	5.6	1.4				11.3	2.1		6.4
		Titre reduced ≥8-fold	127	133	140	140	141	103	129		22
		%	90.1	94.3	99.3	99.3	100.0	73.1	91.5		15.6
											16.7
1A(Δ2)	No tested	No with	7	7	7	7	7	7	0	7	7
		Titre reduced ≤2-fold					5	6	2		
		Titre reduced =4-fold					1	1	4		
		Titre reduced ≥8-fold	7	7	7	7	1	1	1	7	7
Unknown	No tested	No with	1	1	1	1	1	1	1	0	1
		Titre reduced ≤2-fold									
		Titre reduced =4-fold									
		Titre reduced ≥8-fold	1	1	1	1	1	1	1	1	1

Vaccine
NH 2018-19
SH 2019
NH 2019-20

Vaccine
SH 2020

Reference virus results are taken from individual tables as examples. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Figure 11. Phylogenetic comparison of influenza B (Victoria-lineage) HA genes

Vaccine viruses
Reference viruses

Collection date
Sep 2019
Oct 2019
Nov 2019
Dec 2019
Jan 2020

HA2 numbering

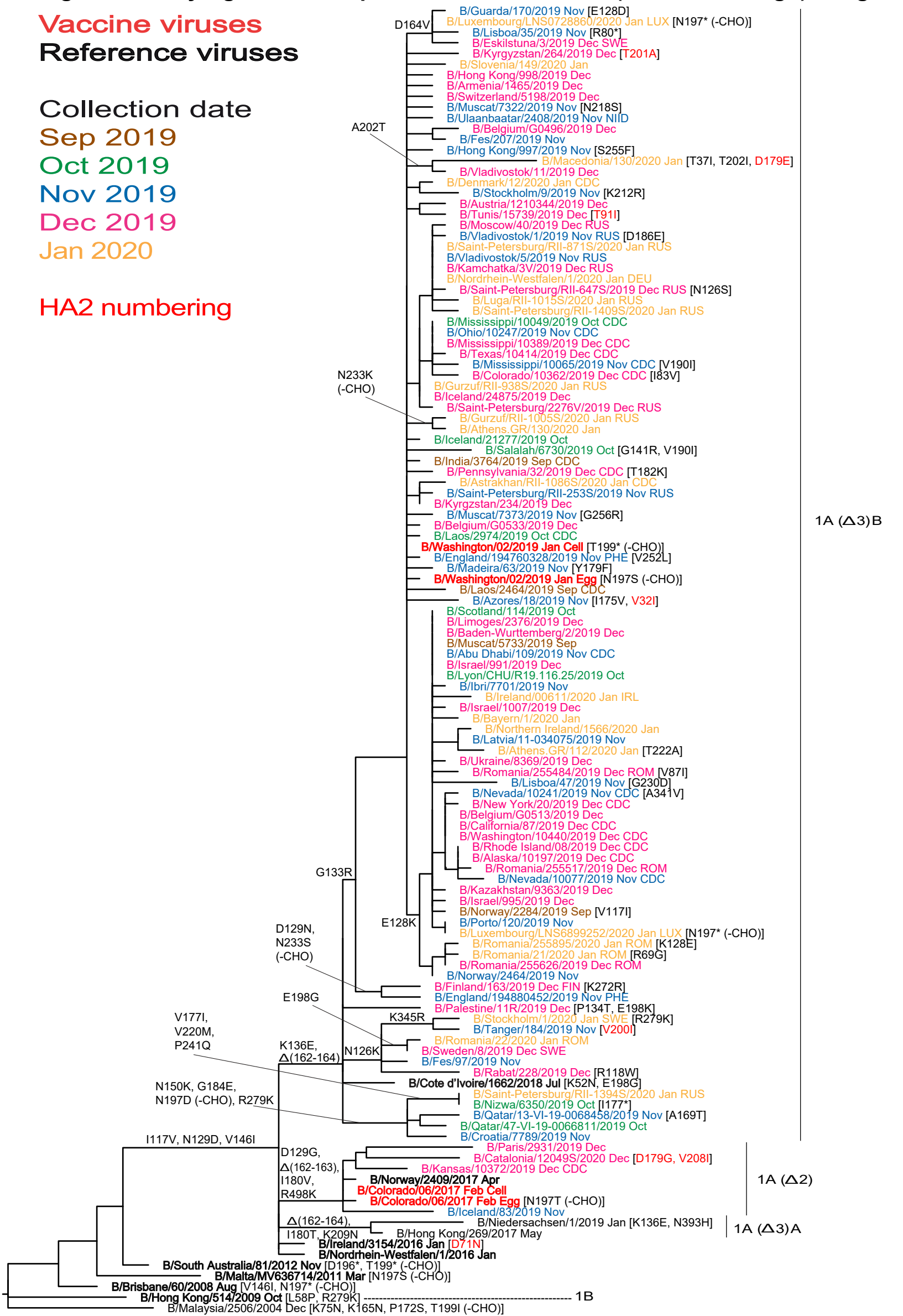
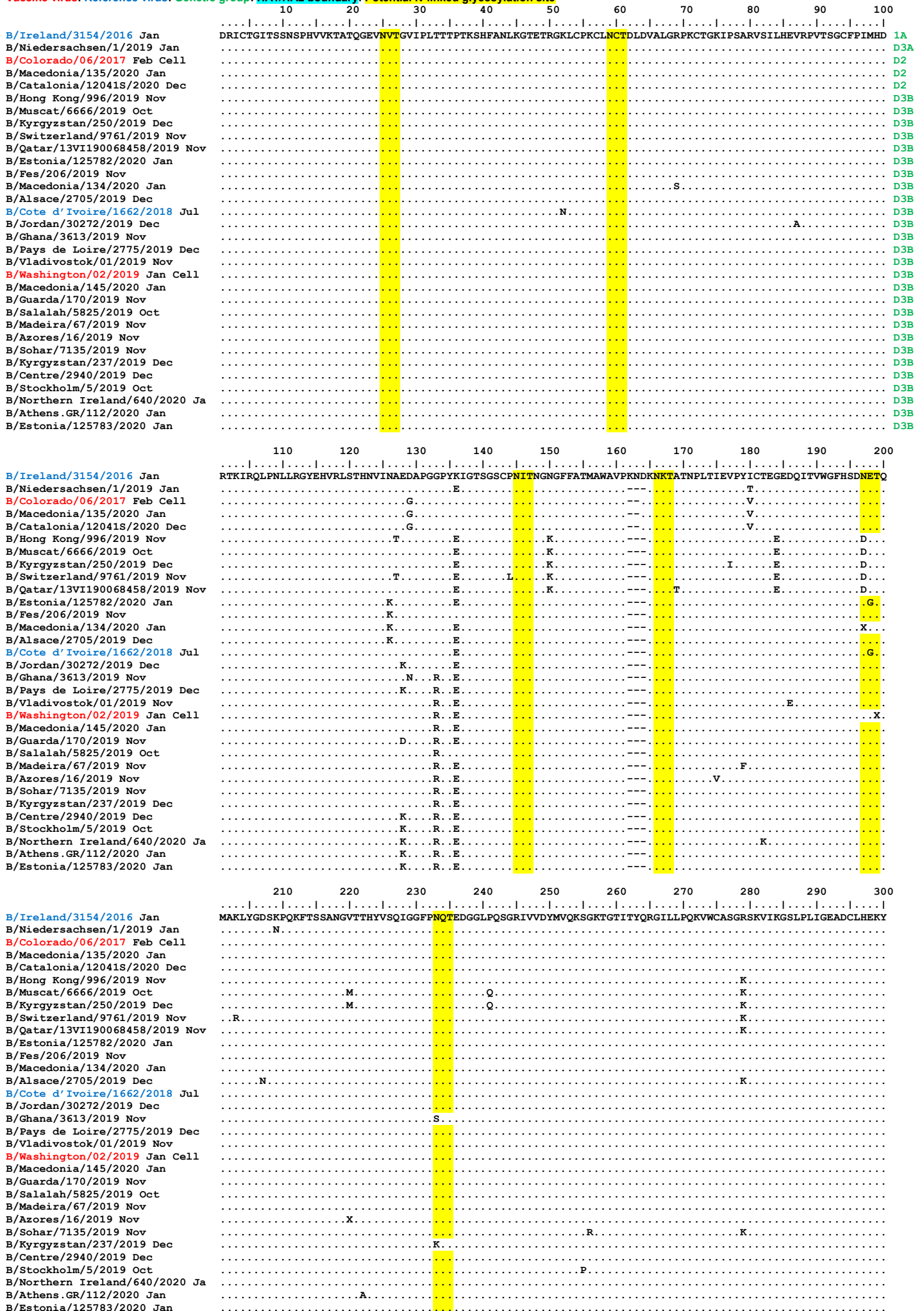


Figure 12. HA protein alignment for B (Victoria-lineage) viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site



	310	320	330	340	350	360	370	380	390	400
B/Ireland/3154/2016 Jan	G	G	L	N	K	S	K	P	Y	T
B/Niedersachsen/1/2019 Jan	G	G	L	N	K	S	K	P	Y	T
B/Colorado/06/2017 Feb Cell	G	G	L	N	K	S	K	P	Y	T
B/Macedonia/135/2020 Jan	G	G	L	N	K	S	K	P	Y	T
B/Catalonia/120418/2020 Dec	G	G	L	N	K	S	K	P	Y	T
B/Hong Kong/996/2019 Nov	G	G	L	N	K	S	K	P	Y	T
B/Muscat/6666/2019 Oct	G	G	L	N	K	S	K	P	Y	T
B/Kyrgyzstan/250/2019 Dec	G	G	L	N	K	S	K	P	Y	T
B/Switzerland/9761/2019 Nov	G	G	L	N	K	S	K	P	Y	T
B/Qatar/13VII90068458/2019 Nov	G	G	L	N	K	S	K	P	Y	T
B/Estonia/125782/2020 Jan	G	G	L	N	K	S	K	P	Y	T
B/Fes/206/2019 Nov	G	G	L	N	K	S	K	P	Y	T
B/Macedonia/134/2020 Jan	G	G	L	N	K	S	K	P	Y	T
B/Alsace/2705/2019 Dec	G	G	L	N	K	S	K	P	Y	T
B/Cote d'Ivoire/1662/2018 Jul	G	G	L	N	K	S	K	P	Y	T
B/Jordan/30272/2019 Dec	G	G	L	N	K	S	K	P	Y	T
B/Ghana/3613/2019 Nov	G	G	L	N	K	S	K	P	Y	T
B/Pays de Loire/2775/2019 Dec	G	G	L	N	K	S	K	P	Y	T
B/Vladivostok/01/2019 Nov	G	G	L	N	K	S	K	P	Y	T
B/Washington/02/2019 Jan Cell	G	G	L	N	K	S	K	P	Y	T
B/Macedonia/145/2020 Jan	G	G	L	N	K	S	K	P	Y	T
B/Guarda/170/2019 Nov	G	G	L	N	K	S	K	P	Y	T
B/Salalah/5825/2019 Oct	G	G	L	N	K	S	K	P	Y	T
B/Madeira/67/2019 Nov	G	G	L	N	K	S	K	P	Y	T
B/Azores/16/2019 Nov	G	G	L	N	K	S	K	P	Y	T
B/Sohar/7135/2019 Nov	G	G	L	N	K	S	K	P	Y	T
B/Kyrgyzstan/237/2019 Dec	G	G	L	N	K	S	K	P	Y	T
B/Centre/2940/2019 Dec	G	G	L	N	K	S	K	P	Y	T
B/Stockholm/5/2019 Oct	G	G	L	N	K	S	K	P	Y	T
B/Northern Ireland/640/2020 Ja	G	G	L	N	K	S	K	P	Y	T
B/Athens.GR/112/2020 Jan	G	G	L	N	K	S	K	P	Y	T
B/Estonia/125783/2020 Jan	G	G	L	N	K	S	K	P	Y	T

	410	420	430	440	450	460	470	480	490	500
B/Ireland/3154/2016 Jan	S	L	S	E	L	V	K	N	L	Q
B/Niedersachsen/1/2019 Jan	S	L	S	E	L	V	K	N	L	Q
B/Colorado/06/2017 Feb Cell	S	L	S	E	L	V	K	N	L	Q
B/Macedonia/135/2020 Jan	S	L	S	E	L	V	K	N	L	Q
B/Catalonia/120418/2020 Dec	S	L	S	E	L	V	K	N	L	Q
B/Hong Kong/996/2019 Nov	S	L	S	E	L	V	K	N	L	Q
B/Muscat/6666/2019 Oct	S	L	S	E	L	V	K	N	L	Q
B/Kyrgyzstan/250/2019 Dec	S	L	S	E	L	V	K	N	L	Q
B/Switzerland/9761/2019 Nov	S	L	S	E	L	V	K	N	L	Q
B/Qatar/13VII90068458/2019 Nov	S	L	S	E	L	V	K	N	L	Q
B/Estonia/125782/2020 Jan	S	L	S	E	L	V	K	N	L	Q
B/Fes/206/2019 Nov	S	L	S	E	L	V	K	N	L	Q
B/Macedonia/134/2020 Jan	S	L	S	E	L	V	K	N	L	Q
B/Alsace/2705/2019 Dec	S	L	S	E	L	V	K	N	L	Q
B/Cote d'Ivoire/1662/2018 Jul	S	L	S	E	L	V	K	N	L	Q
B/Jordan/30272/2019 Dec	S	L	S	E	L	V	K	N	L	Q
B/Ghana/3613/2019 Nov	S	L	S	E	L	V	K	N	L	Q
B/Pays de Loire/2775/2019 Dec	S	L	S	E	L	V	K	N	L	Q
B/Vladivostok/01/2019 Nov	S	L	S	E	L	V	K	N	L	Q
B/Washington/02/2019 Jan Cell	S	L	S	E	L	V	K	N	L	Q
B/Macedonia/145/2020 Jan	S	L	S	E	L	V	K	N	L	Q
B/Guarda/170/2019 Nov	S	L	S	E	L	V	K	N	L	Q
B/Salalah/5825/2019 Oct	S	L	S	E	L	V	K	N	L	Q
B/Madeira/67/2019 Nov	S	L	S	E	L	V	K	N	L	Q
B/Azores/16/2019 Nov	S	L	S	E	L	V	K	N	L	Q
B/Sohar/7135/2019 Nov	S	L	S	E	L	V	K	N	L	Q
B/Kyrgyzstan/237/2019 Dec	S	L	S	E	L	V	K	N	L	Q
B/Centre/2940/2019 Dec	S	L	S	E	L	V	K	N	L	Q
B/Stockholm/5/2019 Oct	S	L	S	E	L	V	K	N	L	Q
B/Northern Ireland/640/2020 Ja	S	L	S	E	L	V	K	N	L	Q
B/Athens.GR/112/2020 Jan	S	L	S	E	L	V	K	N	L	Q
B/Estonia/125783/2020 Jan	S	L	S	E	L	V	K	N	L	Q

	510	520	530	540	550	560	570
B/Ireland/3154/2016 Jan	A	G	T	F	D	A	G
B/Niedersachsen/1/2019 Jan	A	G	T	F	D	A	G
B/Colorado/06/2017 Feb Cell	A	G	T	F	D	A	G
B/Macedonia/135/2020 Jan	A	G	T	F	D	A	G
B/Catalonia/120418/2020 Dec	A	G	T	F	D	A	G
B/Hong Kong/996/2019 Nov	A	G	T	F	D	A	G
B/Muscat/6666/2019 Oct	A	G	T	F	D	A	G
B/Kyrgyzstan/250/2019 Dec	A	G	T	F	D	A	G
B/Switzerland/9761/2019 Nov	A	G	T	F	D	A	G
B/Qatar/13VII90068458/2019 Nov	A	G	T	F	D	A	G
B/Estonia/125782/2020 Jan	A	G	T	F	D	A	G
B/Fes/206/2019 Nov	A	G	T	F	D	A	G
B/Macedonia/134/2020 Jan	A	G	T	F	D	A	G
B/Alsace/2705/2019 Dec	A	G	T	F	D	A	G
B/Cote d'Ivoire/1662/2018 Jul	A	G	T	F	D	A	G
B/Jordan/30272/2019 Dec	A	G	T	F	D	A	G
B/Ghana/3613/2019 Nov	A	G	T	F	D	A	G
B/Pays de Loire/2775/2019 Dec	A	G	T	F	D	A	G
B/Vladivostok/01/2019 Nov	A	G	T	F	D	A	G
B/Washington/02/2019 Jan Cell	A	G	T	F	D	A	G
B/Macedonia/145/2020 Jan	A	G	T	F	D	A	G
B/Guarda/170/2019 Nov	A	G	T	F	D	A	G
B/Salalah/5825/2019 Oct	A	G	T	F	D	A	G
B/Madeira/67/2019 Nov	A	G	T	F	D	A	G
B/Azores/16/2019 Nov	A	G	T	F	D	A	G
B/Sohar/7135/2019 Nov	A	G	T	F	D	A	G
B/Kyrgyzstan/237/2019 Dec	A	G	T	F	D	A	G
B/Centre/2940/2019 Dec	A	G	T	F	D	A	G
B/Stockholm/5/2019 Oct	A	G	T	F	D	A	G
B/Northern Ireland/640/2020 Ja	A	G	T	F	D	A	G
B/Athens.GR/112/2020 Jan	A	G	T	F	D	A	G
B/Estonia/125783/2020 Jan	A	G	T	F	D	A	G

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Substitutions defining subclade 1A(Δ 3)B viruses shown in Black

Viruses displaying low reactivity with antisera raised against B/Washington/02/2019:

Virus group with 4 substitutions

**Individual viruses with
N126K or D129R substitution,
sometimes with additional
substitutions**

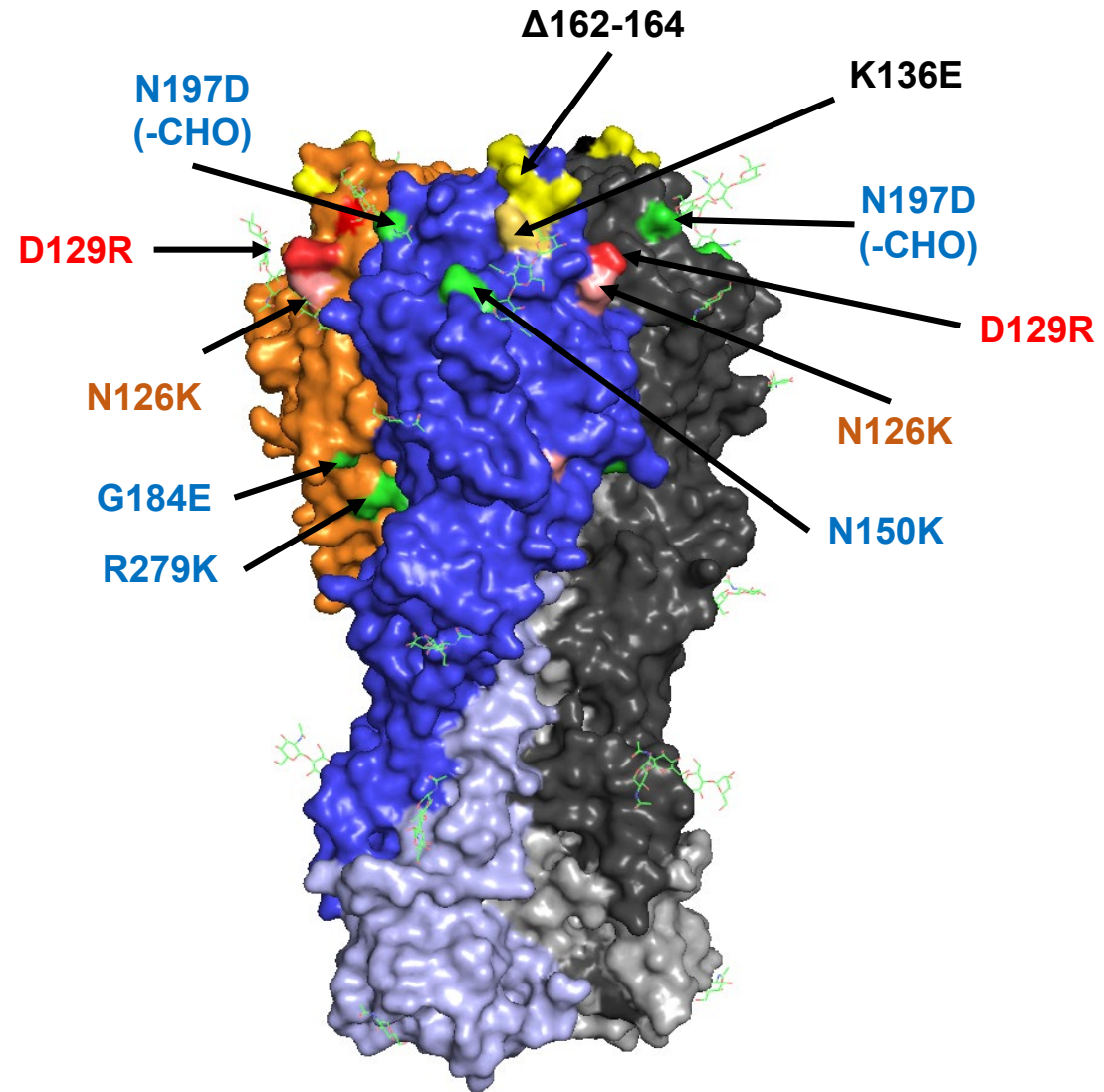


Figure 13. Amino acid substitutions and deletions in the HA of recently circulating B/Victoria lineage viruses mapped onto B/Brisbane/60/2008

Figure 14. Phylogenetic comparison of influenza B (Victoria-lineage) NA genes

Vaccine viruses
Reference viruses

Collection date

Oct 2019
Nov 2019
Dec 2019
Jan 2020

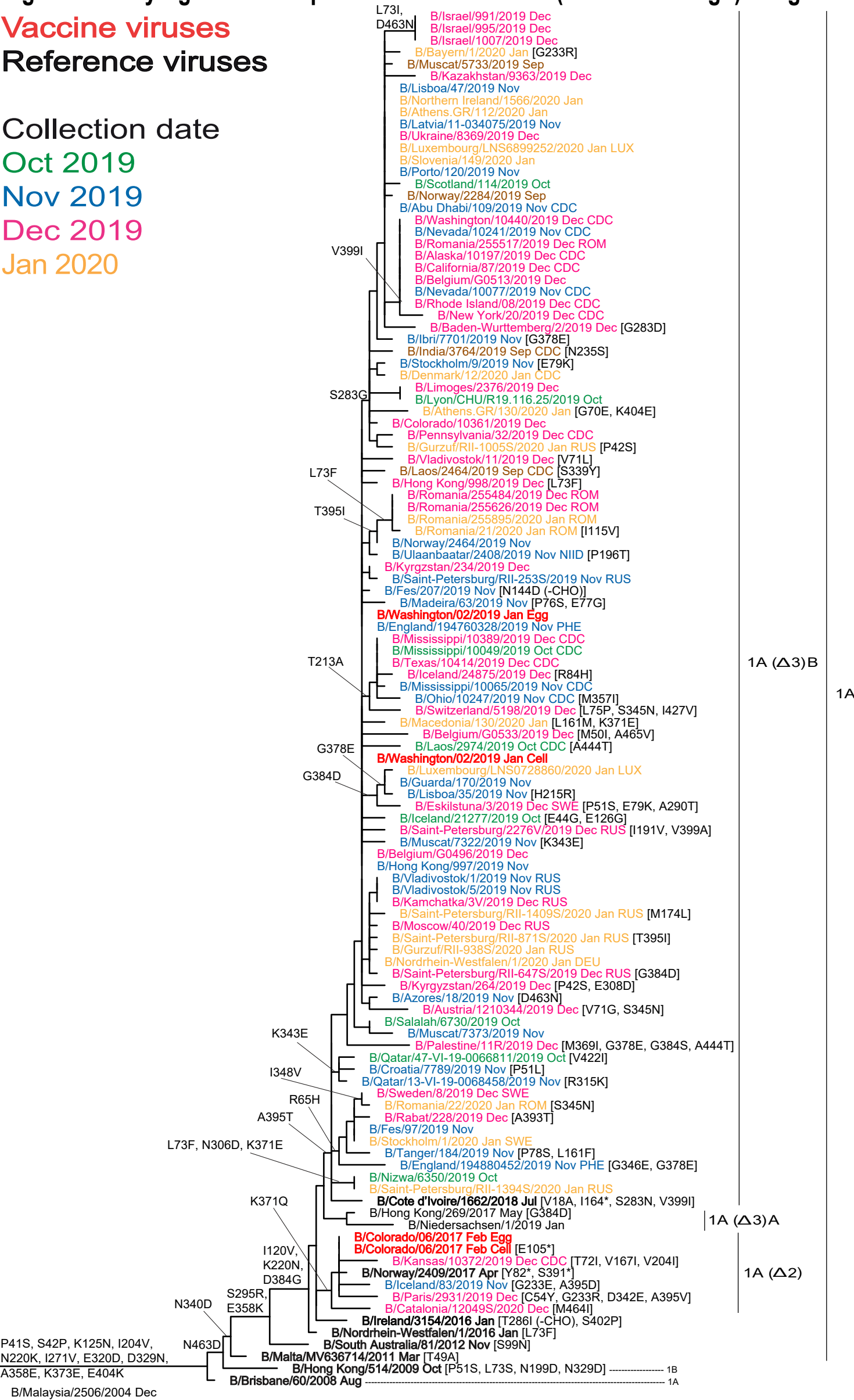
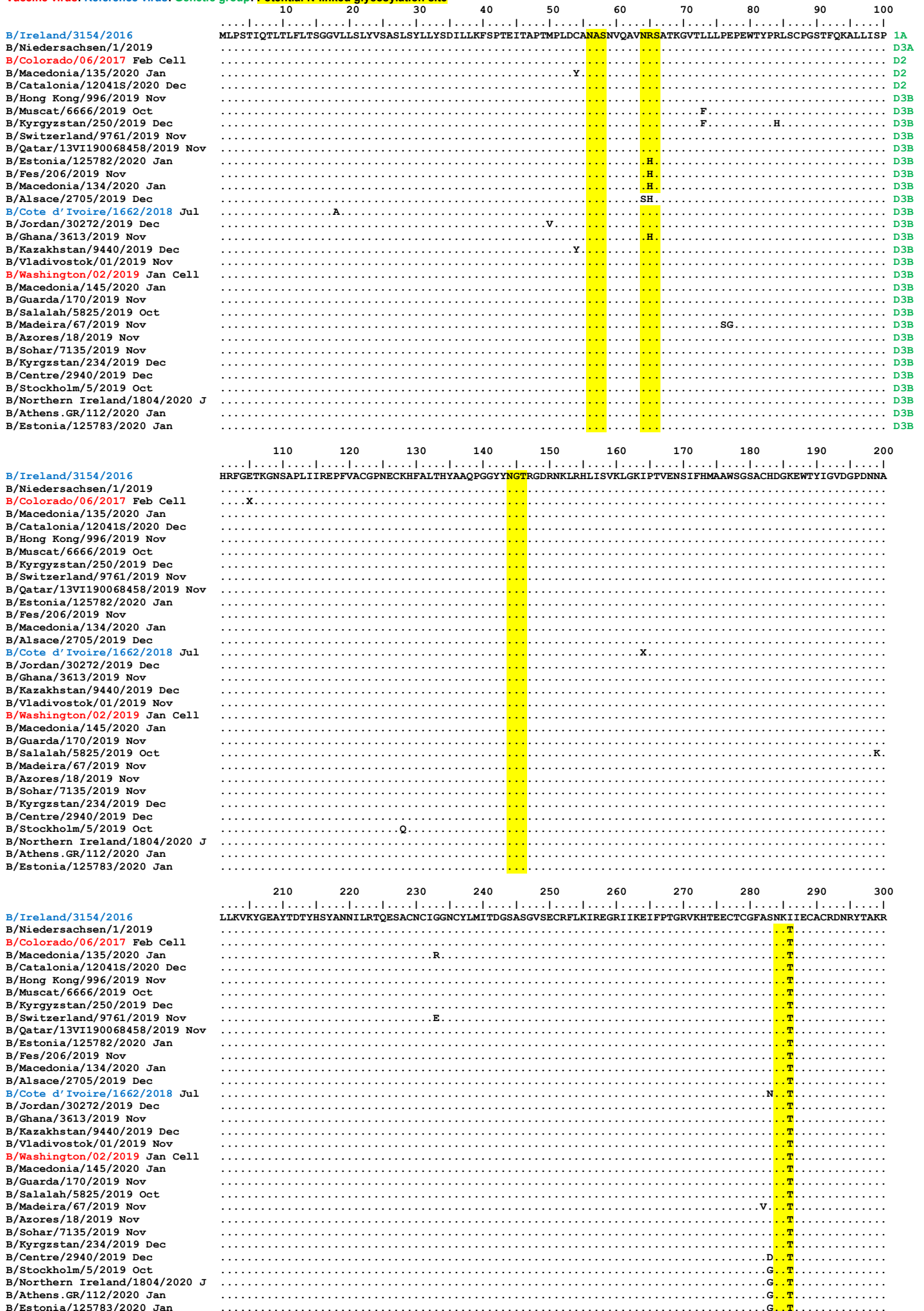


Figure 15. NA protein alignment for B (Victoria-lineage) viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site



	310	320	330	340	350	360	370	380	390	400
B/Ireland/3154/2016	PFVKLN	VDTDAE	IRLMCTD	TYLDT	PRPNDG	SITGPCE	SDGDKG	SGGKGGF	VHQRMK	SKI
B/Niedersachsen/1/2019										
B/Colorado/06/2017 Feb Cell								Q		
B/Macedonia/135/2020 Jan				E				Q		V
B/Catalonia/120418/2020 Dec								Q		
B/Hong Kong/996/2019 Nov									H	T
B/Muscat/6666/2019 Oct		D					E			T
B/Kyrgyzstan/250/2019 Dec		D					E			T
B/Switzerland/9761/2019 Nov										T
B/Qatar/13VI190068458/2019 Nov				K		E				T
B/Estonia/125782/2020 Jan						V				T
B/Fes/206/2019 Nov										T
B/Macedonia/134/2020 Jan		A			F					T
B/Alsace/2705/2019 Dec										T
B/Cote d'Ivoire/1662/2018 Jul										T
B/Jordan/30272/2019 Dec										T
B/Ghana/3613/2019 Nov					E			E		T
B/Kazakhstan/9440/2019 Dec										T
B/Vladivostok/01/2019 Nov										T
B/Washington/02/2019 Jan Cell										T
B/Macedonia/145/2020 Jan								E	D	T
B/Guarda/170/2019 Nov								E	D	T
B/Salalah/5825/2019 Oct						N				T
B/Madeira/67/2019 Nov										T
B/Azores/18/2019 Nov										T
B/Sohar/7135/2019 Nov			N							T
B/Kyrgyzstan/234/2019 Dec										T
B/Centre/2940/2019 Dec					N					T
B/Stockholm/5/2019 Oct										T
B/Northern Ireland/1804/2020 J										T
B/Athens.GR/112/2020 Jan										T
B/Estonia/125783/2020 Jan										T
	410	420	430	440	450	460				
B/Ireland/3154/2016	VPMKEP	GWYSFG	FEIKDK	KCDVPC	IGIEMV	HGGKET	WHSAA	TAIYCL	MGSGQL	LWDVT
B/Niedersachsen/1/2019										
B/Colorado/06/2017 Feb Cell	S									
B/Macedonia/135/2020 Jan	S									
B/Catalonia/120418/2020 Dec	S							I		
B/Hong Kong/996/2019 Nov	S									
B/Muscat/6666/2019 Oct	S									
B/Kyrgyzstan/250/2019 Dec	S									
B/Switzerland/9761/2019 Nov	S									
B/Qatar/13VI190068458/2019 Nov	S									
B/Estonia/125782/2020 Jan	S			I						
B/Fes/206/2019 Nov	S									
B/Macedonia/134/2020 Jan	S									
B/Alsace/2705/2019 Dec	S									
B/Cote d'Ivoire/1662/2018 Jul	S									
B/Jordan/30272/2019 Dec	ST									
B/Ghana/3613/2019 Nov	S									
B/Kazakhstan/9440/2019 Dec	S									
B/Vladivostok/01/2019 Nov	S									
B/Washington/02/2019 Jan Cell	S									
B/Macedonia/145/2020 Jan	S									
B/Guarda/170/2019 Nov	S									
B/Salalah/5825/2019 Oct	S									
B/Madeira/67/2019 Nov	S									
B/Azores/18/2019 Nov	S						N			
B/Sohar/7135/2019 Nov	S									
B/Kyrgyzstan/234/2019 Dec	S									
B/Centre/2940/2019 Dec	S					E	I			
B/Stockholm/5/2019 Oct	S									
B/Northern Ireland/1804/2020 J	S									
B/Athens.GR/112/2020 Jan	S									
B/Estonia/125783/2020 Jan	S									

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Table 11. Summary of influenza B (Yamagata-lineage) viruses analysed, collected after 2019-08-31

MONTH	B Yamagata lineage				
	Number received¹	Number propagated²	≤2 fold³	4 fold³	≥8 fold³
2019					
SEPTEMBER					
OCTOBER					
Hong Kong	1	1	1		
Ireland	1	0			
Norway	1	0			
NOVEMBER					
Austria	1	1	1		
France	2	2	2		
Norway	3	1			1
DECEMBER					
Norway	5	2	2		
2020					
JANUARY					
	14	7	6	0	1
5 Countries			85.7%	0.0%	14.3%

1. 9/14 were received as clinical specimens, only 2 of which yielded virus isolates

2. Propagated to sufficient titre to perform HI assay

3. Fold reduction in HI titre with ferret antiserum raised against B/Phuket/3073/2013 (egg-grown vaccine virus)

Table 12-1. Antigenic analyses of influenza B viruses (Yamagata lineage) 2019-09-18

Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera									
				B/Phuket	B/Estonia	B/Mass	B/Mass	B/Wis	B/Stock	B/Phuket	B/Phuket	B/Maur	B/Maur
				3073/13	55669/11	02/12	02/12	1/10	12/11	3073/13	3073/13	1791/17	I-762
				Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK
	Passage history			SH614 ^{*1,3}	F39/17 ^{*2}	F10/16 ^{*2}	F06/17 ^{*2}	F36/15 ^{*2}	F05/17 ^{*2}	F27/15 ^{*2}	F25/17 ^{*2}	F04/18 ^{*2}	F05/19 ^{*2}
	Ferret number			3	2	2	2	3	3	3	3	3	3
	Genetic Group												
REFERENCE VIRUSES													
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	80	40	160	20	20	<	80	40	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	2560	160	320	640	160	160	40	320	40	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	320	10	10	160	10	20	<	40	<	20
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	<	10	160	40	40	<	160	40	40
B/Stockholm/12/2011	3	2011-03-28	E4/E2	640	<	10	80	20	80	<	160	20	40
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK3	2560	160	320	320	80	160	80	320	320	1280
B/Phuket/3073/2013	3	2013-11-21	E4/E3	640	<	10	80	20	40	<	160	40	40
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	1280	<	20	80	20	40	10	160	160	160
B/Mauritius/I-762/2018	3	2018-09-02	MDCK1/MDCK3	1280	20	20	80	20	40	10	160	160	160
TEST VIRUSES													
B/Norway/1469/2019	3	2019-03-23	MDCK1/MDCK1	2560	20	40	80	20	80	10	160	160	320
B/Norway/1494/2019	3	2019-03-25	MDCK1	320	<	<	40	10	20	<	20	20	20
B/Norway/1931/2019	3	2019-04-27	MDCK1	640	<	<	40	10	20	<	80	80	80
B/Norway/1954/2019	3	2019-05-02	MDCK1	640	<	<	40	10	20	<	40	40	80
B/Norway/2051/2019	3	2019-05-13	MDCK1/MDCK1	640	<	<	40	20	20	<	40	40	80
B/Oman/3987/2019	3	2019-06-02	SIAT1/MDCK1	640	<	<	40	10	20	<	40	40	40
B/Oman/4086/2019	3	2019-06-05	SIAT1/MDCK1	640	<	10	40	10	20	<	80	40	80
B/Norway/2134/2019	3	2019-06-06	MDCK1/MDCK1	640	<	<	<	<	<	<	10	<	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 <= <40; 2 <= <10; 3 hyperimmune sheep serum

B/Yamagata-lineage virus recommended for use in quadravalent vaccines SH 2019, NH 2019-20 and NH 2020

Vaccine[#]

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 12-2. Antigenic analyses of influenza B viruses (Yamagata lineage) 2019-10-02

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				B/Phuket 3073/13	B/Estonia 55669/11	B/Mass 02/12	B/Mass 02/12	B/Wis 1/10	B/Stock 12/11	B/Phuket 3073/13	B/Phuket 3073/13	B/Maur 1791/17	B/Maur I-762
				Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK
	Passage history			SH614 ^{*1,3}	F39/17 ^{*2}	F10/16 ^{*2}	F06/17 ^{*2}	F36/15 ^{*2}	F05/17 ^{*2}	F27/15 ^{*2}	F25/17 ^{*2}	F04/18 ^{*2}	F05/19 ^{*2}
	Ferret number			3	2	2	2	3	3	3	3	3	3
	Genetic Group												
REFERENCE VIRUSES													
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	80	40	160	80	40	<	80	40	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	1280	160	320	640	320	160	40	320	160	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	320	10	10	160	40	40	<	80	<	20
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	<	20	160	160	80	<	320	40	80
B/Stockholm/12/2011	3	2011-03-28	E4/E2	640	<	10	80	80	40	<	160	20	40
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK3	2560	320	640	640	640	320	160	320	640	1280
B/Phuket/3073/2013	3	2013-11-21	E4/E3	640	<	<	80	80	40	<	160	40	40
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	1280	<	40	160	160	80	20	160	160	160
B/Mauritius/I-762/2018	3	2018-09-02	MDCK1/MDCK3	1280	<	40	160	160	80	10	160	160	320
TEST VIRUSES													
B/Norway/1896/2019	3	2019-04-28	MDCK2	1280	<	10	80	80	20	<	80	40	160
B/Norway/1948/2019	3	2019-04-29	MDCK2	640	<	10	80	80	20	<	80	40	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum

B/Yamagata-lineage virus recommended for use in quadravalent vaccines SH 2019, NH 2019-20 and SH 2020

Vaccine[#]

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 12-3. Antigenic analyses of influenza B viruses (Yamagata lineage) 2019-12-10

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Phuket 3073/13	B/Estonia 55669/11	B/Mass 02/12	B/Mass 02/12	B/Wis 1/10	B/Stock 12/11	B/Phuket 3073/13	B/Phuket 3073/13	B/Maur 1791/17	B/Maur I-762	B/Nor 2134/19
				Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK	Egg
	Passage history			SH614 ^{*1,3}	F39/17 ^{*2}	F10/16 ^{*2}	F06/17 ^{*4}	F36/15 ^{*2}	F05/17 ^{*2}	F27/15 ^{*2}	F25/17 ^{*2}	F04/18 ^{*2}	F05/19 ^{*2}	F48/19 ^{*2}
	Ferret number			3	2	2	2	3	3	3	3	3	3	3
	Genetic Group													
REFERENCE VIRUSES														
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	40	20	80	40	40	40	80	20	40	<
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	1280	160	320	320	160	160	160	320	160	320	10
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	320	10	<	160	40	80	10	80	<	10	40
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	<	<	160	80	320	40	320	40	40	40
B/Stockholm/12/2011	3	2011-03-28	E4/E2	320	<	<	40	40	80	20	80	10	20	20
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK3	2560	80	320	160	320	160	320	160	160	640	40
B/Phuket/3073/2013	3	2013-11-21	E4/E3	640	<	<	80	80	80	20	160	20	40	20
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	1280	<	40	40	40	80	80	80	80	80	<
B/Mauritius/I-762/2018	3	2018-09-02	MDCK1/MDCK3	1280	<	40	40	40	80	80	80	80	160	<
B/Norway/2134/2019	3	2019-06-06	E2(am1/al1)	160	<	<	<	<	<	<	<	<	<	2560
TEST VIRUSES														
B/Armenia/16/2019	3	2019-02-08	MDCK1/MDCK1	640	<	10	40	20	40	40	40	40	80	ND
B/Armenia/17/2019	3	2019-02-13	MDCK1/MDCK1	640	<	10	40	20	20	40	40	20	80	ND
B/Armenia/18/2019	3	2019-02-19	MDCK1/MDCK1	640	<	10	40	20	40	40	40	20	80	ND
B/Armenia/21/2019	3	2019-02-23	MDCK1/MDCK1	640	<	20	80	40	80	160	160	80	320	ND
B/Armenia/20/2019	3	2019-02-23	MDCK1/MDCK1	640	<	10	40	40	40	40	40	40	80	ND
B/Norway/2465/2019	3	2019-11-14	MDCK1	160	<	<	<	<	<	<	<	<	<	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum

B/Yamagata-lineage virus recommended for use in quadrivalent vaccines SH 2019, NH 2019-20 and SH 2020

Vaccine#

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 12-4. Antigenic analyses of influenza B viruses (Yamagata lineage) 2020-02-05

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Phuket 3073/13	B/Estonia 55669/11	B/Mass 02/12	B/Mass 02/12	B/Wis 1/10	B/Stock 12/11	B/Phuket 3073/13	B/Phuket 3073/13	B/Maur 1791/17	B/Maur I-762	B/Nor 2134/19
				Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	MDCK	MDCK	Egg
	Passage history			SH614 ^{1,3}	F39/17 ²	F10/16 ²	F06/17 ⁴	F36/15 ²	F05/17 ²	F27/15 ²	F25/17 ²	F04/18 ²	F05/19 ²	F48/19 ²
	Ferret number			3	2	2	2	3	3	3	3	3	3	3
	Genetic Group													
REFERENCE VIRUSES														
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	40	40	80	40	40	40	40	20	40	<
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK4	1280	160	320	320	320	160	160	160	160	320	10
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	320	10	<	160	40	40	10	40	<	10	20
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	<	<	160	80	160	40	320	40	40	40
B/Stockholm/12/2011	3	2011-03-28	E4/E2	640	<	<	80	80	160	40	80	20	40	20
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK3	2560	160	320	160	320	160	160	160	320	640	20
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	<	<	80	80	80	20	80	20	40	20
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	1280	<	20	40	40	80	40	80	80	80	<
B/Mauritius/I-762/2018	3	2018-09-02	MDCK1/MDCK3	1280	<	20	80	80	80	80	80	80	160	<
B/Norway/2134/2019	3	2019-06-06	E2/E2	160	<	<	<	<	10	<	<	<	<	640
TEST VIRUSES														
B/Hong Kong/960/2019	3	2019-10-04	MDCK1/MDCK1	1280	<	<	40	80	40	40	80	40	80	<
B/Austria/1199466/2019	3	2019-11-09	SIAT1/MDCK1	1280	<	<	80	80	80	80	160	80	160	10
B/Lyon/2087/2019	3	2019-11-25	MDCK2/MDCK1	1280	<	10	40	40	40	40	80	40	80	10
B/Paris/2569/2019	3	2019-11-26	MDCK1	640	<	<	40	40	40	40	40	40	80	<
B/Norway/2992/2019	3	2019-12-30	MDCK1/MDCK1	320	<	10	40	40	40	40	40	40	80	<
B/Norway/3123/2019	3	2019-12-30	MDCK1/MDCK1	640	<	20	40	40	40	40	80	40	80	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum

B/Yamagata-lineage virus recommended for use in quadrivalent vaccines SH 2019, NH 2019-20 and SH 2020

Vaccine#

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Figure 16. Influenza B (Yamagata-lineage) HA genes

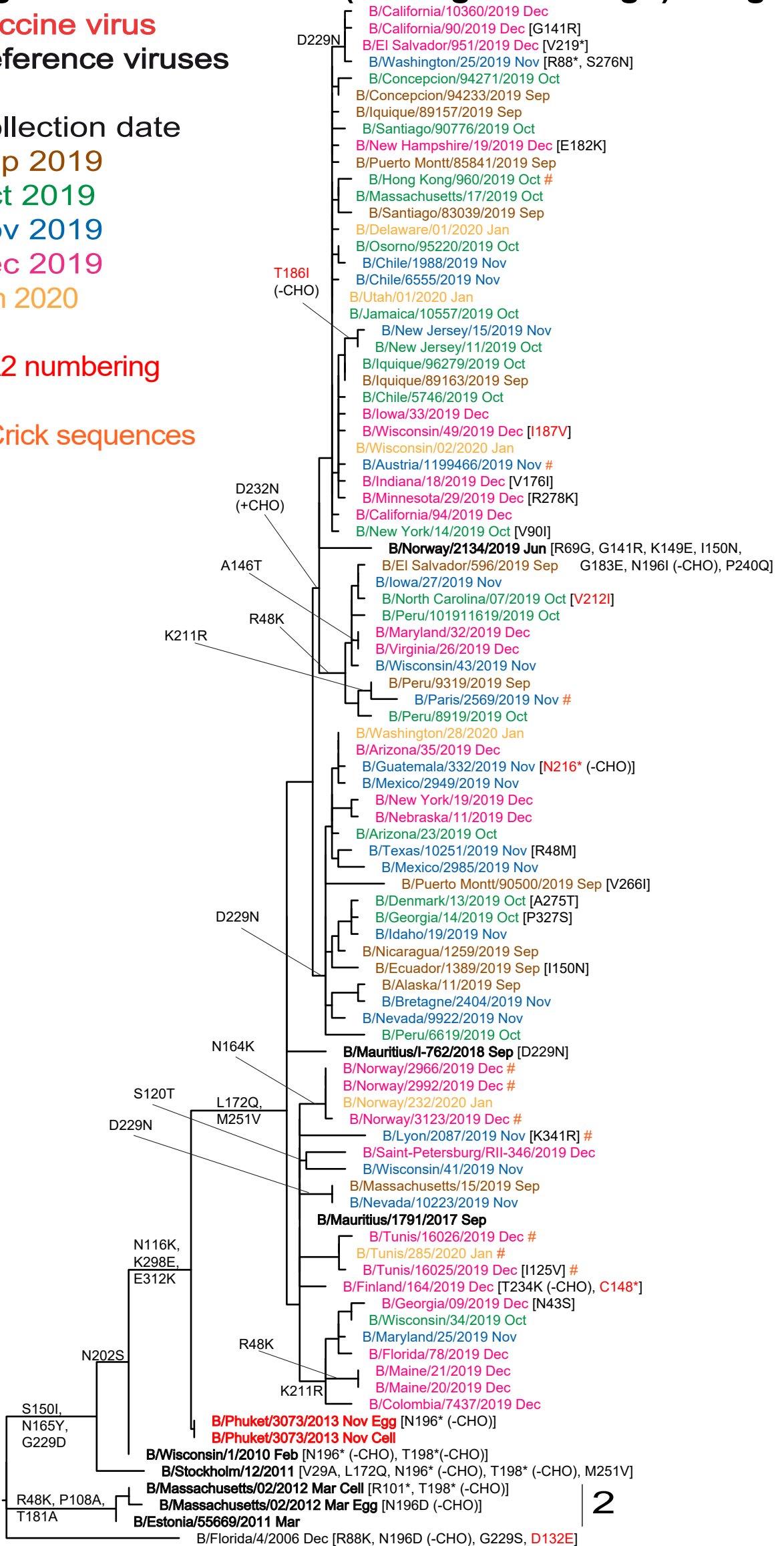
Vaccine virus
Reference viruses

Collection date

Sep 2019
Oct 2019
Nov 2019
Dec 2019
Jan 2020

HA2 numbering

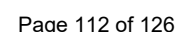
Crick sequences



3

2

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site



	310	320	330	340	350	360	370	380	390	400
B/Phuket/3073/2013 Nov Cell	GL	NS	KP	Y	T	G	K	H	A	I
B/Mauritius/1791/2017 Sep	GL	NS	KP	Y	T	G	K	H	A	I
B/Lyon/2087/2019 Nov										
B/S. Petersburg/RII-346/2019 Dec										
B/Nevada/10223/2019 Nov										
B/Tunis/16026/2019 Dec										
B/Tunis/16025/2019 Dec										
B/Norway/3123/2019 Dec										
B/Mauritius/I-762/2018 Sep										
B/Bretagne/2404/2019 Nov										
B/Nebraska/11/2019 Dec										
B/Arizona/35/2019 Dec										
B/Georgia/14/2019 Oct										
B/Denmark/13/2019 Oct										
B/Paris/2569/2019 Nov										
B/Virginia/26/2019 Dec										
B/Peru/101911619/2019 Oct										
B/North Carolina/07/2019 Oct										
B/Iowa/27/2019 Nov										
B/Norway/2134/2019 Jun										
B/Puerto Montt/85841/2019 Sep										
B/California/79/2019 Dec										
B/Santiago/83039/2019 Sep										
B/Hong Kong/960/2019 Oct										
B/Delaware/01/2020 Jan										
B/California/94/2019 Dec										
B/Utah/01/2020 Jan										
B/Chile/1988/2019 Nov										
B/Minnesota/01/2020 Jan										
B/Austria/1199466/2019 Nov										
B/Iquique/96280/2019 Oct										

	410	420	430	440	450	460	470	480	490	500
B/Phuket/3073/2013 Nov Cell	L	S	E	L	V	K	N	L	Q	R
B/Mauritius/1791/2017 Sep	L	S	E	L	V	K	N	L	Q	R
B/Lyon/2087/2019 Nov										
B/S. Petersburg/RII-346/2019 Dec										
B/Nevada/10223/2019 Nov										
B/Tunis/16026/2019 Dec										
B/Tunis/16025/2019 Dec										
B/Norway/3123/2019 Dec										
B/Mauritius/I-762/2018 Sep										
B/Bretagne/2404/2019 Nov										
B/Nebraska/11/2019 Dec										
B/Arizona/35/2019 Dec										
B/Georgia/14/2019 Oct										
B/Denmark/13/2019 Oct										
B/Paris/2569/2019 Nov										
B/Virginia/26/2019 Dec										
B/Peru/101911619/2019 Oct										
B/North Carolina/07/2019 Oct										
B/Iowa/27/2019 Nov										
B/Norway/2134/2019 Jun										
B/Puerto Montt/85841/2019 Sep										
B/California/79/2019 Dec										
B/Santiago/83039/2019 Sep										
B/Hong Kong/960/2019 Oct										
B/Delaware/01/2020 Jan										
B/California/94/2019 Dec										
B/Utah/01/2020 Jan										
B/Chile/1988/2019 Nov										
B/Minnesota/01/2020 Jan										
B/Austria/1199466/2019 Nov										
B/Iquique/96280/2019 Oct										

	510	520	530	540	550	560
B/Phuket/3073/2013 Nov Cell	G	T	F	N	A	G
B/Mauritius/1791/2017 Sep	G	T	F	N	A	G
B/Lyon/2087/2019 Nov						
B/S. Petersburg/RII-346/2019 Dec						
B/Nevada/10223/2019 Nov						
B/Tunis/16026/2019 Dec						
B/Tunis/16025/2019 Dec						
B/Norway/3123/2019 Dec						
B/Mauritius/I-762/2018 Sep						
B/Bretagne/2404/2019 Nov						
B/Nebraska/11/2019 Dec						
B/Arizona/35/2019 Dec						
B/Georgia/14/2019 Oct						
B/Denmark/13/2019 Oct						
B/Paris/2569/2019 Nov						
B/Virginia/26/2019 Dec						
B/Peru/101911619/2019 Oct						
B/North Carolina/07/2019 Oct						
B/Iowa/27/2019 Nov						
B/Norway/2134/2019 Jun						
B/Puerto Montt/85841/2019 Sep						
B/California/79/2019 Dec						
B/Santiago/83039/2019 Sep						
B/Hong Kong/960/2019 Oct						
B/Delaware/01/2020 Jan						
B/California/94/2019 Dec						
B/Utah/01/2020 Jan						
B/Chile/1988/2019 Nov						
B/Minnesota/01/2020 Jan						
B/Austria/1199466/2019 Nov						
B/Iquique/96280/2019 Oct						

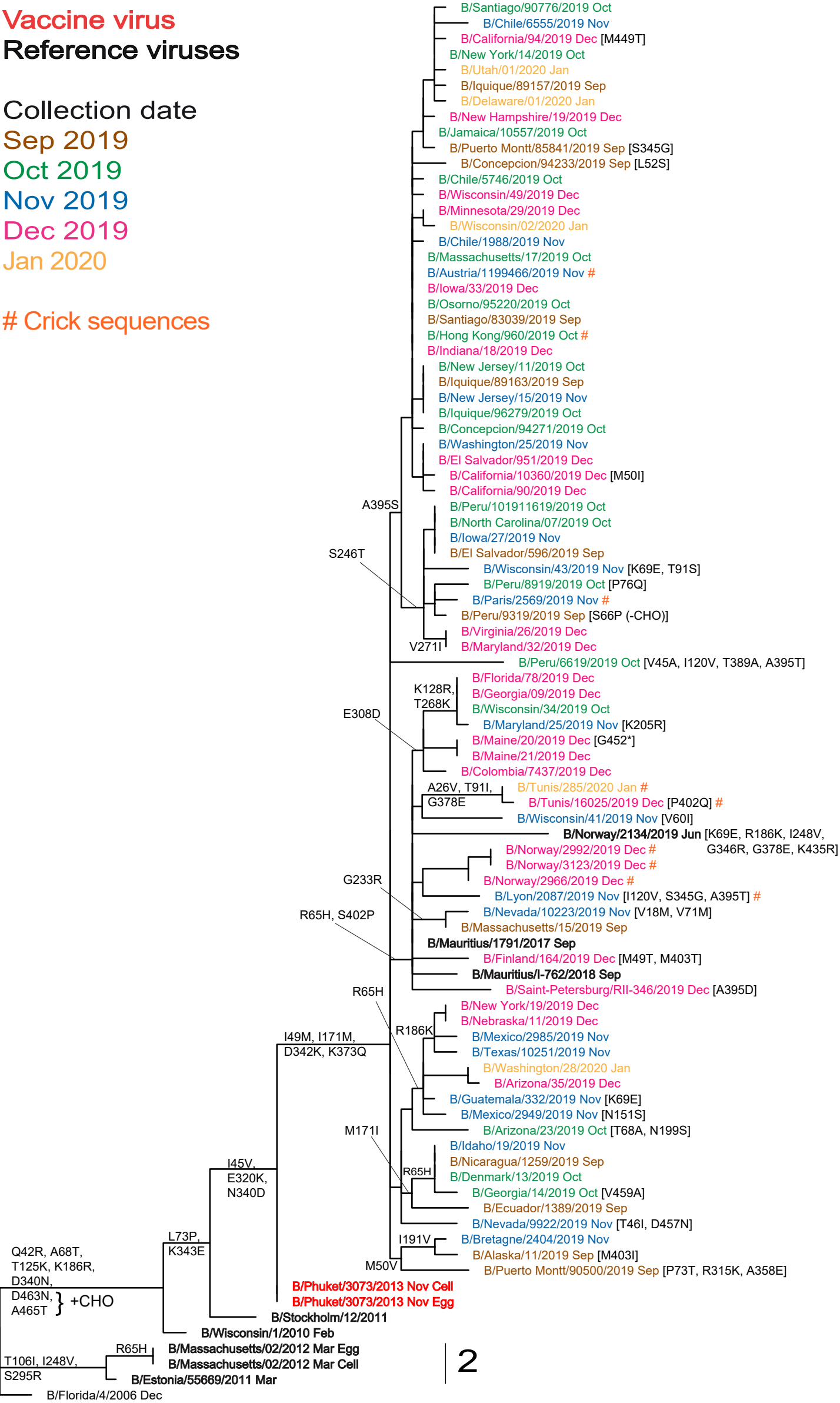
Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 18. Phylogenetic comparison of influenza B (Yamagata-lineage) NA genes

Vaccine virus
Reference viruses

Collection date
Sep 2019
Oct 2019
Nov 2019
Dec 2019
Jan 2020

Crick sequences



2

Figure 19. NA protein alignment for B (Yamagata-lineage) viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100
B/Phuket/3073/2013 Nov Cell	MLPSTIQTLTLFLTS	GGVLLSLYSASLS	YLLYSDDLKFSR	TEVTAPIMPLDCAN	ASNVQAVNRSATK	GVTPLLPEPEWT	YPRLS	CPGSTFQK	ALLISP	3
B/Mauritius/1791/2017 Sep				M		H				3
B/Lyon/2087/2019 Nov				M		H				3
B/S. Petersburg/RII-346/2019 Dec				M		H				3
B/Nevada/10223/2019 Nov		M		M		H	M			3
B/Tunis/16025/2019 Dec			V	M		H			I	3
B/Tunis/285/2020 Jan			V	M		H			I	3
B/Norway/3123/2019 Dec				M		H				3
B/Mauritius/I-762/2018 Sep				M		H				3
B/Bretagne/2404/2019 Nov				MV						3
B/Nebraska/11/2019 Dec				M		H				3
B/Arizona/35/2019 Dec				M		H				3
B/Georgia/14/2019 Oct				M		H				3
B/Denmark/13/2019 Oct				M		H				3
B/Paris/2569/2019 Nov				M						3
B/Virginia/26/2019 Dec				M						3
B/Peru/101911619/2019 Oct				M						3
B/North Carolina/07/2019 Oct				M						3
B/Iowa/27/2019 Nov				M						3
B/Norway/2134/2019 Jun				M		H	E			3
B/Puerto Montt/85841/2019 Sep				M						3
B/California/79/2019 Dec				M						3
B/Santiago/83039/2019 Sep				M						3
B/Hong Kong/960/2019 Oct				M						3
B/Delaware/01/2020 Jan				M						3
B/California/94/2019 Dec				M						3
B/Utah/01/2020 Jan				M						3
B/Chile/1988/2019 Nov				M						3
B/Minnesota/01/2020 Jan				M						3
B/Austria/1199466/2019 Nov				M						3
B/Iquique/96280/2019 Oct				M						3
	110	120	130	140	150	160	170	180	190	200
B/Phuket/3073/2013 Nov Cell	HRFGETKGNSAPLI	IREPFIACGPKCK	HFALTHYAAQPG	GYNGTREDRNL	RHLISVKLGKI	PTVENSIFHMA	AWSGSACHD	GREWTYIGV	DGPDSNA	
B/Mauritius/1791/2017 Sep							M			
B/Lyon/2087/2019 Nov		V					M			
B/S. Petersburg/RII-346/2019 Dec							M			
B/Nevada/10223/2019 Nov							M			
B/Tunis/16025/2019 Dec							M			
B/Tunis/285/2020 Jan							M			
B/Norway/3123/2019 Dec							M			
B/Mauritius/I-762/2018 Sep							M			
B/Bretagne/2404/2019 Nov							M		V	
B/Nebraska/11/2019 Dec							M		K	
B/Arizona/35/2019 Dec							M			
B/Georgia/14/2019 Oct										
B/Denmark/13/2019 Oct										
B/Paris/2569/2019 Nov					X		M			
B/Virginia/26/2019 Dec							M			
B/Peru/101911619/2019 Oct							M			
B/North Carolina/07/2019 Oct							M			
B/Iowa/27/2019 Nov							M			
B/Norway/2134/2019 Jun							M		K	
B/Puerto Montt/85841/2019 Sep							M			
B/California/79/2019 Dec							M			
B/Santiago/83039/2019 Sep							M			
B/Hong Kong/960/2019 Oct							M			
B/Delaware/01/2020 Jan							M			
B/California/94/2019 Dec							M			
B/Utah/01/2020 Jan							M			
B/Chile/1988/2019 Nov							M			
B/Minnesota/01/2020 Jan							M			
B/Austria/1199466/2019 Nov							M			
B/Iquique/96280/2019 Oct							M			
	210	220	230	240	250	260	270	280	290	300
B/Phuket/3073/2013 Nov Cell	LLKIKYGEAYTD	TYHSYAKNILRT	QESACNIGGDC	YLMITDGPAS	GISECRFLKIRE	GRIKEIFPTGR	VKHTTEECT	CGFASNKT	IEACARDNS	SYAKR
B/Mauritius/1791/2017 Sep										
B/Lyon/2087/2019 Nov										
B/S. Petersburg/RII-346/2019 Dec										
B/Nevada/10223/2019 Nov			R							
B/Tunis/16025/2019 Dec										
B/Tunis/285/2020 Jan										
B/Norway/3123/2019 Dec										
B/Mauritius/I-762/2018 Sep										
B/Bretagne/2404/2019 Nov										
B/Nebraska/11/2019 Dec										
B/Arizona/35/2019 Dec										
B/Georgia/14/2019 Oct										
B/Denmark/13/2019 Oct										
B/Paris/2569/2019 Nov				T						
B/Virginia/26/2019 Dec				T			I			
B/Peru/101911619/2019 Oct				T						
B/North Carolina/07/2019 Oct				T						
B/Iowa/27/2019 Nov				T						
B/Norway/2134/2019 Jun				V						
B/Puerto Montt/85841/2019 Sep										
B/California/79/2019 Dec										
B/Santiago/83039/2019 Sep										
B/Hong Kong/960/2019 Oct										
B/Delaware/01/2020 Jan										
B/California/94/2019 Dec										
B/Utah/01/2020 Jan										
B/Chile/1988/2019 Nov										
B/Minnesota/01/2020 Jan										
B/Austria/1199466/2019 Nov										
B/Iquique/96280/2019 Oct										

	310	320	330	340	350	360	370	380	390	400							
B/Phuket/3073/2013 Nov Cell	PFVKLN	VEDTDA	EIRLMCT	KTYLDT	PRPNDG	SITGPCE	SDGDEG	SGGKGG	FVHQRM	ASKIGRW	YSRTMS	KTKRMG	MGLYVK	YDGD	PWTDSE	ALALSG	VM
B/Mauritius/1791/2017 Sep							K					Q					
B/Lyon/2087/2019 Nov							K	G				Q				T	
B/S. Petersburg/RII-346/2019 Dec							K					Q				D	
B/Nevada/10223/2019 Nov							K					Q					
B/Tunis/16025/2019 Dec							K					Q		E			
B/Tunis/285/2020 Jan							K					Q		E			
B/Norway/3123/2019 Dec							K					Q					
B/Mauritius/I-762/2018 Sep							K					Q					
B/Bretagne/2404/2019 Nov							K					Q					
B/Nebraska/11/2019 Dec							K					Q					
B/Arizona/35/2019 Dec							K					Q					
B/Georgia/14/2019 Oct							K					Q					
B/Denmark/13/2019 Oct							K					Q					
B/Paris/2569/2019 Nov							K					Q				S	
B/Virginia/26/2019 Dec							K					Q				S	
B/Peru/101911619/2019 Oct							K					Q				S	
B/North Carolina/07/2019 Oct							K					Q				S	
B/Iowa/27/2019 Nov							K					Q				S	
B/Norway/2134/2019 Jun							K	R				Q		E			
B/Puerto Montt/85841/2019 Sep							K	G				Q				S	
B/California/79/2019 Dec							K					Q				S	
B/Santiago/83039/2019 Sep							K					Q				S	
B/Hong Kong/960/2019 Oct							K					Q				S	
B/Delaware/01/2020 Jan							K					Q				S	
B/California/94/2019 Dec							K					Q				S	
B/Utah/01/2020 Jan							K					Q				S	
B/Chile/1988/2019 Nov							K					Q				S	
B/Minnesota/01/2020 Jan							K					Q				S	
B/Austria/1199466/2019 Nov							K					Q				S	
B/Iquique/96280/2019 Oct							K					Q				S	

	410	420	430	440	450	460
B/Phuket/3073/2013 Nov Cell	VSMEEP	GWYSFG	FEIKDK	KCDVPC	IGIEMV	HGGKTT
B/Mauritius/1791/2017 Sep	P					
B/Lyon/2087/2019 Nov	P					
B/S. Petersburg/RII-346/2019 Dec	P					
B/Nevada/10223/2019 Nov	P					
B/Tunis/16025/2019 Dec	Q					
B/Tunis/285/2020 Jan	P					
B/Norway/3123/2019 Dec	P					
B/Mauritius/I-762/2018 Sep	P					
B/Bretagne/2404/2019 Nov						
B/Nebraska/11/2019 Dec						
B/Arizona/35/2019 Dec						
B/Georgia/14/2019 Oct					A	
B/Denmark/13/2019 Oct						
B/Paris/2569/2019 Nov						
B/Virginia/26/2019 Dec						
B/Peru/101911619/2019 Oct						
B/North Carolina/07/2019 Oct						
B/Iowa/27/2019 Nov						
B/Norway/2134/2019 Jun	P			R		
B/Puerto Montt/85841/2019 Sep						
B/California/79/2019 Dec						
B/Santiago/83039/2019 Sep						
B/Hong Kong/960/2019 Oct						
B/Delaware/01/2020 Jan						
B/California/94/2019 Dec					T	
B/Utah/01/2020 Jan						
B/Chile/1988/2019 Nov						
B/Minnesota/01/2020 Jan						
B/Austria/1199466/2019 Nov						
B/Iquique/96280/2019 Oct						

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Table 13. Antiviral (NAI) testing of isolates with collection dates after 2019-08-31*

Month of Collection	Number of viruses tested for phenotype ¹								Totals
	A(H1N1)pdm09		A(H3N2)		Influenza B-Victoria		Influenza B-Yamagata		
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	
2019									
September	10		19		8				37
October	53	3	62		32		1		151
November	87	2	49		62		4		204
December	100		80	1	65	1	2		249
2020									
January	28		26		16	1			71
Totals	278	5	236	1	183	2	7	0	712

¹ Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369-374

* As of 2020-02-18

Virus name [§]	Subtype/ Lineage	Collection date	Fold difference		NA substitution	
			Oseltamivir	Zanamivir		
NA sequence result					Clinical Sample	Virus
A/Salalah/6780/2019	A(H1N1)pdm09	2019-10-23	30.34 (RI)		H275(Y90: H10)	H275(Y55: H45)
A/Sur/6980/2019	A(H1N1)pdm09	2019-10-27	232.10 (HRI)		H275(Y40: H60)	H275(Y85: H15)
A/Muscat/6997/2019	A(H1N1)pdm09	2019-10-29	369.97 (HRI)		H275Y	H275Y
A/Denmark/3295/2019	A(H1N1)pdm09	2019-11-15		142.56 (HRI)	NR [#]	Q136K
A/Denmark/3311/2019	A(H1N1)pdm09	2019-11-16		280.06 (HRI)	NR	Q136K
B/Beirut/570/2019	B Vic	2019-12-02		7.58 (RI)	NR	In Process
B/Estonia/125782/2020	B Vic	2020-01-13		8.50 (RI)	NR	I348V, V430I
Outliers						
A/Limoges/2326/2019	A(H3N2)	2019-12-19		10.29 (RI)	NR	T148I (-CHO)

[§] No information regarding vaccination status or antiviral treatment of patients was provided

[#] NR = Not received by the Francis Crick Institute

WORLDWIDE INFLUENZA CENTRE
WHO CC for Reference & Research on Influenza
THE FRANCIS CRICK INSTITUTE
LONDON NW1 1AT
United Kingdom

**Report prepared for the WHO annual consultation on the
composition of influenza vaccines for the Northern
Hemisphere 2020-2021**

24th – 27th February 2020

Annex

**Phylogenetic analyses of HA and NA genes of all
types A and B* seasonal influenza viruses, with
collection dates after 2019-08-31, characterised at
the Worldwide influenza Centre as of 2020-02-10.**

*** As so few B/Yamagata-lineage viruses were received (very low numbers detected
worldwide), these trees include all sequences available in GISAID for viruses with
collection dates after 2019-08-31 as of 2020-02-07.**

Vaccine virus

Reference viruses

Jan 2020



Reference viruses

Collection date

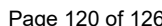
Sep 2019

Oct 2019

Nov 2019

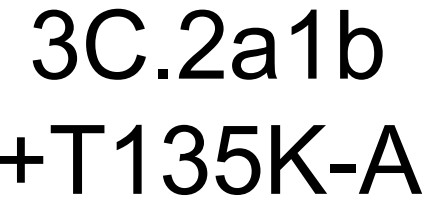
Dec 2019

Jan 2020



Reference viruses

Jan 2020



Reference viruses

Jan 2020



Jan 2020



Reference viruses

Jan 2020

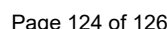


Figure A-7. Influenza B (Yamagata-lineage) HA genes

Vaccine virus

Reference viruses

Collection date

Sep 2019

Oct 2019

Nov 2019

Dec 2019

Jan 2020

Crick sequences

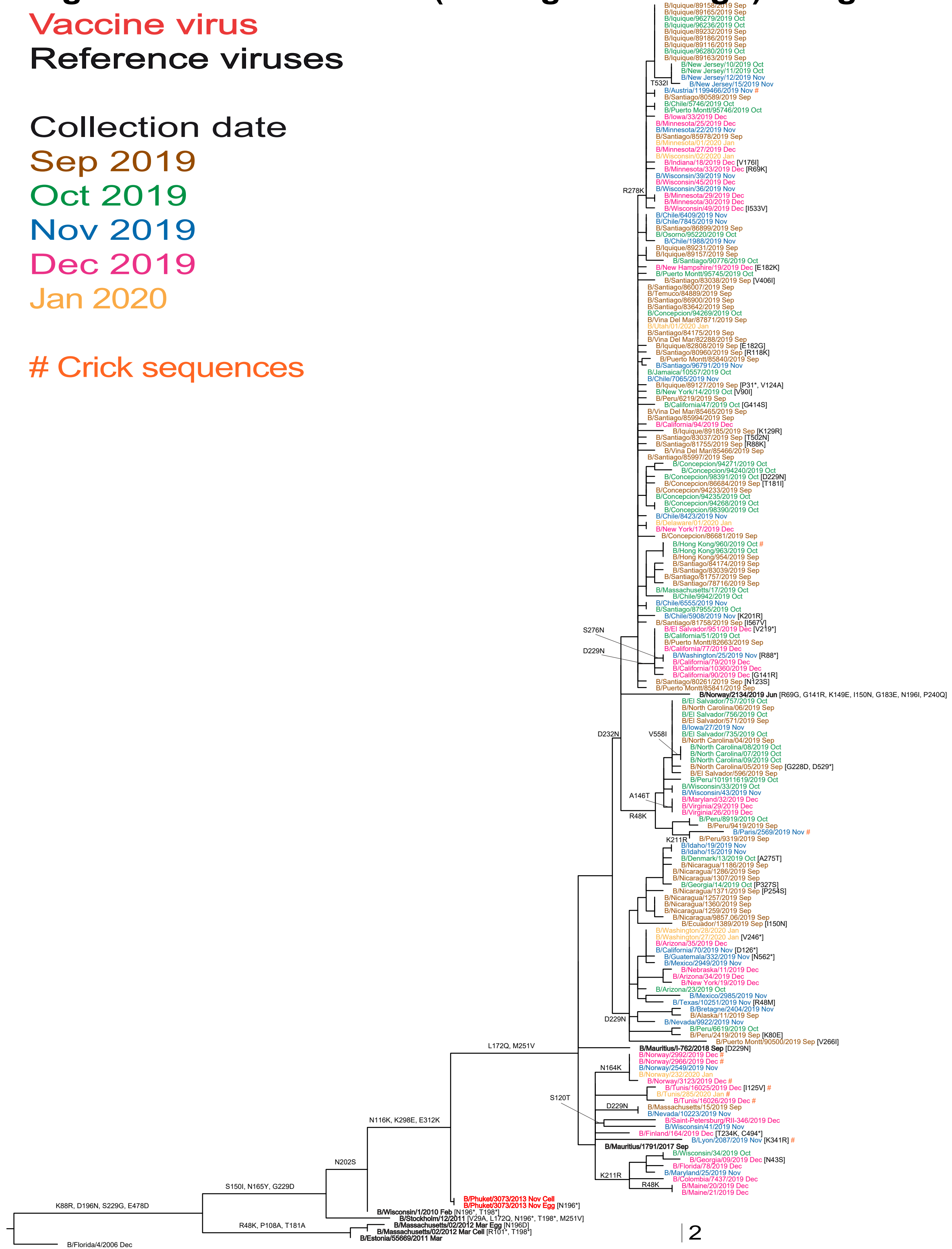


Figure A-8. Influenza B (Yamagata-lineage) NA genes

Vaccine virus

Reference viruses

Collection date

Sep 2019

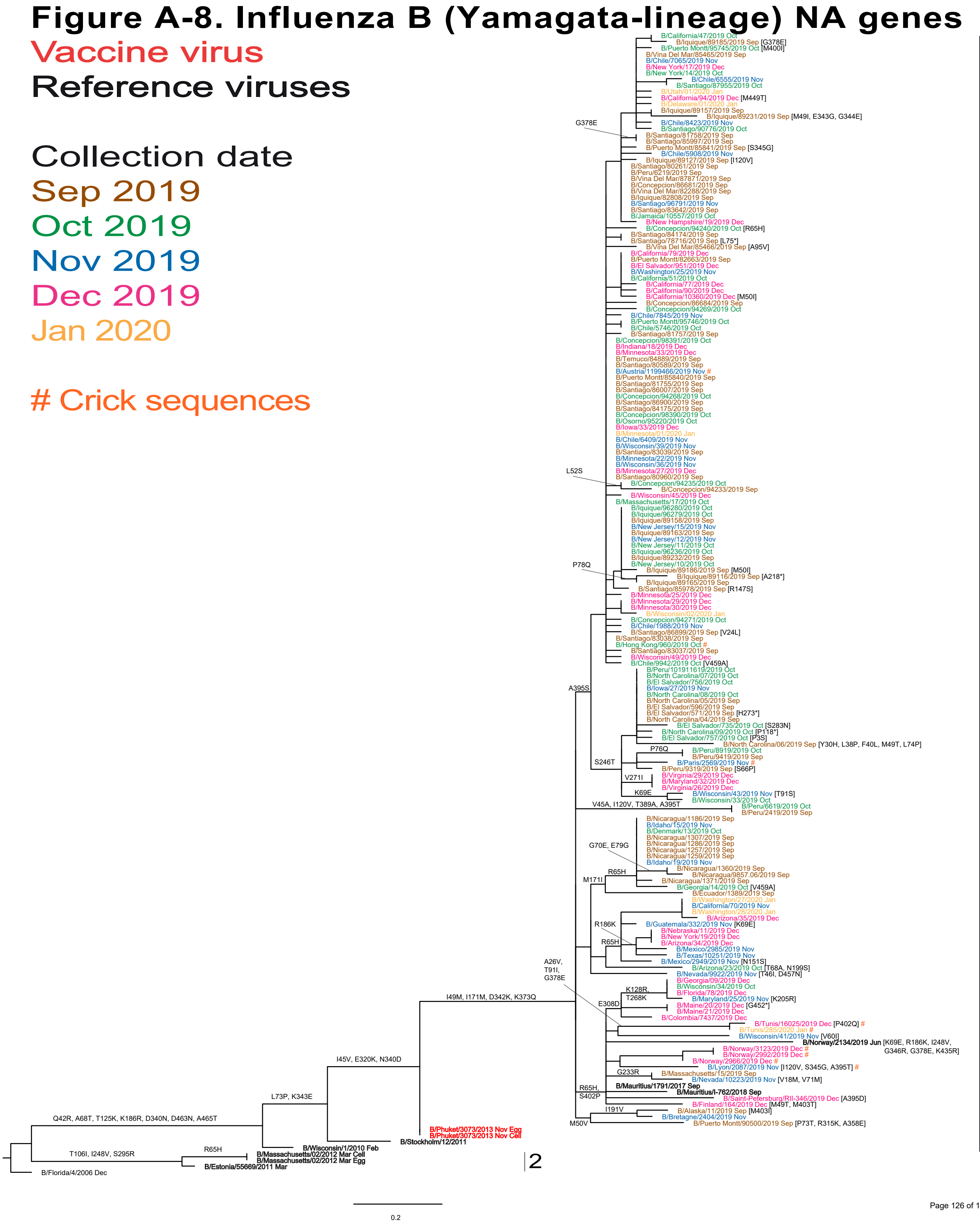
Oct 2019

Nov 2019

Dec 2019

Jan 2020

Crick sequences



3

2